

ACADEMIC REGULATIONS

R22

Bachelor of Technology (B.Tech)

INFORMATION TECHNOLOGY

B. Tech. - Regular Four Year Degree Programme
(For batches admitted from the academic year 2022-2023)
&
(For batches admitted Lateral Entry Scheme from the
academic year 2023-2024)



**MARRI
LAXMAN
REDDY**

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(Autonomous)

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ACADEMIC REGULATIONS (R22)

**B. Tech. - Regular Four Year Degree Programme
(For batches admitted from the academic year 2022-23)**

1.0 For pursuing four year Under Graduate Degree Programme of study in Engineering & Technology (UGP in E&T) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified herein approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2022-23 onwards. Any reference to “Institute” or “College” in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman Academic Council.

2.0 Eligibility for Admission

2.1 Admission to the undergraduate(UG) programme shall be made either on the basis of the merit rank obtained by the qualified student in entrance test conducted by the Telangana State Government (EAMCET) or the College or on the basis of any other order of merit approved by the University/TSHE, subject to reservations as prescribed by the government from time to time.

2.2 The medium of instructions for the entire undergraduate programme in Engineering & Technology will be **English** only.

3.0 B.Tech. Programme Structure

3.1 A student after securing admission shall complete the B.Tech. programme in a minimum period of **four** academic years (8 semesters), and a maximum period of **eight** academic years (16 semesters) starting from the date of commencement of first year first semester, failing which student shall forfeit seat in B.Tech course. Each student shall secure 160 credits (with CGPA ≥ 5) required for the completion of the undergraduate programme and award of the B.Tech. degree.

3.2 **UGC/ AICTE** specified definitions/ descriptions are adopted appropriately for various terms and abbreviations used in these academic regulations/ norms, which are listed below.

3.2.1 Semester Scheme

Each undergraduate programme is of 4 academic years (8 semesters) with the academic year divided into two semesters of 22 weeks (≥ 90 instructional days) each and in each semester - ‘Continuous Internal Evaluation (CIE)’ and ‘Semester End Examination (SEE)’ under Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) indicated by UGC, and curriculum/course structure suggested by AICTE are followed.

3.2.2 Credit Courses

All subjects/ courses are to be registered by the student in a semester to earn credits which shall be assigned to each subject/ course in an L: T: P: C (lecture periods: tutorial periods: practical periods: credits) structure based on the following general pattern.

- One credit for one hour/ week/ semester for Theory/ Lecture (L) courses or Tutorials.

- One credit for two hours/ week/ semester for Laboratory/ Practical (P) courses.

Courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab are mandatory courses. These courses will not carry any credits.

3.2.3 Subject Course Classification

All subjects/ courses offered for the undergraduate programme in E&T (B.Tech. degree programmes) are broadly classified as follows. The College has followed almost all the guidelines issued by AICTE/UGC.

S. No.	Broad Course Classification	Course Group/ Category	Course Description
1	Foundation Courses (FnC)	BS–Basic Sciences	Includes Mathematics, Physics and Chemistry Subjects.
2		ES–Engineering Sciences	Includes Fundamental Engineering Subjects.
3		HS–Humanities and Social Sciences	Includes subjects related to Humanities, Social Sciences and Management.
4	Core Courses (CoC)	PC–Professional Core.	Includes core subjects related to the parent discipline/ department/ branch of Engineering.
5	Elective Courses (ElC)	PE–Professional Electives	Includes elective subjects related to the parent discipline/ department/ branch of Engineering.
6		OE–Open Electives	Elective subjects which include inter-disciplinary subjects or subjects in an area outside the parent discipline/ department/ branch of Engineering.
7	Core Courses	Project Work	B.Tech. Project or UG Project or UG Major Project or Project Stage I & II.
8		Industry Training/ Internship / Industry Oriented Mini- project/ Mini- Project/ Skill Development Courses.	Industry Training/ Internship/ Industry Oriented Mini-Project/ Mini-Project/ Skill Development Courses.
9		Seminar	Seminar/ Colloquium based on core contents related to parent discipline/ department/ branch of Engineering.
10	Minor Courses	-	1 or 2 Credit Courses (subset of HS)
11	Mandatory Courses (MC)	-	Mandatory Courses (non-credit)

4.0 Course Registration

- 4.1** A ‘faculty advisor or counselor’ shall be assigned to a group of 20 students, who will advise the students about the undergraduate programme, its course structure and curriculum, choice/option for subjects/ courses, based on their competence, progress, pre-requisites and interest.
- 4.2** The academic section of the college invites ‘registration forms’ from students before the beginning of the semester through ‘on-line registration’, ensuring ‘date and time

stamping'. The on-line registration requests for any 'current semester' shall be **completed before the commencement of SEEs (Semester End Examinations) of the 'preceding semester'**.

- 4.3 A student can apply for **on-line** registration, **only after** obtaining the '**written approval**' from faculty advisor/counselor, which should be submitted to the college academic section through the Head of the Department. A copy of it shall be retained with the Head of the Department, Faculty Advisor/ Counselor and the student.
- 4.4 A student may be permitted to register for all the subjects/ courses in a semester as specified in the course structure with maximum additional subject(s)/course(s) limited to 6 Credits (any 2 elective subjects), based on **progress** and SGPA/ CGPA, and completion of the '**pre-requisites**' as indicated for various subjects/ courses, in the department course structure and syllabus contents.
- 4.5 Choice for '**additional subjects/ courses**', not more than any 2 elective subjects in any Semester, must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Mentor/HOD.
- 4.6 If the student submits ambiguous choices or multiple options or erroneous entries during **on-line** registration for the subject(s) / course(s) under a given/ specified course group/ category as listed in the course structure, only the first mentioned subject/ course in that category will be taken into consideration.
- 4.7 Subject/ course options exercised through **on-line** registration are final and **cannot** be changed or inter-changed; further, alternate choices also will not be considered. However, if the subject/ course that has already been listed for registration by the Head of the Department in a semester could not be offered due to any inevitable or unexpected reasons, then the student shall be allowed to have alternate choice either for a new subject (subject to offering of such a subject), or for another existing subject (subject to availability of seats). Such alternate arrangements will be made by the Head of the Department, with due notification and time-framed schedule, within **a week** after the commencement of class-work for that semester.
- 4.8 Dropping of subjects/ courses may be permitted, only after obtaining prior approval from the faculty advisor/ counselor 'within a period of 15 days' from the beginning of the current semester.
- 4.9 **Open Electives:** The students have to choose three Open Electives (OE-I, II & III) from the list of Open Electives given by other departments. However, the student can opt for an Open Elective subject offered by his own (parent) department, if the student has not registered and not studied that subject under any category (Professional Core, Professional Electives, Mandatory Courses etc.) offered by parent department in any semester. Open Elective subjects already studied should not repeat/should not match with any category (Professional Core, Professional Electives, Mandatory Courses etc.) of subjects even in the forthcoming semesters.
- 4.10 **Professional Electives:** The students have to choose six Professional Electives (PE-I to VI) from the list of professional electives given.
- 5.0 **Subjects/ courses to be offered**
- 5.1 A subject/ course may be offered to the students, **only if** a minimum of 15 students opt for it.
- 5.2 More than **one faculty member** may offer the **same subject** (lab/ practical may be included with the corresponding theory subject in the same semester) in any semester. However, selection of choice for students will be based on - '**first come first serve** basis and CGPA criterion' (i.e. the first focus shall be on early **on-line entry** from the student

for registration in that semester, and the second focus, if needed, will be on CGPA of the student).

- 5.3 If more entries for registration of a subject come into picture, then the Head of the Department concerned shall decide, whether or not to offer such a subject/ course for **two (or multiple) sections**.
- 5.4 In case of options coming from students of other departments/ branches/ disciplines (not considering **open electives**), first **priority** shall be given to the student of the '**parent department**'.

6.0 Attendance requirements:

- 6.1 A student shall be eligible to appear for the semester end examinations, if the student acquires a minimum of 75% of attendance in aggregate of all the subjects/ courses (including attendance in mandatory courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab) for that semester. **Two periods** of attendance for each theory subject shall be considered, if the student appears for the mid-term examination of that subject.
- 6.2 Shortage of attendance in aggregate upto 10% (65% and above, and below 75%) in each semester may be condoned by the college academic committee on genuine and valid grounds, based on the student's representation with supporting evidence.
- 6.3 A stipulated fee shall be payable for condoning of shortage of attendance.
- 6.4 Shortage of attendance below 65% in aggregate shall in **NO** case be condoned.
- 6.5 **Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examinations of that semester. They get detained and their registration for that semester shall stand cancelled**, including all academic credentials (internal marks etc.) of that semester. **They will not be promoted to the next semester.** They may seek re-registration for all those subjects registered in that semester in which the student is detained, by seeking re-admission into that semester as and when offered; if there are any professional electives and/ or open electives, the same may also be re-registered if offered. However, if those electives are not offered in later semesters, then alternate electives may be chosen from the **same** set of elective subjects offered under that category.
- 6.6 A student fulfilling the attendance requirement in the present semester shall not be eligible for readmission into the same class.

7.0 Academic Requirements

The following academic requirements have to be satisfied, in addition to the attendance requirements mentioned in Item No. 6.

- 7.1 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course, if student secures not less than 35% (21 marks out of 60 marks) in the semester end examination, and a minimum of 40% (40 marks out of 100 marks) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of letter grades, this implies securing '**C**' grade or above in that subject/ course.
- 7.2 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship (or) Seminar, if the student secures not less

than 40% marks (i.e. 40 out of 100 allotted marks) in each of them. The student is deemed to have failed, if he (i) does not submit a report on Industry Oriented Mini Project/Internship, or (ii) not make a presentation of the same before the evaluation committee as per schedule, or (iii) secures less than 40% marks in Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship evaluations.

A student may reappear once for each of the above evaluations, when they are scheduled again; if the student fails in such 'one reappearance' evaluation also, the student has to reappear for the same in the next subsequent semester, as and when it is scheduled.

7.3 Promotion Rules

S. No.	Promotion	Conditions to be fulfilled
1	First year first semester to first year second semester	Regular course of study of first year first semester.
2	First year second semester to Second year first semester	(i) Regular course of study of first year second semester. (ii) Must have secured at least 20 credits out of 40 credits i.e., 50% credits up to first year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3.	Second year first semester to Second year second semester	Regular course of study of second year first semester.
4	Second year second semester to Third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Third year first semester to Third year second semester	Regular course of study of third year first semester.
6	Third year second semester to Fourth year first semester	(i) Regular course of study of third year second semester. (ii) Must have secured at least 72 credits out of 120 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
7	Fourth year first semester to Fourth year second semester	Regular course of study of fourth year first semester.

7.4 A student (i) shall register for all courses/subjects covering 160 credits as specified and listed in the course structure, (ii) fulfills all the attendance and academic requirements for 160 credits, (iii) earn all 160 credits by securing SGPA ≥ 5.0 (in each semester), and CGPA ≥ 5 (at the end of 8 semesters), (iv) **passes all the mandatory courses**, to successfully complete the undergraduate programme. The performance of the student in these 160 credits shall be considered for the calculation of the final CGPA (**at the end of undergraduate programme**), and shall be indicated in the grade card / marks memo of

IV-year II semester.

- 7.5 If a student registers for **‘extra subjects’** (in the parent department or other departments/branches of Engg.) other than those listed subjects totaling to 160 credits as specified in the course structure of his department, the performances in those **‘extra subjects’** (although evaluated and graded using the same procedure as that of the required 160 credits) will not be considered while calculating the SGPA and CGPA. For such **‘extra subjects’** registered, percentage of marks and letter grade alone will be indicated in the grade card / marks memo as a performance measure, subject to completion of the attendance and academic requirements as stated in regulations Items 6 and 7.1 – 7.4 above.
- 7.6 A student eligible to appear in the semester end examination for any subject/ course, but absent from it or failed (thereby failing to secure **‘C’** grade or above) may reappear for that subject/ course in the supplementary examination as and when conducted. In such cases, internal marks (CIE) assessed earlier for that subject/ course will be carried over, and added to the marks to be obtained in the SEE supplementary examination for evaluating performance in that subject.
- 7.7 A student **detained in a semester due to shortage of attendance may be re-admitted in the same semester in the next academic year for fulfillment of academic requirements.** The academic regulations under which a student has been re-admitted shall be applicable. Further, no grade allotments or SGPA/ CGPA calculations will be done for the entire semester in which the student has been detained.
- 7.8 A student detained **due to lack of credits, shall be promoted to the next academic year only after acquiring the required number of academic credits.** The academic regulations under which the student has been readmitted shall be applicable to him.

8.0 Evaluation - Distribution and Weightage of Marks

- 8.1 The performance of a student in every subject/course (including practicals and Project Stage – I & II) will be evaluated for 100 marks each, with 40 marks allotted for CIE (Continuous Internal Evaluation) and 60 marks for SEE (Semester End-Examination).
- 8.2 In CIE, for theory subjects, during a semester, there shall be two mid-term examinations. Each Mid-Term examination consists of two parts i) **Part – A** for 10 marks, ii) **Part – B** for 20 marks with a total duration of 2 hours as follows:

1. Mid Term Examination for 30 marks:
 - a. Part - A : Objective/quiz paper for 10 marks.
 - b. Part - B : Descriptive paper for 20 marks.

The objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 10 marks. The descriptive paper shall contain 6 full questions out of which, the student has to answer 4 questions, each carrying 5 marks. The **average of the two Mid Term Examinations** shall be taken as the final marks for Mid Term Examination (for 30 marks).

The remaining 10 marks of Continuous Internal Evaluation are distributed as:

2. Assignment for 5 marks. (**Average of 2 Assignments** each for 5 marks)
3. Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the concerned subject for 5 marks.

While the first mid-term examination shall be conducted on 50% of the syllabus, the second mid-term examination shall be conducted on the remaining 50% of the syllabus.

Five (5) marks are allocated for assignments (as specified by the subject teacher concerned). The first assignment should be submitted before the conduct of the first mid-term examination, and the second assignment should be submitted before the conduct of the second mid-term examination. The average of the two assignments shall be taken as the final marks for assignment (for 5 marks).

Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the subject concerned for 5 marks before II Mid-Term Examination.

- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled in spite of appearing the SEE.

There is NO Computer Based Test (CBT) for R22 regulations.

The details of the end semester question paper pattern are as follows:

- 8.2.1** The semester end examinations (SEE), for theory subjects, will be conducted for 60 marks consisting of two parts viz. i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.
- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
 - Part-B consists of five questions (numbered from 2 to 6) carrying 10 marks each. Each of these questions is from each unit and may contain sub-questions. For each question there will be an “either” “or” choice, which means that there will be two questions from each unit and the student should answer either of the two questions.
 - The duration of Semester End Examination is 3 hours.
- 8.3** For practical subjects there shall be a Continuous Internal Evaluation (CIE) during the semester for 40 marks and 60 marks for semester end examination. Out of the 40 marks for internal evaluation:
1. A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks
 2. **10 marks for viva-voce** (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
 3. Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 10 marks.

4. The remaining 10 marks are for Laboratory Project, which consists of the Design (or) Software / Hardware Model Presentation (or) App Development (or) Prototype Presentation submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

The Semester End Examination shall be conducted with an external examiner and the laboratory teacher. The external examiner shall be appointed from the other colleges which will be decided by the principal of the College.

In the Semester End Examination held for 3 hours, total 60 marks are divided and allocated as shown below:

1. 10 marks for write-up
 2. 15 for experiment/program
 3. 15 for evaluation of results
 4. 10 marks for presentation on another experiment/program in the same laboratory course and
 5. 10 marks for viva-voce on concerned laboratory course.
- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled inspite of appearing the SEE.

8.4 The evaluation of courses having ONLY internal marks in I Year I Semester and II Year II Semester is as follows:

1. I Year I, II Semester course (*ex., Elements of CE/ME/EEE/ECE/CSE/Seminar & etc*): The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course. Seminar evaluation conducted through day to day evaluation of the students in presentation/abstract.
2. II Year II Semester *Real-Time (or) Field-based Research Project* course: The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two Mid-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the internal committee as per schedule, or (ii) secures less than 40% marks in this course.

- 8.5 There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization. Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without effecting regular course work. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be **NO internal marks** for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.
- 8.6 The UG project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Stage – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start his project work.
- 8.7 UG project work shall be carried out in two stages: Project Stage – I for approval of project before Mid-II examinations in IV Year I Semester and Project Stage – II during IV Year II Semester. Student has to submit project work report at the end of IV Year II Semester. The project shall be evaluated for 100 marks before commencement of SEE Theory examinations.
- 8.8 For Project Stage – I, the departmental committee consisting of Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he does not submit a report on Project Stage - I or does not make a presentation of the same before the evaluation committee as per schedule.
- A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such ‘one reappearance’ evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.
- 8.9 For Project Stage – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/ Internship/SDC etc. and the main Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

For conducting viva-voce of project, Principal/COE selects an external examiner from the

list of experts in the relevant branch submitted by the Principal of the College.

A student who has failed, may reappear once for the above evaluation, when it is scheduled again; if student fails in such ‘one reappearance’ evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

8.10 A student shall be given only one time chance to re-register for a maximum of two subjects in a semester:

- If the internal marks secured by a student in the Continuous Internal Evaluation marks for 40 (Sum of average of two mid-term examinations consisting of Objective & descriptive parts, Average of two Assignments & Subject Viva- voce/PPT/ Poster presentation/ Case Study on a topic in the concerned subject) are less than 35% and failed in those subjects.

A student must re-register for the failed subject(s) for 40 marks within four weeks of commencement of the classwork in next academic year.

In the event of the student taking this chance, his Continuous Internal Evaluation marks for 40 and Semester End Examination marks for 60 obtained in the previous attempt stand cancelled.

8.11 For mandatory non-credit Audit courses, The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course.

8.12 For Mandatory Non-Credit Courses offered in a Semester, after securing $\geq 65\%$ attendance and has secured not less than 40% of marks in the sum Total. Then the student is **PASS** and will be qualified for the award of the degree. No marks or Letter Grade shall be allotted for these courses/activities. However, for non credit courses ‘**Satisfactory**’ or ‘**Unsatisfactory**’ shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.

9.0 Grading Procedure

9.1 Grades will be awarded to indicate the performance of students in each Theory Subject, Laboratory/Practicals/ Industry-Oriented Mini Project/Internship/SDC and Project Stage. Based on the percentage of marks obtained (Continuous Internal Evaluation plus Semester End Examination, both taken together) as specified in item 8 above, a corresponding letter grade shall be given.

9.2 As a measure of the performance of a student, a 10-point absolute grading system using the following letter grades (as per UGC/AICTE guidelines) and corresponding percentage of marks shall be followed:

% of Marks Secured in a Subject/Course (Class Intervals)	Letter Grade (UGC Guidelines)	Grade Points
Greater than or equal to 90%	O (Outstanding)	10
80 and less than 90%	A ⁺ (Excellent)	9
70 and less than 80%	A (Very Good)	8

60 and less than 70%	B⁺ (Good)	7
50 and less than 60%	B (Average)	6
40 and less than 50%	C (Pass)	5
Below 40%	F (FAIL)	0
Absent	Ab	0

- 9.3 A student who has obtained an ‘F’ grade in any subject shall be deemed to have ‘**failed**’ and is required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered. In such cases, internal marks in those subjects will remain the same as those obtained earlier.
- 9.4 To a student who has not appeared for an examination in any subject, ‘**Ab**’ grade will be allocated in that subject, and he is deemed to have ‘**Failed**’. A student will be required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered next. In this case also, the internal marks in those subjects will remain the same as those obtained earlier.
- 9.5 A letter grade does not indicate any specific percentage of marks secured by the student, but it indicates only the range of percentage of marks.
- 9.6 A student earns Grade Point (GP) in each subject/ course, on the basis of the letter grade secured in that subject/ course. The corresponding ‘Credit Points’ (CP) are computed by multiplying the grade point with credits for that particular subject/ course.

Credit Points (CP) = Grade Point (GP) x Credits For a course

- 9.7 A student passes the subject/ course only when **GP ≥ 5 (‘C’ grade or above)**
- 9.8 The Semester Grade Point Average (SGPA) is calculated by dividing the sum of credit points (ΣCP) secured from all subjects/ courses registered in a semester, by the total number of credits registered during that semester. SGPA is rounded off to **two** decimal places. SGPA is thus computed as

$$\text{SGPA} = \{ \sum_{i=1}^N C_i G_i \} / \{ \sum_{i=1}^N C_i \} \dots \text{For each Semester,}$$

where ‘i’ is the subject indicator index (considering all subjects in a semester), ‘N’ is the no. of subjects ‘**registered**’ for the semester (as specifically required and listed under the course structure of the parent department), C_i is the no. of credits allotted to the i^{th} subject, and G_i represents the grade points (GP) corresponding to the letter grade awarded for that i^{th} subject.

- 9.9 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student in all semesters considered for registration. The CGPA is the ratio of the total credit points secured by a student in **all** registered courses (of 160) in **all** semesters, and the total number of credits registered in **all** the semesters. CGPA is rounded off to **two** decimal places. CGPA is thus computed from the I year II semester onwards at the end of each semester as per the formula

$$\text{CGPA} = \{ \sum_{j=1}^M C_j G_j \} / \{ \sum_{j=1}^M C_j \} \dots \text{for all S Semesters registered}$$

(i.e., up to and inclusive of S Semesters, $S \geq 2$),

where ‘M’ is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has ‘REGISTERED’ from the 1st Semester onwards up to and inclusive of the Semester S (obviously $M > N$), ‘j’ is the Subject indicator index (takes into account all Subjects from 1 to S Semesters), C_j is the no. of Credits allotted to the j^{th} Subject, and G_j represents the Grade Points (GP)

corresponding to the Letter Grade awarded for that jth Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

Illustration of calculation of SGPA:

Course/Subject	Credits	Letter Grade	Grade Points	Credit Points
Course 1	4	A	8	$4 \times 8 = 32$
Course 2	4	O	10	$4 \times 10 = 40$
Course 3	4	C	5	$4 \times 5 = 20$
Course 4	3	B	6	$3 \times 6 = 18$
Course 5	3	A+	9	$3 \times 9 = 27$
Course 6	3	C	5	$3 \times 5 = 15$
	21			152

$$\text{SGPA} = 152/21 = 7.24$$

Illustration of Calculation of CGPA up to 3rd Semester:

Semester	Course/ Subject Title	Credits Allotted	Letter Grade Secured	Corresponding Grade Point (GP)	Credit Points (CP)
I	Course 1	3	A	8	24
I	Course 2	3	O	10	30
I	Course 3	3	B	6	18
I	Course 4	4	A	8	32
I	Course 5	3	A+	9	27
I	Course 6	4	C	5	20
II	Course 7	4	B	6	24
II	Course 8	4	A	8	32
II	Course 9	3	C	5	15
II	Course 10	3	O	10	30
II	Course 11	3	B+	7	21
II	Course 12	4	B	6	24
II	Course 13	4	A	8	32
II	Course 14	3	O	10	30
III	Course 15	2	A	8	16
III	Course 16	1	C	5	5
III	Course 17	4	O	10	40
III	Course 18	3	B+	7	21
III	Course 19	4	B	6	24
III	Course 20	4	A	8	32
III	Course 21	3	B+	7	21
	Total Credits	69		Total Credit Points	518

$$\text{CGPA} = 518/69 = 7.51$$

The calculation process of CGPA illustrated above will be followed for each subsequent semester until 8th semester. The CGPA obtained at the end of 8th semester will become the final CGPA secured for entire B.Tech. programme.

- 9.10 For merit ranking or comparison purposes or any other listing, **only the ‘rounded off’** values of the CGPAs will be used.
- 9.11 SGPA and CGPA of a semester will be mentioned in the semester Memorandum of Grades if all subjects of that semester are passed in first attempt. Otherwise the SGPA and CGPA shall be mentioned only on the Memorandum of Grades in which sitting he passed his last exam in that semester. However, mandatory courses will not be taken into consideration.

10.0 Passing Standards

- 10.1 A student shall be declared successful or ‘passed’ in a semester, if he secures a $GP \geq 5$ (‘C’ grade or above) in every subject/course in that semester (i.e. when the student gets an $SGPA \geq 5.0$ at the end of that particular semester); and he shall be declared successful or ‘passed’ in the entire undergraduate programme, only when gets a CGPA ≥ 5.00 (‘C’ grade or above) for the award of the degree as required.
- 10.2 After the completion of each semester, a grade card or grade sheet shall be issued to all the registered students of that semester, indicating the letter grades and credits earned. It will show the details of the courses registered (course code, title, no. of credits, grade earned, etc.) and credits earned. **There is NO exemption of credits in any case.**

11.0 Declaration of results

- 11.1 Computation of SGPA and CGPA are done using the procedure listed in 9.6 to 9.9.
- 11.2 For final percentage of marks equivalent to the computed final CGPA, the following formula may be used.
- $$\% \text{ of Marks} = (\text{final CGPA} - 0.5) \times 10$$

12.0 Award of Degree

- 12.1 A student who registers for all the specified subjects/ courses as listed in the course structure and secures the required number of 160 credits (with $CGPA \geq 5.0$), within 8 academic years from the date of commencement of the first academic year, shall be declared to have ‘**qualified**’ for the award of B.Tech. degree in the branch of Engineering selected at the time of admission.
- 12.2 A student who qualifies for the award of the degree as listed in item 12.1 shall be placed in the following classes.
- 12.3 A student with final CGPA (at the end of the undergraduate programme) > 8.00 , and fulfilling the following conditions - shall be placed in ‘**First Class with Distinction**’.
- However, he
- (i) Should have passed all the subjects/courses in ‘**First Appearance**’ within the first 4 academic years (or 8 sequential semesters) from the date of commencement of first year first semester.
 - (ii) Should not have been detained or prevented from writing the semester end examinations in any semester due to shortage of attendance or any other reason.

A student not fulfilling any of the above conditions with final CGPA > 8 shall be placed in

‘First Class’.

- 12.4 Students with final CGPA (at the end of the undergraduate programme) ≥ 7.0 but < 8.00 shall be placed in **‘First Class’**.
- 12.5 Students with final CGPA (at the end of the undergraduate programme) ≥ 6.00 but < 7.00 , shall be placed in **‘Second Class’**.
- 12.6 All other students who qualify for the award of the degree (as per item 12.1), with final CGPA (at the end of the undergraduate programme) ≥ 5.00 but < 6 , shall be placed in **‘pass class’**.
- 12.7 A student with final CGPA (at the end of the undergraduate programme) < 5.00 will not be eligible for the award of the degree.
- 12.8 Students fulfilling the conditions listed under item 12.3 alone will be eligible for award of **‘Gold Medal’**.
- 12.9 **Award of 2-Year B.Tech. Diploma Certificate**

1. A student is awarded 2-Year UG Diploma Certificate in the concerned engineering branch on completion of all the academic requirements and earned all the 80 credits (with in 4 years from the date of admission) upto B. Tech. – II Year – II Semester, if the student want to exit the 4-Year B. Tech. program. The student **once opted and awarded for 2-Year UG Diploma Certificate, the student will not be permitted to join** in B. Tech. III Year – I Semester and continue for completion of remaining years of study for 4-Year B. Tech. Degree.
2. A student may be permitted to take one year break after completion of II Year – II Semester or B. Tech. – III Year – II Semester (with Principal permission through the HOD of the department well in advance) and can re-enter the course in **next Academic Year in the same college** and complete the course on fulfilling all the academic credentials within a stipulated duration i.e. double the duration of the course (Ex. within 8 Years for 4-Year program).

13.0 Withholding of results

- 13.1 If the student has not paid the fees to the College at any stage, or has dues pending due to any reason whatsoever, or if any case of indiscipline is pending, the result of the student may be withheld, and the student will not be allowed to go into the next higher semester. The award or issue of the degree may also be withheld in such cases.

14.0 Transitory Regulations**A. For students detained due to shortage of attendance:**

1. A Student who has been detained in I year of R18 Regulations due to lack of attendance, shall be permitted to join I year I Semester of AR22 Regulations and he is required to complete the study of B.Tech./B. Pharmacy programme within the stipulated period of eight academic years from the date of first admission in I Year.
2. A student who has been detained in any semester of II, III and IV years of R18 regulations for want of attendance, shall be permitted to join the corresponding semester of AR22 Regulations and is required to complete the study of B.Tech./B. Pharmacy within the stipulated period of eight academic years from the date of first admission in I Year. The AR22 Academic Regulations under which a student has been readmitted shall be applicable to that student from that semester. See rule (C) for further Transitory

Regulations.

3. For students detained due to shortage of credits: A student of MLR18 Regulations who has been detained due to lack of credits, shall be promoted to the next semester of AR22 Regulations only after acquiring the required number of credits as per the corresponding regulations of his/her first admission. The total credits required are 160 including both MLR18 & AR22 regulations. The student is required to complete the study of B.Tech. within the stipulated period of eight academic years from the year of first admission. The AR22 Academic Regulations are applicable to a student from the year of readmission. See rule (C) for further Transitory Regulations.

B. For readmitted students in AR22 Regulations:

4. A student who has failed in any subject under any regulation has to pass those subjects in the same regulations.
5. The maximum credits that a student acquires for the award of degree, shall be the sum of the total number of credits secured in all the regulations of his/her study including AR22 Regulations. **There is NO exemption of credits in any case.**
6. If a student is readmitted to AR22 Regulations and has any subject with 80% of syllabus common with his/her previous regulations, that particular subject in AR22 Regulations will be substituted by another subject to be suggested by the Concerned BOS.

15.0 Student Transfers

- 15.1 There shall be no branch transfers after the completion of admission process.
- 15.2 There shall be no transfers from one college/stream to another within the constituent colleges and units of Jawaharlal Nehru Technological University Hyderabad.
- 15.3 The students seeking transfer to colleges affiliated to JNTUH from various other Universities/institutions have to pass the failed subjects which are equivalent to the subjects of JNTUH, and also pass the subjects of JNTUH which the students have not studied at the earlier institution. Further, though the students have passed some of the subjects at the earlier institutions, if the same subjects are prescribed in different semesters of JNTUH, the students have to study those subjects in JNTUH in spite of the fact that those subjects are repeated.
- 15.4 The transferred students from other Universities/Institutions to College who are on rolls are to be provided one chance to write the CBT (for internal marks) in the **equivalent subject(s)** as per the clearance letter issued by the University.
- 15.5 The college have to provide one chance to write the internal examinations in the **equivalent subject(s)** to the students transferred from other universities/institutions to JNTUH autonomous affiliated colleges who are on rolls, as per the clearance (equivalence) letter issued by the University.

16.0 Malpractice Prevention Committee

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice / indiscipline in examinations. The committee shall consist of:

- a) Controller of examinations - Chairman
- b) Addl. Controller of examinations.- Member Convener
- c) Subject expert - member
- d) Head of the department of which the student belongs to. - Member
- e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and

punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing / keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

17.0 Scope

- 17.1 The academic regulations should be read as a whole, for the purpose of any interpretation.
- 17.2 In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of Academic Council is final.
- 17.3 The College may change or amend the academic regulations, course structure or syllabi at any time, and the changes or amendments made shall be applicable to all students with effect from the dates notified by the college authorities.
- 17.4 Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.

ACADEMIC REGULATIONS FOR B.TECH (LATERAL ENTRY SCHEME)
FROM THE AY 2023-24

1. Eligibility for the award of B.Tech Degree (LES)

The LES students after securing admission shall pursue a course of study for not less than three academic years and not more than six academic years.

2. The student shall register for 120 credits and secure 120 credits with CGPA ≥ 5 from II year to IV-year B.Tech. programme (LES) for the award of B.Tech. degree.
3. The students, who fail to fulfil the requirement for the award of the degree in six academic years from the year of admission, shall forfeit their seat in B.Tech.
4. The attendance requirements of B. Tech. (Regular) shall be applicable to B.Tech. (LES).

5. Promotion rule

S. No	Promotion	Conditions to be fulfilled
1	Second year first semester to second year second semester	Regular course of study of second year first semester.
2	Second year second semester to third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 24 credits out of 40 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	Third year first semester to third year second semester	Regular course of study of third year first semester.
4	Third year second semester to fourth year first semester	i) Regular course of study of third year second semester. ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Fourth year first semester to fourth year second semester	Regular course of study of fourth year first semester.

6. All the other regulations as applicable to B. Tech. 4-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).

7. LES students are not eligible for 2-Year B. Tech. Diploma Certificate.

Malpractices Rules

Disciplinary Action For / Improper Conduct in Examinations

	Nature of Malpractices/ Improper conduct	Punishment
	If the student:	
1. (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which student is appearing but has not made use of (material shall include any marks on the body of the student which can be used as an aid in the subject of the examination).	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other student orally or by any other body language methods or communicates through cell phones with any student or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the students involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the student is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The hall ticket of the student is to be cancelled
3.	Impersonates any other student in connection with the examination.	The student who has impersonated shall be expelled from examination hall. The student is also debarred and forfeits the seat. The performance of the original student who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.

4.	Smuggles in the answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
6.	Refuses to obey the orders of the chief superintendent/COE/ACoE/any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the college campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the student(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The students also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7.	Leaves the exam hall taking away answer script or intentionally tears off the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.

8.	Possesses any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.
9.	If student of the college, who is not a student for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat. Person(s) who do not belong to the college will be handed over to the police and, a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared for including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the student has appeared for including practical examinations and project work of that semester/year examinations.
12	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the Principal for further action to award a suitable punishment.	

Malpractices identified by squad or special invigilators

1. Punishments to the students as per the above guidelines.
2. Punishment for staff: (if the squad reports that the staff is also involved in encouraging malpractices)
 - a. A show-cause notice shall be issued to the staff.
 - b. Impose a suitable fine on the staff.

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COURSE STRUCTURE

**INFORMATION TECHNOLOGY
COURSE STRUCTURE – R22**

I YEAR I SEMESTER									
Induction program for one weeks									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal	External	Total
A6BS01	Linear Algebra and Calculus	BSC	3	1	0	4	40	60	100
A6BS07	Applied Physics	BSC	3	1	0	4	40	60	100
A6CS02	Programming for Problem Solving	ESC	3	0	0	3	40	60	100
A6ME02	Engineering Drawing	ESC	1	0	3	2.5	40	60	100
A6CS03	Programming for Problem Solving Lab	ESC	0	0	3	1.5	40	60	100
A6BS08	Applied Physics Lab	BSC	0	0	3	1.5	40	60	100
A6ME04	Engineering workshop	ESC	0	0	3	1.5	40	60	100
A6HS04	Seminar	HSMC	0	0	2	1	50	-	50
A6IT01	Basics of Information Technology	ESC	0	0	2	1	50	-	50
TOTAL			11	2	14	20	380	420	800
I YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal	External	Total
A6BS02	Numerical Methods and Integral Transforms	BSC	3	1	0	4	40	60	100
A6HS01	English for skill Enhancement	HSMC	3	0	0	3	40	60	100
A6BS09	Engineering Chemistry	BSC	3	1	0	4	40	60	100
A6EE60	Basic Electrical and Electronics Engineering	ESC	3	0	0	3	40	60	100
A6EC03	Electronic Devices and Applications	ESC	2	0	0	2	40	60	100
A6HS02	English Language and Communication Skills Lab.	HSMC	0	0	3	1.5	40	60	100
A6CS04	Python Programming Lab	ESC	0	0	3	1.5	40	60	100
A6EC04	Introduction to Internet of Things	ESC	0	0	2	1	40	60	100
A6BS11	Environmental Science	MC	3	0	0	0	50	-	50
TOTAL			17	2	8	20	370	480	850

II YEAR I SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6BS03	Computer Oriented Statistical Methods	BSC	3	1	0	4	40	60	100
A6CS08	Discrete Mathematics	BSC	3	0	0	3	40	60	100
A6IT02	Object Oriented Programming through Java	PCC	3	0	0	3	40	60	100
A6CS09	Database Management Systems	PCC	3	0	0	3	40	60	100
A6HS08	Business Economics and Financial Analysis	HSMC	3	0	0	3	40	60	100
A6IT03	Object Oriented Programming through Java Lab	PCC	0	0	3	1.5	40	60	100
A6CS10	Database Management Systems Lab	PCC	0	0	3	1.5	40	60	100
A6IT04	Skill Development Course (HTML, CSS, JavaScript, Bootstrap)	PCC	0	0	2	1	-	100	100
TOTAL			15	1	8	20	280	520	800
Mandatory Course									
A6HS05	Gender Sensitization	MC	0	0	2	0	50	-	50
II YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6CS28	Digital Electronics and Computer Organization	ESC	3	0	0	3	40	60	100
A6CS11	Operating Systems	PCC	3	0	0	3	40	60	100
A6IT05	Software Engineering and Design	PCC	3	0	0	3	40	60	100
A6CS15	Design and Analysis of Algorithms	PCC	3	0	0	3	40	60	100
A6IT06	Data Structures through Java	PCC	3	0	0	3	40	60	100
A6IT07	Data Structures through Java Lab	PCC	0	0	3	1.5	40	60	100
A6IT08	Software Engineering and Design Lab	PCC	0	0	3	1.5	40	60	100
A6IT09	Real Time Research Project/Societal related project	PWC	0	0	4	2	50	-	50
TOTAL			15	0	10	20	330	420	750
Mandatory Course									
A6HS06	Constitution of India	MC	2	0	0	0	50	-	50

III YEAR I SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6IT10	Full Stack Development	PCC	3	0	0	3	40	60	100
A6IT11	Automata and Compiler Design	PCC	3	0	0	3	40	60	100
A6IT12	Data Communication & Computer Networks	PCC	3	0	0	3	40	60	100
	Professional Elective – I	PEC	3	0	0	3	40	60	100
A6IT13	Cloud & DevOps	PCC	3	0	0	3	40	60	100
A6IT14	Full Stack Development Lab	PCC	0	0	3	1.5	40	60	100
A6IT15	Linux Programming Lab	PCC	0	0	2	1	40	60	100
A6IT16	Cloud & DevOps Lab	PCC	0	0	3	1.5	40	60	100
A6IT17	MOOCS/Independent Study	PCC	0	0	2	1	-	100	100
TOTAL			15	0	10	20	320	580	900
Mandatory Course									
A6HS10	Human Values and Professional Ethics	MC	2	0	0	0	50	-	50
III YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6AI06	Machine Learning	PCC	3	0	0	3	40	60	100
A6IT18	Testing Automation	PCC	3	0	0	3	40	60	100
	Professional Elective – II	PEC	3	0	0	3	40	60	100
	Professional Elective – III	PEC	3	0	0	3	40	60	100
	Open Elective-I	OEC	3	0	0	3	40	60	100
A6AI09	Machine Learning Lab	PCC	0	0	3	1.5	40	60	100
A6IT19	Testing Automation Lab	PCC	0	0	3	1.5	40	60	100
A6IT20	Mini Project	PWC	0	0	4	2	-	100	100
TOTAL			15	0	10	20	280	520	800
Mandatory Course									
A6BS11	Environmental Sciences (only for LE s)	MC	2	0	0	0	50	-	50

IV YEAR I SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6DS06	Big Data Technologies	PCC	3	0	0	3	40	60	100
A6CS36	Cyber Security and Cyber Laws	PCC	3	0	0	3	40	60	100
	Professional Elective – IV	PEC	3	0	0	3	40	60	100
	Professional Elective – V	PEC	3	0	0	3	40	60	100
	Open Elective-II	OEC	3	0	0	3	40	60	100
A6DS07	Big Data Technologies Lab	PCC	0	0	2	1	40	60	100
A6IT21	Research Project Phase – 1	PWC	0	0	8	4	100	-	100
TOTAL			15	0	10	20	340	360	700
IV YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6HS15	Organizational Behavior	HSMC	3	0	0	3	40	60	100
	Professional Elective – VI	PEC	3	0	0	3	40	60	100
	Open Elective-III	OEC	3	0	0	3	40	60	100
A6IT22	Research Project Phase – 2	PWC	0	0	22	11	40	60	100
TOTAL			9	0	22	20	160	240	400

PROFESSIONAL ELECTIVES

PROFESSIONAL ELECTIVE – I			PROFESSIONAL ELECTIVE - II		
S. No.	Code	Subject	S. No.	Code	Subject
1	A6AI11	Data Mining	1	A6IT39	Introduction to Artificial Intelligence
2	A6IT23	Introduction to Data Science	2	A6DS13	Data Wrangling
3	A6IT24	Information Security	3	A6IT27	Network Security
4	A6IT25	Mobile Application Development	4	A6IT28	Adhoc & Sensor Networks
PROFESSIONAL ELECTIVE – III			PROFESSIONAL ELECTIVE - IV		
S. No.	Code	Subject	S. No.	Code	Subject
1	A6IT29	Soft Computing	1	A6AI28	Pattern Recognition
2	A6IT26	Information Retrieval Systems	2	A6DS21	Data Visualization Using Tableau
3	A6CY25	Blockchain Technology	3	A6IT30	Security Testing
4	A6AI14	Natural Language Processing	4	A6IT31	High Performance Computing
PROFESSIONAL ELECTIVE – V			PROFESSIONAL ELECTIVE - VI		
S. No.	Code	Subject	S. No.	Code	Subject
1	A6AI17	Deep Learning	1	A6IT33	E-Commerce
2	A6DS28	Predictive Analytics	2	A6CS22	Distributed Computing
3	A6IT32	Software Project Management	3	A6IT34	Network Administration
4	A6CY16	Crime Investigation & Digital Forensics	4	A6AI12	Image Processing

OPEN ELECTIVES

OPEN ELECTIVE – I			
S. No.	Course Code	Course Name	Offering Department
1	A6AE19	Fundamentals of Avionics	Aeronautical Engineering
2	A6AE20	Introduction to Aerospace Technology	
3	A6CS48	Software Project Management Fundamentals	Computer Science and Engineering
4	A6CS49	Data Analytics	
5	A6EC55	Microprocessors and Interfacing	Electronics & Communication Engineering
6	A6EC56	Principles of Communications	
7	A6EE54	Electrical Wiring and Safety Measures	Electrical & Electronics Engineering
8	A6EE55	Electrical Materials	
9	A6IT28	Adhoc & Sensor Networks	Information Technology
10	A6IT35	Object Oriented Programming	
11	A6ME55	Basic Mechanical Engineering	Mechanical Engineering
12	A6ME60	Fundamentals of Engineering Materials	
13	A6HS08	Business Economics and Financial Analysis	HSM
14	A6HS09	Basics of Entrepreneurship	

OPEN ELECTIVE – II			
S. No.	Course Code	Course Name	Offering Department
1	A6AE25	Introduction to Jets and Rockets	Aeronautical Engineering
2	A6AE26	Non-Destructive Testing Methods	
3	A6CS09	Database Management System	Computer Science and Engineering
4	A6CS43	Data Warehousing and Business Intelligence	
5	A6EC57	Microcontrollers and Applications	Electronics & Communication Engineering
6	A6EC58	Fundamentals of Image processing	
7	A6EE56	Neural Networks and Fuzzy Logic	Electrical & Electronics Engineering
8	A6EE57	Energy Storage Systems	
9	A6IT30	Security Testing	Information Technology
10	A6IT36	Human Computer Interaction	
11	A6ME65	Fundamentals of Operations Research	Mechanical Engineering
12	A6ME70	Industrial Engineering & Management	
13	A6HS11	Advanced Entrepreneurship	HSM

OPEN ELECTIVE - III			
S. No.	Course Code	Course Name	Offering Department
1	A6AE40	Introduction to Aircraft Industry	Aeronautical Engineering
2	A6AE41	Unmanned Aerial Vehicles	
3	A6CS51	Artificial Intelligence and Applications	Computer Science and Engineering
4	A6CS36	Cyber Security & Cyber laws	
5	A6EC59	Introduction to Sensors and Actuators	Electronics & Communication Engineering
6	A6EC60	Introduction to Computer Vision	
7	A6EE58	Solar Energy and Applications	Electrical & Electronics Engineering
8	A6EE59	Instrumentation and Control	
9	A6IT37	Introduction to Computer Networks	Information Technology
10	A6IT31	High Performance Computing	
11	A6ME75	Renewable Energy Sources	Mechanical Engineering
12	A6ME80	Elements of Electric and Hybrid vehicles	
13	A6HS12	Management Science	HSM

I B.TECH I SEMESTER SYLLABUS

LINEAR ALGEBRA & CALCULUS

I B.Tech - I Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6BS01	BSC	3	1	0	4	40	60	100

Contact Classes: 44 Tutorial Classes: 08 Practical Classes:-- 00 Total Classes: 52

COURSE OBJECTIVES:

To learn

1. Concept of Rank of a matrix, Consistency and solving system of linear equations.
2. Concept of eigen values, eigen vectors and diagonalization of the matrix.
3. The concept of differential equations and solve them using appropriate methods.
4. Evaluate multiple integrals and improper integrals
5. The partial derivatives of several variable functions.

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Solve the system of linear equations using rank of the matrices.
2. Find the Eigen values and Eigen vectors of a matrix
3. Identify the different types of differential equations and solve them using appropriate methods.
4. Evaluate the improper integrals using beta and gamma functions.
5. Find the Maxima and Minima of several variable functions.

UNIT-I MATRICES AND THEIR APPLICATIONS

**Classes:
08**

Real matrices: Symmetric-skew-symmetric and orthogonal matrices –Complex matrices: Hermitian, Skew –Hermitian and Unitary matrices –Elementary row and column transformations –Elementary matrix-Finding rank of a matrix by reducing to Echelon form and Normal form-Finding the inverse of a matrix using elementary row/column transformations (Gauss-Jordan method)-Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix –Solving $m \times n$ and $n \times n$ linear system of equations by Gauss Elimination, Gauss seidel Method

UNIT-II EIGEN VALUES, EIGEN VECTORS

**Classes:
08**

Eigen values and Eigen vectors and its properties (without proof), Cayley-Hamilton theorem (Statement and verification)-Finding inverse and powers of a matrix by Cayley-Hamilton theorem, Diagonalization of matrices. Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to Canonical forms by Orthogonal Transformation.

UNIT-III ORDINARY DIFFERENTIAL EQUATIONS AND THEIR APPLICATIONS

**Classes:
10**

Introduction- Exact and reducible to Exact differential equations-Newton's Law of cooling-Law of Growth and Decay. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$, $\sin ax$, $\cos ax$, $e^{ax}v(x)$, $x^n v(x)$ - Method of variation of parameters L-C-R Circuits.

UNIT-IV MULTIPLE INTEGRALS, BETA AND GAMMA FUNCTIONS**Classes:
10**

Double and triple integrals (Cartesian and polar), Change of order of integration in double integrals, Change of variables (Cartesian to polar) in double integrals. Finding the area and volume of a region using double and triple integral.

Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions.

UNIT-V CALCULUS OF SEVERAL VARIABLES**Classes:
08**

Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative - Chain rule, Jacobians-functional dependence & independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange's method (with constraints).

Text Books:

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.

Reference Books:

1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006.
2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008.
3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.

Web references:

1. https://www.efunda.com/math/math_home/math.cfm
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

E -Text Books:

- 1.<https://www.e-booksdirectory.com/details.php?ebook=10166>

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

APPLIED PHYSICS

I B.Tech - I Semester: Common for all Branches

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6BS07	BSC	3	1	0	4	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Learn the basic principles of quantum physics and its applications
2. Understand the formation of energy bands and atomic structure in solids for material classification
3. Understand the underlying mechanism involved in construction and working properties of different types of semiconductor devices
4. Learn the basic principles of laser and optical fiber in information technology
5. Identify the importance of nanoscale and various fabrication and characterization techniques and quantum computations in engineering applications

COURSE OUTCOMES:

At the end of the course students will be able to:

1. **Analyze** the microscopic properties of materials using principles of quantum physics for engineering applications
2. **Explain** the behavior of different electronic materials based on the concepts of band theory
3. **Apply** the knowledge of Solar PV cells for choice of materials in efficient alternate energy generation
4. **Gain** the knowledge of production of laser and usage of fibers in fiber optic communication technology
5. **Comprehend** the knowledge of quantum physics in quantum computation for secure information Technology.

UNIT-I Quantum Physics Classes: 10

Black body radiation, Stefan-Boltzmann's law, Planck's radiation law (Qualitative treatment), Photoelectric effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation, Physical Significance of the wave Function, Particle in One Dimensional Potential Box.

UNIT-II Electronic properties of Materials & Band theory of solids Classes: 10

Electronic properties of Materials: Classical free electron theory and Quantum free electron theories of metals, success and drawbacks, Bloch theorem, Kronig-Penny model (Qualitative treatment), E-k diagram, effective mass of electron.

Band theory of solids: Origin of energy band formation in solids, Fermi energy level, Fermi-Dirac distribution law, Classification of materials as conductors, insulators and semiconductors.

UNIT-III Semi-conductors & Semiconductor Devices Classes: 08

Semiconductors: Intrinsic and Extrinsic Semiconductors, formation of PN junction diode and its V-I characteristics, Direct and Indirect band gap semiconductors, Hall effect and its applications.

Semiconductor Devices: Construction, working and V-I characteristics of PIN Diode, LED, Solar cell and their applications.

UNIT-IV

Laser & Fiber Optics

Classes: 12

Laser: Characteristics of Laser, Absorption, Spontaneous and Stimulated emission of radiations. Lasing actions-Pumping mechanism, Meta stable state and Population inversion, Nd-YAG laser, CO2 laser, Applications of lasers in different fields.

Fiber Optics: Structure of fibers, Total Internal Reflection, Acceptance angle – Numerical Aperture, Types of fibers- SI and GI fibers, Single and Multimode fibers - SMSI, MMSI, MMGI, Fiber Optic Communication system, Signal Degradation - Attenuation mechanism, Dispersion, Applications of fibers in different fields.

UNIT-V

Physics of Quantum computing & Quantum gates

Classes: 10

Physics of Quantum computing: Idea of classical bits and qubits, advantages with qubits over classical bits, Bloch vector representation of state of qubits.

Quantum gates-Single qubit logic gates- Pauli X, Y, Z and Hadmard gate in matrix form, Two level gates-CNOT and SWAP gates and representation in matrix form, Comments on No cloning theorem, Entanglement, Quantum Teleportation – Basic Idea, Quantum Key distribution protocol - BB84 protocol.

Text Books:

1. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVth Ed.
2. Nielsen M. A., I. L Chung, Quantum Computation & Quantum Information, Cambridge Univ. Press.
3. M. N. Avadhanulu, P.G. Kshirsagar & TVS Arun Murthy" A Text book of Engineering Physics"- S. Chand Publications, 11th Edition 2019.

Reference Books:

1. Quantum Physics, H.C. Verma, TBS Publication, 2nd Edition 2012
2. Fundamenttals of Physics – Halliday, Resnick and Walker, John Wiley & Sons, 11th Edition, 2018.
3. Engineering Physics by Shatendra Sharma and Jyotsna Sharma, Pearson Publication, 2019
4. B.K Pandey and Chaturvedi, Engineering Physics, Cengage Learning, 2nd Edition, 2022

PROGRAMMING FOR PROBLEM SOLVING

I B.Tech - I Semester: Common for all Branches

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6CS02	ESC	3	0	0	3	40	60	100

COURSE OBJECTIVES :

1. To familiarize with the syntax and semantics of C programming language.
2. To learn the usage of structured programming approach in solving problems.
3. To use arrays, pointers, strings and structures in solving problems.
4. To understand how to solve problems related to matrices, Searching and sorting.
5. To understand how to use files to perform read and write operations.

COURSE OUTCOMES :

1. Apply algorithmic thinking to understand, define and solve problems
2. Develop computer programs using programming constructs and control structures and to use arrays to develop C programs
3. Decompose a problem into functions to develop modular reusable code and to use pointers to solve complex problems.
4. Use Strings and structures to formulate algorithms and programs.
5. Use FILE to perform read and write operations.

UNIT-I INTRODUCTION - PROBLEM SOLVING AND ALGORITHMIC THINKING & INTRODUCTION TO C LANGUAGE

Classes: 12

Algorithm -Definition, Characteristics of Algorithm.Constituents of algorithms: - Sequence, Selection and Repetition. Algorithm with Example: Roots Of a Quadratic Equations, Minimum And Maximum Numbers of a Given Set, Given number is prime number or not, given integer is palindrome or not, etc. Flowchart/Pseudo Code with examples.

Introduction To C Language: Structure of C Program, Data Types, data input and output statements, Operators, Precedence and Associativity of operators, Evaluation of Expressions, Type Conversions In Expressions.

UNIT-II CONTROL STRUCTURES AND ARRAYS

Classes: 15

Control structures: Decision statements; if and switch statement; **Loop control statements:** while, for and do while loops, Jump statements: break, continue, goto statements.

Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays

UNIT-III FUNCTIONS AND POINTERS

Classes: 17

Functions: Function definition, Types of Functions: User defined and built-in Functions, Advantages of User Defined Functions. Parameter passing in functions: Call by value, Call by reference, Passing arrays to functions, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Towers of Hanoi etc.. Storage classes.

Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, Functions returning pointers, Dynamic memory allocation.

UNIT-IV STRINGS AND USER DEFINED DATA TYPES

Classes: 10

Strings: Arrays of characters, variable length character strings, inputting character strings, character library functions, String Handling Functions, Arrays Of Strings

Structures and Unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, Structures and functions, Self-referential structures, unions, typedef, enumerations.

UNIT-V FILE HANDLING , SEARCHING AND SORTING**Classes: 10**

File Handling: Command Line Arguments, File Modes, And Basic File Operations Read, Write And Append, Example Programs. Random Access Using fseek, ftell and rewind Functions.

Basic Searching And Sorting Algorithms: Linear and Binary Search, Bubble Sort, Insertion Sort, Quick Sort.

Text Books:

1. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)
2. Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3rd edition, 2017.

Reference Books:

- 1.W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2nd Edition, 1988.
2. Yashavant Kanetkar, "Exploring C", BPB Publishers, 2nd Edition, 2003.
- 3.Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4th Edition, 2014.4. R. S. Bichkar, "Programming with C", Universities Press, 2nd Edition, 2012.
- 4.Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2nd Edition, 2006.
5. Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4th Edition, 2014.

Web References:

1. https://en.wikipedia.org/wiki/Computational_thinking
2. <https://nptel.ac.in/courses/106/104/106104128/>
3. <https://en.cppreference.com/w/c/language>
4. <https://www.learn-c.org/>

E-Text Books:

1. https://slidelegend.com/queue/computational-thinking-for-the-modern-problem-solver_59d6f01e1723ddb0c7a0df47.html
2. http://flowgorithm.altervista.org/#elf_l1_Lw
3. <http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm>

MOOC Course

1. <https://www.coursera.org/learn/computational-thinking-problem-solving>
2. https://onlinecourses.nptel.ac.in/noc18_cs33/preview
3. <https://www.alison.com/courses/Introduction-to-Programming-in-c>
4. <http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.html>

ENGINEERING DRAWING

I B.Tech – I Semester (CSE, CSC, CSD, CSM, CSIT, IT & ECE)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME02	ESC	1	0	3	2.5	40	60	100

COURSE OUTCOMES:

At the end of the course the student should be able to:

1. Understand various commands and create drawing in AutoCAD.
2. Construct various engineering curves and know their importance.
3. Prepare orthographic projections of objects by visualizing them in different positions.
4. Solve the problem of projections of planes and solids in different positions.
5. Construct the isometric view into orthographic views and vice versa.

UNIT-I INTRODUCTION TO ENGINEERING DRAWING

Introduction to Engineering Drawing: Principles and their significance.

Introduction to Computer Aided Drafting: Initial Setup Commands, Draw Commands, modify commands, 2D Drawings-Simple Exercises.

UNIT-II ENGINEERING CURVES

Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only), Involute.

UNIT-III ORTHOGRAPHIC PROJECTIONS OF POINTS AND LINES

Principles of Orthographic Projections –Projections of points.

Projection of lines inclined to both planes (First angle projection only).

UNIT-IV PROJECTIONS OF PLANES AND SOLIDS

Projections of Planes: Projections of regular planes inclined to one plane.

Projection of Solids: Solids inclined to one plane (Regular solids).

UNIT-V ISOMETRIC VIEW AND ORTHOGRAPHIC VIEWS – CONVERSION

Isometric view: Drawing Isometric circles, Dimensioning Isometric Objects.

Conversion of Isometric view to Orthographic views and Orthographic to isometric view.

Text Books:

1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing House.
2. Agrawal B. & Agrawal C. M., Engineering Graphics, TMH Publication.

Reference Books:

1. D.M. Kulkarni, A.P.Rastogi, A.K. Sarka “Engineering Graphics with AutoCAD” PHI publications.
2. Narayana, K.L. & P Kanniah, Text book on Engineering Drawing, Scitech Publishers.
3. Shah, M.B. & Rana B.C., Engineering Drawing and Computer Graphics, Pearson Education.

PROGRAMMING FOR PROBLEM SOLVING LAB

I B.Tech – I Semester: Common to all branches

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6CS03	ESC	0	0	3	1.5	40	60	100

COURSE OBJECTIVES:

1. To be familiarize with flowgorithm to solve simple problems
2. To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc.
3. To develop modular, reusable and readable C Programs using the concepts like functions, arrays, strings, pointers and structures.

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Solve simple mathematical problems using Flowgorithm.
2. Correct syntax errors as reported by the compilers and logical errors encountered at run time Develop programs by using decision making and looping constructs.
3. Implement real time applications using the concept of array, pointers, functions and structures. Solve real world problems using matrices, searching and sorting.

LIST OF EXPERIMENTS

Week-1 INTRODUCTION TO FLOGORITHM

1. Installation and working of Flowgorithm Software.
2. Write and implement basic arithmetic operations using Flowgorithm – sum, average, product, difference, quotient and remainder of given numbers etc.
3. Draw a flowchart to calculate area of Shapes (Square, Rectangle, Circle and Triangle).
4. Draw a flowchart to find the sum of individual digits of a 3 digit number
5. Draw a flowchart to read input name, marks of 5 subjects of a student and display the name of the student, the total marks scored, percentage scored.
6. Draw a flowchart to find roots of a quadratic equation.
7. Draw a flowchart to find the largest and smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd
8. Draw a flowchart to check whether the triangle is equilateral, isosceles or scalene triangle
9. Draw a flowchart to check whether a given number is palindrome or not.

Week-2 BASIC DATA TYPES

- 1) Write a C program to find division of 1 integer 1 float numbers.
- 2) Write a C program to find division of 2 integer numbers.
- 3) Write a C program to find average of (1int, 1float) numbers.
- 4) Write a C Program to Swap Numbers without Using Temporary Variable
- 5) Write a C Program to Swap two Numbers Using Temporary Variable
- 6) Write a C Write a program to read the values of x, y and z and print the results of the following expressions in one line.
 - a) $(x+y+z) / (x-y-z)$
 - b) $(x+y+z) / 3$
 - c) $(x+y) * (x-y) * (y-z)$

Week-3 OPERATORS

- 1) Write a C program to convert temperature from Fahrenheit to Celsius and vice versa ($c = (f - 32) / 1.8$)
- 2) Write a C program to find area and perimeter of a circle. ($area = \pi r^2$ perimeter $= 2\pi r$)
- 3) Write a C program to calculate area and perimeter of a right angled triangle.
 - a. ($Area = 1/2 * b * h$ perimeter $= w + h + \sqrt{w^2 + h^2}$)
- 4) Find the sum of natural numbers 1 to n. (read n as input) (Use formula $sum = n(n+1)/2$)
- 5) Write a C program to calculate Simple interest ($SI = PTR/100$)
- 6) Write a C program to calculate area and perimeter of a rectangle. Area $= l * b$ Perimeter $= 2 * (l + b)$
- 7) Write a C program to calculate the value of the third angle of a triangle if two angles are given as input. ($a + b + c = 180$)
- 8) Write a C program to read the consumer number and number of units consumed and the cost per unit and print the amount to be paid. ($Amt = num \text{ of units} * cost$)
- 9) Write a C Program to calculate area and perimeter of a triangle.
 - a. Perimeter $= (a + b + c)$
 - b. $s = (a + b + c) / 2$
 - c. Area $= \sqrt{s * (s - a) * (s - b) * (s - c)}$
- 10) Write a C program to read five Subject marks and find the average.
- 11) Write a C program to Calculate Compound interest ($CI = p(1 + r/100)^n$)

Week-4 CONDITIONAL STATEMENTS

1. Write a C program to find largest and smallest of given numbers.
2. Write a C program which takes two integer operands and one operator from the user (+, -, *, /, %) use switch)
3. Write a program to compute grade of students using if else ladder. The grades are assigned as followed:

$marks < 50$ F
 $50 \leq marks < 60$ C
 $60 \leq marks < 70$ B
 $70 \leq marks$ B+
 $80 \leq marks < 90$ A
 $90 \leq marks \leq 100$ A+
4. Write a C program to whether given year leap year or not.
5. Write a C program to find whether given triangle is scalene or isosceles or equilateral.

Week-5 LOOPING STATEMENTS

- 1) Write a C program to find Sum of individual digits of given integer
- 2) Write a C program to generate first n terms of Fibonacci series
- 3) Write a C Program to find the Sum of Series $SUM = 1 - x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$
- 4) Write a C program to print the Fibonacci sequence up to given value of n.
- 5) Write a C program to print the multiplication table to the given value of n.
- 6) Write a C program to check whether given number is palindrome or not
- 7) Write a C program to check whether given number is perfect or not

Week-6 NESTED LOOPING STATEMENTS

- 1) Write a C program to generate prime numbers between 1 and n

- 2) Write a C program to generate Pascal's triangle.
- 3) Write a C program to generate the following pyramid of numbers.

```
      1
     1 3 1
    1 3 5 3 1
```

- 4) Write a C program to generate the pattern

```
*****
*****
*****
****
***
**
*
```

Week-7 ARRAYS

- 1) Write a C Program to implement following searching methods
 - i. Binary Search
 - ii. Linear Search
- 2) Write a C program to find largest and smallest number in a list of integers
- 3) Write a C program
 - i. To add two matrices
 - ii. To multiply two matrices
- 4) Write a C program to find Transpose of a given matrix

Week-8 FUNCTIONS

- 1) Write a C program to find the factorial of a given integer using non recursive functions
- 2) Write a C program to find GCD of given integers using non recursive functions
- 3) Write a C Program to find the power of a given number using non recursive functions
- 4) Write a C program to find sum of natural numbers using non recursive function.
- 5) Write a C program to reverse a given integer number using non recursive functions
- 6) Write a C Program to find binary equivalent of a given decimal number using recursive functions.
- 7) Write a C Program to print Fibonacci sequence using recursive functions.
- 8) Write a C Program to find LCM of 3 given numbers using recursive functions
- 9) Write a C program to find the factorial of a given integer using recursive functions
- 10) Write a C program to print fibonacci series till n terms using recursion.

Week-9 STRINGS

- 1) Write a C program using to Insert a sub string into a given main string from a given position
- 2) Write a C program using to Delete n characters from a given position in a string
- 3) Write a C program to determine if given string is palindrome or not
- 4) Write C Programs to demonstrate the following string handling functions.
 - a. strcat()
 - b. strcmp()
 - c. strrev()
 - d. strcpy()
 - e. strlen()
 - f. strstr()
 - g. strncpy()

- h. `strncat()`
- i. `strncmp()`

Week-10 POINTERS

- 1) Write a C program to read the elements of 1-d array using pointers and print them in reverse order using pointers.
- 2) Write a C Program to read two elements dynamically using `malloc ()` function and interchange the two numbers using call by reference.
- 3) Write a C Program to read and print the elements of 1-D array using `calloc()` memory allocation function and reallocate memory for the array by increasing the size of the array, read and print the elements of reallocated array.
- 4) Write a C Program to print 2-D array using pointers.

Week-11 STRUCTURES

- 1) Write a C Program using functions to
 - a) Reading a complex number
 - b) Writing a complex number
 - c) Add two complex numbers
 - d) Multiply two complex numbersNote: represent complex number using structure
- 2) Write a C program to read employee details employee number, employee name, basic salary, hra and da of n employees using structures and print employee number, employee name and gross salary of n employees.

Week-12 FILES

- 1) Write a C program to read and print the content of a file.
- 2) Write a C program copy the content of one file to another file
- 3) Write a C program to merge two file into third file.
- 4) Write a C Program to find the number of lines in a text file

Text Books:

- 1) Riley DD, Hunt K.A. Computational Thinking for the Modern Problem Solver. CRC press, 2014 Mar 27.
- 2) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13th Edition, 2012.

Reference Books:

1. Ferragina P, Luccio F. Computational Thinking: First Algorithms, Then Code. Springer; 2018
2. King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2nd Edition, 2015.
3. Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3rd Edition, 2004.
4. Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1st Edition, 1994.

APPLIED PHYSICS LAB

I B.Tech – I Semester: Common for all Branches

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6BS08	BSC	0	0	3	1.5	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the temperature and other dependent properties of semiconductor devices and their usages in different applications
2. To know the Laser and Fibre optic technologies and its applications in real time scenario
3. Understand the electromagnetic properties using experimental knowledge
4. Understand the method of least squares fitting

COURSE OUTCOMES:

By the end of the course students will be able to:

1. **Analyze** the electric properties of semiconductor materials by determining energy gap of semiconductors, charge carrier concentration in Semiconductors using Hall effect and threshold voltage of LEDs, photo current in Photo diodes, solar cell, and temperature effect on resistance using thermistor.
2. **Identify** the optical properties of light such as diffraction phenomenon using grating material for calculation of the wavelength of Laser and acceptance angle, NA of optical fiber using OFC and determine the value of Plank's constant using a light source and interference by using Newton's rings
3. **Analyze** the electromagnetic properties of a current carrying coil by using Stewart Gee's experiment
4. **Analyze** the least squares fitting method for data analysis using experimental data of Tensional Pendulum

LIST OF EXPERIMENTS

Experiment-1	ENERGY GAP OF P-N JUNCTION DIODE: To determine the energy gap of a given semiconductor diode
Experiment-2	SOLAR CELL: To study the V-I and V-P characteristics and determine the fill factor of solar cell
Experiment-3	LIGHT EMITTING DIODE: To study the characteristics of LED by plotting V-I graph and determine the threshold value of given LEDs
Experiment-4	HALL EFFECT: To determine Hall co-efficient and charge carrier concentration of a given semiconductor
Experiment-5	PIN PHOTO DIODE: To study the V-I Characteristics of Photo Diode with respect to intensity of light
Experiment-6	OPTICAL FIBRE: To determine the numerical aperture and acceptance angle of an optical fiber
Experiment-7	LASER: To determine the wavelength of a given laser source by using diffraction grating method
Experiment-8	NEWTON'S RINGS: To determine the radius of curvature of a given Plano convex lens by forming Newton's rings
Experiment-9	THERMISTOR: To study the variation of resistance with respect to temperature using thermistor
Experiment-10	Understanding the method of least squares - Torsional pendulum as an example

Experiment-11 **PLANCK'S CONSTANT:** To determine value of planck's constant by measuring radiation in fixed spectral range

Experiment-12 **STEWART GEE'S EXPERIMENT:** To study the variation of magnetic field along the axis of a circular coil and calculation of magnetic flux

Note: Students have to perform any 8 experiments

Reference Books:

1. "Applied Physics Lab Manual"- Dr. Radhika Devi, Mr. A V Laxman Rao, N. Noel
2. S. Balasubramanian, M.N Srinivasan "A Text book of practical Physics" – S Chand Publishers, 2017.

ENGINEERING WORKSHOP

I B.Tech – I Semester: (CSC , CSD , CSE , CSM , CSIT, IT & ECE)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME04	ESC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100

COURSE OUTCOMES:

At the end of the course the student should be able to:

1. Assemble the different components
2. Identify and apply suitable tools for different trades of engineering processes.
3. Practice on manufacturing of components using workshop trades including Soldering, Carpentry, Fitting and Tin smithy & Fabricate Components with their own hands.
4. Apply basic electrical engineering knowledge for house wiring practice.
5. Learn the safety precautions for various operations in basic trades.

WEEKS	TRADES FOR EXERCISES
	Fitting
Week 1	Filing Four Sides of MS Work piece
Week 2	L Fit
	Carpentry
Week 3	Half Lap Joint
Week 4	Dove Tail Joint
	Tin Smithy
Week 5	Prepare a Rectangular Tray
Week 6	Prepare A Square Tin
	Electrical
Week 7	Series and Parallel Connection One Way Switch
Week 8	Two Way Switch Connection Stair Case Wiring
	Electronics
Week 9	Soldering - Series Connection & Parallel Connection
Week 10	Desoldering & Construction of Wheat stone bridge
	TRADES FOR DEMONSTRATION AND EXPOSURE
Week 11	Introduction to Black smithy
Week 12	Introduction to Plumbing

TEXT BOOKS

1. Work shop Manual - P. Kannaiah/ K.L. Narayana/ Scitech.
2. Workshop Manual / K. Venugopal / Anuradha.

REFERENCE BOOKS:

1. Hajra Choudhury S.K., Hajra Choudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010.
2. Workshop Manual / Venkat Reddy/ BSP

SEMINAR

I B.Tech – I Semester: Common for all Branches

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6HS04	HSMC	0	0	1	1	100	0	100

COURSE OBJECTIVE:

Seminar is an important component of learning in an Engineering College, where the student gets acquainted with preparing a report & presentation on a topic.

PERIODICITY / FREQUENCY OF EVALUATION : Twice

PARAMETERS OF EVALUATION:

1. The seminar shall have topic allotted and approved by the faculty.
2. The seminar is evaluated for 50 marks for internal only.
3. The students shall be required to submit the rough drafts of the seminar outputs within one week of the commencement of the class work.
4. Faculty shall make suggestions for modification in the rough draft. The final draft shall be presented by the student within a week thereafter.
5. Presentation schedules will be prepared by Department in line with the academic calendar.

THE SEMINARS SHALL BE EVALUATED IN TWO STAGES AS FOLLOWS:

A. ROUGH DRAFT

In this stage, the student should collect information from various sources on the topic and collate them in a systematic manner. He/ She may take the help of the concerned faculty.

The report should be typed in “MS-Word” file with “calibri” font, with font size of 16 for main heading, 14 for sub-headings and 11 for the body text. The contents should also be arranged in Power Point Presentation with relevant diagrams, pictures and illustrations. It should normally contain 10 to 15 slides, consisting of the followings:

1.	Topic, name of the student & faculty	1 Slide
2.	List of contents	1 Slide
3.	Introduction	1 Slides
4.	Descriptions of the topic (point-wise)	6 - 10 Slides
5.	Conclusion	1 - 2 Slides
6.	References/Bibliography	1 Slide

The soft copy of the rough draft of the seminar presentation in MS Power Point format along with the draft report should be submitted to the concerned faculty, with a copy to the concerned HOD within stipulated time.

The evaluation of the rough draft shall generally be based upon the following.

1	Punctuality in submission of rough draft	4
2	Dress Code	6
3	Resources from which the seminar have been based	4
4	Report , and content of Presentation	10
5	Depth of the students knowledge in the subject	10
6	Reception from Questions	10
7	Time Management, Classroom Dynamic	6
	Total Marks	50

After evaluation of the first draft the supervisor shall suggest further reading, additional work and fine tuning, to improve the quality of the seminar work. Within 7 days of the submission of the rough draft, the students are to submit the final draft incorporating the suggestions made by the faculty.

BASICS OF INFORMATION TECHNOLOGY

I B.Tech – I Semester: IT /CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT01	ESC	L	T	P	C	CIE	SEE	Total
		0	0	2	1	50	0	50

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Know the working principles of functional units of a basic Computer
2. Understand program development, the use of data structures and algorithms in problem solving.
3. Know the need and types of operating system, database systems.
4. Understand the significance of networks, internet, WWW and cyber security.
5. Understand Autonomous systems, the application of artificial intelligence.

UNIT-I Computer Applications & Basics

Classes: 6

Introduction, Basic Applications of Computer, Components of Computer, Computer Hardware & Software, Input Devices, Output Devices, Central Processing Unit, Modern's Processor, Computer Chip, Computer Ports

Computer Memory: Introduction to computer Memory, Types of Memory, Register memory, Cache memory, Primary memory, Secondary memory, Memory Units

UNIT-II Computer Operating System

Classes: 6

Basics of Operating System, Linux, Windows, MacOS, Android, Task Icons, Bars, System Settings, Setting Date & Time, File Management, Process, Multitasking, Graphical User Interfaces, and Command Line Interface, MS-DOS Commands.

Windows based applications: MS Office.

Graphics and Multimedia: Getting Images into Your Computer, Graphics Software Paint Programs, Draw Programs, Photo-Editing Program's, Computer-Aided Design Programs, 3-D and Animation Software.

UNIT-III Introduction to Computer Networks

Classes: 6

Internet, Internet's History, www, URL, DNS, Layers of OSI Model, Routers, Types of Networks, TCP/IP, UDP/IP IP address, Dial-up Connections, High-Speed Broadband Connections, Integrated Services Digital Network (ISDN) Service, Cable Modem Service, Firewalls, Networks Interface cards, Servers in Computer Network, Applications of Servers.

UNIT-IV Programming and Databases

Classes: 6

Program: Introduction, Purpose of Program, Level of programming languages, Types of Programming languages, HTML, Fundamentals of Scripting languages

Data Structures: Introduction to Data Structures, categories of Data structures, Basic Operations of Data structures.

UNIT-V Fundamentals of Design and Technologies

Classes: 6

WordPress: Creating a WordPress Site (Installing WordPress)

Information Technology: Elements of Information Technology, Telecommunication, Fiber Optics, Integrated Service Digital Network (ISDN), Blue-Gene Computer, Cloud Computing, Cyber Crime & Cyber Security, Artificial Intelligence, Machine Learning, IoT, Block Chain, Electronic Commerce.

Text Books:

1. Introduction to computers, Peter Norton, 8th Edition, Tata McGraw Hill, 2017.

Reference Books:

1. Invitation to Computer Science, G. Michael Schneider, Macalester College, Judith L. Gersting University of Hawaii, Hilo, Contributing author: Keith Miller University of Illinois, Springfield. Fundamentals of Computers, Reema Thareja, Oxford Higher Education, Oxford University Press.
2. Computer Fundamentals, Anita Goel, Pearson Education India, 2010.
3. Elements of computer science, Cengage.

I B.TECH II SEMESTER SYLLABUS

NUMERICAL METHODS AND INTEGRAL TRANSFORMS

I B.Tech – II Semester: Common for all Branches

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6BS02	BSC	3	1	0	4	40	60	100

COURSE OBJECTIVES:

To learn

1. Curve fitting and Interpolation techniques.
2. Numerical techniques.
3. Fourier series for periodic function
4. Laplace transforms
5. Concept and application of Fourier Transforms and Vector differentiation.

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Apply Curve fitting and Interpolation techniques.
2. Apply various numerical techniques
3. Find the Fourier series of the periodic functions.
4. Obtain the Laplace transforms of functions
5. Find Fourier transforms and apply vector differentiation techniques.

UNIT-I INTERPOLATION AND CURVE FITTING

Classes:
08

INTERPOLATION: Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Difference of a polynomial – Missing terms - Newton's forward interpolation, Newton's backward interpolation, Gauss's forward and backward interpolation formulae. Interpolation with unequal intervals – Lagrange's interpolation.

CURVE FITTING: Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of the form $y = a e^{bx}$, $y = a x^b$, $y = a b^x$ by the method of least squares.

UNIT-II NUMERICAL TECHNIQUES

Classes:
08

ROOT FINDING TECHNIQUES:

Bisection method Regula falsi method, Iteration method and Newton Raphson method.

NUMERICAL INTEGRATION :

Trapezoidal rule - Simpson's one-third rule - Simpson's three-eighth rule.

NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS:

Taylor's series method – Euler's - modified Euler's Method – Runge-Kutta method.

UNIT-III FOURIER SERIES

Classes:10

Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Half range Fourier sine and cosine expansions.

UNIT-IV LAPLACE TRANSFORMS

Classes:
10

Laplace transforms of elementary functions- First shifting theorem - Change of scale property –

Multiplication by t^n - Division by t – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations.

UNIT-V FOURIER TRANSFORMS AND VECTOR DIFFERENTIATION

Classes:
10

Fourier Integral theorem (Statement only)-Fourier Sine and Cosine Integrals, Fourier Transforms, Cosine and Sine transforms, properties, Inverse transforms.

Vector functions, vector differentiation, gradient, directional derivative, divergence, curl and scalar potential.

Text Books:

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.

Reference Books:

1. G.B.Thomas, calculus and analytical geometry, 9th Edition, Pearson Reprint 2006.
2. N.P Bali and Manish Goyal, A Text of Engineering Mathematics, Laxmi publications, 2008.
3. E.L.Ince, Ordinary differential Equations, Dover publications,1958.

Web references:

1. https://www.efunda.com/math/math_home/math.cfm
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

E -Text Books:

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

ENGLISH FOR SKILL ENHANCEMENT

I B.Tech – II Semester: Common for all Branches

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6HS01	HSMC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course will enable the students to:

1. Develop language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academicsubjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/ compose clearly and creatively, and adjust writing style appropriately to the content,the context, and nature of the subject.
5. Develop language components to communicate effectively in formal and informal situations.

COURSE OUTCOMES:

1. Will be able to acquire language proficiency with emphasis on Vocabulary, Grammar,Reading andWriting skills.
2. Apply the theoretical and practical components of English syllabus to study academicsubjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/ compose clearly and creatively, and adjust writing style appropriately to the content,the context, and nature of the subject.
5. Use language components to communicate effectively in formal and informal situations

UNIT-I

‘Toasted English’ by R. K. Narayan

Classes: 8

Vocabulary	: The Concept of Word Formation -The Use of Prefixes and Suffixes - Acquaintance with Prefixes and Suffixes from Foreign Languages to form Derivatives - Synonyms and Antonyms
Grammar	: Identifying Common Errors in Writing with Reference to Articles and Prepositions
Reading:	: Reading and Its Importance- Techniques for Effective Reading
Writing	: Sentence Structures – Use of Phrases and Clauses in Sentences – Importanceof Proper Punctuation – Techniques for Writing precisely – Paragraph Writing – Types, Structures and Features of a Paragraph – Creating Coherence – Organizing Principles of Paragraphs in Documents.

UNIT-II

‘Appro JRD’ by Sudha Murthy

Classes: 9

Vocabulary	: Words Often Misspelt – Homophones, Homonyms and Homographs
Grammar	: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement
Reading	: Sub-Skills of Reading – Skimming and Scanning – Exercises for Practice
Writing	: Nature and Style of Writing- Defining /Describing People, Objects, Places and Events – Classifying- Providing Examples or Evidence.

UNIT-III **‘Lessons from Online Learning’ by F.Haider Alvi, Deborah Hurst et al** **Classes: 9**

Vocabulary	: Words Often Confused – Words from Foreign Languages and their Use in English.
Grammar	: Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses
Reading	: Sub-Skills of Reading – Intensive Reading and Extensive Reading – Exercises for Practice
Writing	: Format of a Formal Letter-Writing Formal Letters E.g., Letter of Complaint, Letter of Requisition, Email Etiquette, Job Application with CV/Resume

UNIT-IV **Art and Literature** **Classes: 8**

Vocabulary	: Standard Abbreviations in English
Grammar	: Redundancies and Clichés in Oral and Written Communication
Reading	: Survey, Question, Read, Recite and Review (SQ3R Method) - Exercises for Practice
Writing	: Writing Practices- Essay Writing-Writing Introduction and Conclusion -Précis Writing

UNIT-V **Go, Kiss the World’ by Subroto Bagchi** **Classes: 8**

Vocabulary	: Technical Vocabulary and their Usage
Grammar	: Common Errors in English (<i>Covering all the other aspects of grammar which were not covered in the previous units</i>)
Reading	: Reading Comprehension-Exercises for Practice
Writing	: Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports - Writing a Report

Text Books:

1. “English: Language, Context and Culture” by Orient BlackSwan Pvt. Ltd, Hyderabad.2022. Print.

Reference Books:

1. Effective Academic Writing by Liss and Davis (OUP)
2. Richards, Jack C. (2022) Interchange Series. Introduction, 1,2,3. Cambridge University Press
3. Wood, F.T. (2007). Remedial English Grammar. Macmillan.
4. Chaudhuri, Santanu Sinha. (2018). Learn English: A Fun Book of Functional Language, Grammar and Vocabulary. (2nd ed.,). Sage Publications India Pvt. Ltd.
5. (2019). Technical Communication. Wiley India Pvt. Ltd.
6. Vishwamohan, Aysha. (2013). English for Technical Communication for Engineering Students. Mc Graw-Hill Education India Pvt. Ltd.
7. Swan, Michael. (2016). Practical English Usage. Oxford University Press. Fourth Edition

Web References:

1. <http://www.bbc.co.uk/learningenglish>
2. <http://learnenglish.britishcouncil.org>
3. <https://www.cambridgeenglish.org/learning-english/>
4. <https://study.com/academy/subj/english.html>

E-Text Books:

1. <https://www.pdfdrive.com/advanced-english-books.html>

MOOC Courses:

1. <http://nptel.ac.in/courses/109/106/109106067>
2. <https://www.britishcouncil.org/tr/en/english/mooc>

ENGINEERING CHEMISTRY

I B.TECH - II SEMESTER: Common for all Branches

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6BS09	BS C	3	1	0	4	40	60	100

COURSE OBJECTIVES:

1. To bring adaptability to new developments in Engineering Chemistry and to acquire the skills required to become a perfect engineer.
2. To include the importance of water in industrial usage, fundamental aspects of battery chemistry, significance of corrosion it's control to protect the structures.
3. To imbibe the basic concepts of petroleum and its products.
4. To acquire required knowledge about engineering materials like cement, smart materials and Lubricants.

COURSE OUTCOMES:

At the end of the course student will be able to:

1. Understand the basic properties of water and its usage in domestic and industrial purpose.
2. Acquire the basic knowledge of electrochemical procedures related to corrosion
3. Learn the fundamentals and general properties of polymers and other engineering materials. Predict potential applications of chemistry and practical utility in order to become good engineers and entrepreneurs.

UNIT-I WATER AND ITS TREATMENT

Introduction to hardness of water – Estimation of hardness of water by complexometric method and related numerical problems. Specifications and Disinfection of potable water by chlorination and break - point chlorination. Boiler troubles: Sludges, Scales and Caustic embrittlement. Internal treatment of Boiler feed water- Calgon conditioning - Phosphate conditioning - Colloidal conditioning, External treatment methods -Softening of water by ion- exchange processes. Desalination of water – Reverse osmosis.

UNIT-II BATTERY CHEMISTRY AND CORROSION

Introduction - Classification of batteries- primary, secondary and reserve batteries with examples. Basic requirements for commercial batteries. Construction, working and applications of: Zn-air and Lithium ion battery.

Fuel Cells- Differences between battery and a fuel cell, Construction and applications of Methanol Oxygen fuel cell and Solid oxide fuel cell.

Corrosion: Causes and effects of corrosion. Factors affecting rate of corrosion, Mechanism of dry and electrochemical corrosion. Types of corrosion: Galvanic, water-line corrosion. Corrosion control methods- Cathodic protection – Sacrificial anode and impressed current methods.

UNIT-III POLYMERIC MATERIALS

Definition – Classification of polymers with examples – Types of polymerization – addition (free radical addition) and condensation polymerization with examples – Nylon 6:6, Terylene

Plastics: Definition and characteristics- thermoplastic and thermosetting plastics, Preparation, Properties and engineering applications of PVC and Bakelite.

Elastomers: Characteristics –preparation – properties and applications of Buna-S, Butyl and Thiokol rubber.

Conducting polymers: Characteristics and Classification with examples-mechanism of conduction in trans-polyacetylene and applications of conducting polymers.

Biodegradable polymers: Concept and advantages - Polylactic acid and poly vinyl alcohol and their applications.

UNIT-IV ENERGY SOURCES

Introduction, Calorific value of fuel – HCV, LCV- Dulong's formula. Liquid fuels – petroleum and its refining, cracking types – moving bed catalytic cracking. Knocking – octane and cetane rating, synthetic petrol - Fischer-Tropsch's process; Gaseous fuels – composition and uses of natural gas, LPG and CNG, Biodiesel – Transesterification, advantages.

UNIT-V ENGINEERING MATERIALS

Smart materials and their engineering applications Shape memory materials- Poly L- Lactic acid. Thermo response materials- Polyacryl amides, Poly vinyl amides

Lubricants: Classification of lubricants with examples-characteristics of a good lubricants – mechanism of lubrication (thick film, thin film and extreme pressure)- properties of lubricants: viscosity, cloud point, pour point, flash point and fire point.

TEXT BOOKS:

1. Engineering Chemistry by P.C. Jain and M. Jain, Dhanpatrai Publishing Company, 2010
2. Engineering Chemistry by Rama Devi, Venkata Ramana Reddy and Rath, Cengage learning, 2016
3. A text book of Engineering Chemistry by M. Thirumala Chary, E. Laxminarayana and K. Shashikala, Pearson Publications, 2021.
4. Textbook of Engineering Chemistry by Jaya Shree Anireddy, Wiley Publications

REFERENCE BOOKS:

1. Engineering Chemistry by Shikha Agarwal, Cambridge University Press, Delhi (2015)
2. Engineering Chemistry by Shashi Chawla, Dhanpatrai and Company (P) Ltd. Delhi (2011)

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

I B.TECH - II SEMESTER: (Common to CSE, IT, AIML, DS, CS and CSIT, Aero, Mech)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EE60	ESC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. Develop fundamentals, including Ohm's law, Kirchhoff's laws and be able to solve for currents, voltages and power in electrical circuits.
2. Develop EMF equation and analyze the operation of DC Machines.
3. Analyze the working principle of Transformer.
4. Discuss the operation of AC Machines.
5. Analyze the operation of PN junction diode and rectifiers.
6. Discuss the operation and characteristics of Transistors.

COURSE OUTCOMES:

Upon successful completion of this course, student will be able to:

1. Evaluate current and voltage values in resistive circuits with independent sources.
2. Explain the working of DC machines and solve the numerical problems..
3. Explain the working of AC electrical machines and solve the numerical problems.
4. Analyze the V-I characteristics of PN – junction diode and describe the operation of rectifiers.
5. Analyze the different configurations of Transistors and obtain its characteristics.

UNIT I : ELECTRICAL CIRCUITS

Classes : 12

Basic definitions-Ohm's Law, types of elements, types of sources, Kirchhoff's Laws – simple problems.Series& parallel resistive networks with DC excitation, star to delta and delta to star transformations.

UNIT II : DC MACHINES

Classes : 12

Principle of Operation of DC Motor, types of DC motor, Torque equation & Losses and problems.

DC Generator construction and working Principle, EMF Equation types of generators and problems.

UNIT III : AC MACHINES

Classes : 12

Working principle and Construction of transformer, Emf Equation & problems. Principle operation of

3-phase induction motor, slip and torque Equation, Torque –slip characteristics

& problems.

UNIT IV : DIODE AND ITS CHARACTERISTICS

Classes : 12

PN Junction Diode: Operation of PN junction Diode: forward bias and reverse bias, Characteristics of PN Junction Diode – Zener Effect – Zener Diode and its Characteristics. Rectifiers, Half wave, Full wave and bridge Rectifiers –capacitor filters, inductor filters

UNIT V : TRANSISTORS

Classes : 10

Bipolar Junction Transistor - NPN & PNP Transistor, CB, CE, CC Configurations and Characteristics.

Text Books:

1. Basic Electrical Engineering by *M.S.Naidu and S.Kamakshaiah* TMH
2. Electronic Devices and circuits by *J.Millman, C.C.Halkias and Satyabrata Jit* 2ed.,

Reference Books:

1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, "Basic Electrical, Electronics and Computer Engineering", Tata McGraw Hill, Second Edition, (2006).
2. Nagsarkar T K and Sukhija M S, "Basics of Electrical Engineering", Oxford press (2005).
3. Mehta V K, "Principles of Electronics", S.Chand & Company Ltd, (1994).
4. Mahmood Nahvi and Joseph A. Edminister, "Electric Circuits", Schaum' Outline Series, McGraw Hill, (2002).

ELECTRONIC DEVICES AND APPLICATIONS

I B.TECH – II SEMESTER: IT/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EC03	ESC	2	0	0	2	40	60	100

COURSE OBJECTIVES:

1. To introduce components such as diodes, BJTs and FETs.
2. To know the applications of devices.
3. To know the switching characteristics of devices.

COURSE OUTCOMES:

Upon completion of the Course, the students will be able to:

1. Acquire the knowledge of various electronic devices and their use on real life.
2. Know the applications of various devices.
3. Acquire the knowledge about the role of special purpose devices and their applications.

UNIT-I

Diodes: Diode- Static and Dynamic resistances, Equivalent circuit, Diffusion and Transition Capacitances, V-I Characteristics, Diode as a switch-switching times.

UNIT-II

Diode Applications: Rectifier - Half Wave Rectifier, Full Wave Rectifier, Bridge Rectifier, Rectifiers with Capacitive and Inductive Filters, Clippers-Clipping at two independent levels, Types of Clampers.

UNIT-III

Bipolar Junction Transistor (BJT): Principle of Operation, Common Emitter, Common Base and Common Collector Configurations, Transistor as a switch, Need for Biasing, BJT as Amplifier.

UNIT-IV

Junction Field Effect Transistor (FET): Construction, Principle of Operation, Pinch-Off Voltage, Volt-Ampere Characteristic, Comparison of BJT and FET, FET as Voltage Variable Resistor, FET as amplifier.

UNIT-V

Special Purpose Devices: Zener Diode-Characteristics, Zener diode as Voltage Regulator, Operation of - SCR, UJT, Photo diode, Solar cell, LED.

TEXTBOOKS:

1. Jacob Millman-Electronic Devices and Circuits, McGraw Hill Education
2. Robert L. Boylestad, Louis Nashelsky-Electronic Devices and Circuits theory, 11th Edition, 2009, Pearson.

REFERENCE BOOKS:

1. Horowitz-Electronic Devices and Circuits, David A. Bell-5th Edition, Oxford.
2. Chinmoy Saha, Arindam Halder, Debaati Ganguly – Basic Electronics- principles and Applications, Cambridge, 2018.

ENGLISH LANGUAGE AND COMMUNICATION SKILLS

I B.TECH – II SEMESTER: IT

CourseCode	Category	Hours /			Credits	MaximumMarks		
		L	T	P		CIE	SEE	Total
A6HS02	HSMC	0	0	3	1.5	40	60	100

COURSE OBJECTIVES:

The course should enable the students:

1. To facilitate computer-assisted multimedia instruction enabling individualized and independent language learning.
2. To sensitize the students to the nuances of English speech sounds, word accent, intonation and rhythm
3. To bring about a consistent accent and intelligibility in students' pronunciation of English by providing an opportunity for practice in speaking.
4. To improve the fluency of students in spoken English and neutralize the impact of dialects.
5. To train students to use language appropriately for public speaking, group discussions and interviews.

COURSE OUTCOMES:

Students will be able to:

1. Acquire the skill of independent language learning
2. Overcome with the nuances of English speech sounds, word accent, intonation and rhythm.
3. Obtain the consistent accent and intelligibility in pronunciation.
4. Neutralize the impact of dialects.
Apply language appropriately public speaking, group discussions and interviews

English Language and Communication Skills Lab (ELCS) shall have two parts:

a. Computer Assisted Language Learning(CALL)Lab

b. Interactive Communication Skills (ICS)Lab

Listening Skills

Objectives

1. To enable students develop their listening skills to appreciate its role in the LSRW skills approach to language and improve their pronunciation.
2. To equip students with necessary training in listening so that they can comprehend the speech of people of different backgrounds and regions.

Students should be given practice in listening to the sounds of the language, to be able to recognize them and find the distinction between different sounds, to be able to mark stress and recognize and use the right into nation in sentences.

- Listening for general content
- Listening to fill up information
- Intensive listening
- Listening for specific information

Speaking Skills

Objectives

1. To involve students in speaking activities in various contexts
2. To enable students express themselves fluently and appropriately in social and professional contexts
 - Oral practice
 - Describing objects/situations/people
 - Role play–Individual/Group activities
 - Just A Minute(JAM) Sessions

Exercise-I

CALLLab:

Understand: Listening Skill- Its importance – Purpose- Process- Types- Barriers- Effective Listening.

Practice: Introduction to Phonetics – Speech Sounds – Vowels and Consonants – Minimal Pairs- Consonant Clusters- Past Tense Marker and Plural Marker-*Testing Exercises*

ICSLab:

Understand: Spoken vs. Written language-Formal and Informal English.

Practice: Ice – Breaking Activity and JAM Session-Situational Dialogues –Greetings–Taking Leave–Introducing One self and Others.

Exercise-II

CALLLab:

Understand: Structure of Syllables – Word Stress– Weak Forms and Strong Forms – Stress pattern in sentences– Intonation.

Practice: Basic Rules of Word Accent - Stress Shift - Weak Forms and Strong Forms- Stress pattern in sentences – Intonation -*Testing Exercises*

ICSLab:

Understand: Features of Good Conversation–Strategies for Effective Communication.

Practice: Situational Dialogues – Role Play- Expressions in Various Situations –Making Requests and Seeking Permissions –Telephone Etiquette.

Exercise-III

CALLLab:

Understand: Errors in Pronunciation-Neutralising Mother Tongue Interference (MTI).

Practice: Common Indian Variants in Pronunciation – Differences between British and AmericanPronunciation-TestingExercises

ICSLab:

*Understand:*Descriptions-Narrations-Giving DirectionsandGuidelines–BlogWriting

Practice: Giving Instructions – Seeking Clarifications – Asking for and Giving Directions – ThankingandResponding–AgreeingandDisagreeing –SeekingandGivingAdvice–MakingSuggestions.

Exercise-IV

CALLLab:

Understand: Listening forGeneral Details.

Practice: Listening ComprehensionTests-Testing Exercises

ICSLab:

Understand: Public Speaking – Exposure to Structured Talks - Non-verbal Communication- Presentation Skills.

Practice: Making a Short Speech–Extempore-Making a Presentation.

Exercise-V

CALLLab:

Understand: Listening for Specific Details.

Practice: Listening Comprehension Tests-Testing Exercises

ICSLab:

Understand: Group Discussion

Practice: Group Discussion

Reference Books:

1. (2022). *English Language Communication Skills – Lab Manual cum Workbook*. Cengage Learning India Pvt. Ltd.
2. Shobha, KN & Rayen, J. Lourdes. (2019). *Communicative English – A workbook*. Cambridge University Press
3. Kumar, Sanjay & Lata, Pushp. (2019). *Communication Skills: A Workbook*. Oxford University Press
4. Board of Editors. (2016). *ELCSLab Manual: A Workbook for CALL and ICSLab Activities*. Orient Black Swan Pvt. Ltd.
5. Mishra, Veerendra et al. (2020). *English Language Skills: A Practical Approach*. Cambridge University Press

Websites:

<https://www.britishcouncil.org>

<https://www.bbc.co.uk>

<https://www.grammarly.com>

<https://www.fluentu.com>

<https://www.cambridgeenglish.org/exams-and-tests/business-preliminary>

<https://www.cambridgeenglish.org/exams-and-tests/business-vantage>

PYTHON PROGRAMMING LAB

I B. TECH- II SEMESTER: IT/CSE

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS04	ESC	0	0	3	1.5	40	60	100

COURSE OBJECTIVES:

1. To understand the problem solving approaches.
2. To learn the basic programming constructs in Python.
3. To practice various computing strategies for Python-based solutions to real world problems.
4. To use Python data structures – lists, tuples, dictionaries.

To do input/output with files in Python

COURSE OUTCOMES:

1. Write, test, and debug simple Python programs.

LIST OF PROGRAMS

WEEK -1

- a) Write a program to perform different Arithmetic Operations on numbers in Python
- b) Write a Python program which accepts the radius of a circle from the user and compute the area.
- c) Write a Python program to get the Python version you are using.

WEEK -2

- a) Write a Python program to convert temperatures to and from Celsius, Fahrenheit.
[Formula: $c/5 = f-32/9$]
- b) Write a python script to print the current date in the following format "Sun May 29 02:26:23 IST 2017"
- c) A library charges a fine for every book returned late. For first 6 days the fine is 50 paise, for 10-15 days fine is one rupee and above 15 days fine is 5 rupees. If you return the book after 30 days your membership

WEEK -3

- a) Write a python function to find largest of three numbers.
- b) Write a Python function that prints prime numbers in between 50 and 100.
- c) Write a python program to find factorial of a number using Recursion.

WEEK -4

Write a program to demonstrate working with Strings and string operations.

WEEK -5

Write a program to demonstrate working with dictionaries in python

WEEK -6

Write a program to demonstrate working with tuples and List in python.

WEEK -7

a) Write a script named hellow.py. This script should prompt the user for the names of two text files. The contents of the first file should be input and written to the second file.

b) Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order

WEEK -8

Write python programs to demonstrate class & object, static and instance method implementation.

WEEK -9

Write python programs to demonstrate Inheritance and Polymorphism

WEEK -10

Write python programs to demonstrate Exception Handling in python

WEEK -11

Write python programs to demonstrate NumPy library and supporting functions.

WEEK -12 SORTING

- a) Draw an Olympic Symbol in Python using Turtle
- b) Develop a simple login page using GUI Tkinter

TEXT BOOKS:

1. Allen B. Downey, Think Python : How to Think like a Computer Scientist, 2nd Edition, Oâ€™Reilly Publishers, 2016.
2. Karl Beecher, Computational Thinking: A Beginners Guide to Problem Solving and Programming, 1st Edition, BCS Learning and Development Limited, 2017.

REFERENCE BOOKS:

1. Paul Deitel and Harvey Deitel, Python for Programmers, Pearson Education, 1st Edition, 2021.
2. G Venkatesh and Madhavan Mukund, Computational Thinking: A Primer for Programmers and Data Scientists, 1st Edition, Notion Press, 2021.
3. John V Guttag, Introduction to Computation and Programming Using Python: With Applications to Computational Modeling and Understanding Data, Third Edition, MIT Press, 2021
4. Eric Matthes, Python Crash Course, A Hands – on Project Based Introduction to Programming, 2nd

INTRODUCTION TO INTERNET OF THINGS

I B.Tech - II Semester: CSE/IT/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EC04	ESC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

COURSE OBJECTIVES:

1. To develop basic programming skills through graphical programming
2. To learn hardware interfacing and debugging techniques

COURSE OUTCOMES:

At the end of the course, student will be able to the algorithms for simple problems

CO 1: Able to demonstrate various sensor interfacing using Visual Programming Language.

CO 2: Able to analyze various Physical Components.

CO 3: Able to demonstrate Wireless Control of Remote Devices.

CO 4: Able to design and develop Mobile Application which can interact with Sensors

Introduction to IOT

1. Introduction to basic electronic components and digital electronic
2. Introduction to sensors and Actuators
3. Introduction to microcontroller

LIST OF EXPERIMENTS

Week-1	1. Blinking of LED with different delays
Week-2	Digital I/O Interface [IR Sensor, PIR Sensor]
Week-3	Analog Interface [ADC, Temperature Sensor]
Week-4	Motor speed And Direction control
Week-5	Serial Communication
Week-6	Wireless Interface –Bluetooth & Wi-Fi Technologies
Week-7	Wireless Control of wheeled robot
Week-8	Smart Home Android App Development

References:

1. Sylvia Libow Martinez, Gary S Stager, Invent To Learn: Making, Tinkering, and Engineering in the Classroom, Constructing Modern Knowledge Press, 2016
2. Michael Margolis, Arduino Cookbook, Oreilly, 2011

ENVIRONMENTAL SCIENCE

I B.TECH - II SEMESTER: Common for all Branches

CourseCode:	Category	Hours/Week			Credits	MaximumMarks		
		L	T	P	C	CIE	SEE	Total
A6BS11	MC	3	0	0	0	40	60	100

COURSE OUTCOMES:

On Successful completion of this course, Students will be able to

1. Identify the consequences of human actions on the web of life, global economy and quality of human life.
2. Evaluate the strategies for scientific, social, economic and legal environmental protection.
3. Study the impact of conservation of biodiversity.
4. Analyze the reasons for environmental pollution.
5. Assess the environmental impact of air, water, biological and socio-economical aspects and risk assessment towards sustainable future.

UNIT-I ECOSYSTEMS

CLASSES:7

Eco systems: Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food web and ecological pyramids.

Flow of energy, Bio geochemical cycles, Bio accumulation, Bio magnification, eco system value, services and carrying capacity.

UNIT-II NATURAL RESOURCES

CLASSES:8

Natural Resources: Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems-case studies. **Mineral resources:** use and exploitation, environmental effects of extracting and using mineral resources. **Land resources:** Forest resources.

Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.

UNIT-III BIODIVERSITY AND BIOTIC RESOURCES

CLASSES:5

Bio diversity and Biotic Resources: Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values.

India as a mega diversity nation: Hot spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation. National Biodiversity Act-Case studies.

UNIT-IV ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES

CLASSES:5

Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Air Pollution, Water Pollution drinking water quality standards. Soil Pollution Impacts of modern agriculture, Noise Pollution Health hazards, standards. Concepts of bioremediation.

Global Environmental Problems and Global Efforts: Ozone depletion and Ozone depleting substances (ODS) Concepts of Bioremediation International conventions / Protocols: Earth summit, Kyoto protocol and Montréal Protocol; NAPCC-GoI Initiatives, COP 24, COP25.

UNIT-V ENVIRONMENTAL POLICY, LEGISLATION & EIA

CLASSES:5

Environmental Policy, Legislation & Environmental Impact Assessment (EIA): Environmental Protection act, Legal aspects Air Act1981, Water Act, Forest Act, Wild life Act, Municipal solid waste management, biomedical waste management hazardous waste management and handling rules.

Environmental Impact Assessment: EIA structure, methods of baseline data acquisition. Strategies for risk assessment, Towards Sustainable Future: Concept of Sustainable Development, Urban Sprawl, Concept of Green Building.

TEXT BOOKS:

1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission. Environmental Studies by R.Rajagopalan, Oxford University Press.

II B.TECH I SEMESTER SYLLABUS

COMPUTER ORIENTED STATISTICAL METHODS

II B.TECH – I SEMESTER: CSE/IT/CSD/CSC/CSI

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS03	BSC	L	T	P	C	CIA	SEE	Total
		3	1	0	4	40	60	100

COURSE OBJECTIVES:

1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments.
2. The concept of correlation and regression, covariance and sampling distribution
3. Evaluation of the given data for appropriate test of hypothesis for large samples.
4. Evaluation of the given data for appropriate test of hypothesis for small samples and one way ANOVA
5. To learn the concept of Markov chain, transition probabilities in discrete & continuous time and Stochastic simulation techniques.

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments.
2. Apply the concept of correlation and regression to find covariance and Sampling distribution
3. Evaluate the given data for appropriate test of hypothesis for Large samples.
4. Evaluate the given data for appropriate test of hypothesis for small samples and one way ANOVA
5. Recognize if a given stochastic system with finite number of states is a Markov chain or not and also identify classes of states in Markov chains and Characterize the states.

UNIT-I RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

Classes :8

Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation.

Binomial, Poisson, Normal distributions -their Properties. Moments about origin, central moments, skewness, Kurtosis and find the mean and variance.

UNIT-II CORRELATION & REGRESSION AND SAMPLING DISTRIBUTIONS

Classes:9

Coefficient of correlation, the rank correlation, Covariance of two random variables. Regression- Regression Coefficient, The lines of regression.

Sampling: Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance. Parameter estimation- Point estimation and interval estimation

UNIT-III TESTING OF HYPOTHESIS - I

Classes:9

Testing of hypothesis: Null hypothesis, Alternate hypothesis, Type I & Type II errors – critical region, confidence interval, and Level of significance. One sided test, Two sided test.

Large sample tests:

- (i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances)
- (ii) Tests of significance of difference between sample S.D and population S.D.
- (iii) Tests of significance difference between sample proportion and population proportion, difference between

two sample proportions.

UNIT-IV TESTING OF HYPOTHESIS-II**Classes: 8**

Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit. One way ANOVA

UNIT-V STOCHASTIC PROCESSES AND MARKOV CHAINS**Classes: 8**

Introduction to Stochastic processes- Markov process. Transition Probability, Transition Probability Matrix, First order and Higher order Markov process, n-step transition probabilities, Markov chain, Steady state condition, Markov analysis.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.
2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall.
3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning

Reference Books:

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. Fundamentals of Mathematical Statistics by S.C. Gupta & V.K. Kapoor, S. Chand
3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press

Web references:

1. https://www.efunda.com/math/math_home/math.cfm
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

E -Text Books:

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

DISCRETE MATHEMATICS

II B. TECH- I SEMESTER: IT/CSIT/CSE

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS08	BSC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To help students to analyze and examine the validity of argument by using propositional and predicate calculus
2. To impart basics of relations and functions
3. To facilitate students in applying sets relations and digraphs to solve applied problems
4. To help students to solve the given recurrence relation using different methods such as substitution, Generating function and characteristics roots equation.
5. To acquire knowledge in graph theory Concepts and Apply those concepts in related theoretical problems

COURSE OUTCOMES

At the end of the course, student will be able to

1. Analyze and examine the validity of argument by using propositional and predicate calculus
2. Apply sets relations and digraphs to solve applied problems
3. Apply basic counting techniques to solve the combinatorial problems
4. Formulate problems and solve recurrence relations.
5. Model and solve real-world problems using graphs and trees

UNIT - I MATHEMATICAL LOGIC

CLASSES: 15

Statements and notations, Connectives, Well formed formulas, Truth Tables, Tautology, Equivalence implication, Normal forms, Logical Inference, Rules of inference, Direct Method, Direct Method using CP(Conditional Proof), Consistency, Proof of contradiction, Automatic Theorem Proving.

UNIT - II RELATIONS

CLASSES: 11

Introduction to set theory, Relations, Properties of Binary Relations, Equivalence Relation, Transitive closure, Compatibility and Partial ordering relations, Lattices, Hasse diagram. Functions: inverse Function, Composition of functions. Algebraic Systems, Semi groups and Monoids

UNIT - III ELEMENTARY COMBINATORICS

CLASSES: 12

Basis of counting, Combinations & Permutations, Enumeration of Combinations and Permutations, Enumeration of Combinations and Permutations With repetitions, Enumerating Permutations with Constrained repetitions, Binomial Coefficients, Binomial and Multinomial theorems, The principles of Inclusion – Exclusion, Pigeon- hole principles and its applications.

UNIT - IV RECURRENCE RELATION

CLASSES: 11

Generating Functions, Function of Sequences, Calculating Coefficient of generating function, Recurrence relations, Solving recurrence relation by substitution and Generating functions, The method of Characteristics roots

UNIT - V GRAPHS

CLASSES: 10

Basic Concepts, Isomorphism and Subgraphs, Trees and their properties, Spanning Trees- DFS,BFS, Minimal Spanning Trees- Prim's, Kruskal's Algorithm, Planar Graphs, Euler's Formula, Multi graph and Euler circuits, Hamiltonian Graphs, Chromatic number.

TEXT BOOKS

1. Discrete Mathematics for computer scientists & Mathematicians, *J.L. Mott, A. Kandel, T.P. Baker* PHI
2. Discrete Mathematical Structures With Applications to Computer Science, JP Tremblay, R Manohar

REFERENCE BOOKS

1. Logic and Discrete Mathematics, *Grass Man & Trembley*, Pearson Education. Joyner

OBJECT ORIENTED PROGRAMMING THROUGH JAVA

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT02	ESC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the basic object oriented programming concepts and apply them in problem solving.
2. Illustrate inheritance and polymorphism concepts for reusing the program.
3. Demonstrate on the exception handling mechanism
4. Develop multi-threading and data-centric applications using JDBC.
5. Understand the basics of java collection framework

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Understand the basic programming concepts and apply them in problem solving
2. Implement object-oriented principles, such as encapsulation, inheritance, polymorphism, and abstraction.
3. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally).
4. Develop java application to interact with database by using relevant software component (JDBC Driver).
5. Build real world applications using Collection framework

UNIT-I JAVA BASICS

Classes: 12

JAVA BASICS: Review of Object oriented concepts, History of Java, Java buzzwords, JVM architecture, Data types, Variables, Scope and life time of variables, arrays, operators, control statements, type conversion and casting, simple java program, constructors, methods, Static block, Static Data, Static Method, String and String Buffer Classes, Using Java API Document.

UNIT-II INHERITANCE, POLYMORPHISM, PACKAGES AND INTERFACES

Classes: 11

INHERITANCE AND POLYMORPHISM: Basic concepts, Types of inheritance, Member access rules, Usage of this and Super key word, Method Overloading, Method overriding, Abstract classes, Encapsulation, Need for encapsulation in java, Data hiding vs Encapsulation, getter and setter methods, Dynamic method dispatch, Usage of final keyword.

PACKAGES AND INTERFACES: Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces

UNIT-III EXCEPTION HANDLING AND FILES

Classes: 10

EXCEPTION HANDLING: Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes.

I / O STREAMS AND FILES: Concepts of streams, Stream classes- Byte and Character stream, Reading consoleInput and Writing Console output, File Handling

UNIT-IV MULTITHREADING AND JDBC

Classes: 10

MULTI THREADING: Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication.

JDBC-Connecting to Database - JDBC Type 1 to 4 drives, connecting to a database, querying a Database and processing the results, updating data with JDBC

UNIT-V COLLECTION FRAMEWORK

Classes: 10

COLLECTION FRAMEWORK: Introduction to Java Collections, Overview of Java Collection frame work, Generics, Commonly used Collection classes- Array List, Vector, Hash table, Stack, Enumeration, Iterator, String Tokenizer, Random, Scanner, calender and Properties

Text Books:

1. Herbert Schildt and Dale Skrien, *Java Fundamentals – A comprehensive Introduction*, McGraw Hill, 1st Edition, 2013.
2. Herbert Schildt, *Java the complete referencell*, McGraw Hill, Osborne, 7th Edition, 2011.
3. T.Budd, *Understanding Object- Oriented Programming with Javall*, Pearson Education, Updated Edition (New Java 2 Coverage), 1999.

Reference Books:

1. P.J.Dietel and H.M.Dietel , *Java How to program*, Prentice Hall, 6th Edition, 2005.
2. P.Radha Krishna , *Object Oriented programming through Javall*, CRC Press, 1st Edition, 2007.
3. S.Malhotra and S. Choudhary, *Programming in Javall*, Oxford University Press, 2nd Edition, 2014.

DATABASE MANAGEMENT SYSTEMS

II B.Tech I Semester: IT/CSE/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS09	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the concepts of database management systems and ER Diagrams.
2. Gain knowledge on relational model and integrity constraints.
3. Understand SQL queries and Normalization of relations.
4. Understand with the basics of transaction processing and concurrency control.
5. Understand the database storage structures and access techniques

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Recognize the basic concepts and the applications of database systems.
2. Design ER-models to represent simple database application scenarios and convert ER-Model to Relational Model.
3. Demonstrate SQL queries and apply Normalization techniques.
4. Be acquainted with the basics of transaction processing and concurrency control.
5. Familiarity with database storage structures and access techniques

UNIT-I INTRODUCTION TO DATABASE AND DATABASE DESIGN

Classes: 9

Database System Applications: A Historical Perspective, File Systems versus a DBMS, the Data Model, Levels of Abstraction in a DBMS, Data Independence, Structure of a DBMS

Introduction to Database Design: Database Design and ER Diagrams, Entities, Attributes, and Entity Sets, Relationships and Relationship Sets, Additional Features of the ER Model, Conceptual Design With the ER Model

UNIT-II RELATIONALMODEL , RELATIONAL ALGEBRA AND CALCULUS

Classes: 9

Introduction to the Relational Model: Integrity constraint over relations, enforcing integrity constraints, querying relational data, logical database design, introduction to views, destroying/altering tables and views.

Relational Algebra, Tuple relational Calculus, Domain relational calculus.

UNIT-III INTRODUCTION TO SQL & SCHEMA REFINEMENT

Classes: 14

SQL: QUERIES, CONSTRAINTS, TRIGGERS: form of basic SQL query, UNION, INTERSECT, and EXCEPT, Nested Queries, aggregation operators, NULL values, complex integrity constraints in SQL, triggers and active databases.

Schema Refinement: Problems caused by redundancy, decompositions, problems related to decomposition, reasoning about functional dependencies, First, Second, Third normal forms, BCNF, lossless join decomposition, multivalued dependencies, Fourth normal form, Fifth normal form

UNIT-IV TRANSACTION MANAGEMENT AND CONCURRENCY CONTROL

Classes: 9

Transaction Concept, Transaction State, Implementation of Atomicity and Durability, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for serializability, Lock Based Protocols, Timestamp Based Protocols, Validation- Based Protocols, Multiple Granularity, Recovery and Atomicity, Log–

Based Recovery, Recovery with Concurrent Transactions.

UNIT-V OVERVIEW OF STORAGE AND INDEXING. PL/SQL.

Classes: 9

Data on External Storage, File Organization and Indexing, Cluster Indexes, Primary and Secondary Indexes, Index data Structures, Hash Based Indexing, Tree based Indexing, Comparison of File Organizations, Indexes- Intuitions for tree Indexes, Indexed Sequential Access Methods (ISAM),

B+ Trees: A Dynamic Index Structure.

Text Books:

1. Database System Concepts, Silberschatz, Korth, McGraw hill, V edition.3rd Edition
2. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Tata Mc Graw Hill

Reference Books:

1. Database Systems design, Implementation, and Management, Peter Rob & Carlos Coronel 7th Edition.
2. Fundamentals of Database Systems, Elmasri Navrate, Pearson Education
3. Introduction to Database Systems, C. J. Date, Pearson Education
4. Oracle for Professionals, The X Team, S.Shah and V. Shah, SPD.
5. Database Systems Using Oracle: A Simplified guide to SQL and PL/SQL, Shah, PHI.
6. Fundamentals of Database Management Systems, M. L. Gillenson, Wiley Student Edition

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

II B. TECH – I SEMESTER: IT

Course Code	Category	Hours /Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6HS08	HSMC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.

COURSE OUTCOMES:

At the end of the course, the student will

1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures.
2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis.
3. Analyze how capital budgeting decisions are carried out.
4. Understanding the framework for both manual and computerized accounting process
5. Know how to analyze and interpret the financial statements through ratio analysis.

UNIT-I Introduction & Demand Analysis

CLASSES: 10

Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.

UNIT-II Production & Cost Analysis: Production Function

CLASSES: 10

Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance

UNIT-III Markets & New Economic Environment

CLASSES: 12

Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario

UNIT-IV Capital Budgeting

CLASSES: 14

Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).

UNIT-V Introduction to Financial Accounting & Financial Analysis**CLASSES: 14**

Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, and Trial Balance - Final Accounts (Tracing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.

TEXTBOOKS

1. Varshney&Maheswari: Managerial Economics, Sultan Chand, 2009.
2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013.
3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.

REFERENCEBOOKS

1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012.
2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012.
3. Lipsey&Chrystel, Economics, Oxford University Press, 2012.
4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012.
5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012.
6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012.
7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012.
8. Dwivedi: Managerial Economics, Vikas, 2012.
9. Shailaja&Usha: MEFA, University Press, 2012.
10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012.
11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011.
12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.

OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB

II B.Tech - I Semester –IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT03	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100

COURSE OUTCOMES:

1. Implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity.
2. Understand the use of different exception handling mechanisms and concept of multithreading for robust and efficient application development.
3. Understand and implement concepts on file streams and operations in java programming for a given application programs.
4. Develop java application to interact with database by using relevant software component (JDBC Driver).

LIST OF EXPERIMENTS

WEEK-1 JAVABASICS

- a. Write a java program that prints all real solutions to the quadratic equation $ax^2+bx+c=0$. Read in a, b, c and use the quadratic formula.
- b. The Fibonacci sequence is defined by the following rule. The first two values in the sequence are 1 and 1. Every subsequent value is the sum of the two values preceding it. Write a java program that uses both recursive and non-recursive functions.

WEEK-2 ARRAYS

- a. Write a java program to sort given list of integers in ascending order.
- b. Write a java program to multiply two given matrices

WEEK-3 STRINGS

- a. Write a java program to check whether a given string is palindrome.
- b. a. Write a java program for sorting a given list of names in ascending order.

WEEK-4 OVERLOADING & OVERRIDING

- a. Write a java program to implement method overloading and constructors overloading.
- b. Write a java program to implement method overriding.

WEEK-5 INHERITANCE

- a. Write a java program to create an abstract class named Shape that contains two integers and an empty method named print Area (). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method print Area () that prints the area of the given shape.

WEEK-6 INTERFACES

- a. Write a program to create interface A in this interface we have two method meth1 and meth2. Implements this interface in another class named MyClass.
- b. Write a program to give example for multiple inheritance in Java.

WEEK-7 EXCEPTION HANDLING

- a. Write a program that reads two numbers Num1 and Num2. If Num1 and Num2 were not integers, the program would throw a Number Format Exception. If Num2 were zero, the

program would throw an Arithmetic Exception Display the exception..

WEEK-8 I/O STREAMS

- a. Write a java program that reads a file name from the user, and then displays information about whether the file exists, whether the file is readable, whether the file is writable, the type of file and the length of the file in bytes.
- b. Write a java program that displays the number of characters, lines and words in a text file.

WEEK-9 MULTI THREADING

- a. Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number

WEEK-10 GENERICS

- a. Write a Java program to swap two different types of data using Generics.
- b. Write a Java program to find maximum and minimum of two different types of data using Generics

WEEK-11 COLLECTIONS

Create a linked list of elements.

- a. Delete a given element from the above list.
- b. Display the contents of the list after deletion

WEEK-12 CONNECTING TO DATABASE

- a. Write a java program that connects to a database using JDBC and does add, delete, modify and retrieve operations.

TEXT BOOKS:

1. P. J. Deitel, H. M. Deitel, "Java for Programmers", Pearson Education, PHI, 4th Edition, 2007.
2. P. Radha Krishna, "Object Oriented Programming through Java", Universities Press, 2nd Edition, 2007
3. Bruce Eckel, "Thinking in Java", Pearson Education, 4th Edition, 2006.
4. Sachin Malhotra, Saurabh Chaudhary, "Programming in Java", Oxford University Press, 5th Edition, 2010.

DATABASE MANAGEMENT SYSTEMS LAB

II B.Tech - I Semester: CSE/IT/CSIT

Course Code	Category	Hours / Week			Credit	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS10	PCC	0	0	3	1.5	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Apply the basic concepts of Database Systems and Applications.
2. Use the basics of SQL and construct queries using SQL in database creation and interaction
3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
4. Analyze and Select storage and recovery techniques of database system.

COURSE OUTCOMES:

The course should enable the students to:

1. Apply the basic concepts of Database Systems and Applications.
2. Develop an ER model for a given database.
3. Use the basics of SQL and construct queries using SQL in database creation and interaction.
4. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
5. Analyze and Select storage and recovery techniques of database system.

LIST OF EXPERIMENTS

Week-1 DDL Commands

- a. Creation of Tables using SQL- Overview of using SQL tool and Data types in SQL
- b. Altering Tables and
- c. Dropping Tables

Week-2 Create Table with Primary key and Foreign Key& DML Commands

- Creating Tables (along with Primary and Foreign keys), Practicing DML commands-
 - a. Insert,
 - b. Update
 - c. Delete.

Week-3 Selection Queries

- Practicing Select command using following operations
 - a. AND, OR
 - b. ORDER BY
 - c. BETWEEN
 - d. LIKE

Week-4 AGGREGATE FUNCTIONS and Views

- Practice Queries using following functions
 - a. COUNT,
 - b. SUM,
 - c. AVG,
 - d. MAX,
 - e. MIN,
- Apply constraint on aggregation using
 - a. GROUP BY,
 - b. HAVING,
- VIEWS Create , Modify and Drop

Week-5 Nested QUERIES

- Practicing Nested Queries using
 - a. UNION,
 - b. INTERSECT,
 - c. CONSTRAINTS
 - d. IN

Week-6 CO- RELATED NESTED QUERIES

- Practicing Co – Related Nested Queries using
 - a. EXISTS,
 - b. NOT EXISTS. ANY, ALL
 -

Week-7 JOIN QUERIES

- Practicing Join Queries using
 - a. Inner join
 - b. Outer join
 - c. Equi join

Week- TRIGGERS

- Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.

Week-9 PROCEDURES

- Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure

Week-10 CURSORS

- Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.

Week-11 PL/SQL Part 1

- Practice PL/SQL –
 - a. block structure,
 - b. variables,
 - c. data types,

Week-12 PL/SQL Part 2

- Practice PL/SQL –
 - a. operators,
 - b. control structures;
- aseca

Case study 1: College Management

Case study 2 : An Enterprise/Organization

Case study 3 : Library Management system

Case study 4: Sailors and shipment system

Reference Books:

1. Database System Concepts, by Silberschatz, Sudarshan, and Korth, 6th edition.
2. Database management System by RaghuRamaKrishna, 3rd edition

Web References:

1. <http://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database>
2. https://www.w3schools.com/sql/sql_create_table.asp
3. http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm_export=print
4. <http://ssyu.im.ncnu.edu.tw/course/CSDB/chap14.pdf>
5. <http://web.cs.ucdavis.edu/~green/courses/ecs165a-w11/8-query.pdf>

SKILL DEVELOPMENT COURSE (HTML/CSS/JAVASCRIPT/BOOTSTRAP)

II B.Tech - I Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT04	PCC	0	0	2	1	-	100	100

COURSE OBJECTIVES:

The course should enable the students to:

1. To familiarize with the syntax and semantics of HTML
2. To learn the usage of CSS for styling web pages.
3. To use Javascript for creating dynamic and interactive web content like applications and browsers.
4. To use bootstrap for enabling responsive development of mobile-first websites
5. To develop a complete website by using HTML, CSS, JavaScript and bootstrap with some case studies.

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Apply basic HTML for creating websites.
2. Apply CSS for styling web pages.
3. Demonstrate the usage of JavaScript for creating dynamic and interactive web content
4. Use Bootstrap for front-end development of creating websites and web apps.
5. Develop an interactive website using, HTML, CSS, JavaScript and bootstrap.

UNIT-I

Classes: 6

Introduction to HTML, Basic Formatting Tags, Grouping using DIV-Span, Headers, List, Images, Hyperlink, Table, iframes, miscellaneous Tags.

UNIT-II

Classes: 6

CSS Introduction, Types of CSS, Syntax-Selectors, Font, Box Model, Display Positioning, Background cursor, List Tables, Audio- video file creation

UNIT-III

Classes: 6

Introduction to Java Scripts, data types and variables, Operators, Statements, functions, Objects in Java Script, event handling, forms with Javascript, DOM, Dynamic HTML with Java Script.

UNIT-IV

Classes: 6

Introduction to Bootstrap, Syntax, Container, Bootstrap component, Advanced Component, Utilities

UNIT-V

Classes: 6

Case study 1: Create a Dynamic Web page for College Library Management System

Case Study 2: Create a website using animation

Text Books:

1. Web Programming, building internet applications, Chris Bates 2nd edition, WILEY Dreamtech (Unit 1,2,3,5)
2. Learning Bootstrap, Aravind Shenoy, Ulrich Sossou, December 2014, Packt Publishing, ISBN: 9781782161844 (Unit 4,5)

GENDER SENSITIZATION

II B. TECH - I SEMESTER: IT/CSIT

Course Code	Category	Hours /Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6HS05	MC	0	0	2	0	50	-	50

COURSE OBJECTIVES

1. To provide a critical perspective on the socialization of men and women.
2. To introduce students to information about some key biological aspects of genders.
3. To expose the students to debates on the politics and economics of work.
4. To help students reflect critically on gender violence.
5. To expose students to more egalitarian interactions between men and women.

COURSE OUTCOMES

1. Develop a better understanding of important issues related to gender in contemporary India.
2. Sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender
3. Attain a finer grasp of how gender discrimination works in our society and how to counter it
4. Men and Women students and professionals will be better equipped to work and live together as equals.
5. Create a sense of appreciation of women in all walks of life.

UNIT-I UNDERSTANDING GENDER

CLASSES: 03

Introduction: Introduction to Gender, What is Gender, Why should we study it. Socialization: Making Women, Making Men-Preparing for Woman hood. Growing up Male. First lessons in Caste: Different Masculinities.

UNIT-II GENDER ROLES AND RELATIONS

CLASSES: 03

Two or Many-Struggles with Discrimination-Missing Women-Sex Selection and Its Consequences
Declining Sex Ratio. Demographic Consequences-Gender Spectrum: Beyond the Binary

UNIT-III GENDER AND LABOUR

CLASSES: 03

Devaluation of Labour-Housework: The Invisible Labor- "My Mother doesn't Work." "Share the Load."-Work
Its Politics and Economics-Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading:
Wages and Conditions of Work.

UNIT-IV GENDER-BASED VIOLENCE

CLASSES: 04

Sexual Harassment: Say No! -Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment-
Further Reading: "Chupulu". Domestic Violence: Speaking Out Is Home a Safe Place? -When Women
Unite [Film]. Rebuilding Lives. Thinking about Sexual Violence Blaming the Victim-"I Fought for my
Life...." Additional Reading: The Caste Face of Violence.

UNIT-V GENDER AND COEXISTENCE

CLASSES: 03

Gender Issues- Just Relationships: Being Together as Equals Mary Kom and Onler. Love and Acid just do
not Mix. Love Letters. Mothers and Fathers. Rosa Parks The Brave Heart.

TEXT BOOKS

All the five Units in the Textbook,

1. "Towards a World of Equals: A Bilingual Textbook on Gender" written by A. Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by Telugu Akademi, Hyderabad, Telangana State in the year 2015.

REFERENCE BOOKS

1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012
2. Abdulali Sohaila. "I Fought For My Life...and Won." Available online at: <http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/>

WEB REFERENCES

1. <http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/>

E-TEXTBOOKS

1. Abdul ali Sohaila. "I Fought For My Life...and Won." Available online at: <http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/>

II B.TECH II SEMESTER SYLLABUS

DIGITAL ELECTRONICS AND COMPUTER ORGANIZATION

II B. TECH- II SEMESTER: IT/CSE

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS28	ESC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand different number systems and Boolean Algebra.
2. Design of combinational and sequential logic circuits.
3. Understand different Computer Instructions and addressing modes.
4. Understand concepts of register transfer logic and arithmetic operations.
5. Learn different types of memory hierarchy.

COURSE OUTCOMES:

1. Apply minimization techniques to simplify Boolean expressions.
2. Apply the principles of Digital electronics to design combinational and sequential logic circuits.
3. Understand the basics of instruction set and their impact on processor design.
4. Illustrate register transfer operations.
5. Analyze memory hierarchy and I/O Communication.

UNIT – I NUMBER THEORY and BOOLEAN ALGEBRA

CLASSES: 14

Representation of numbers of different radix, conversion of numbers from one radix to another radix, r-1's complement and r's complement. Basic theorems and properties of Boolean algebras, Canonical and Standard forms, Digital Logic Gates, Universal Logic Gates. K- Map Method.

UNIT - II DESIGN OF COMBINATIONAL AND SEQUENTIAL LOGIC CIRCUITS

CLASSES: 12

Design of full adder, full subtractor. Decoder, Encoder, Multiplexer, De-multiplexer and comparator. Latches, Flip-Flops, Design of Ripple counters, Johnson counter, Shift registers.

UNIT - III BASIC COMPUTER ORGANIZATION AND DESIGN

CLASSES: 12

Instruction codes, Computer Registers, Computer Instructions, Timing and Control, Instruction cycle, Memory-Reference Instructions, Input-Output and interrupt. Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation.

UNIT – IV REGISTER TRANSFER AND MICRO-OPERATIONS

CLASSES: 11

Register Transfer Language, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations. Pipeline and Parallel Processing: Parallel processing, Pipelining, Arithmetic pipeline, Instruction pipeline.

UNIT – V MEMORY SYSTEM

CLASSES: 11

INPUT OUTPUT ORGANIZATION: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA.

MEMORY ORGANIZATION: Memory Hierarchy, Main memory, Auxiliary memory, Associate memory, Cache memory

TEXT BOOKS

1. Digital Design- Morris Mano, PHI, 3rd Edition.
2. Digital Principles and Applications by Leach, Paul Malvino, 5th Edition.
3. Computer System Architecture- M. Morris Mano, 3rd edition, Pearson/PHI, India.

REFERENCE BOOKS

1. Switching and Finite Automata Theory by Zvi.Kohavi, Tata McGraw Hill.
2. Fundamentals of Digital circuits, A. Anand Kumar, Third Edition, 2013, PHI.
3. Computer Organization and Architecture_William Stallings Sixth Edition, Pearson/PHI
4. Structured Computer Organization_Andrew S. Tanenbaum, 4th Edition PHI/Pearson

OPERATING SYSTEMS

II B. TECH- II SEMESTER: IT/CSE

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS11	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES

1. To explain main components of OS and their working
2. To familiarize the operations performed by OS as a resource Manager
3. To familiarize with Dead Locks
4. To impart various scheduling policies of OS
5. To teach the different memory management techniques.

COURSE OUTCOMES

1. Analyze the different structures and services of operating system.
2. Analyze various algorithms used for OS services with respect to defined/chosen criteria.
3. Solve the resource allocation and sharing problems.
4. Assess different methods to solve OS problems.
5. Analyze the memory management approaches of operating systems.

UNIT-I OPERATING SYSTEMS OVERVIEW

CLASSES: 14

OPERATING SYSTEMS OVERVIEW: Introduction, operating system operations, process management, memory management, storage management, protection and security, distributed systems.

OPERATING SYSTEMS STRUCTURES: Operating system services and systems calls, system programs, operating system structure, operating systems generations

UNIT - II PROCESS MANAGEMENT

CLASSES: 16

PROCESS MANAGEMENT: Process concepts, process state, process control block, scheduling queues, process scheduling, Scheduling algorithms, and multithreaded programming.

CONCURRENCY AND SYNCHRONIZATION: Process synchronization, critical section problem, Peterson's solution, synchronization hardware, semaphores, classic problems of synchronization, readers and writers problem, dining philosophers problem, monitors, synchronization examples(Solaris).

UNIT - III DEAD LOCKS

CLASSES: 14

DEADLOCKS: System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker's algorithm.

MEMORY MANAGEMENT: Swapping, contiguous memory allocation, paging, structure of the page table, segmentation, virtual memory, demand paging, page-replacement algorithms, allocation of frames, thrashing.

UNIT - IV FILE SYSTEM

CLASSES: 12

FILE SYSTEM: Concept of a file, access methods, directory structure, file system mounting, file sharing, protection.

File system implementation: file system structure, file system implementation, directory implementation, allocation methods, free-space management, efficiency and performance.

UNIT - V MASS STORAGE STRUCTURE

CLASSES: 14

MASS STORAGE STRUCTURE: overview of mass storage structure, disk structure, disk attachment,

disk scheduling algorithms, swap space management, stable storage implementation, tertiary storage structure.

I/O: Hardware, application I/O interface, kernel I/O subsystem, transforming I/O requests to hardware operations, streams, performance

TEXT BOOKS

1. Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne (2006).
2. Operating System Principles, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne (2006), 7th edition, Wiley India Private Limited, New Delhi.

REFERENCE BOOKS

1. Operating Systems, Internals and Design Principles, Stallings (2006), 5th edition, Pearson Education, India.
2. Modern Operating Systems, 2nd edition, Andrew S. Tanenbaum (2007), Prentice Hall of India, India
3. Operating systems, Deitel&Deitel (2008), 3rd edition, Pearson Education, India.

WEB LINKS

1. <https://nptel.ac.in/courses/106/102/106102132/>
2. https://onlinecourses.nptel.ac.in/noc21_cs88/preview

SOFTWARE ENGINEERING AND DESIGN

II B.Tech - II Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT05	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. To define software engineering process and practices, and demonstrate various process models.
2. To develop software requirements and the SRS document.
3. To make use of structural UML diagrams.
4. To make use of structural UML diagrams.
5. To learn levels of software testing and Methodologies

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Choose appropriate process model depending on the user requirements
2. Demonstrate the principles and requirements at various phases of software development
3. Model structural UML Diagrams.
4. Model behavioural UML Diagrams.
5. Distinguish different testing strategies

UNIT-I PROCESS MODELS

Classes: 9

Software Process: Process and project, Software development process models - Waterfall model, relational unified process, Time boxing model, Agile Manifesto, Extreme programming Model, Using process models in a project, Project management process.

UNIT-II FUNCTIONAL REQUIREMENTS

Classes: 9

Functional Requirements, Non-functional Requirements - Software requirement analysis and specification: Value of good SRS, requirement process, requirement specification using Data flow Diagram. Case study - ATM, Library Management System.

UNIT-III STRUCTURAL UML DIAGRAMS

Classes: 9

Introduction to UML - Structural UML diagrams - Class Diagram, Package Diagram, -Component Diagram, and Deployment Diagram. Case studies - Point of Sale System, Online Ticket Reservation System.

UNIT-IV BEHAVIOURAL UML DIAGRAMS

Classes: 9

Use-case diagram, Interaction Diagram, Activity diagram, State Machine diagram. Case studies - Point of Sale System, Online Ticket Reservation System.

UNIT-V SOFTWARE TESTING

Classes: 9

Software Testing: Testing Concepts, Levels of Testing, Testing Process, Black-Box Testing, White- Box Testing Techniques, Art of Debugging.

Text Books:

1. A Concise Introduction to Software Engineering (Undergraduate Topics in Computer Science), Pankaj Jalote, Springer International Edition, 2008. (Unit 1, 2, &5)
2. Craig Larman," Applying UML and Patterns: An Introduction to object-oriented Analysis and Design and iterative developmentII, Third Edition, Pearson Education, 2005. (Unit 3&4)

Reference Books:

1. Roger S, —Software Engineering – A Practitioner's ApproachII, seventh edition, Pressman, 2010.
2. Pearson Edu, —Software Engineering by Ian sommervilleII, 9 th edition, 2010.
3. Mike O'Docherty, —Object-Oriented Analysis & Design: Understanding System Development with

DESIGN AND ANALYSIS OF ALGORITHMS

II B. TECH- II SEMESTER: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS15	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. To demonstrate performance of algorithms with respect to time and space complexity.
2. To explain graph and tree traversals.
3. To explain the concepts greedy method and dynamic programming. Applying for several applications like knapsack problem, job sequencing with deadlines, and optimal binary search tree, TSP and so on respectively.
4. To illustrate the methods of backtracking and branch bound techniques to solve the problems like n-queens problem, graph colouring and TSP respectively.
5. To familiarize the concepts of deterministic and non-deterministic algorithms.

COURSE OUTCOMES:

At the end of this course students will be able to:

1. Identify various Time and Space complexities of various algorithms
2. Understand Tree Traversal method and Greedy Algorithms
3. Apply Dynamic Programming concept to solve various problems
4. Apply Backtracking, Branch and Bound concept to solve various problems
5. Implement different performance analysis methods for non-deterministic algorithms

UNIT-I

Classes: 9

INTRODUCTION: Algorithm, pseudo code for expressing algorithms, performance analysis-space complexity, time complexity, asymptotic notation- big (O) notation, omega notation, theta notation and little (o) notation, recurrences, probabilistic analysis, disjoint set operations, union and find algorithms.

UNIT-II

Classes: 9

DIVIDE AND CONQUER: General method, applications-analysis of binary search, quick sort, merge sort, AND OR Graphs.

GREEDY METHOD: General method, Applications-job sequencing with deadlines, Fractional knapsack problem, minimum cost spanning trees, Single source shortest path problem.

UNIT-III

Classes: 9

GRAPHS (Algorithm and Analysis): Breadth first search and traversal, Depth first search and traversal, Spanning trees, connected components and bi-connected components, Articulation points.

DYNAMIC PROGRAMMING: General method, applications - optimal binary search trees, 0/1 knapsack problem, All pairs shortest path problem, Travelling sales person problem, Reliability design.

UNIT-IV

Classes: 9

BACKTRACKING: General method, Applications- n-queen problem, Sum of subsets problem, Graph coloring and Hamiltonian cycles.

BRANCH AND BOUND: General method, applications - travelling sales person problem, 0/1 knapsack problem- LC branch and bound solution, FIFO branch and bound solution.

UNIT-V

Classes: 9

NP-HARD AND NP-COMPLETE PROBLEMS: Basic concepts, non-deterministic algorithms, NP-hard and NP-complete classes.

TEXT BOOKS

1. Ellis Horowitz, Satraj Sahni, Rajasekharam (2007), Fundamentals of Computer Algorithms, 2nd edition, University Press, New Delhi.

REFERENCE BOOKS

1. R. C. T. Lee, S. S. Tseng, R.C. Chang and T. Tsai (2006), Introduction to Design and Analysis of Algorithms A strategic approach, McGraw Hill, India.
2. Allen Weiss (2009), Data structures and Algorithm Analysis in C++, 2nd edition, Pearson education, New Delhi.
3. Aho, Ullman, Hopcroft (2009), Design and Analysis of algorithms, 2nd edition, Pearson education, New Delhi

DATA STRUCTURES THROUGH JAVA

II B.Tech - II Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT06	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. The student should be able to choose appropriate data structures, understand the performance, and use it to design algorithms for a specific problem.
2. Students should be able to understand the linear data structures and their usage.
3. Students will learn the concept of searching, sorting and hashing concepts.
4. Students will learn different types of trees and their implementation
5. Student should be able to learn the concepts of Graphs and their applications along with text processing algorithms

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Write the algorithms for various operations on list using arrays and linked list and analyze the time complexity of its operations.
2. Apply linear data structures like stack and queue in problem solving and also learn the construction of heaps.
3. Demonstrate various searching and sorting techniques and compare their computational complexities in terms of space and time and also learn the concepts of hashing techniques.
4. Construct and analyze different search trees.
5. design graphs and their traversals and also learn text processing algorithms

UNIT-I

Classes: 12

Data structures: Linear Data Structures and its operations, Non-linear data structures, List using arrays and linked list, Single linked lists - Insertion, Deletion, Search operations, Double linked lists- Insertion, Deletion, Search Operations, Circular Linked List - Insertion, Deletion, Search operations.

UNIT-II

Classes: 10

Stack: -Stack ADT, Implementation using arrays and linked list. Applications of stacks: Infix to postfix expression conversion, Evaluation of Postfix expressions and balancing the symbols.

Queue: Queue ADT, Implementation of Queue using arrays and linked list, Circular Queue, DEQUE

Priority Queue: Introduction, Representation, Heap- properties, Max heap and Min heap – Build heap, insertion and deletion.

UNIT-III

Classes: 10

Searching: Linear and Binary search methods

Sorting: Bubble sort, Insertion sort, Quick sort, Merge sort, Heap sort, comparison of sorting methods.

Hashing: Introduction, Hash functions, Collision Resolution Techniques- Separate Chaining and Open Addressing. Rehashing, Extendible Hashing.

UNIT-IV

Classes: 10

Trees: Introduction, Binary Trees: properties, representation, Tree traversals: In order, preorder, Post order, Binary Search Trees: Properties, Insertion, Deletion and Search operations, Balanced Binary search tree - AVL Trees and its operations, B-Trees-definition, insertion and searching operations.

UNIT-V

Classes: 12

Graphs: Fundamentals, Representation of graphs, Graph Traversals: BFS, DFS, Applications of Graphs-Minimum cost spanning tree using Kruskal's and Prim's Algorithms.

Pattern matching algorithms-Brute force, Knuth Morris-Pratt algorithm

Text Books:

1. Data structures, Algorithms and Applications in Java, S.Sahni, Universities Press. Second Edition, 2004.
2. Data structures and Algorithm Analysis in Java, M.A.Weiss, 2nd edition, Addison-Wesley (Pearson Education).

Reference Books:

1. Java for Programmers, Deitel and Deitel, Pearson education.
2. Data structures and Algorithms in Java, R.Lafore, Pearson education.
3. Java: The Complete Reference, 8th edition, Herbert Schildt, TMH.
4. Data structures and Algorithms in Java, M.T.Goodrich, R.Tomassia, 3rd edition, Wiley India

DATA STRUCTURES THROUGH JAVA LAB

II B.Tech - II Semester – IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT07	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100

COURSE OUTCOMES:

1. Ability to select the data structures that efficiently model the information in a problem.
2. Ability to assess efficiency trade-offs among different data structure implementations or combinations.
3. Implement and know the application of algorithms for sorting and pattern matching.
4. Design programs using a variety of data structures, including hash tables, binary and general tree structures, search trees, tries, heaps, graphs, and B-trees.
5. Design and analysis of text processing algorithms

LIST OF EXPERIMENTS

WEEK-1

1. Write a program that implements the following operations on Single linked list
 - a) Insertion at begin
 - b) Insertion at end
 - c) Insertion at specific position
 - d) Deletion at begin
 - e) deletion at end
 - f) Deletion at specific position

WEEK-2

1. Write a program that implements the following operations on double linked list
 - a) Insertion at begin
 - b) Insertion at end
 - c) Insertion at specific position
 - d) Deletion at begin
 - e) deletion at end
 - f) Deletion at specific position

WEEK-3

1. Write a program to implement the following using an array and linked list
 - a) Stack ADT
 - b) Queue ADT

WEEK-4

1. Write a program to implement the following applications of stack.
 - a) Conversion of Infix Expression to postfix Expression.
 - b) Evaluation of Postfix Expression
 - c) Implementation of Balancing Symbols.

WEEK-5

1. Write a program to implement the DEQUE (double ended queue) ADT using
 - a) Array
 - b) Singly linked list
 - c) Doubly linked list

WEEK-6

1. Write a program to implement Linear search and binary search
2. Write a program to implement the following sorting techniques
 - a) Bubble sort
 - b) Selection sort
 - c) Insertion sort
 - d) Heap sort.

WEEK-7

1. Write a program to perform the following collision resolution techniques
 - a) Linear Probing
 - b) Quadratic Probing
 - c) Double Hashing
 - d) separate chaining

WEEK-8

1. Write a program to implement the following operations.
 - a. Create a binary tree of integers
 - b. Traverse the above Binary tree recursively in Pre-order, In-order and Post-order

WEEK-9

1. Write a program to create a binary search tree and perform the following operations
 - a) Insertion b) deletion c) search

WEEK-10

1. Write a program to perform the following operations on B-Trees
 - a) Insertion b) deletion c) Search

WEEK-11

1. Write a program to implement Graph Representations
 - a) Adjacency Matrix b) Adjacency List

WEEK-12

1. Write a program for implementing the following graph traversal algorithms
 - a). Depth first traversal b). Breadth first traversal.

WEEK-13

1. Write a program for generating minimum cost spanning tree using the following algorithms
 - a) Prim's b) Krushkal's

WEEK-14

1. Write a program that implements KMP algorithm for pattern matching

TEXT BOOKS:

1. Data Structures and Algorithms in java, 3rd edition, A.Drozdek, Cengage Learning.
2. Data Structures with Java, J.R.Hubbard, 2nd edition, Schaum's Outlines, TMH.
3. Data Structures and algorithms in Java, 2nd Edition, R.Lafore, Pearson Education.
4. Data Structures using Java, D.S.Malik and P.S. Nair, Cengage Learning.
5. Data structures, Algorithms and Applications in java, 2nd Edition, S.Sahani, UniversitiesPress.
6. Design and Analysis of Algorithms, P.H.Dave and H.B.Dave, Pearson education.
7. Data Structures and java collections frame work, W.J.Collins, McGraw Hill.
8. Java: the complete reference, 7th editon, Herbert Schildt, TMH.
9. Java for Programmers, P.J.Deitel and H.M.Deitel, Pearson education / Java: How toProgramP.J.Deitel and H.M. Deitel , 8th edition, PHI.
10. Java Programming, D.S.Malik, Cengage Learning.
11. A Practical Guide to Data Structures and Algorithms using Java, S.Goldman&K.Goldman,Chapman& Hall/CRC, Taylor & Francis Group.(Note: Use packages like java.io, java.util, etc).

SOFTWARE ENGINEERING AND DESIGN LAB

II B.Tech - II Semester – IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT08	PCC	0	0	3	1.5	40	60	100

COURSE OUTCOMES:

1. To write the problem statement for the given system.
2. To capture the requirements specification for an intended software system using DFD & use case Modeling.
3. To draw the structural UML diagrams for the given specification
4. To draw the behavioral UML diagrams for the given specification
5. To perform Different testing Methodologies on the given specification.

LIST OF EXPERIMENTS

WEEK-1

- a. Identify a software system that needs to be developed.
- b. Document the Software Requirements Specification (SRS) for the identified system.

WEEK-2

- a. Draw level 0, level 1, and level 2 Data Flow Diagram (DFD) for the identified System

WEEK-3

- a. Draw the Class diagrams and show various class relationships
- b. Draw Package diagram.

WEEK-4

- a. Draw UML component and deployment diagram for the identified system.

WEEK-5

- a. Identify use cases and develop the Use Case model with include and extend relationships.

WEEK-6

- a. Using the identified scenarios, find the interaction between objects and represent them using Sequence diagram.

WEEK-7

- a. Using the identified scenarios, find the interaction between objects and represent them using Sequence diagram.

WEEK-8

- a. Draw relevant Activity Diagrams for the same system.

WEEK-9

- a. Draw relevant State chart diagram for the same system

WEEK-10

- a. Case Study - I

WEEK-11

- a. Case Study - II

WEEK-12

1. Case Study - III.

Suggested Domains:

1. Online course reservation system
2. Airline/Railway reservation system
3. Exam registration
4. Stock maintenance system
5. Recruitment system.
6. Library management system
7. Student information system
8. ATM

TEXT BOOKS:

1. A Concise Introduction to Software Engineering (Undergraduate Topics in Computer Science), PankajJalote, Springer International Edition, 2008.
2. Craig Larman," Applying UML and Patterns: An Introduction to object-oriented Analysis and Design and iterative developmentll, Third Edition, Pearson Education, 2005.
3. Mike O'Docherty, —Object-Oriented Analysis & Design: Understanding System Development withUML 2.0ll, John Wiley & Sons, 2005

CONSTITUTION OF INDIA

II B. TECH – II SEMESTER: IT

Course Code	Category	Hours/Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6HS06	MC	2	0	0	0	50	-	50

COURSE OBJECTIVES:

Students will be able to:

1. Understand the need for constitution
2. Appreciate the fundamental duties and rights of the citizens of India.
3. Explain the role and amendments of constitution in a democratic society.
4. Describe the directive principles of state policy and their significance.
5. List the key features of the constitution, union government and state government.

COURSE OUTCOMES:

Students will be able to:

1. Create awareness about the constitutional values and objectives written in the Indian constitution.
2. List fundamental rights and fundamental duties of Indian citizens.
3. Identify the division of legislative, executive and financial powers between the union and state governments.
4. Understand the working of Indian democracy, its institutions and processes at the local, state and union levels.
5. Explain the functions and responsibilities of election commission of India and union public service commission.

UNIT-I HISTORY OF MAKING OF THE INDIAN CONSTITUTION Classes:12

Introduction to the constitution of India, the making of the constitution and salient features of the constitution.

UNIT-II PHILOSOPHY OF THE INDIAN CONSTITUTION Classes:12

Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures.

UNIT-III UNION GOVERNMENT Classes:12

Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and Functions), Union judiciary (Supreme court), Jurisdiction of the supreme court.

UNIT-IV STATE GOVERNMENT Classes:12

State Government, State legislature (Legislative Assembly/ Vidhan Sabha, Legislative council/ Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (Highcourts)

UNIT-V ELECTION COMMISSION Classes:12

Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.

TEXTBOOKS:

1. M.V.Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional
2. Law, Prentice–Hall of India Pvt. Ltd. New Delhi
3. Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Rights, Challenges to Civil Rights Guarantees in India, Oxford University Press, 2012
4. P.M. Bakshi, The constitution of India, Universal Law Publishing Co.,
5. The Constitution of India, 1950 (Bare Act), Government Publication.
6. Dr. S.N. Busi, Dr. B.R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.

REFERENCE BOOKS:

1. M.P. Jain, Indian Constitution Law, 7th Edn. LexisNexis, 2014.
2. D.D. Basu, Introduction to the Constitution of India, LexisNexis, 2015.
3. Indian constitution at work, NCERT
4. Subash Kashyap, Indian Constitution, National Book Trust
5. J.A. Siwach, Dynamics of Indian Government & Politics
6. D.C. Gupta, Indian Government and Politics
7. H.M. Sreevai, Constitutional Law of India, 4th edition in 3 volumes (Universal Law Publication)
8. J.C. Johari, Indian Government and Politics Hans
9. J. Raj Indian Government and Politics

Web References:

1. www.hss.iitb.ac.in/en/lecture-details
2. www.iitb.ac.in/en/event/2nd-lecture-institute-lecture-series-indian-constitution

MOOC Course

1. nptel.ac.in/courses/109104074/8
2. nptel.ac.in/courses/109104045/
3. nptel.ac.in/courses/101104065/

III B.TECH I SEMESTER SYLLABUS

FULL STACK DEVELOPMENT

III Year I Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT10	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. To become knowledgeable about the most recent web development technologies
2. Idea for creating two tier and three tier architectural web applications
3. Design and Analyze real time web applications.
4. Constructing suitable client and server side applications.
5. To learn core concept of both front end and back end programming.

COURSE OUTCOMES:

The students can able to learn:

1. Develop a fully functioning website and deploy on a web server.
2. Gain Knowledge about the front end and back end tools
3. Find and use code packages based on their documentation to produce working results in a project
4. Create web pages that function using external data
5. Implementation of web application employing efficient database access

UNIT I WEB DEVELOPMENT BASICS

Classes: 13

HTML-Introduction, Presentation tags-Bold, Italic, Headings, Paragraph, Line Break, form, iframe, image tag

CSS3- Syntax, Terminology, Properties-color and Background, Text, Font, Layouts – Flexbox Layout,Grid Layout ,Positioning

Version control - Git&GitHub, Basic Git Commands-Add, Commit, log, Push, Pull, Delete, Forking

UNIT II FRONT END DEVELOPMENT

Classes: 15

Java script basics Introduction-Variables, Data Types, Operators, Control Structures, Java script OOPS concepts, JS Validation, JS Functions-Function Declaration, Function Expression, Arrow Functions

JQuery Introduction- Basic JQuery Syntax, JQuery events-Click Event, Mouse Events, Form Events, Keyboard Events

UNIT III REACT JS

Classes: 12

REACT JS: Introduction to React -Key Features, Basic example, React Router and Single Page Applications- Installation, Basic Setup, React Components-Functional Components, Class Components, React Forms, React JS Flow Architecture, Introduction to Redux, Client-Server Communication -Fetch and Axios

UNIT IV JAVA WEB DEVELOPMENT

Classes: 12

Model View Controller(MVC) Overview- Model, View, Controller

Spring basics-Inversion of Control (IoC),Dependency Injection (DI) ,Aspect-Oriented Programming (AOP), MVC Architecture using Spring-Setting up Spring MVC, Controller, View, Application Configuration

RESTful API using Spring Framework- Setting up Spring REST, Controller, model, Building an application using Maven -Setting up Maven, Directory Structure,Running the Application

UNIT V DATABASES & DEPLOYMENT

Classes: 13

SQL Basic commands>Create, Alter, Drop, Select, Insert, Update, Delete, Join

Data persistence using Spring, JDBC: Spring JDBC Setup, Database Configuration, Define the Data Source Bean, Using Jdbc Template

Deploying application in Cloud.-Prepare the Application, Choose a cloud Provider, Create a Cloud Account and Set Up Services, Deploy the Application

TEXT BOOKS:

1. MASTERING HTML, CSS & Java Script Web Publishingby Laura Lemay,RafeColburn , Jennifer Kyrnin by BPB Publications.
2. Version Control with Git :Powerful Tools and Techniques for Collaborative Software Development By Jon Loeliger, Matthew McCullough
3. Professional JavaScript for Web Developers by Matt Frisbie · 2019
4. ReactJS by Example - Building Modern Web Applications with ReactVipul A M, PrathameshSonpatki · 2016
5. Spring MVC: Designing Real-World Web Applications by ShameerKunjumohamed, HamidrezaSattari, Alex Bretet · 2016

REFERENCES:

1. Full-Stack JavaScript Development by Eric Bush, 2016.
2. Mastering Full Stack React Web Development Paperback – April 28, 2017 by Tomasz Dyl,Kamil Przeorski, Maciej Czarnecki

AUTOMATA AND COMPILER DESIGN

III B.TECH – I Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT11	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the fundamental concepts of formal languages and regular expressions.
2. Identify the need of compiler and stages of compiler.
3. Enumerating top down and bottom up parsing techniques used in compilation process.
4. Understand Syntax directed translation scheme and different ways of representing Intermediate code.
5. Issues in design of Code generation and different code optimization techniques.

COURSE OUTCOMES

The course should enable the students to:

1. Able to employ finite state machines and regular expressions for modelling and solving computing problems.
2. Classify compilation phases and construction of Derivation trees for the grammar.
3. Construct the parsing tables using different types of parsing approaches.
4. Able to apply Syntax directed translation for the given expression and different ways of representing Intermediate code.
5. Analyze the issues in code generation and applying different code optimization techniques..

UNIT– I

Classes: 13

Introduction to Finite Automata: Central Concepts of Finite Automata – Alphabets, Strings, Languages. Deterministic Finite Automata - Formal Definition, Design of DFA, Nondeterministic Finite Automata - Formal Definition, Design of NFA, Conversion from NFA with ϵ -transitions to NFA without ϵ -transitions. Conversion from NFA to DFA.

Regular Expressions: Introduction to Regular Expressions, Identity rules of Regular Expression, Algebraic Laws for Regular Expressions, Conversion of Finite Automata to Regular Expressions.

UNIT– II

Classes: 12

Introduction to Compilers: Language Processing system, Phases of Compiler, Pass and Phases of Translation, Boot Strapping.

Lexical Analysis: Role of Lexical analyzer, Recognition of Tokens, Context Free Grammar, Derivations- Leftmost and Rightmost Derivations, Parse Trees, Ambiguity, Elimination of Left Recursion, Elimination of Left Factoring.

UNIT–III

Classes: 13

Top Down Parsing: Types of Parsers, Calculation of First and Follow, Construction of LL(1) Parsing table, Recursive Descent Parser.

Bottom up Parsing: Introduction, Classification of Bottom up parsing- LR Parser, Operator precedence Parser. Shift Reduce parser, Construction of parsing tables- LR(0), SLR(1), CLR(1), LALR(1).

UNIT– IV

Classes: 10

Syntax-Directed Translation: Syntax-Directed Definition, Types of Attributes, Annotated parse tree, Syntax Directed Translation Scheme, SDT for Infix to Postfix.

Intermediate-Code Generation: Introduction, Types of Intermediate code, Implementation of Three Address Code.

UNIT– V

Classes: 12

Run-Time Environments: Source Language Issues, Storage Allocation Techniques, Activation Record.

Code Optimization: Principle sources of Code optimization, Basic Blocks, Optimization of Basic Blocks, Loop optimization, DAG representation of Basic Block.

Code Generation: Issues in design of Code Generation, Object code forms, Peephole Optimization.

Text Books:

1. Introduction to Automata Theory, Languages, and Computation, 3rd Edition, John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, Pearson Education.
2. Compilers: Principles, Techniques and Tools By Aho, Lam, Sethi, and Ullman, Second Edition, Pearson, 2014.
3. Aho, Sethi & Ullman, "Compilers: Principles, Techniques and Tools", Pearson Education

Reference Books:

1. Compilers: Principles, Techniques and Tools By Aho, Sethi, and Ullman, Addison-Wesley, 1986
2. Introduction to Formal languages Automata Theory and Computation, Kamala Krithivasan, Rama R, Pearson.
3. V Raghvan, " Principles of Compiler Design", McGraw-Hill.

DATA COMMUNICATION & COMPUTER NETWORKS

III Year I Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT12	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. To introduce the Fundamentals of data communication networks
2. To demonstrate the Functions of various protocols of Data link layer.
3. To demonstrate Functioning of various Routing protocols.
4. To introduce the Functions of various Transport layer protocols.
5. To understand the significance of application layer protocols

COURSE OUTCOMES:

The course should enable the students to:

1. Know the Categories and functions of various Data communication Networks
2. Design and analyze various error detection techniques.
3. Demonstrate the mechanism of routing the data in network layer
4. Know the significance of various Flow control and Congestion control Mechanisms
5. Know the Functioning of various Application layer Protocols

UNIT I

Classes: 14

Introduction to Data Communications: Components, Data Representation, Data Flow, Networks- Distributed Processing, Network Criteria, Physical Structures, Network Models, Categories of Networks Interconnection of Networks, The Internet - A Brief History, The Internet Today, Protocol and Standards - Protocols, Standards, Standards Organizations, Internet Standards. Network Models, Layered Tasks, OSI model, Layers in OSI model, TCP/IP Protocol Suite, Addressing Introduction, Wireless Links and Network Characteristics, WiFi: 802.11 Wireless LANs -The 802.11 Architecture.

UNIT II

Classes: 10

Data Link Layer: Introduction to the Link Layer, The Services Provided by the Link Layer, Types of errors, Redundancy, Detection vs Correction, Error-Detection and Correction Techniques, Parity Checks, Cyclic Redundancy Check (CRC) , Framing, Flow Control and Error Control protocols ,HDLC, Multiple Access Protocols, Random Access ,ALOHA, Controlled access,

UNIT III

Classes: 12

The Network Layer: Introduction to network layer, the services provided by network layer, Routing, Network Service Models, Virtual Circuit and Datagram Networks-Virtual-Circuit Networks, Datagram Networks, Switching, Output Processing, Queuing, The Routing Control Plane, The Internet Protocol(IP):Forwarding and Addressing in the Internet- Datagram format, Ipv4 Addressing, Internet Control Message Protocol(ICMP), IPv

UNIT IV

Classes: 11

Transport Layer: Introduction and Transport Layer Services : Relationship Between Transport and Network Layers, Overview of the Transport Layer in the Internet, Multiplexing and De-multiplexing, Connectionless Transport: UDP -UDP Segment Structure, UDP Checksum, Go- Back-N(GBN), Selective Repeat(SR), Connection Oriented Transport: TCP - The TCP Connection, TCP Segment Structure, Flow Control, TCP Connection Management.

UNIT V

Classes: 11

Application Layer: Principles of Networking Applications – Network Application Architectures, Processes Communicating, Transport Services Available to Applications, Transport Services Provided by the File Transfer: FTP Commands and Replies, Electronic Mail in the Internet- STMP, Comparison with HTTP, DNS- The Internet's Directory Service – Service Provided by DNS, Overview of How DNS Works, DNS Records and messages.

TEXTBOOKS:

1. Computer Networking A Top-Down Approach – Kurose James F, Keith W, 6th Edition, Pearson.
2. Data Communications and Networking Behrouz A. Forouzan 4th Edition McGraw-Hill Education

REFERENCES:

1. Data communication and Networks - Bhusan Trivedi, Oxford university press, 2016
2. Computer Networks -- Andrew S Tanenbaum, 4th Edition, Pearson Education
3. Understanding Communications and Networks, 3rd Edition, W. A. Shay, Cengage Learning

DATA MINING (PROFESSIONAL ELECTIVE – I)

III B. TECH- I SEMESTER: IT

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6AI11	PEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Learn data mining concepts understand association rules mining.
2. Discuss classification algorithms learn how data is grouped using clustering techniques.
3. Develop the abilities of critical analysis to data mining systems and applications.
4. Implement practical and theoretical understanding of the technologies for data mining
5. Understand the strengths and limitations of various data mining models

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Perform the pre-processing of data and apply mining techniques on it.
2. Identify the association rules, classification and clusters in large data sets.
3. Solve real world problems in business and scientific information
4. Classify web pages, extracting knowledge from the web.
5. Apply supervised learning techniques on given data sets

UNIT - I INTRODUCTION TO DATA MINING

CLASSES: 14

Introduction to Data Mining: Introduction, What is Data Mining, Definition, Knowledge Discovery from Data (KDD), What Kinds of Data can be Mined, Data Mining Tasks, Integration of Data Mining System With A Database or Data Warehouse System, Types of Data Sets and Attribute Values

Preprocessing: Data Quality, Major Tasks in Data Pre-processing , Data Cleaning and Data Integration, Data Reduction, Data Transformation and Data Discretization , Measures of Similarity and Dissimilarity- Basics

UNIT - II ASSOCIATION RULES

CLASSES: 12

Association Rules: Problem Definition, Frequent Item Set Generation, The APRIORI Principle, Support and Confidence Measures, Association Rule Generation; APRIORI Algorithm, The Partition Algorithms, FP-Growth Algorithms, Compact Representation of Frequent Item Set- Maximal Frequent Item Set, Closed Frequent Item Set.

UNIT - III CLASSIFICATION

CLASSES: 14

Classification: Problem Definition, General Approaches to solving a classification problem ,Evaluation of Classifiers , Classification techniques, Decision Trees-Decision tree Construction , Methods for Expressing attribute test conditions, Measures for Selecting the Best Split, Algorithm for Decision tree Induction ; Naive-Bayes Classifier, Bayesian Belief Networks; K- Nearest neighbor classification-Algorithm and Characteristics.

UNIT - IV CLUSTERING

CLASSES: 10

Clustering: Problem Definition, Clustering Overview, Evaluation of Clustering Algorithms, Partitioning Clustering-K-Means Algorithm, K-Means Additional issues, PAM Algorithm; Hierarchical Clustering-Agglomerative Methods and divisive methods, Basic Agglomerative Hierarchical Clustering Algorithm, Specific techniques, Key Issues in Hierarchical Clustering, Strengths and Weakness; Outlier Detection.

UNIT - V WEB AND TEXT MINING

CLASSES: 10

Web And Text Mining: Introduction, web mining, web content mining, web structure mining, we usage mining, Text mining –unstructured text, episode rule discovery for texts, hierarchy of categories, text clustering.

TEXT BOOKS

1. Data Mining: Concepts and Techniques - Jiawei Han, Micheline Kamber, Jian Pei (2012), 3rd edition, Elsevier, United States of America.
2. Introduction to Data Mining, Pang-Ning Tan, Vipin Kumar, Michael Steinbach, Pearson Education

E -Text Books:

1. https://books.google.co.in/books?id=pQws07tdpjoC&printsec=copyright&redir_esc=y#v=onepage&q&f=false

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

INTRODUCTION TO DATA SCIENCE

(PROFESSIONAL ELECTIVE – I)

III B. TECH- I SEMESTER - IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT23	PEC	3	0	0	3	40	60	100

Pre-requisites: Probability and Statistics, Database Management Systems

Course Objectives: The Students will learn the following:

1. Practice data analytics and machine learning approaches.
2. Adapt big data solutions in different domains including social media, transportation, finance and medicine.
3. Apply the capability and the functionality of No-SQL systems
4. Optimize business decisions and create competitive advantage with Big Data analytics.
5. Solve problems using Hadoop and introducing map reduce paradigm.

Course Outcomes: After the successful completion of the course, the students would be able to:

1. Apply mathematical logic to solve problems and Identify the characteristics of datasets and compare the trivial data and big data for various applications.
2. Implement machine learning techniques and computing environments that are suitable for the applications under consideration.
3. Solve problems associated with batch learning and online learning, and the big data characteristics such as high dimensionality, dynamically growing data, and in particular scalability issues.
4. Apply scaling up machine learning techniques and associated computing techniques and technologies.
5. Recognize and implement various ways of selecting suitable model parameters for different machine learning techniques.

UNIT I:

Classes: 09

SQL using R & Correlation and Regression Analysis: Introduction to NoSQL, Connecting R to NoSQL databases, Excel and R integration with R connector, Regression Analysis, Assumptions of OLS Regression, Regression Modelling, Correlation, ANOVA, Forecasting, Heteroscedasticity, Autocorrelation, Introduction to Multiple Regression, etc.

UNIT II:

Classes: 09

Understand the Verticals - Engineering, Financial and others: Understanding systems viz. Engineering Design, Manufacturing, Smart Utilities, Production lines, Automotive, Technology, etc., Understanding Business problems related to various businesses.

UNIT III:

Classes: 12

Manage your work to meet requirements: Understanding Learning objectives, Introduction to work & meeting requirements, Time Management, Work management & prioritization, Quality & Standards Adherence. Work effectively with Colleagues, Introduction to work effectively, Team Work, Professionalism, Effective Communication skills, etc.

UNIT IV:

Classes: 10

Data Management & Introduction to Big Data Tools: Design Data Architecture and manage the data for analysis, Understand various sources of Data like Sensors/signal/GPS etc. Export all the data onto Cloud ex. AWS/Rackspace etc. Introduction to Big Data tools like Hadoop, Spark, Impala, etc., Data ETL process, identify gaps in the data and follow-up for decision making.

UNIT V:

Classes: 10

Big Data Analytics & Machine Learning Algorithms: Run descriptive to understand the nature of the available data, collate all the data sources to suffice business requirements, Run descriptive statistics for all the variables and observe the data ranges, Outlier detection and elimination, Hypothesis testing, and determining the multiple analytical methodologies, Train Model on 2/3 sample data using various Statistical/Machine learning algorithms, Test model on 1/3 sample for prediction, etc.

Textbooks:

1. Bill Franks, Taming the Big Data Tidal Wave, 1st Edition, Wiley, 2012.
2. Dalgaard, Peter, —Introductory statistics with Rll , Springer Science & Business Media, 2008.

References:

1. Richard Cotton, —Learning Rll , O_Reilly, 2013.
2. Frank J. Ohlhorst, Big Data Analytics, 1st Edition, Wiley, 2012.
3. Brain S. Everitt, —A Handbook of Statistical Analysis Using Rll , Second Edition, LLC, 2014.
4. T. Hastie, R. Tibshirani, and J. Friedman, —Elements of Statistical Learningll , Springer, 2009.

INFORMATION SECURITY (PROFESSIONAL ELECTIVE – I)

III Year I Semester: CSE/IT/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT24	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Explain the risks faced by computer systems and conventional techniques to protect information.
2. Identify and analyse standard symmetric cryptography Techniques to secure the information.
3. Use asymmetric cryptography Techniques to secure the information.
4. Develop security mechanisms to Authenticate Information and users.
5. Implement cryptography algorithms for Authentication Services and Email Security.

UNIT – I Introduction to Information Security

Classes: 10

Attacks, Vulnerability, Security Goals, Security Services and mechanisms,

Conventional Cryptographic Techniques: Conventional substitution and transposition ciphers, Block cipher and Stream Cipher, Steganography

UNIT – II Symmetric Cryptography

Classes: 12

Symmetric Cryptography Techniques: Block Cipher Design Principles, DES, Strength of DES, Triple DES, AES, Block Cipher Modes of Operation, Stream Cipher and RC4.

UNIT – III Asymmetric Cryptography

Classes: 10

Asymmetric Cryptography Techniques: Principles Public key crypto Systems, RSA algorithms Diffie Hellman Key Exchange., Elliptic Curve Cryptography

UNIT – IV Message Authentication

Classes: 12

MESSAGE AUTHENTICATION AND HASH FUNCTIONS: Authentication Function, Message Authentication Code, Hash Function, Security of Hash Function and MACs.

HASH AND MAC ALGORITHM: Secure Hash Algorithm, Whirlpool, HMAC, CMAC.

DIGITAL SIGNATURE: Digital Signature, Authentication Protocol, Digital Signature Standard

UNIT – V Authentication Application Email Security

Classes: 09

AUTHENTICATION APPLICATION: Kerberos, X.509 Authentication Service, Public Key Infrastructure.

EMAIL SECURITY: Pretty Good Privacy (PGP) and S/MIME.

Text Books:

1. William Stallings (2017), Cryptography and Network Security: Principles and Practice, 7th edition, Pearson Education, India.
2. Security in Computing, Fourth Edition, by Charles P. P fleeger, Pearson Education

Reference Books:

1. Charlie Kaufman (2002), Network Security: Private Communication in a Public World, 2nd edition, Prentice Hall of India, New Delhi.
2. Atul Kahate (2008), Cryptography and Network Security, 2 nd edition, Tata Mc Grawhill, India.
3. Robert Bragg, Mark Rhodes (2004), Network Security: The complete reference, Tata McGrawhill, India.

MOBILE APPLICATION DEVELOPMENT (PROFESSIONAL ELECTIVE – I)

III B. TECH- I SEMESTER - IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT25	PEC	3	0	0	3	40	60	100

COURSE OBJECTIVE

1. Produce apps for Android platform devices.
2. Gain a basic understanding of computer architecture and object-oriented programming
3. Develop a working knowledge of Android Studio Application.
4. Understand mobile design principles.
5. Identify need and opportunity in app markets.

COURSE OUTCOMES

By the conclusion of this course, students will be able to:

1. Describe those aspects of mobile programming that make it unique from programming for other platforms.
2. Critique mobile applications on their design pros and cons,
3. Utilize rapid prototyping techniques to design and develop sophisticated mobile interfaces,
4. Program mobile applications for the Android operating system that use basic and advanced phone features, and
5. Deploy applications to the Android marketplace for distribution.

UNIT I

Introduction to Android: The Android Platform, Android SDK, Eclipse Installation, Android Installation, Building your First Android application, Understanding Anatomy of Android Application, Android Manifest file.

UNIT II

Android Application Design Essentials: Anatomy of an Android applications, Android terminologies, Application Context, Activities, Services, Intents, Receiving and Broadcasting Intents, Android Manifest File and its common settings, Using Intent Filter, Permissions.

UNIT III

Android User Interface Design Essentials: User Interface Screen elements, Designing User Interfaces with Layouts,

UNIT IV

Testing Android applications, Publishing Android application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resources.

UNIT V

Using Common Android APIs: Using Android Data and Storage APIs, Managing data using SQLite, Sharing Data between Applications with Content Providers

TEXT BOOKS:

1. Neil Smyth, "Android Studio Development Essentials" - Android 7 Edition (2016).
2. J F DiMarzio, "Beginning Android Programming with Android Studio", 4th Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978-8126565580.

CLOUD & DEVOPS

III Year I Semester: CSE/IT/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT13	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. To understand the fundamentals of cloud computing, including its benefits and different service models.
2. To understand the different Version control tools like Git,
3. To understand the concepts of Continuous Integration/ Continuous Testing/ Continuous Deployment)
4. To understand Configuration management using Ansible
5. Illustrate the benefits and drive the adoption of cloud-based Devops tools to solve real world problems

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Implement and manage core cloud services like IAM, EC2, S3, VPC, and RDS effectively.
2. Understand different actions performed through Version control tools like Git.
3. Perform Continuous Integration and Continuous Testing and Continuous Deployment using Jenkins by building and automating test cases using Maven & Gradle.
4. Ability to Perform Automated Continuous Deployment and to do configuration management using Ansible.
5. Build, test, and deploy a simple sample application through the CI/CD pipeline, demonstrating automation and efficiency in software delivery.

UNIT – I Cloud Computing Introduction

Classes: 10

Cloud Computing, Why Cloud Computing, Advantages of Cloud Computing, Virtualization, cloud service models, Cloud deployment models, Public clouds: Amazon Web Services, Microsoft Azure, and Google Cloud Services.

AWS Cloud services: IAM Roles, and Policies, EC2, S3, VPC, RDS

UNIT – II DevOps Introduction

Classes: 10

DevOps definition, History of DevOps, DevOps Roles, DevOps Need, DevOps Problems & Solution, DevOps Main Objectives, DevOps: Continuous Delivery and Benefits, DevOps and Software Development Life Cycle: Waterfall Model, Agile Model, DevOps on Cloud.

Version control systems: Git and Github

GIT Features, 3-Tree Architecture, GIT – Clone /Commit / Push, GIT Hub Projects, GIT Hub Management, GIT Rebase & Merge, GIT Stash, Reset, Checkout, GIT Clone, Fetch, Pull.

UNIT – III Compile, Build & Continuous Integration

Classes: 12

Introduction, Installation of Maven, POM files, Maven Build lifecycle, Build phases(compile build, test, package) Maven Profiles, Maven repositories(local, central, global),Maven plugins, Maven create and build Artifacts, Dependency management, Installation of Gradle, Understand build using Gradle

CONTINUOUS INTEGRATION/CONTINUOUS DELIVERY USING JENKINS

Definition of Jenkins, Install & Configure Jenkins, Jenkins Architecture Overview, Creating a Jenkins Job,

Configuring a Jenkins job, Introduction to Plugins, Commonly used plugins (Git Plugin,). Jenkins Build Pipe Line.

UNIT – IV Configuration management and containerization

Classes: 10

Containerization: Docker, What is Docker Image, Docker Installation, Working with Docker Containers, Container, Docker Engine, Crating Containers with an Image, Working with Images, Docker Command Line Interface, Docker Compose, Docker Hub, Docker Trusted Registry, Docker swarm, Docker attach, Docker File & Commands.

Configuration: Ansible Introduction, Installation, Ansible master/slave configuration, YAML basics, Ansible modules, Ansible Inventory files, Ansible playbooks, Ansible Roles.

UNIT – V BUILDING DEVOPS PIPELINES USING AWS

Classes: 08

Create Github Account, Repository, Create AWS Organization, Create a new pipeline, Build a sample code, Modify AWS-pipelines.

Text Books:

1. Stephen Baron, "AWS: The Complete Beginner's Guide", 2020
2. "The DevOps Adoption Playbook: A Guide to Adopting DevOps in a Multi-Speed IT Enterprise" by Sanjeev Sharma
3. "DevOps. Building CI/CD Pipelines with Jenkins, Docker Container, AWS ECS, JDK 11, Git and Maven 3" by John Edward Cooper Berg

Reference Books:

1. Hands-On Azure Devops: Cidc Implementation For Mobile, Hybrid, And Web Applications Using Azure Devops And Microsoft Azure: CICD Implementation for ... DevOps and Microsoft Azure (English Edition) Paperback – 1 January 20202. by Mitesh Soni
2. Jeff Geerling, "Ansible for DevOps: Server and configuration management for humans", First Edition, 2015.
3. David Johnson, "Ansible for DevOps: Everything You Need to Know to Use Ansible for DevOps", Second Edition, 2016.
4. Mariot Tsitoara, "Ansible 6. Beginning Git and GitHub: A Comprehensive Guide to Version Control, Project Management, and Teamwork for the New Developer", Second Edition, 2019.
5. <https://www.jenkins.io/user-handbook.pdf>
6. <https://maven.apache.org/guides/getting-started/>
7. "Cloud Computing: From Beginning to End" by Ray J. Rafaels
8. Douglas E. Comer , "The Cloud Computing Book", Edition: 1st, 2023

FULL STACK DEVELOPMENT LAB

III Year I Semester: IT/CSE/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT14	PCC	0	0	3	1.5	40	60	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

The course should enable the students to:

1. Usage of various front and back end Tools
2. They can understand and create applications on their own
3. Demonstrate and Designing of Websites can be carried out.
4. Develop web based application using suitable client side and server sidecode.
5. Implement web based application using effective database access

LIST OF EXPERIMENTS

Week 1:

Write a program to create a simple webpage using HTML.

Week 2:

Write a program to create a website using HTML CSS and JavaScript?.

Week 3:

Write a program to build a Chat module using HTML CSS and JavaScript?

Week 4:

Write a program to create a simple calculator Application using React JS.

Week 5:

Write a program to create a voting application using React JS.

Week 6:

Write a program to create and Build a Password Strength Check using JQuery.

Week 7:

Write a program to create and Build a star rating system using JQuery

Week8:

Create a project on Project Management application. This is going to be the toughest of all with a lot of features. They should have a framework of a basic social media site with users being able to communicate with each other. Users should be able to create projects and tasks within projects. Users must be able to assign tasks to other users and must be able to comment on it just like on a social media post. To start with, each user can have a calendar view and a kanban style board. Users must be able to close and archive tasks as well as projects when completed.

Week 9:

Create a project on Content Management System for a blog Using the CMS users must be able to design a web page using the drag and drop method. Users should be able to add textual or media content into placeholders that are attached to locations on the web page using drag and drop method.

Week 10:

Create a project on Grocery delivery application Assume this project is for a huge online departmental store. Assume that they have a myriad of grocery items at their godown. All items must be listed on the website, along with their quantities and prices. Users must be able to sign up and purchase groceries. The system should present him with delivery slot options, and the user must be able to choose his preferred slot. Users must then be taken to the payment page where he makes the payment with his favourite method.

Week11:

Create a project e-commerce portal for used furniture sales Assume this project is for a startup that is acquiring used furniture from users at a price, refurbishing them and selling them off at a margin. The website must display second-hand furniture that is currently being sold.

TEXTBOOKS:

1. Web Design with HTML, CSS, JavaScript and JQuery Set Book by Jon Duckett
ProfessionalJavaScript for Web Developers Book by Nicholas C. Zakas
2. Learning PHP, MySQL, JavaScript, CSS & HTML5: A Step-by- Step Guide to CreatingDynamic Websites by Robin Nixo
3. Full Stack JavaScript: Learn Backbone.js, Node.js and MongoDB. Copyright © 2015 BYAZAT MARDAN

REFERENCES:

1. Full-Stack JavaScript Development by Eric Bush.
2. Mastering Full Stack React Web Development Paperback – April 28, 2017 by TomaszDyl , Kamil Przeorski , Maciej Czarnecki

LINUX PROGRAMMING LAB

III Year I Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT15	PCC	L	T	P	C	CIE	SEE	Total
		0	0	2	1	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Apply the basic commands to handle the Linux Environment
2. Use the Shell/C scripting constructs to modify the file system content.
3. Implement the process management concepts using C language.
4. Design Client-Server Applications using Sockets and TCP/UDP protocols.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Write and use the Shell Scripts in managing Linux Environment
2. Construct C Scripts to handle the File system in Linux.
3. Implement the IPC Mechanisms in Linux operating system using C language
4. Build and Analyze the Client-Server Environment using various protocols.

LIST OF EXPERIMENTS

Week 1:

Study and Practice on various commands like man, passwd, tty, script, clear, date, cal, cp, mv, ln, rm, unlink, mkdir, rmdir, du, df, mount, umount, find, unmask, ulimit, ps, who, w.

Week 2:

Study and Practice on various commands like cat, tail, head, sort, nl, uniq, grep, egrep, fgrep, cut, paste, join, tee, pg, comm, cmp, diff, tr, awk, tar, cpio.

Week 3:

- A. Write a shell script that accepts a file name, starting and ending line numbers as arguments and displays all the lines between the given line numbers.
- B. Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.

Week 4:

- A. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
- B. Write a shell script that receives any number of file names as arguments checks if every argument supplied is a file or a directory and reports accordingly. Whenever the argument is a file, the number of lines on it is also reported.

Week 5:

- A. Write a shell script that accepts a list of file names as its arguments, counts and reports the occurrence of each word that is present in the first argument file on other argument files.
- B. Write a shell script to list all of the directory files in a directory.

Week 6:

- A. Practice awk commands
- B. Write an awk script to count the number of lines in a file that do not contain vowels.
- C. Write an awk script to find the number of characters, words and lines in a file.

Week 7:

- A. Write a c program that makes a copy of a file using standard I/O and system calls.
- B. Implement in C the following UNIX commands using System calls.
 - A. cat B. mv
- C. Implement in C the following UNIX commands using System calls.
 - A. ls B. ls -l C.ls -a

Week8:

- A. Write a C program to create a child process and allow the parent to display “parent” and the child to display “child” on the screen.
- B. Write a C program to create a Zombie process.
- C. Write a C program that illustrates how an orphan is created.

Week 9:

- A. Write a C program that illustrates suspending and resuming processes using signals.
- B. Write a program to handle the signals like **SIGINT**, **SIGQUIT**, **SIGFPE**.

Week 10:

- A. Write C programs that illustrate communication between two unrelated processes using named pipe (FIFO).
- B. Write a C program in which a parent writes a message to a pipe and the child reads the message.

Week11:

- A. Implement **message queue** form of IPC.
- B. Implement **shared memory** form of IPC.

Week 12:

- A. Write Client server program in C for connection oriented communication between server and client process using unix domain sockets to perform the following
- B. Write Client server program in C for connection oriented communication between server and client process using internet domain sockets.

TEXT BOOKS:

1. Unix System Programming using C++, T.Chan, PHI.
2. Unix Concepts and Applications, 4th Edition, Sumitabha Das, TMH,2006.

REFERENCES:

1. Beginning Linux Programming, 4th Edition, N.Matthew, R.Stones,Wrox, Wiley India Edition,rp-2008
2. Linux System Programming, Robert Love, O'Reilly, SPD.
3. Advanced Programming in the Unix environment, 2nd Edition, W.R.Stevens, Pearson Education.
4. Unix Network Programming ,W.R.Stevens,PHI.
5. Unix for programmers and users, 3rd Edition, Graham Glass, King Ables, Pearson Education.

CLOUD & DEVOPS LAB

III Year I Semester: IT/CSE/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT16	PCC	0	0	3	1.5	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Apply the concepts of AWS and its cloud services.
2. Deploy and use virtual instances
3. Implement continuous integration using AWS CodeBuild.
4. Implement end-to-end continuous integration and continuous deployment (CI/CD) using AWS CodePipeline.

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Deploy secured virtual instances in Amazon AWS.
2. Deploy various cloud services like S3, Databases
3. Proficiency in basic Git and GitHub commands.
4. Deploy automated applications

LIST OF EXPERIMENTS

Week 1:

Create Amazon AWS EC2 Linux instance with conceptual understanding of SSH client software protocol and keys.

Week 2:

Create Amazon AWS EC2 Windows server instance with conceptual understanding of RDP (Remote Desktop Protocol).

Week 3:

Create cloud storage Bucket using Amazon Simple Storage Service (S3). Perform the following operations:

- A. Create a folder within a S3 Bucket
- B. Upload content to S3
- C. Change permissions to allow public access of contents.
- D. Set Meta Data on an S3 Bucket
- E. Delete an S3 Bucket and its content

Week 4:

Launch and connect to an Amazon Relational Data Base (RDS) Service, Use any one from MySQL, Oracle, SQL Server and postgresQLDataBase engines

Week 5:

Study and practice Git and GitHub commands.

Week 6:

Install and configure Jenkins to build and deploy Java or Web Applications.

Week 7:

Install and configure Docker tool and perform Docker commands for content management

Week8:

Create a Docker image and push Docker image into AWS ECR.

Week 9:

Create Repositories, Cloning, and Pushing Code Changes using AWS Code Commit.

Week 10:

Implement continuous integration using AWS Code build.

Week11:

Automate application deployment into EC2 using AWS Code Deploy.

Week 12:

Implement CI/CD using AWS Code pipeline that automates the build, test, and deployment phases.

Text Books:

1. Thomas Erl , "Cloud Computing: Concepts, Technology & Architecture", Edition: 1st or later
2. Douglas E. Comer , "The Cloud Computing Book", Edition: 1st, 2023
3. Stephen Baron, "AWS: The Complete Beginner's Guide", Kindle Edition 2020

Reference Books:

1. Hands-On Azure Devops: Cidc Implementation For Mobile, Hybrid, And Web Applications Using Azure Devops And Microsoft Azure: CICD Implementation for ... DevOps and Microsoft Azure (English Edition) Paperback – 1 January 2020. by Mitesh Soni
2. Jeff Geerling, "Ansible for DevOps: Server and configuration management for humans", First Edition, 2015.
3. David Johnson, "Ansible for DevOps: Everything You Need to Know to Use Ansible for DevOps", Second Edition, 2016.
4. Mariot Tsitoara, "Ansible 6. Beginning Git and GitHub: A Comprehensive Guide to Version Control, Project Management, and Teamwork for the New Developer", Second Edition, 2019.
5. <https://www.jenkins.io/user-handbook.pdf>
6. <https://maven.apache.org/guides/getting-started/>

HUMAN VALUES AND PROFESSIONAL ETHICS

III Year I Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6HS10	MC	2	0	0	0	50	-	50

COURSE OBJECTIVES:

1. To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2. To facilitate the development of a Holistic perspective among students towards life, profession and happiness, based on a correct understanding of the Human reality and the rest of Existence. Such a holistic perspective forms the basis of Value based living in a natural way.
3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually satisfying human behaviour and mutually enriching interaction with Nature.

COURSE OUTCOMES:

It ensures students sustained happiness through identifying the essentials of human values and skills.

1. It facilitates a correct understanding between profession and happiness
2. It helps students understand practically the importance of trust, mutually satisfying human behaviour and enriching interaction with nature.
3. Ability to develop appropriate technologies and management patterns to create harmony in professional and personal life.

UNIT-I

Introduction

Classes: 12

Need, basic Guidelines, Content and Process for Value Education: Understanding the need, basic guidelines, content and process for Value Education. Self Exploration - what is it? - its content and process; 'Natural Acceptance' and Experiential Validation - as the mechanism for self exploration. Continuous Happiness and Prosperity - A look at basic Human Aspirations. Right understanding, Relationship and Physical Facilities - the basic requirements for fulfillment of aspirations of every human being with their correct priority. Understanding Happiness and Prosperity correctly - A critical appraisal of the current scenario. Method to fulfill the above human aspirations: understanding and living in harmony at various levels.

UNIT-II Understanding Harmony in the Human Being - Harmony in Myself! **Classes: 12**

Understanding human being as a co-existence of the sentient 'I' and the material 'Body'. Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha. Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer). Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail. Programs to ensure Sanyam and Swasthya.

UNIT-III Understanding Harmony in the Family and Society - Harmony in Human - Human Relationship **Classes: 12**

Understanding harmony in the Family the basic unit of human interaction. Understanding values in human - human relationship; meaning of Nyaya and program for its fulfilment to ensure Ubhay-tripti; Trust (Vishwas) and Respect (Samman) as the foundational values of relationship. Understanding the meaning of Vishwas; Difference between intention and competence. Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in relationship. Understanding the harmony in the society (society being an extension of family): Samadhan, Samridhi, Abhay, Sah-astiva as comprehensive Human Goals. Visualizing a universal harmonious order in society - Undivided Society (AkhandSamaj), Universal Order (SarvabhaumVyawastha) - from family to world family!

UNIT-IV **Understanding Harmony in the nature and Existence - Whole existence as Co-existence** **Classes: 12**

Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature - recyclability and self-regulation in nature. Understanding Existence as Co-existence (Sah-astiva) of mutually interacting units in all-pervasive space. Holistic perception of harmony at all levels of existence.

UNIT-V **Implications of the above Holistic Understanding of Harmony on Professional Ethics** **Classes: 12**

Natural acceptance of human values, Definitiveness of Ethical Human Conduct, Basic for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics:

- A. Ability to utilize the professional competence for augmenting universal human order,
- B. Ability to identify the scope and characteristics of people-friendly and eco-friendly production systems,
- C. Ability to identify and develop appropriate technologies and management patterns for above production systems.
- D. Case studies of typical holistic technologies, management models and production systems. Strategy for transition from the present state to Universal Human Order.
- E. At the level of individual: as socially and ecologically responsible engineers, technologists and managers
- F. At the level of society: as mutually enriching institutions and organizations.

TEXT BOOKS:

1. R. R. Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.
2. Prof. K. V. Subba Raju, 2013, Success Secrets for Engineering Students, Smart Student Publications, 3rd Edition.

REFERENCE BOOKS:

1. Ivan Illich, 1974, Energy & Equity, The Trinity Press, Worcester, and HarperCollins, USA
2. E. F. Schumacher, 1973, Small is Beautiful: a study of economics as if people mattered. Blond & Briggs, Britain.
3. A Nagraj, 1998 Jeevan Vidya ekParichay, Divya Path Sansthan, Amarkantak.
4. Sussan George, 1976, How the Other Half Dies, Penguin Press, Reprinted 1986, 1991.
5. P. L. Dhar, R. R. Gaur, 1990, Science and Humanism, Commonwealth Publishers.
6. A. N. Tripathy, 2003, Human Values, New Age International Publishers.
7. Subhas Palekar, 2000, How to practice Natural Farming, Pracheen(Vaidik) Krishi Tantra Shodh, Amravati.
8. Donella H. Meadows, Dennis L. Meadows, Jorgen Randers, William W. Behrens III, 1972, Limits to Growth - Club of Rome's report, Universe Books.
9. E G Seebauer & Robert L. Berry, 2000, Fundamentals of Ethics for Scientists & Engineers, Oxford University Press.
10. M Govindrajan, S Natrajan & V. S Senthil kumar, Engineering Ethics (including Human Values), Eastern Economy Edition, Prentice Hall of India Ltd.

WEB REFERENCES:

1. Value Education website, <http://www.uptu.ac.in>
2. Story of Stuff, <http://www.storyofstuff.com>
3. Al Gore, An Inconvenient Truth, Paramount Classics, USA
4. Charle Chaplin, Modern Times, United Artists, USA
5. IIT Delhi, Modern Technology - the Untold Story

III B.TECH II SEMESTER SYLLABUS

MACHINE LEARNING

III B. TECH- II SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6AI06	PCC	3	0	0	3	40	60	100

Course Objectives:

1. Ability to comprehend the concept of supervised and unsupervised learning techniques
2. Differentiate regression, classification and clustering techniques and to implement their algorithms.
3. analyze the performance of various machine learning techniques and to select appropriate features for training machine learning algorithms.
4. Be familiar with basic machine learning algorithms with classification and clustering.
5. Gain experience of doing independent study and research

Course Outcomes:

Upon successful completion of the course, the student is able to

1. Recognize the characteristics of machine learning that makes it useful to solve real-world problems.
2. Evaluate exploratory data analysis and Data preparation and preprocessing on different datasets.
3. Design and implement machine learning solutions of classification, regression problems.
4. Choose an appropriate clustering technique to solve real world problems.
5. Choose a suitable machine learning model, implement and examine the performance of the chosen model for a given real world problems.

UNIT-I INTRODUCTION

Classes: 12

Machine Learning Foundations: Introduction to machine learning, learning problems and scenarios, need for machine learning, types of learning, standard learning tasks, the Statistical Learning Framework, Probably Approximately Correct (PAC) learning.

UNIT-II DESCRIPTIVE STATISTICS

Classes: 12

Data representation, types of data: Quantitative data-nominal, ordinal, Qualitative data- interval and continuous, central tendency- calculating mean mode median, mean vs median, variability, variance, standard deviation, Mean Absolute Deviation using sample dataset, finding the percentile, interquartile range, Box Plot, Outlier, whisker, calculating correlation, covariance, causation. Exploratory data analysis, Data preparation, Data preprocessing and Data visualization

UNIT-III SUPERVISED LEARNING

Classes: 14

Learning a Class from Examples, types of Classification- Linear, Non-linear, Multi-class and Multi-label classification.

Perceptron: – multilayer neural networks – back propagation - learning neural networks structures – support vector machines: – soft margin SVM and hard margin – going beyond linearity – generalization and over fitting – regularization – validation.

Decision Trees: ID3, Classification and Regression Trees (CART), Regression: Linear Regression, Multiple Linear Regression, Logistic Regression.

UNIT-IV UNSUPERVISED LEARNING

Classes: 12

Introduction to clustering, Hierarchical: AGNES, DIANA, Partitional: K-means clustering, K-Mode Clustering, Self-Organizing Map, Expectation Maximization, Gaussian Mixture Models, Principal Component Analysis (PCA), Locally Linear Embedding (LLE), Factor Analysis,

Neural Networks: Introduction, Perceptron, Multilayer Perceptron, Support vector machines: Linear and NonLinear, Kernel Functions, K-Nearest Neighbors.

UNIT-V ENSEMBLE AND PROBABILISTIC LEARNING

Classes: 12

Ensemble Learning Model Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Gradient Boosting, Xg boost, Stacking,

Bayesian Learning, Bayes Optimal Classifier, Naïve Bayes Classifier, Bayesian Belief Networks, Mining Frequent Patterns

Text Books:

1. Giuseppe Bonaccorso, —Machine Learning Algorithms, 2nd Edition, Packt, 2018
2. Tom Mitchel —Machine Learning, Tata McGraw Hill, 2017.

Reference Books:

1. Ethem Alpaydin, —Introduction to Machine Learning, PHI, 2004
 2. Stephen Marshland, —Machine Learning: An Algorithmic Perspective, CRC Press Taylor & Francis, 2nd Edition, 2015
 3. Abhishek Vijavargia —Machine Learning using Python, BPB Publications, 1st Edition, 2018
- INTRODUCTION TO MACHINE LEARNING
1. Introduction to Machine Learning The Wikipedia Guide
 2. Ethem Alpaydin, Introduction to machine learning, second edition, MIT press.

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

TESTING AUTOMATION

III B. TECH- II SEMESTER – IT

Course Code:	Category	Hours/Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT18	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

Verifying the functionality of the software.

1. Gain a basic understanding of measuring the quality and coverage of testing
2. Develop a working knowledge of Selenium Application.
3. Understand Web Elements with tools and technique.
4. learn about the latest tools and technologies that will help to get rid of bugs and create excellent software

COURSE OUTCOMES:

By the conclusion of this course, students will be able to:

1. Gain in-depth knowledge of quality assurance by understanding different methodologies, types, and levels of testing.
2. Understanding the History of complete Selenium versions and how Increase the accuracy and speed of tests
3. Understand the major concepts of Selenium such as IDE, RC, Grid and Web Driver.
4. Understand all the parameters of automating testing with Selenium.
5. Learn about additional tools such as TestNG.

UNIT I

Classes: 10

Introduction to Testing : Introduction to Stages of Testing, History of Automation Testing, Test Cases to Automate ,What is Framework, Different Automation tools, Challenges Automation & Mitigations.

Junit: Introduction, features, Annotations, Applications of Junit.

UNIT II

Classes: 12

Introduction to Selenium: History and various versions of selenium, Introduction to Basics of OOP concepts, Selenium IDE, Features of selenium IDE, Selenium RC, Selenium components, Advantages of using Selenium over other tools.

UNIT III

Classes: 10

Selenium Web Driver: Introduction to Selenium Web Driver 3.0, Automation Setup for Selenium Web Driver, Configuration of Selenium Jar using Eclipse, Fire path for Mozilla Firefox browser. Web Driver Interface Methods.

UNIT IV

Classes: 10

Introduction to Web Elements: Basics of HTML, Locators, locators available in Selenium, Web Element Methods.Handling Web Tables: What is Web Tables, Extracting values from Web Tables, Static and Dynamic Web Tables.

UNIT V

Classes: 10

TestNG: TestNG & TestNG features, How to integrate TestNG with Eclipse, TestNG Annotations Page Object Model, Advantages of POM, implement POM using Page Object and Page Factory by using Selenium Web Driver. Maven Tool: Installation create a maven project in Eclipse Maven Build Cycles.

TEXT BOOKS:

1. Automated Software Testing: Foundations, Applications and Challenges (Services and Business Process Reengineering)- 1st ed. 2020 Edition.by Ajay Kumar Jena (Editor), Himansu Das (Editor), Durga Prasad Mohapatra (Editor).
2. Automated Software Engineering: A Deep Learning-Based Approach (Learning and Analytics in Intelligent Systems) 1st ed. 2020 Edition-by Suresh Chandra Satapathy (Author), Ajay Kumar Jena (Contributor), Jagannath Singh (Contributor), & 1 more

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

PROFESSIONAL ELECTIVE – II

III B.Tech - II Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT39	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. State Space Searching techniques to solve a real-world problem
2. Heuristic Searching techniques to optimize a problem iteratively
3. Randomized Search and Emergent Systems to determine the best solution for the developed model
4. Optimal Path algorithms to determine shortest paths in different circuits
5. Problem Decomposition to decompose a problem into manageable sub-components is a necessity in complex problem-solving tasks

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Apply appropriate State Space Searching techniques to solve a real-world problem.
2. Apply appropriate Heuristic Searching techniques to solve a real-world problem.
3. Analyze the problem and infer new knowledge using Randomized Search and Emergent Systems.
4. Apply Optimal Paths Algorithms to real world problems.
5. Analyze the Problem Decomposition Methods and their test cases.

UNIT-I

Classes: 9

Introduction: Artificial Intelligence, Historical Backdrop, What is Artificial Intelligence, AI Agents –types of Agents, Applications of AI.

State Space Search: Generate and Test, Simple Search, Depth First Search (DFS), Breadth First Search (BFS), Comparison of BFS and DFS, Depth Bounded DFS (DBDFS), Depth First Iterative Deepening (DFID).

UNIT-II

Classes: 10

Heuristic Search: Heuristic Functions, Best First Search, Hill Climbing, Local Maxima, Solution Space Search, Variable Neighbourhood Descent, Beam Search, Tabu Search, Peak to Peak Methods..

UNIT-III

Classes: 10

Randomized Search and Emergent Systems: Iterated Hill Climbing, Simulated Annealing, Genetic Algorithms, the Travelling Salesman Problem, Neural Networks, Emergent Systems.

UNIT-IV

Classes: 10

Finding Optimal Paths: Brute Force, Branch and Bound, Refinement Search, Dijkstra's Algorithm, Algorithm A*, Admissibility of A*, Iterative Deepening A* (IDA*), Recursive Best First Search (RBFS), Pruning the Closed List, Pruning the Open List, Divide and Conquer Beam Stack Search.

UNIT-V

Classes: 9

Problem Decomposition: SAINT, Dendral, Goal Trees, Rule based Systems, XCON, Rule Based Expert Systems.

Case Studies: Autonomous Vehicle, Medical Image Analysis.

Text Books:

1. A First Course in Artificial Intelligence, Deepak Khemani, Tata Mc-Graw Hill Education, 2013.

Reference Books:

1. AI – A Modern Approach, Stuart Russel, Peter Norvig, 4th Edition, Pearson Education, 2010.
2. Artificial Intelligence, Kevin Knight, Elaine Rich, Shivshankar B. Nair, 3rd Edition, Tata Mc-Graw Hill Education, 2019.

DATA WRANGLING (PROFESSIONAL ELECTIVE - II)

III B.Tech - II Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6DS13	PEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Provide with the necessary knowledge of Data Science Concepts.
2. Learn how to collect, store and manage data from different sources.
3. Provide the concepts and needs of Data Visualization.
4. Introduce the important data science modules NumPy, and Matplotlib
5. Introduce the input/output with files in Python and statistical processing of a data using Pandas

COURSE OUTCOMES

At the end of the course, students will be able to:

1. Inspect and execute the basic data format.
2. Determine the computations with Excel and pdf files
3. Prioritize the concepts of data cleanup
4. Originate and analyze the Image and video data
5. Elaborate the concepts web scraping

UNIT - I INTRODUCTION TO DATA WRANGLING

CLASSES: 10

What Is Data Wrangling?- Importance of Data Wrangling -How is Data Wrangling performed?- Tasks of Data Wrangling-Data Wrangling Tools-Introduction to Python-Python Basics-Data Meant to Be Read by Machines-CSV Data-JSON Data-XML Data.

UNIT – II WORKING WITH EXCEL FILES AND PDFS

CLASSES: 12

Installing Python Packages-Parsing Excel Files-Parsing Excel Files -Getting Started with Parsing-PDFs and Problem Solving in Python-Programmatic Approaches to PDF Parsing-Converting PDF to Text-Parsing PDFs Using pdf miner-Acquiring and Storing Data-Databases: A Brief Introduction-Relational Databases: MySQL and PostgreSQL-Non-Relational Databases: NoSQL-When to Use a Simple File-Alternative Data Storage.

UNIT – III DATA CLEANUP

CLASSES: 12

Combining and Merging Data Sets - Reshaping and Pivoting - Data Transformation - String manipulations - Regular Expression

UNIT – IV DATA EXPLORATION AND ANALYSIS

CLASSES: 12

Exploring Data-Importing Data-Exploring Table Functions-Joining Numerous Datasets-Identifying Correlations-Identifying Outliers-Creating Groupings-Analyzing Data-Separating and Focusing the Data Presenting Data-Visualizing the Data-Charts-Time-Related Data-Maps-Interactives-Words-Images, Video, and Illustrations-Presentation Tools-Publishing the Data-Open Source Platforms.

UNIT – V WEB SCRAPING

CLASSES: 14

What to Scrape and How-Analyzing a Web Page-Network/Timeline-Interacting with JavaScript-In-Depth Analysis of a Page-Getting Pages-Reading a Web Page-Reading a Web Page with lxml-XPath-Advanced Web Scraping-Browser-Based Parsing-Screen Reading with Selenium-Screen Reading with Ghost. PySpidering the Web-Building a Spider with Scrapy-Crawling Whole Websites with Scrapy.

TEXT BOOKS:

1. Jacqueline Kazil and Katharine Jarmul. 2016. Data Wrangling with Python: Tips and Tools to Make Your Life Easier (1st. ed.). O'Reilly Media, Inc.
2. McGregor, Susan E. Practical Python Data Wrangling and Data Quality. " O'Reilly Media, Inc.", 2021.

NETWORK SECURITY (PROFESSIONAL ELECTIVE – II)

III Year II Semester: CSE/IT/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT27	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Identify the risks faced by computer networks and network security.
2. Explain access controls in computer networks and cloud security.
3. Use the different web and wireless security protocols.
4. Analyse the different protocols used in IP and SNMP to secure the computer networks.
5. Identify different types of viruses, intruders and use suitable firewall, IDS systems to prevent intruders.

UNIT – I Introduction

Classes: 10

INTRODUCTION: Security trends, OSI Security Architecture, Fundamental Security Design Principles, Attack Surfaces and Attack Trees, A model for network security.

UNIT – II Network Access Control and Cloud Security

Classes: 12

NETWORK ACCESS CONTROL: Network Access control, Extensible Authentication Protocol, IEEE 802.1x Port-Based Network Access Control.

CLOUD SECURITY: Cloud Computing, Cloud Security risks and countermeasures, Data Protection in cloud, cloud security as a services.

UNIT – III Asymmetric Cryptographic

Classes: 10

WEB SECURITY: Secure Socket Layer (SSL) and Transport Layer Security (TLS), Secure Electronic Transaction (SET), HTTPS, SSH

WIRELESS SECURITY: Wireless Security, Mobile Device Security, IEEE802.11i Wireless Lan Security

UNIT – IV Message Authentication

Classes: 12

IP SECURITY: Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload, Combining Security Associations and Key Management.

NETWORK MANAGEMENT SECURITY: Basic Concepts of SNMP, SNMPv1, SNMPv3

UNIT – V Intruders and Firewall

Classes: 09

INTRUDERS: Intruders, Viruses and related threats. Strong Authentication, Access Controls, Wireless Security, Honeypots, Traffic flow security

FIREWALL: Design and Types of Firewalls, Personal Firewalls, IDS, Trusted Systems.

Text Books:

1. William Stallings (2017), Network Security Essentials (Applications and Standards), 6th edition, Pearson Education, India.
2. William Stallings (2017), Cryptography and Network Security: Principles and Practice, 7th edition, Pearson Education, India.

Reference Books:

1. Charlie Kaufman (2002), Network Security: Private Communication in a Public World, 2nd edition, Prentice Hall of India, New Delhi.
2. Atul Kahate (2008), Cryptography and Network Security, 2nd edition, Tata Mc Grawhill, India.
3. Robert Bragg, Mark Rhodes (2004), Network Security: The complete reference, Tata Mc Grawhill, India.

AD-HOC & SENSOR NETWORKS (PROFESSIONAL ELECTIVE – II)

III B. TECH- II SEMESTER - IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT28	PEC	3	0	0	3	40	60	100

Course Objectives:

1. To understand the concepts of sensor networks
2. To understand the MAC and transport protocols for ad hoc networks
3. To understand the security of sensor networks
4. To understand the applications of adhoc and sensor networks

Course Outcomes:

At the end of the course, student will be able to:

1. Know the principles and characteristics of wireless sensor networks.
2. Use proactive routing protocols function and their implications on data transmission delay and bandwidth consumption.
3. Find issues in MAC Layer Protocols, transport layer protocols in MANETS and WSNs.
4. Find security issues and familiar with security key management mechanisms MANETS and WSNs.
5. Practice network simulations on various simulators.

UNIT – I

Classes: 10

Introduction to Ad Hoc Networks - Characteristics of MANETs, Applications of MANETs and Challenges of MANETs.

Routing in MANETs - Criteria for classification, Taxonomy of MANET routing algorithms, Topology- based routing algorithms-**Proactive**: DSDV; **Reactive**: DSR, AODV; Hybrid: ZRP; Position-based routing algorithms-**Location Services**-DREAM, Quorum-based; **Forwarding Strategies**: Greedy Packet, Restricted Directional Flooding-DREAM, LAR.

UNIT – II

Classes: 10

Data Transmission - Broadcast Storm Problem, **Rebroadcasting Schemes**-Simple-flooding, Probability-based Methods, Area-based Methods, Neighbor Knowledge-based: SBA, Multipoint Relaying, AHBP. **Multicasting**: **Tree-based**: AMRIS, MAODV; **Mesh-based**: ODMRP, CAMP; **Hybrid**: AMRoute, MCEDAR.

UNIT – III

Classes: 10

Geocasting: Data-transmission Oriented-LBM; Route Creation Oriented-GeoTORA, MGR. TCP over Ad Hoc TCP protocol overview, TCP and MANETs, Solutions for TCP over Ad hoc

UNIT – IV

Classes: 10

Basics of Wireless, Sensors and Lower Layer Issues: Applications, Classification of sensornetworks, Architecture of sensor network, Physical layer, MAC layer, Link layer, Routing Layer.

UNIT – V

Classes: 10

Upper Layer Issues of WSN: Transport layer, High-level application layer support, Adapting to the inherent dynamic nature of WSNs, Sensor Networks and mobile robots.

TEXT BOOKS:

1. Ad Hoc and Sensor Networks – Theory and Applications, Carlos Corderio Dharma P. Aggarwal, World Scientific Publications, March 2006, ISBN – 981–256–681–3.

2. Wireless Sensor Networks: An Information Processing Approach, Feng Zhao, Leonidas Guibas, Elsevier Science, ISBN – 978-1-55860-914-3 (Morgan Kaufman).

REFERENCE BOOKS:

1. Advanced Selenium Web Accessibility Testing, Software Automation Testing Secrets Revealed, Narayanan Palani(You?)|2019
2. Automated Software Testing A Complete Guide - 2020 Edition-by Gerardus Blokdyk (Author).
3. Software Automation Testing Secrets Revealed Part 2 Selenium Webdriver
Mastering Web Automated Testing -Use of Selenium with Practical Answers to Testing Interview Questions
Narayanan Palani(You?)|2016.

WEB LINKS

1. <https://www.guru99.com/selenium-tutorial.html>.
2. <https://www.browserstack.com/guide/automation-testing-tutorial>

SOFT COMPUTING

(PROFESSIONAL ELECTIVE – III)

III B. TECH- II SEMESTER - IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT29	PEC	3	0	0	3	40	60	100

Course Objective:

The course should enable the students to:

1. Teach basic neural networks, fuzzy systems, and optimization algorithms concepts and their relations
2. Provide knowledge of Neuron model, and Applications of NN to discuss their work.
3. Provide the graduate the better understanding of Fuzzy Logic and Evolutionary Computations.

Course Outcomes:

Upon successful completion of the course, the student is able to

1. Identify and describe soft computing techniques and their roles in building intelligent machines
2. Analyze fuzzy logic and reasoning to handle uncertainty and solve various engineering problems.
3. Apply neural networks and genetic algorithms to combinatorial optimization problems.
4. Evaluate and compare solutions by various soft computing approaches for a given problem.
5. Identify various Fuzzy oriented Databases and genetic algorithms.

UNIT – I

CLASSES: 12

BASICS OF ARTIFICIAL NEURAL NETWORK: Characteristics of Neural Networks, Structure and working of a biological neural network, artificial neural network: terminology, models of neurons: McCulloch Pitts model, Perceptron model, Adaline model, topology, Basic learning laws.

FUNCTIONAL UNITS FOR ANN FOR PATTERN RECOGNITION TASK: Pattern recognition problem, Basic functional units, PR by functional units.

UNIT – II

CLASSES: 12

FEEDFORWARD NEURAL NETWORKS:

SUPERVISED LEARNING - I: Perceptrons - Learning and memory, Learning algorithms, Error correction and gradient decent rules, Perceptron learning algorithms.

SUPERVISED LEARNING-II: Backpropagation, Multilayered network architectures, Back propagation learning algorithm, Example applications of feed forward neural networks.

UNIT – III

CLASSES: 12

FEEDBACK NEURAL NETWORKS & SELF ORGANIZING FEATURE MAP: Introduction, Associative learning, Hopfield network, Error performance in Hopfield networks, simulated annealing, Boltzmann machine and Boltzmann learning, state transition diagram and false minima problem, stochastic update, simulated annealing, Boltzmann machine, bidirectional associative memory, bam stability analysis. Self organization, generalized learning laws, competitive learning, vector quantization, self organizing feature map, applications of self organizing feature map.

UNIT – IV

CLASSES: 12

FUZZY LOGIC: Fuzzy set theory, crisp sets, operations on crisp set, fuzzy sets, fuzzy versus crisp, operations, fuzzy relations, crisp relations, properties. Fuzzy logic Application: Fuzzy Control of Blood Pressure.

UNIT – V

CLASSES: 12

FUZZY LOGIC IN DATABASE AND INFORMATION SYSTEMS: Fuzzy Information, Fuzzy Logic in database Systems, Fuzzy Relational data Models, operations in Fuzzy Relational data Models, Design theory for Fuzzy

Relational databases, Fuzzy information Retrieval and Web search, Fuzzy Object Oriented databases.

GENETIC ALGORITHMS: Introduction to Genetic Algorithms, Evolutionary Algorithms.

TEXT BOOKS:

1. Satish Kumar (2004), Neural Networks A classroom Approach, Tata McGraw Hill Publication, New Delhi.
2. Lotfi A. Zadeh(1997), Soft computing and Fuzzy Logic, World Scientific Publishing Co., Inc. River Edge, NJ, USA.

REFERENCE BOOKS:

1. B. Yegnanarayana (2006), Artificial Neural Networks, Prentice Hall of India, New Delhi, India.
 2. John Yen, Reza Langari (2006), Fuzzy Logic, Pearson Education, New Delhi, India.
- S. Rajasekaran, Vijaylakshmi Pari (2003), Neural networks, Fuzzy Logic and Genetic Algorithms Synthesis and Applications, Prentice Hall of India, New Delhi, India.

INFORMATION RETRIEVAL SYSTEMS (PROFESSIONAL ELECTIVE – III)

III B. TECH- II SEMESTER - IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT26	PEC	3	0	0	3	40	60	100

Pre-requisites: Relational Database Management Systems, Data Mining

Course Objectives: The Students will learn the following:

1. To use different information retrieval techniques in various application areas
2. To apply IR principles to locate relevant information large collections of data
3. To analyze performance of retrieval systems when dealing with unmanaged datasources
4. To implement retrieval systems for web search tasks.

Course Outcomes:

After the successful completion of the course, the students would be able to:

1. Understand the basics of web search
2. Identify the data structures like Inverted Indices used in Information retrieval systems
3. Apply the different techniques on compression of index including the dictionary and its posting list
4. Understand and Apply various text search algorithms
5. Identify the evaluation metrics of information system

Unit I: Introduction

Classes: 09

Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses. Information Retrieval System Capabilities - Search, Browse, Miscellaneous

Unit II: Cataloging and Indexing

Classes: 09

Objectives, Indexing Process, Automatic Indexing, Information Extraction.

Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N - gram data structure, PAT data structure, Signature file structure, Hypertext data structure.

Automatic Indexing: Classes of automatic indexing, Statistical indexing, Natural language, Concept indexing, Hypertext linkages

Unit III: Document and Term Clustering

Classes: 10

Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters.

User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, Weighted searches of Boolean systems, Multimedia Searching.

Unit IV: Information presentation and Text Search Algorithms

Classes: 10

Information presentation Introduction, Presentation of the hits, Display of the Item, Collaborative Filtering, Multimedia presentation, Human Perception and Presentation.

Text Search Algorithms

Introduction, Software text search algorithms, Hardware text search systems.

Unit V: Information System Evaluation

Classes: 08

Introduction, Measures used in system evaluations, Multimedia Information retrieval Evaluation, Measurement example – TREC Evolution.

Text books:

1. Gerald Kowalski, Information Retrieval Architecture and Algorithms, Springer Science, 2011.
2. Ricardo Baeza-Yates, Modern Information Retrieval, Pearson Education, 2007.
3. David A Grossman and Ophir Frieder, Information Retrieval: Algorithms and Heuristics, 2nd Edition, Springer, 2004

References:

1. William B Frakes and Ricardo Baeza - Yates, Information Retrieval Data Structures and Algorithms, Pearson Education, 1992.
2. Robert Korfhage, John Wiley & Sons, Information Storage and Retrieval.
3. Christopher D. Manning and Prabhakar Raghavan, Introduction to Information Retrieval, Cambridge University Press, 2008

BLOCK CHAIN TECHNOLOGY (PROFESSIONAL ELECTIVE – III)

III B. TECH- II SEMESTER – IT/DS

Course Code	Category	Hours/Week			Credits	Maximum Marks		
		L	T	P		CI E	SE E	Total
A6CY25	PEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand how block chain systems (mainly Bitcoin and Ethereum) work
2. Knowledge on cryptocurrency
3. Design smart contracts and distributed applications
4. Builds mart contracts and distributed applications
5. Integrate ideas from block chain technology into their own projects

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Explain design principles of Bitcoin and Ethereum.
2. Design, build, and deploy a distributed application.
3. Analyze the working of Smart Contracts
4. Analyze and apply the working of smart contracts
5. Created – centralized appson Ethereum

UNIT-I UNDERSTANDINGBLOCKCHAIN

CLASSES:10

Introduction to Block chain, network and mechanism, history of block chain, types of block chain, blocks and transactions, Peer to Peer systems, the block structure in block chain, Distributed Ledger Technology (DLT), Digital Signatures, Hashes as addresses, Using a key as identity.

UNIT-II BLOCKCHAIN1.0:CRYPTOCURRENCY

CLASSES:10

Bitcoin Basics, Tokens and ICOs, Distributed Consensus, Consensus in Bitcoin, The Double – Spend and Byzantine Generals' Computing Problems, How a Cryptocurrency Works.

UNIT-III BLOCKCHAINSMARTCONTRACTS

CLASSES:10

Introduction to Ethereum, Smart Contracts, Transactions in Ethereum, Consensus Mechanism in Ethereum, Programming using Solidity, Ethereum accounts, Ether, Gas, structuring a contract, ERC20 Standards.

UNIT-IV BUILDINGSMARTCONTRACTS

CLASSES:10

Wallet Development Projects, Block chain Development Platforms and APIs, Best Practices and Auditing, Smart Contract Mini Project.

UNIT-V BUILDINGDAPPS

CLASSES:08

Introduction to Hyper ledger, Key Frame works and Tools, Programming using Golang, DApp Mini Project.

TEXT BOOKS

1. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, Princeton University Press

REFERENCE BOOKS

1. Antonopoulos, Mastering Bitcoin: Unlocking Digital Cryptocurrencies
2. Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System
3. Dr. Gavin Wood, "ETHEREUM: A Secure Decentralized Transaction Ledger," Yellow

NATURAL LANGUAGE PROCESSING (PROFESSIONAL ELECTIVE – III)

III B.TECH - II SEMESTER: IT/AI

Course Code	Category	Hours/Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6AI14	PEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Provide theoretical concepts of language processing that shows how to explore interesting bodies of text.
2. Familiarize with fundamental topics in language processing that include tagging, classification, and information extraction using tiny Python programs.
3. Facilitate understanding of formal grammar to describe the structure of an unlimited set of sentences.
4. Acquaint with methods to parse a sentence, recognize its syntactic structure and construct - representations of meaning.
5. Familiarize with design of existing corpora, the typical workflow for creating a corpus and the life cycle of a corpus

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Discuss basic concepts and applications of Natural Language processing
2. Analyze words using prefix, suffix, grammar, PoS.
3. Analyze structures of Semantic and Discourse to determine what words and phrases mean in relationship
4. Analyze Natural Language Generation and apply machine translation
5. Implement levels of NLP system using the Components or lexical resources to demonstrate Morphology syntax of a language.

UNIT-I OVERVIEW AND LANGUAGE MODELING CLASSES:10

Overview: Origins and challenges of NLP Language and Grammar-Processing Indian Languages-NLP Applications-Information Retrieval. Language Modeling: Introduction-Various Grammar-based Language Models-Statistical Language Model.

UNIT-II WORD LEVEL AND SYNTACTIC ANALYSIS CLASSES:12

Word Level Analysis: Introduction Morphological Parsing-Spelling Error Detection and correction-Words and Word Classes-Part-of Speech Tagging. Parsing: Constituency Parsing - Probabilistic Parsing.

UNIT-III SEMANTIC ANALYSIS AND DISCOURSE PROCESSING CLASSES:10

Semantic Analysis: Introduction- Meaning Representation-Lexical Semantics Ambiguity-Word Sense Disambiguation. Discourse Processing: Introduction- cohesion-Reference Resolution Discourse Coherence and Structure.

UNIT-IV NATURAL LANGUAGE GENERATION AND MACHINETRANSLATION CLASSES:12

Natural Language Generation: Introduction-Architecture of NLG Systems Generation Tasks and Representations-Application of NLG. Problems in Machine Translation, Characteristics of Indian Languages- Machine Translation Approaches Translation involving Indian Languages.

UNIT -V APPLICATIONS AND LEXICAL RESOURCES CLASSES:12

Information Extraction, Automatic Text Categorization and Text Summarization, Question-Answering System. LEXICAL RESOURCES: Introduction - WordNet- FrameNet – Stemmers - POS Tagger, ResearchCorpora, NLTK.

TEXTBOOKS

1. Tanveer Siddiqui, U.S. Tiwary, "Natural Language Processing and Information Retrieval", Oxford University Press, 2008.

REFERENCE BOOKS

1. Christopher Manning "Foundations of Statistical Natural Language Processing," MIT Press, July 1999.
2. Dan Jurafsky, James H Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, Pearson Education India, 2nd edition (2013).

WEB REFERENCES

1. <https://huggingface.co/learn/nlp-course/chapter1/3?fw=pt>
2. https://www.tutorialspoint.com/natural_language_processing/index.htm

MACHINE LEARNING LAB

III B.TECH – II SEMESTER: IT/AI

Course Code	Category	Hours/Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6AI09	PCC	0	0	3	1.5	40	60	100

COURSE OBJECTIVES

1. Make use of Data sets in implementing the machine learning algorithms
2. Implement the machine learning concepts and algorithms in any suitable language of choice.
3. To impart knowledge on the basic concepts underlying machine learning.
4. To acquaint with the process of selecting features for model construction.
5. To familiarize different types of machine learning techniques.

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Design and implement machine learning solutions to classification, regression problems.
2. Analyze the complexity of Machine Learning algorithms and their limitations.
3. Apply appropriate data sets to the Machine Learning algorithms.
4. Identify and apply Machine Learning algorithms to solve real world problems
5. Apply supervised and unsupervised techniques on various data sets.

LIST OF EXPERIMENTS

WEEK-1 BASICS

Write a program to demonstrate the following

- a) Operation of data types in Python.
- b) Different Arithmetic Operations on numbers in Python.
- c) Create, concatenate and print a string and access substring from a given string.
- d) Append, and remove lists in python.
- e) Demonstrate working with tuples in python.
- f) Demonstrate working with dictionaries in python.

WEEK-2 STATISTICAL OPERATIONS

Using python write a NumPy program to compute the

- a) Expected Value
- b) Mean
- c) Standard deviation
- d) Variance
- e) Covariance
- f) Covariance Matrix of two given arrays.

WEEK-3 DATA PREPROCESSING – CONTINUOUS / DISCRETE DATA

For a given set of training data examples stored in a .CSV file, demonstrate Data Preprocessing in Machine learning with the following steps

- a) Getting the dataset.
- b) Importing libraries.
- c) Importing datasets.
- d) Finding Missing Data.
- e) Finding Outliers
- f) Splitting dataset into training and test set.
- g) Feature scaling.

WEEK-4 DATA PREPROCESSING – CATEGORICAL DATA

For a given set of training data examples stored in a .CSV file, demonstrate Data Preprocessing

In Machine learning with the following steps

- Getting the dataset.
- Importing libraries.
- Importing datasets.
- Finding Missing Data.
- Encoding Categorical Data.
- Splitting dataset into training and test set.
- Feature scaling.

WEEK-5 DECISION TREE

Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

WEEK-6 LINEAR REGRESSION

Build a linear regression model using python for a particular data set by

- Splitting Training data and Test data.
- Evaluate the model (intercept and slope).
- Visualize the training set and testing set
- predicting the test set result
- compare actual output values with predicted values.

WEEK-7 MULTIPLE LINEAR REGRESSION

Build a multiple linear regression model using python for a particular data set by

- Splitting Training data and Test data.
- Evaluate the model (intercept and slope).
- Visualize the training set and testing set
- predicting the test set result
- compare actual output values with predicted values.

WEEK-8 LOGISTIC REGRESSION

The dataset contains information of users from a company's database. It contains information about UserID, Gender, Age, EstimatedSalary, and Purchased. Use this dataset for predicting that user will purchase the company's newly launched product or not by Logistic Regression model.

User ID	Gender	Age	EstimatedSalary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	0
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	0
15728773	Male	27	58000	0
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	0
15727311	Female	35	65000	0
15570769	Female	26	80000	0
15606274	Female	26	52000	0
15746139	Male	20	86000	0
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1
15617482	Male	45	26000	1
15704583	Male	46	28000	1

WEEK-9 CLUSTERING

A python program to implement K-Means, Hierarchical Clustering and PCA.

WEEK-10 **KNN & SVM**

Write a Python program to implement KNN and SVM.

WEEK-11 **NAIVE BAYES**

Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.

WEEK-12 **RANDOM FOREST ALGORITHM**

Implement Random Forest Algorithm using Python.

TEXTBOOKS

1. Aurelien Géron - Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition. September 21019, O'Reilly Media, Inc., ISBN: 9781492032649.
2. Tom Mitchel "Machine Learning", Tata McGraW Hill, 2017.

TESTING AUTOMATION LAB

III- B. TECH - II SEMESTER - IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT19	PCC	0	0	3	1.5	40	60	100

COURSE OBJECTIVE

To teach the students basics Testing and Selenium programs and its execution

1. To teach the students the differences between Manual Testing and Automation Testing
2. To make the students learn concepts like Web Elements and Basic of Frame Work
3. To make the students understand Tools and techniques to test the web application and its Functionality.
4. To make the students understand the usage TestNG and Maven Working Technique.

COURSE OUTCOMES:

1. Understand the use of different Types of Testing and Installation of Software Supporting for Testing.
2. Implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity.
3. Understand and implement concepts on Web Application Testing and tools need to test those application.
4. Develop java application to interact with TestNg and Maven Component (JDBC Driver).

LIST OF EXPERIMENTS

Week -1

Program-1: Installation of Selenium IDE and Installation of Firebug
Supporting Software Maven

- Mozilla Firefox
- Install Java (jdk)
- Active internet connection

Week-2

Use the Gamil Login page Web Application to Test manually (Function Testing, Integration Testing and UAT).

Week-3

Use Facebook Registration page to Test for Beta ,alpha and Smoke Testing.to valid its input fields Properly

Week-4

Use the Mercury Tours website as our web application under test. It is an online flight reservation system that contains all the elements we need for this tutorial. Its URL is <http://newtours.demoaut.com/> and this will be our Base URL.

Week-5

Locators in Selenium IDE

- Locating by ID
- Locating by Name
- Locating by XPath
- Locating by DOM (Document Object Model)

Week-6

Write a Program for Accessing Links & Tables - Selenium Web driver.

Week-7

Write a program for Keyboard Mouse Events & Uploading Files - Web driver

Week 8

Uploading Files :use <http://www.megafileupload.com/> as our test application. This site easily allows any visitor to upload and download files without requiring them to sign up.

Week 9

Installing TestNG in Eclipse:

Step 1

- Launch Eclipse.
- On the menu bar, click Help.
- Choose the “Install New Software...” option

Step 2

- In the Install dialog box, click the Add button

Step 3

1. In “Name”, type TestNG.
2. In “Location”, type <http://beust.com/eclipse>.
3. Click OK

Step 4

- Notice that “TestNG - <http://beust.com/eclipse>” was populated onto the “Work with:” textbox.
- Check the “TestNG” check box as shown below, then click Next.

Step 5

- Click Next again on the succeeding dialog box until you reach the License Agreement dialog. Click “I accept the terms of the license agreement” then click Finish

Week 10

Develop an Small Web Application which Contains

- Home Page
- Login or Register page
- Contact information page .

Do testing of each Page using Selenium Testing

REFERENCE BOOKS:

1. Advanced Selenium Web Accessibility Testing, Software Automation Testing Secrets Revealed, Narayanan Palani (You?)|2019
2. Automated Software Testing A Complete Guide - 2020 Edition-by Gerardus Blokdyk (Author).
3. Software Automation Testing Secrets Revealed Part 2 Selenium Webdriver Mastering Web Automated Testing -Use of Selenium with Practical Answers to Testing Interview Questions Narayanan Palani(You?)|2016.

WEB LINKS

1. <https://www.guru99.com/selenium-tutorial.html>.
2. <https://www.browserstack.com/guide/au>

ENVIRONMENTAL SCIENCE

III B. TECH- II SEMESTER: IT (only for LEs)

Course Code:	Category	Hours / Week				Credits			Maximum Marks	
		L	T	P	C	CIE	SEE	Total		
A6BS11	MC	2	0	0	0	50	-	50		

COURSE OUTCOMES:

On Successful completion of this course, Students will be able to

1. Identify the consequences of human actions on the web of life, global economy and quality of human life.
2. Evaluate the strategies for scientific, social, economic and legal environmental protection.
3. Study the impact of conservation of biodiversity.
4. Analyze the reasons for environmental pollution.
5. Assess the environmental impact of air, water, biological and socio-economical aspects and risk assessment towards sustainable future.

UNIT-I

ECOSYSTEMS

CLASSES: 7

Ecosystems: Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food web and ecological pyramids.

Flow of energy, Biogeochemical cycles, Bioaccumulation, Bio magnification, ecosystem value, services and carrying capacity.

UNIT-II

NATURAL RESOURCES

CLASSES: 8

Natural Resources: Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems-case studies. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources. Land resources: Forest resources.

Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.

UNIT-III

BIODIVERSITY AND BIOTIC RESOURCES

CLASSES: 5

Biodiversity and Biotic Resources: Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values.

India as a mega diversity nation: Hot spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation. National Biodiversity Act-Case studies.

UNIT-IV

ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES

CLASSES: 5

Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Air Pollution, Water Pollution drinking water quality standards. Soil Pollution Impacts of modern agriculture, Noise Pollution Health hazards, standards. Concepts of bioremediation.

Global Environmental Problems and Global Efforts: Ozone depletion and Ozone depleting substances (ODS) Concepts of Bioremediation International conventions / Protocols: Earth summit, Kyoto protocol and Montréal Protocol; NAPCC-Gol Initiatives, COP 24, COP25.

UNIT-V

ENVIRONMENTAL POLICY, LEGISLATION & EIA

CLASSES: 5

Environmental Policy, Legislation & Environmental Impact Assessment (EIA):

Environmental Protection act, Legal aspects Air Act1981, Water Act, Forest Act, Wild life Act, Municipal solid waste management, biomedical waste management hazardous waste management and handling rules.

Environmental Impact Assessment: EIA structure, methods of baseline data acquisition. Strategies for risk assessment, Towards Sustainable Future: Concept of Sustainable Development, Urban Sprawl, Concept of Green Building.

TEXT BOOKS:

1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission.
1. Environmental Studies by R. Rajagopalan, Oxford University Press.

IV B.TECH I SEMESTER SYLLABUS

BIG DATA TECHNOLOGIES

IV B. TECH- I SEMESTER:

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6DS06	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES

1. Understand what big data is and how Big Data Technologies can help organizations achieve a competitive advantage.
2. Provide an overview of Apache Hadoop and its ecosystem components.
3. Understand Map Reduce Jobs
4. Processing of Big Data with advanced architectures like Spark.
5. To understand practical machine learning scalable and easy.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Understand fundamentals of Big Data Technologies.
2. Investigate Hadoop framework and Hadoop Distributed File system.
3. Demonstrate the MapReduce programming model to process the big data along with Hadoop tools.
4. Implement Big Data code in Apache Spark (in PySpark).
5. Run Supervised and Unsupervised machine learning on Large-Scale Data.

UNIT-I Introduction to Big Data Analytics

Classes :12

Introduction to Big Data Analytics: Big Data, Scalability and Parallel Processing, Designing Data Architecture, Data Sources, Quality, Pre-Processing and Storing, Data Storage and Analysis, Big Data Analytics Applications and Case Studies.

UNIT-II Introduction to Hadoop

Classes-12

Introduction to Hadoop: Introduction, Hadoop and its Ecosystem, Hadoop Distributed File System, MapReduce Framework and Programming Model, Hadoop Yarn, Hadoop Ecosystem Tools.

Hadoop Distributed File System Basics: HDFS Design Features, Components, HDFS User Commands.

Hadoop Ecosystem Components: Using Apache Pig, Hive, Sqoop, Flume, Oozie, HBase.

UNIT-III MapReduce, Hive and Pig

Classes:12

MapReduce, Hive and Pig: Introduction, MapReduce Map Tasks, Reduce Tasks and MapReduce Execution, Composing MapReduce for Calculations and Algorithms, Hive, HiveQL, Pig.

UNIT-IV Large-Scale Data Processing with PySpark

Classes:12

Apache Spark, Spark programming. (Python and PySpark), RDDs, Data Frames, Spark SQL, PySpark, NumPy, SciPy, Code Optimization, Cluster Configurations, Linear Algebra Computation in Large Scale, Distributed File Storage Systems.

UNIT-V Large Scale Machine Learning with Spark

Classes:13

Basic statistics, Data sources, Pipelines, Extracting, transforming and selecting features, Classification and Regression, Clustering, Collaborative filtering, Frequent Pattern Mining, Model selection and tuning.

Text Books:

1. Raj Kamal and Preeti Saxena, "Big Data Analytics Introduction to Hadoop, Spark, and Machine-Learning", McGraw Hill Education, 2018 ISBN: 9789353164966, 9353164966
2. Douglas Eadline, "Hadoop 2 Quick-Start Guide: Learn the Essentials of Big Data Computing in the Apache Hadoop 2 Ecosystem", 1st Edition, Pearson Education, 2016. ISBN13: 978-9332570351

Reference Books:

1. Tom White, "Hadoop: The Definitive Guide", 4th Edition, O'Reilly Media, 2015.ISBN-13: 978- 9352130672
2. Perrin, J. (2020). Spark in action (2nd ed.). (Covers Apache Spark 3 with examples in Java, Python, and Scala) O'Reilly Media Inc.
3. Arshdeep Bahga, Vijay Madisetti, "Big Data Analytics: A Hands-On Approach", 1st Edition, VPT Publications, 2018. ISBN-13: 978-0996025577
4. Damji, J., Wenig, B., Das, T., Lee, D. (2020). Learning spark (2nd ed.) O'Reilly Media Inc.
5. Nudurupati, S. (2021). Essential PySpark for scalable data analytics: A beginner's guide to harnessing the power and ease of PySpark 3 Packt Publishing

CYBER SECURITY AND CYBER LAWS

IV B. TECH- I SEMESTER: IT/CSE

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS36	PCC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To learn cybercrime cyber security and cyber law.
2. To understand the cyber attacks.
3. To understand information gathering.
4. To learn how to detect and prevent a cyber attack and tools for mitigating them.
5. Develop Competencies For Dealing With Frauds And Deceptions (Confidence Tricks, Scams) And Other Cyber Crimes

COURSE OUTCOMES

1. Explain the basics of cyber security, cyber crime and cyber law.
2. Classify various types of attacks and learn the tools to launch the attacks.
3. Apply various tools to perform information gathering.
4. Understand Cyber crimes and cyber laws.
5. Explore The Legal And Policy Developments In Various Countries To Regulate Cyberspace

UNIT - I INTRODUCTION

CLASSES: 16

Cyber Security-definition, Need for Cyber Security, Protection, Security, risk, threat, vulnerability, exploit, attack, confidentiality, integrity, availability, non-repudiation, authentication , authorization, codes, plain text, encryption, decryption, cipher text, key, ciphers, Symmetric and asymmetric cryptography, Public key , private key ,Crypt analysis, Cyber forensics.

UNIT - II CYBER ATTACKS

CLASSES:14

CYBER ATTACKS (definitions and examples): Denial-of-service attacks, Man-in-the middle attack, Phishing, spoofing and spam attacks, Driveby attack, Password attack, SQL injection attack, Cross-site scripting attack, Eavesdropping attack, Birthday attack, Malware attacks, Social Engineering attacks

UNIT - III HANDLING THE ATTACKS

CLASSES: 11

Firewalls, logging and intrusion detection systems, e-mail security, security issues in operating systems, ethics of hacking and cracking.

UNIT - IV CYBER CRIMES

CLASSES: 12

Cyber Crime and Information Security – History of Cyber Crime ,classifications of Cyber Crimes – Tools and Methods –Password Cracking, Key loggers, Spywares, SQL Injection – Network Access Control – CloudSecurity – Web Security – Wireless Security

UNIT - V CYBER SPACE AND CYBER LAWS

CLASSES: 16

Cyber Space- Fundamental definitions -Interface of Technology and Law – Jurisprudence and-Jurisdiction in Cyber Space - Indian Context of Jurisdiction - Enforcement agencies – Need for IT act - UNCITRAL – E-Commerce basics .Information Technology Act, 2000 - Aims and Objects — Overview of the Act – Jurisdiction
Cyber Law,A Global Perspective on Cyber Crimes; Cyber Laws – The Indian IT Act – Cybercrime and Punishment.

TEXT BOOKS

1. Anand Shinde, "Introduction to Cyber Security Guide to the World of Cyber Security", Notion Press, 2021.
2. Nina Godbole, Sunit Belapure, "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Wiley Publishers, 2011.
3. Cyber Law- The Indian Perspective, Pavan Duggal
4. Guide to Cyber Law, Rodney D. Ryder

REFERENCE BOOKS

1. David Kim, Michael G. Solomon, "Fundamentals of Information Systems Security", Jones & Bartlett Learning Publishers, 2013
2. Patrick Engebretson, "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made easy", Elsevier, 2011
3. Justice Yatindra Singh: Cyber Laws, Universal Law Publishing Co., New Delhi
4. Farouq Ahmed, Cyber Law in India, New Era publications, New Delhi

PATTERN RECOGNITION PROFESSIONAL ELECTIVE - IV

IV B. TECH- I SEMESTER: IT/AI

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6AI28	PEC	3	0	0	3	40	60	100

Prerequisites

1. Students are expected to have knowledge basic linear algebra, basic probability theory and basic programming techniques
2. A course on "Computational Mathematics"
3. A course on "Computer Oriented Statistical Methods"

Course Objectives

The course should enable the students to:

1. This course introduces fundamental concepts, theories, and algorithms for pattern recognition and machine learning.
2. To Understand Pattern Representation, Nearest Neighbor Based Classifier, Bayes and Classifier.
3. To get knowledge on Hidden Markov Models, Decision Trees.
4. To Acquire Support Vector Machines
5. To Understand Clustering and an application of hand-written digit recognition.

Course Outcomes

At the end of the course, student will be able to:

1. Understand the theory, benefits, inadequacies and possible applications of various machine learning and pattern recognition algorithms
2. Understanding, identify and employ suitable machine learning techniques in classification of pattern recognition
3. To understand hidden markov models, decision tree.
4. Analyze support vector machines, combination of classifiers
5. Acquire partitional clustering, clustering and an application of hand-written digit recognition

UNIT-I INTRODUCTION TO PATTERN RECOGNITION

Classes: 09

Introduction: What is Pattern Recognition, Data Sets for Pattern Recognition, Different Paradigms for Pattern Recognition. Representation: Data Structures for Pattern Representation, Representation of Clusters, Proximity Measures, Size of Patterns, Abstractions of the Data Set, Feature Extraction, Feature Selection, Evaluation of Classifier, Evaluation of Clustering.

UNIT-II CLASSIFIERS BASED ON BAYES DECISION THEORY

Classes: 10

Nearest Neighbor Based Classifier: Nearest Neighbor Algorithm, Variants of the NN Algorithm use of the Nearest Neighbor Algorithm for Transaction Databases, Efficient Algorithms, Data Reduction, Prototype Selection. Bayes Classifier: Bayes Theorem, Minimum Error Rate Classifier, Estimation of Probabilities, Comparison with the NNC, Naïve Bayes Classifier, Bayesian Belief Network.

UNIT-III HIDDEN MARKOV MODELS, DECISION TREE

Classes: 10

Hidden Markov Models: Markov Models for Classification, Hidden Markov Models, Classification using HMMs. Decision Trees: Introduction, Decision Tree for Pattern Classification, Construction of Decision Trees, Splitting at the Nodes, Overfitting and Pruning, Examples of Decision Tree Induction.

UNIT-IV

SUPPORT VECTOR MACHINES, COMBINATION OF CLASSIFIERS

Classes: 08

Support Vector Machines: Introduction, Learning the Linear Discriminant Functions, Neural Networks, SVM for Classification. Combination of Classifiers: Introduction, Methods for Constructing Ensembles of Classifiers, Methods for Combining Classifiers.

UNIT-V

PARTITIONAL CLUSTERING, CLUSTERING

Classes: 11

Clustering: Why is Clustering Important, Hierarchical Algorithms, Partitional Clustering, and Clustering Large Data Sets. An Application-Hand Written Digit Recognition: Description of the Digit Data, Preprocessing of Data, Classification Algorithms, Selection of Representative Patterns, Results.

Text Books:

1. Pattern Recognition: An Algorithmic Approach: Murty, M. Narasimha, Devi, V. Susheela, Springer Pub, 1 st Ed.

Reference Books:

1. Machine Learning - Mc Graw Hill, Tom M. Mitchell.
2. Fundamentals Of Speech Recognition: Lawrence Rabiner and Bing- Hwang Juang. PrenticeHall Pub.

Web references:

1. <https://www.coursera.org/learn/libm-ai-workflow-feature-engineering-bias-detection>
2. <https://www.coursera.org/learn/3d-modeling-rhinoscript>
3. <https://www.coursera.org/learn/text-retrieval>
4. https://onlinecourses.nptel.ac.in/noc23_ee119/preview
5. <https://www.coursera.org/learn/deteccion-objetos>

E -Text Books:

[https://darmanto.akakom.ac.id/pengenalanpola/Pattern%20Recognition%204th%20Ed.%20\(2009\).pdf](https://darmanto.akakom.ac.id/pengenalanpola/Pattern%20Recognition%204th%20Ed.%20(2009).pdf)

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

DATA VISUALIZATION USING TABLEAU (PROFESSIONAL ELECTIVE -IV)

IV B. TECH- I SEMESTER: IT/DS

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6DS21	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

1. Connect to the data and customize a data source.
2. Edit metadata; create groups and hierarchies in field data.
3. Use sets to compare data subsets.
4. Use the Tableau workspace to create visualizations.
5. Build a range of essential chart types for analysis.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Create basic calculations, including arithmetic calculations, custom aggregations, and ratios; and use quick table calculations.
2. Apply analytics using reference lines and box plots.
3. Create parameters to help users dynamically modify values.
4. Combine data sources by creating relationships, joins, unions, and blends.
5. Build interactive dashboards and stories to reveal data insights.

UNIT-I

Classes :12

Connect to and Customize Data: Connecting to Data, customizing a Data Source, working with a Data Extract, **Organize Data and Create Filters:** Creating Groups in Your Data, Creating Hierarchies in Your Data, Understanding Filtering in Tableau, Filtering Your Data, Sorting Your Data, Using Sets to Highlight Data.

UNIT-II

Classes-12

Build Common Views: Working with Dates to Visualize Time-Based Data, Creating Custom Date Fields and Hierarchies, Comparing Multiple Measures in Views, Using Scatter Plots To Show Relationships Between Measures, Creating Spreadsheet-like Views Using Text Tables, Using a Highlight Table to Show Specific Values, Showing Breakdowns of the Whole Using Pie Charts, Showing Breakdowns of the Whole Using Tree Maps, Using Bar-in-Bar Charts and Bullet Graphs to Compare Measures, **Map Geographic Data:** Creating Symbol and Filled Maps, Creating a Density Map

UNIT-III

Classes:12

Create Calculated Fields: Creating Calculated Fields for Deeper Analysis, Working with String and Type Conversion Functions, Working with Date Functions, Working with Aggregate Functions, **Apply Table Calculations:** Using Quick Table Calculations to Analyse Data, **Create Level of Detail (LOD) Expressions:** Using Level of Detail Expressions, Using Filters with LOD Expressions, Using Fixed LOD Expressions for Cohort Analysis

UNIT-IV

Classes:12

Apply Analytics: Highlighting Values with Reference Lines and Bands, Using Parameters to Control Data in the View, Using Histograms and Box & Whisker Plots to Show Distribution, **Work with Multiple Data Sources:** Creating Relationships between Tables, Joining Tables Using a Common Field, Using Unions to Combine Data, Blending Multiple Data Sources Using a Common Field

UNIT-V

Classes:12

Use a Background Image: Module Overview, Concept: Plotting Data on a Custom Background, Demo: Adding an Image and Plotting Data, Activity: Plot Data on an Image, **Create Dashboards and Stories:** Building a Dashboard, Creating Interactive, dashboards Using Actions, refining a Dashboard, Telling Stories with Data, **Share and Publish Content:** Publishing Workbooks,

Exporting and Downloading Content

Text Books:

1. Jones, Ben. *Communicating data with Tableau: Designing, developing, and delivering data visualizations*. " O'Reilly Media, Inc.", 2014.
2. Murray, Daniel G. *Tableau your data!: fast and easy visual analysis with tableau software*. John Wiley & Sons, 2013.

Reference Books:

1. Baldwin, David. *Mastering Tableau*. Packt Publishing Ltd, 2016.
2. Murray, Daniel G. *Tableau your data!: fast and easy visual analysis with tableau software*. John Wiley & Sons, 2013.

Web references:

<https://www.tableau.com/learn/training>

SECURITY TESTING (PROFESSIONAL ELECTIVE – IV)

IV Year I Semester: IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT30	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Explain the basic principles and techniques of how attackers can enter computer systems.
2. Perform analyses of data breaches and audits of information technology security.
3. Evaluate the strengths and weaknesses of various information technology solutions regarding data security.
4. Present and perform demonstrations of pen tests for educational purposes.
5. evaluate the societal role of hacking from a social, ethical and economic standpoint

UNIT – I Introduction to OWASP

Classes: 12

OWASP: Introduction to web applications security, threats and OWASP principles, introduction to secure design, web server: introduction a secure setup of apache, firewalling a server Browser: general concepts, functionalities, browsers war, configuration (HTTP-cookies, contents, scripting etc. attack to browsers, and users tracking/profiling (third party cookies, super cookies, XSS, CSFR, Command Injection), browser security (add-ons, plugins, same-origin policy etc.) & secure browsing.

UNIT – II OWASP Privacy preserving

Classes: 12

OWASP Privacy preserving: attacks to privacy, (spyware & backdoors, browser, email etc.) Tracking techniques: (HTTP cookies, third party cookies, browser fingerprinting, CSP) Advanced browser configuration, anonymity and onion routing (Tor). Internet E-mail: Architecture and infrastructure, functions, agents and standards, MIME & PGP, phishing, spamming & spoofing, DKIM, SPF, introduction to email forensics.

UNIT – III Introduction to Ethical Hacking Terminology

Classes: 10

Introduction to Ethical Hacking Terminology-Five stages of hacking-Vulnerability Research-Legal implication of hacking Impact of hacking. Foot printing & Social engineering.

UNIT – IV Information gathering methodologies

Classes: 10

Information gathering methodologies-Competitive Intelligence- DNS Enumerations- Social Engineering attacks. Scanning & Enumeration Port Scanning-Network Scanning- Vulnerability Scanning- NMAP scanning tool- OS Finger printing Enumeration. System Hacking Password.

UNIT – V Sniffers & SQL Injection

Classes: 10

Sniffers & SQL Injection Active and passive sniffing- ARP Poisoning- Session Hijacking- DNS Spoofing- Conduct SQL Injection attack – Counter measures. Cracking techniques- Key loggers- Escalating privileges- Hiding Files-Steganography technologies- Countermeasures.

Text Books:

1. Kimberly Graves, "CEH: Official Certified Ethical Hacker Review Guide", Wiley Publishing Inc., 2007.ISBN: 978-0-7821-4437-6.

Reference Books:

1. Shakeel Ali & Tedi Heriyanto, "Backtrack -4: Assuring security by penetration testing", PACKT Publishing., 2011. ISBN: 978-1-849513-94-4.

HIGH PERFORMANCE COMPUTING (PROFESSIONAL ELECTIVE – IV)

IV B. TECH- I SEMESTER - IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT31	PEC	3	0	0	3	40	60	100

Course Objectives:

1. To learn about Modern Processors and concepts
2. To understand the concepts of Optimizations
3. To learn about Parallel Computers and Programming
4. To study about Memory Parallel Programming using Open MP and MPI

Course Outcomes:

At the end of the course, student will be able to:

1. Transform algorithms in the computational area to efficient programming code for modern computer architectures
2. Write, organise and handle programs for scientific computations.
3. Use tools for performance optimisation and debugging and Analyse code with respect to performance.
4. Suggest and implement performance improvements
5. Write, organise and handle programs for MIP

UNIT – I MODERN PROCESSORS

Classes: 09

Stored-program computer architecture, General-purpose cache-based microprocessor architecture, Memory hierarchies, Multicore processors, Multi-threaded processors, Vector processors.

UNIT – II BASIC OPTIMIZATION TECHNIQUES FOR SERIAL CODE

Classes: 09

Scalar profiling, Common sense optimizations, Simple measures, large impact, The role of compilers, C++ optimizations

UNIT – III PARALLEL COMPUTERS

Classes: 09

Taxonomy of parallel computing paradigms, Shared-memory computers, Distributed-memory computers, Hierarchical (hybrid) systems, Networks, Need for parallelization, Parallelism, Parallel scalability

UNIT- IV SHARED MEMORY PARALLEL PROGRAMMING WITH OPENMP

Classes: 08

introduction to Open MP, Parallel execution, Data scoping, Open MP work sharing for loops, Synchronization, Reductions, Loop scheduling, Tasking, Miscellaneous, Open MP-parallel Jacobi algorithm, Open MP: Wave front parallelization.

UNIT – V DISTRIBUTED-MEMORY PARALLEL PROGRAMMING WITH MPI

Classes: 09

Message passing, introduction to MPI, A simple example, Messages and point-to-point communication, Collective communication, Nonblocking point-to-point communication, Virtual topologies, MPI parallelization of a Jacobi solver, MPI implementation, Performance properties

TEXT BOOK:

1. Georg Hager, Gerhard Wellein, "Introduction to High Performance Computing for Scientists and Engineers", Chapman & Hall / CRC Computational Science series, 2011.

REFERENCE BOOKS:

1. Charles Severance, Kevin Dowd, "High Performance Computing", O'Reilly Media, 2nd Edition, 1998.
2. Kai Hwang, Faye Alaye Briggs, "Computer Architecture and Parallel Processing", McGraw Hill, 1984

DEEP LEARNING (PROFESSIONAL ELECTIVE – V)

IV B. TECH- I SEMESTER: IT/AI

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6AI17	PEC	3	0	0	3	40	60	100

Course Objectives:

The course should enable the students to:

1. To learn Deep learning techniques and their applications.
2. To acquire the knowledge of neural network architectures, Deep learning methods and algorithms.
3. To understand CNN and RNN algorithms and their applications.

Course Outcomes:

At the end of the course, student will be able to:

1. Understand various learning models.
2. Design and develop various Neural Network Architectures.
3. Understand approximate reasoning using Convolution Neural Networks.
4. Analyze and design Deep learning algorithms in different applications.
5. Ability to apply CNN and RNN techniques to solve different applications.

UNIT-I BASICS

Classes: 12

Introduction: Historical Trends in Deep Learning, McCulloch Pitts Neuron, Thresholding Logic, Perceptrons, Perceptron Learning Algorithm. Representation Power of MLPs, Sigmoid Neurons, Gradient Descent, Feed forward Neural Networks, Representation Power of Feed forward Neural Networks

UNIT-II FEEDFORWARD NEURAL NETWORKS

Classes: 12

Feed Forward Neural Networks: Back propagation, Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMS Prop, Adam, Eigenvalues and eigenvectors, Eigenvalue Decomposition, Basis Principal Component Analysis and its interpretations, Singular Value Decomposition

UNIT-III AUTO ENCODERS & REGULARIZATION

Classes: 12

Auto encoders: Dealing with human-generated data, defining big data, architectural foundation, analytical data warehouses, Hadoop, data in motion and streaming data, integration of big data with traditional data

Regularization: Bias Variance Tradeoff, L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods, Dropout, Greedy Layer wise Pre-training, Better activation functions, Better weight initialization methods, Batch Normalization

UNIT-IV CONVOLUTION NEURAL NETWORKS

Classes: 12

Convolutional Neural Network: The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Innately Strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data Types. LeNet, AlexNet, ZF-Net, VGGNet, GoogLeNet, ResNet, Visualizing Convolutional Neural Networks, Guided Back propagation, Deep Dream, Deep Art, Fooling Convolutional Neural Networks

UNIT-V RECURRENT NEURAL NETWORKS

Classes: 12

Building a cognitive health care application: Recurrent Neural Networks, Back propagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, GRU, LSTMs, Encoder Decoder Models, Attention Mechanism, Attention over images

Text Books:

1. Goodfellow. I., Bengio. Y. and Courville. A., “ Deep Learning”, MIT Press, 2016.

Reference Books:

1. Tom M. Mitchell, “Machine Learning”, MacGraw Hill, 1997.
2. Stephen Marsland, “Machine Learning - An Algorithmic Perspective “, CRC Press, 2009.
3. LiMin Fu, “Neural Networks in Computer Intelligence”, McGraw-Hill edition, 1994.

PREDICTIVE ANALYTICS (PROFESSIONAL ELECTIVE-V)

IV B. TECH - I SEMESTER – IT/DS

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6DS28	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To know the use of the binary classifier and numeric predictor nodes to automate model selection.
2. To advice on when and how to use each model. Also learn how to combine two or more models to improve prediction
3. To advice on when and how to use each model. Also learn how to combine two or more models to improve prediction

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Understand the process of formulating business objectives
2. Compare the underlying predictive modeling techniques
3. Select appropriate predictive modeling approaches to identify cases to progress with.
4. Apply predictive analytics tools to analyze real-life business problems.
5. Demonstrate case-based practical problems using predictive analytics techniques to interpret model outputs.

UNIT-I Introduction to Data Mining

CLASSES:12

Introduction to Data Mining Introduction, what is Data Mining? Concepts of Data mining, Technologies Used, Data Mining Process, KDD Process Model, CRISP – DM, Mining on various kinds of data, Applications of Data Mining, Challenges of Data Mining.

UNIT-II Data Understanding

CLASSES:12

Data Understanding and Preparation Introduction, Reading data from various sources, Data visualization, Distributions and summary statistics, Relationships among variables, Extent of Missing Data. Segmentation, Outlier detection, Automated Data Preparation, Combining data files, Aggregate Data, Duplicate Removal, Sampling DATA, Data Caching, Partitioning data, Missing Values.

UNIT-III Model development

CLASSES:12

Model development & techniques Data Partitioning, Model selection, Model Development Techniques, Neural networks, Decision trees, Logistic regression, Discriminant analysis, Support vector machine, Bayesian Networks, Linear Regression, Cox Regression, Association rules.

UNIT-IV Model Evaluation and Deployment

CLASSES:12

Model Evaluation and Deployment Introduction, Model Validation, Rule Induction Using CHAID, Automating Models for Categorical and Continuous targets, Comparing and Combining Models

UNIT-V Evaluation Charts for Model Comparison

CLASSES:12

Evaluation Charts for Model Comparison, Metalevel Modeling, Deploying Model, Assessing Model Performance, Updating a Model.

TEXT BOOKS

1. Predictive & Advanced Analytics (IBM ICE Publication)

REFERENCEBOOKS

1. "Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die" by Eric Siegel
2. "Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking" by Foster Provost and Tom Fawcett
3. "Applied Predictive Analytics: Principles and Techniques for the Professional Data Analyst" by Dean Abbott

SOFTWARE PROJECT MANAGEMENT (PROFESSIONAL ELECTIVE – V)

IV B. TECH- I SEMESTER - IT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT32	PEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

To learn

1. The main goal of software development projects is to create a software system with a predetermined functionality and quality in a given time frame and with given costs.
2. For achieving this goal. Models are required for determining target values and for continuously controlling these values.
3. This course focuses on principles, techniques, methods & tools for model- based engagement of software projects.
4. Assurance of product quality and process adherence (quality assurance), as well as experience-based creation & improvement of models (process management).

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understanding the specific roles within a software organization as related to project and process management.
2. Understanding the basic infrastructure competences (e.g., process modeling and measurement).
3. Understanding the basic steps of project planning, project management. Quality assurance, and process management and their relationships.
4. Describe and determine the purpose and importance of project management from the perspectives of planning, tracking and completion of project.
5. Compare and differentiate organization structures and project structures to implement a project To manage project schedule, expenses and resources with the suitable protect Management tools.

UNIT – I

Classes: 10

CONVENTIONAL SOFTWARE MANAGEMENT: The waterfall model, conventional software Management performance. Evolution of Software Economics: Software Economics. Pragmatic software cost estimation.

IMPROVING SOFTWARE ECONOMICS: Reducing Software product size, improving software processes, improving team effectiveness. Improving automation, Achieving required quality, peer inspections.

UNIT – II

Classes: 10

THE OLD WAY AND THE NEW WAY - The principles of conventional software engineering. Principles of modern software management, transitioning to an iterative process.

LIFE-CYCLE PHASES AND PROCESS ARTIFACTS: Engineering and Production phases, inception phase, elaboration phase, construction phase, transition phase, The artifact sets, Management artifacts, Engineering artifacts, pragmatic artifacts.

UNIT – III

Classes: 10

WORK FLOWS OF THE PROCESS: Software Process Workflow, Inter Trans Workflows.

CHECKPOINTS OF THE PROCESS: Major Mile Stones, Minor Milestones, Periodic status assessments.

ITERATIVE PROCESS PLANNING: Work breakdown structures, planning guidelines, cost and scheduled estimation, Iteration planning process.

UNIT – IV

Classes: 10

PROJECT ORGANIZATIONS:

Line-of- business organizations, project organizations, evolution of organizations. Process Automation: Automation Building blocks.

PROJECT CONTROL AND PROCESS INSTRUMENTATION: The seven core Metrics, Management indicators, and quality indicators. Life cycle expectations, pragmatic Software Metrics, Metrics automation.

UNIT – V

Classes: 10

FUTURE SOFTWARE PROJECT MANAGEMENT: Modern Project Profiles, Next Generation Software Economics, Modern Process Transitions.

Case Study: The Command Centre Processing and Display System, Replacement (CCPDS. R).

Text Books:

1. Software Project Management. Walker Royce, Pearson Education.
2. Software Project Management, Bob Hughes & Mike Cotterell, fourth edition, Tate McGrawHd.

Reference Books:

1. Applied Software Project Management, Andrew Stelbian & Jennifer Greene, O'Reilly. 2006
2. Head First PMP, Jennifer Greene & Andrew Stelman, O'Reilly. 2007
3. Software Engineering Project Management. Richard H. Thayer & Edward Yourdon, second edition, Wiley India, 2004.
4. A Guide to Project Management, Jim Highsmith. Pearson education, 2004
5. The art of Project management. Scott Berkun. O'Reilly, 2005.
6. Software Project Management in Practice. Pankaj Jalote. Pearson Education, 2002.

CRIME INVESTIGATION & DIGITAL FORENSICS (PROFESSIONAL ELECTIVE - V)

IV B. TECH- I SEMESTER – IT/CY

Course Code:	Category	Hours/Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	TOTAL
A6CY16	PEC	3	0	0	3	40	60	100

Course Objectives:

1. Provides an in-depth study of the rapidly changing and fascinating field of computer forensics.
2. Combines both the technical expertise and the knowledge required to investigate, detect and prevent digital crimes.
3. Knowledge on digital forensics legislations, digital crime, forensics processes and procedures, data acquisition and validation, e-discovery tools
4. E-evidence collection and preservation, investigating operating systems and file systems, network forensics, art of steganography and mobile device forensics

Course Outcomes:

1. Understand the fundamentals of cybercrime and issues.
2. Understand different investigation tools for cybercrime.
3. Understand basics of Forensic Technology and Practices.
4. Analyze different laws, ethics and evidence handling procedures.

UNIT – I

INTRODUCTION

Classes:10

Introduction: Introduction and Overview of Cyber Crime, Nature and Scope of Cyber Crime, Types of Cyber Crime: Social Engineering, Categories of Cyber Crime, Property Cyber Crime.

UNIT – II

CYBER CRIME ISSUES

Classes:10

Cyber Crime Issues: Unauthorized Access to Computers, Computer Intrusions, White collar Crimes, Viruses and Malicious Code, Internet Hacking and Cracking, Virus Attacks, Pornography, Software Piracy, Intellectual Property, Mail Bombs, Exploitation, Stalking and Obscenity in Internet, Digital laws and legislation, Law Enforcement Roles and Responses.

UNIT - III

INVESTIGATION

Classes:08

Investigation: Introduction to Cyber Crime Investigation, Investigation Tools, eDiscovery, Digital Evidence Collection, Evidence Preservation, E-Mail Investigation, E-Mail Tracking, IP Tracking, E-Mail Recovery, Hands on Case Studies. Encryption and Decryption Methods, Search and Seizure of Computers, Recovering Deleted Evidences, Password Cracking.

UNIT – IV

DIGITAL FORENSICS

Classes:12

Digital Forensics: Introduction to Digital Forensics, Forensic Software and Hardware, Analysis and Advanced Tools, Forensic Technology and Practices, Forensic Ballistics and Photography, Face, Iris and Fingerprint Recognition, Audio Video Analysis, Windows System Forensics, Linux System Forensics, Network Forensics.

UNIT - V	LAWS & ACTS	Classes:10
Laws and Acts: Laws and Ethics, Digital Evidence Controls, Evidence Handling Procedures, Basics of Indian Evidence ACT IPC and CrPC, Electronic Communication Privacy ACT, Legal Policies.		
Text Books:		
1. Nelson Phillips and Enfinger Steuart, "Computer Forensics and Investigations", Cengage Learning New Delhi, 2009. 2. Kevin Mandia, Chris Prosise, Matt Pepe, "Incident Response and Computer Forensics ", Tata McGraw -Hill, New Delhi, 2006.		
Reference Books:		
1. Robert M Slade," Software Forensics", Tata McGraw - Hill, New Delhi, 2005. 2. Bernadette H Schell, Clemens Martin, "Cybercrime", ABC – CLIO Inc, California, 2004. 3. "Understanding Forensics in IT ", NIIT Ltd, 2005		

BIG DATA TECHNOLOGIES LAB

IV B. TECH. - I SEMESTER – IT/DS

Course Code	Category	Hours / Week			Credits		Maximum Marks	
		L	T	P	C	CIE	SEE	Total
A6DS07	PCC	0	0	2	1	40	60	100

COURSE OBJECTIVES

1. Get familiar with Hadoop distributions, configuring Hadoop and performing File management tasks.
2. Understand the storage of data in a distributed file system.
3. Introduce the basics required to develop map reduce programs.
4. Introduce programming tools PIG & HIVE in the Hadoop ecosystem.
5. Demonstrate the usage of PySpark to implement ML concepts..

COURSE OUTCOMES

At the end of the course, students will be able to:

1. Connect to hadoop cluster, experiment with various Linux and HDFS commands to store data.
2. Apply the knowledge of MapReduce programming to process the stored data in HDFS.
3. Write scripts using Pig and retrieve data from Hadoop using HiveQL.
4. Create data processing pipelines with Spark
5. Build and tune machine learning models with Spark ML

LIST OF EXPERIMENTS:

WEEK 1

- i) Perform setting up and installing Vmware for Hadoop and Linux.
- ii) Basic Linux Commands
- iii) Run basic HDFS shell commands

WEEK 2

Implement the following file management tasks in Hadoop:

- i) Adding files and directories
- ii) Retrieving files
- iii) Deleting files and directories.

WEEK 3

- i) Develop a MapReduce program to calculate the frequency of a given word in a given file.
- ii) Develop a MapReduce program to find the maximum temperature in each year.

WEEK 4

Design MapReduce algorithms to take a very large file of integers and produce as output:

- i) The largest integer
- ii) The average of all the integers.
- iii) The same set of integers, but with each integer appearing only once. *
- iv) The count of the number of distinct integers in the input.*

WEEK 5

Implement **Matrix Multiplication** on **Hadoop** Using **Map Reduce**.

WEEK 6

- i) Run Pig and perform basic PIG commands.
- ii) Write Pig Latin scripts to sort, group, join, project, and filter your data.

WEEK 7

- i) Practice Basic HiveQL Commands, read data from various File Formats and create Data Definition Statements and Data Manipulation Statements.
- ii) Write Queries using select.

WEEK 8

- i) Interactive Analysis with the Spark Shell
- ii) Writing and running Spark program

WEEK 9

Implement the following algorithms for classification using PySpark.

- i) Logistic Regression
- ii) Decision Tree Classifier
- iii) Naive Bayes

WEEK 10

Implement the following algorithms for clustering using PySpark.

- i) K-Means
- ii) Latent Dirichlet Allocation (LDA)
- iii) Gaussian Mixture Model (GMM)

WEEK 11

Implement collaborative filtering using spark ML library.

WEEK 12

Implement FP-Growth using Spark ML Library.

TEXT BOOKS

1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media
2. Singh, Pramod. *Machine Learning with PySpark: With Natural Language Processing and Recommender Systems*. Apress, 2018.
3. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.

IV B.TECH II SEMESTER SYLLABUS

ORGANIZATIONAL BEHAVIOR

IV B. TECH – II SEMESTER: IT

Course Code	Category	Hours /Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6HS15	HSMC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. To provide understanding of the basic concepts of organizational behavior.
2. To highlight the importance of individual behaviors in an organizational setting.
3. To describe the various aspects of group behaviors in an organization.
4. To learn and appreciate different cultures and innovatively engage in solving organizational challenges.
5. To elaborate on the impact of work–life balance and importance of managing stress and emotions in the workplace.

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Gain the conceptual framework and the theories underlying Organizational Behavior.
2. Describe the different theories related to Individual behavior in the Organization.
3. Analyze Group Behavioral influence in the Organization.
4. Demonstrate skills required for working in groups and explain the essential concepts of organizational culture, climate, change and organizational development.
5. Learn the concepts of work life balance and managing stress and emotions in the work place.

UNIT I: INTRODUCTION

Classes: 12

The concept of Organizational Behavior - Importance of OB to the field of Management – Hawthorne Studies - Approaches to Organizational Theory – Organizational Behavior Models - Opportunities and Challenges for OB – Disciplines that contribute to OB.

UNIT II: INDIVIDUAL BEHAVIOUR

Classes: 12

Learning – Theories, Principles and Process – Perception – Importance - Perception Model - Factors influencing Perception – Personality – Determinants and Theories – Attitudes and Values – Motivation – Nature, Techniques and Theories.

UNIT III: GROUP BEHAVIOUR

Classes: 12

Group Dynamics – Group Formation – Group Norms – Group Decision making – Group Cohesiveness – Team Building – Leadership – Styles and Theories – Power and Politics – Conflict Management – Types and Sources – Conflict Resolution

UNIT IV: ORGANIZATIONAL DYNAMICS

Classes: 12

Organizational Culture – Organizational Climate - Organizational Change: Approaches and Resistance to change– Organizational Development – Organizational Structure - Communication: Meaning, Types, Process and Barriers- Power – Sources of Power – Leadership – Theories and Styles

UNIT V: CONTEMPORARY ISSUES IN ORGANIZATIONAL BEHAVIOUR

Classes: 12

Quality of Work Life - Work life balance – Stress Management – Importance and techniques – Emotional Intelligence.

Suggested Readings:

1. Stephen Robbins, Timothy A. Judge, Organizational Behaviour, PHI Pvt. Ltd.
2. Nelson, Quick, Khandewal, Organizational Behaviour, Cengage Learning
3. Williams, Tripathy, MGMT-Principles of Management, Cengage Learning.
4. L.M.Prasad, Principles and Practice of Management, Sultan and Chand.
5. Udai Pareek and Sushama Khanna, Organization Behavior, Oxford Publishing
6. S.S. Khanka, , Organization Behavior, Oxford Publishing S Chand Publishing.

E-COMMERCE **(PROFESSIONAL ELECTIVE - VI)**

IV B.TECH - II Semester: CSE/IT/CSIT

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT33	PEC	3	0	0	3	40	60	100

Course Objectives:

The course should enable the students to:

1. Become familiar with state of the art Electronic Model, Payment Mechanisms.
2. Understand the basic principal of E-Government, Securities, and Supply Chain Mechanisms.
3. Evaluate and observe various online businesses Management.
4. Provide guiding principles behind the design and strategy of the customer web interface.
5. Understand the traditional and new communication/marketing approaches that create competitive advantage in the New Economy.

Course Outcomes:

Upon successful completion of the course, the student is able to

1. Analyze the principles of E-commerce and basics of World Wide Web.
2. Understand the concept of electronic data interchange and its legal, social and technical aspects.
3. Analyze the security issues over the web, the available solutions and future aspects of e-commerce security.
4. Develop the concept of E-banking, electronic payment system
5. Create different types of payment modes.

Unit I

Classes: 12

Introduction Overview of Electronic Commerce - Definition of Electronic Commerce - E Business - Potential Benefits of E Commerce - Impact of E Commerce on Business Models - E Commerce Applications - Market Forces influencing highway - The global information distribution networks - The regulatory environment for E Commerce.

Unit II

Classes: 12

Electronic Data Interchange (EDI), Electronic Commerce and the Internet Introduction - traditional EDI systems - Benefits - Data Transfer and Standards - Financial EDI – EDI Systems and the Internet - Legal, Security and Private Concerns - Authentication - Internet Trading Relationships Consumer to Business (B2C) Business to Business (B2C), Consumer to Consumer (C2C) Government to Citizen - Features and Benefits - Portal Vs. Website. Impact. - Intra Organizational E Commerce - Supply Chain Management.

Unit III

Classes: 12

Cryptography and Authentication Introduction - Messaging Security Issues - Confidentiality - Integrity - Authentication. Encryption Techniques - Integrity Check Values and Digital Signatures - Good Encryption Practices – Key Management - key management tasks - Additional Authentication Methods. Firewalls - Definition - component - Functionality - securing the firewall - factors considered in securing the firewall - Limitations.

Unit IV

Classes: 12

Electronic Payment Mechanisms Introduction r- The SET protocol - SET vs. SSL - Payment Gateway - Certificate Issuance – Certificate Trust Chain - Cryptography Methods - Dual Signatures - SET Logo - Compliance Testing - Status of Software - Magnetic Strip Cards - Smart Cards - Electronic Cheques -

Electronic Cash - Third party Processors and Credit Cards - Risk and electronic payment system - Designing Electronic payment systems.

Unit V

Classes: 12

E-Commerce Strategy and Implementation Strategic Planning for E-Commerce, Strategy in Action, Competitive Intelligence on the Internet, Implementation; Plans and Execution, Project and Strategy assessment - Managerial Issues – Global Electronic Commerce - Future of Electronic Commerce.

Text Books:

1. Electronic Commerce - Security, Risk Management and Control, Greenstein and Feinman, Irwin Mc.Gra-Hill, 2000.
2. Frontiers of electronic commerce – Kalakata, Whinston, Pearson.
3. E-Commerce fundamentals and applications Hendry Chan, Raymond Lee, Tharam Dillon, Ellizabeth Chang, John Wiley.

Reference Books:

1. E-Commerce, S.Jaiswal – Galgotia.
2. E-Commerce, Efrain Turbon, Jae Lee, David King, H.Michael Chang.
3. Electronic Commerce – Gary P.Schneider – Thomson. 4. E-Commerce – Business, Technology, Society, Kenneth C.Taudon, Carol

DISTRIBUTED COMPUTING (PROFESSIONAL ELECTIVE - VI)

IV B.TECH - II SEMESTER: IT/CSE

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS22	PEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

Students are expected to

1. Understand the foundations of distributed systems.
2. Learn issues related to clock Synchronization and the need for global state in distributed systems
3. Learn distributed mutual exclusion and deadlock detection algorithms
4. Understand the significance of agreement, fault tolerance and recovery protocols in Distributed Systems
5. Learn the characteristics of peer-to-peer and distributed shared memory systems

COURSE OUTCOMES:

By the end of the course, the student will able to

1. Elucidate the foundations and issues of distributed systems
2. Illustrate various synchronization issues and global state for distributed systems.
3. Illustrate Mutual Exclusion and Deadlock detection algorithms in distributed systems
4. Describe the agreement protocols and fault tolerance mechanisms in distributed systems
5. Describe the features of peer-to-peer and distributed shared memory systems.

UNIT - I A model of distributed computations

CLASSES: 12

Definition, Relation to computer system components, Motivation, Relation to parallel systems, Message-passing systems versus shared memory systems, Primitives for distributed communication, Synchronous versus asynchronous executions, Design issues and challenges.

A model of distributed computations: A distributed program, A model of distributed executions, Models of communication networks, Global state, Cuts, Past and future cones of an event, Models of process communications.

Logical Time: A framework for a system of logical clocks, Scalar time, Vector time, Physical clock synchronization: NTP.

UNIT - II Message Ordering & Snapshots

CLASSES: 14

Global state and snapshot recording algorithms: Introduction, System model and definitions, Snapshot algorithms for FIFO channels, Snapshot algorithms for non-FIFO channels,

Message Ordering & Snapshots: Message ordering and group communication: Message ordering paradigms, Asynchronous execution with synchronous communication, Synchronous program order on an asynchronous system, Group communication, Causal order (CO), Total order.

UNIT - III Distributed Mutex& Deadlock

CLASSES: 14

Distributed Mutex& Deadlock: Distributed mutual exclusion algorithms: Introduction, Preliminaries, Lamport's algorithm, Ricart-Agrawala algorithm, Maekawa's algorithm, Suzuki-Kasami's broadcast algorithm.

Deadlock detection in distributed systems: Introduction, System model, Preliminaries, Models of deadlocks, Knapp's classification, Algorithms for the single resource model, the AND model and the OR model.

UNIT - IV Recovery & Consensus

CLASSES: 12

Recovery & Consensus: Check pointing and rollback recovery: Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, Coordinated check pointing algorithm, Algorithm for asynchronous check point in gand recovery.

Consensus and agreement algorithms: Problem definition, Overview of results, Agreement in a failure, free system, Agreement in synchronous systems with failures.

UNIT - V Peer-to-peer computing and overlay graphs

CLASSES: 14

Peer-to-peer computing and overlay graphs: Introduction, Data indexing and overlays, Chord–Content addressable networks, Tapestry.

Distributed shared memory: Abstraction and advantages, Memory consistency models, Shared memory Mutual Exclusion.

TEXT BOOKS

1. Distributed computing: Principles, algorithms, and systems, Ajay Kshemkalyani and MukeshSinghal, Cambridge University Press, 2011.
2. Distributed Systems Concepts and Design, George Coulouris, Jean Dollimore and Tim Kindberg, Fifth Edition, Pearson Education, 2012.

REFERENCE BOOKS

1. Distributed Operating Systems: Concepts and Design, Pradeep K Sinha, Prentice Hall of India, 2007.
2. Advanced concepts in operating systems. MukeshSinghal and Niranjana G. Shivaratri, McGraw-Hill, 1994.
3. Java in Distributed Systems, Concurrency, Distribution and Persistence, Marko Boger, 2001.
4. Distributed Systems: Principles and Paradigms, Tanenbaum A.S., Van Steen M., Pearson Education, 2007.
5. Kshemkalyani Ajay D, Mukesh Singhal, Distributed Computing: Principles, Algorithms and Systems, Cambridge Press, 2011
6. Mukesh Singhal, Niranjana G Shivaratri, Advanced Concepts in Operating systems, McGraw Hill Publishers 1994

Web Links

1. https://onlinecourses.nptel.ac.in/noc21_cs15/preview
2. <https://www.distributedsystemscourse.com/>

NETWORK ADMINISTRATION (PROFESSIONAL ELECTIVE – VI)

IV B. TECH- II SEMESTER - IT

Course Code	Category	Hours / Week			Credits		Maximum Marks	
		L	T	P	C	CIE	SEE	Total
A6IT34	PEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to

1. Demonstrate the need for interoperable network administration and management
2. Understand general concepts and architecture behind standards based network management
3. Understand concepts and terminology associated with SNMP and TMN
4. Appreciate network management as a typical distributed application
5. Understand Advanced Information Processing Techniques such as Distributed Object Technologies, Software Agents and Internet Technologies used for network management

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Identify and define the basic system administration and networking standards (OSI and TCP/IP)
2. Acquire the knowledge about various network management tools and the skill to use them in Monitoring a network.
3. Analyze the challenges faced by Network managers
4. Evaluate various commercial network management systems and open network management systems.
5. Analyze and interpret the data provided by NMS and take suitable actions.

UNIT – I

Classes 12

Introduction Analogy of Telephone Network Management, Data and Telecommunication Network, TCP/IP-Based Networks, The Internet and Intranets. Communications Protocols and Standards Communication Architectures, Protocol Layers and Services; Case Histories of Networking and Management – Challenges of Information Technology Managers.

UNIT – II

Classes 12

Network Management Goals, Organization, and Functions, Goal of Network Management, Network Provisioning, Network Operations and the NOC, Network Installation and Maintenance; Network and System Management, Network Management System platform, Current Status and Future of Network Management.

UNIT – III

Classes 12

Basic Foundations Standards, Models, and Language Network Management Standards, Network Management Model, Organization Model, Information Model, Communication Model, Abstract Syntax Notation one-ASN.1- Terminology, Symbols, and Conventions, Objects and Data Types, Object Names, An Example of ASN.1 from ISO 8824; Encoding Structure, Macros, Functional Model.

UNIT – IV

Classes 12

SNMPv1 Network Management Organization and Information models Managed Network, The History of SNMP Management, Internet Organizations and standards, The SNMP Model, The Organization Model, and Information Model.

Communication and Function models SNMP Communication Model – The SNMP Architecture, Administrative Model, SNMP Specifications, SNMP Operations, Functional Model.

UNIT – V

Classes 12

SNMP Management – RMON Remote Monitoring, RMON SMI and MIB, RMON1- RMON1 Textual Conventions, RMON1 Groups and Functions, Relationship Between Control and Data Tables, RMON1

Common and Ethernet Groups, RMON Token Ring Extension Groups, RMON2 – The RMON2 Management Information Base, RMON2 Conformance Specifications; ATM Remote Monitoring, A Case Study of Internet Traffic Using RMON.

Text Books:

1. Mani Subramanian, —Network Management Principles and Practicell, 2nd Edition, Pearson Education, 2010.

Reference Books:

1. Morris, —Network managementll, 1st Edition, Pearson Education, 2008.
2. Mark Burges, —Principles of Network System Administrationll, 1st Edition, Wiley Dream Tech, 2008

IMAGE PROCESSING (PROFESSIONAL ELECTIVE - VI)

IV B. TECH- II SEMESTER: IT/AI

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6AI12	PEC	3	0	0	3	40	60	100

Course Objectives

The course should enable the students to:

1. Study the image fundamentals and mathematical transforms necessary for image processing.
2. Study Image enhancement techniques.
3. Study image restoration procedures.
4. Study the image compression procedures.

Course Outcomes

At the end of the course, student will be able to:

1. Review the fundamental concepts of a digital image processing system.
2. Perform image enhancement techniques in spatial and frequency domain.
3. Apply the concept of image restoration.
4. Apply the concept of image segmentation.
5. Elucidate the mathematical modeling of image compression.

UNIT-I Digital Image Fundamentals

Classes: 12

Digital Image through Scanner, Digital Camera. Concept of Gray Levels. Gray Level to Binary Image Conversion. Sampling and Quantization. Relationship between Pixels. Imaging Geometry. 2D Transformations-DFT, DCT, KLT and SVD.

UNIT-II Image Enhancement

Classes: 10

Image Enhancement in Spatial Domain Point Processing, Histogram Processing, Spatial Filtering, Enhancement in Frequency Domain, Image Smoothing, Image Sharpening.

UNIT-III Image Restoration

Classes: 10

Image Restoration Degradation Model, Algebraic Approach to Restoration, Inverse Filtering, Least Mean Square Filters, Constrained Least Squares Restoration, Interactive Restoration.

UNIT-IV Image Segmentation

Classes: 10

Image Segmentation Detection of Discontinuities, Edge Linking and Boundary Detection, Thresholding, Region Oriented Segmentation.

UNIT-V Image Compression

Classes: 12

Image Compression Redundancies and their Removal Methods, Fidelity Criteria, Image Compression Models, Source Encoder and Decoder, Error Free Compression, Lossy Compression.

Text Books:

1. Digital Image Processing: R.C. Gonzalez & R. E. Woods, Addison Wesley/ Pearson Education, 2 nd Ed, 2004.

Reference Books:

1. Fundamentals of Digital Image Processing: A. K. Jain, PHI.
2. Digital Image Processing using MATLAB: Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins: Pearson Education India, 2004.
3. Digital Image Processing: William K. Pratt, John Wiley, 3rd Edition, 2004.

OPEN ELECTIVES OFFERED BY AERONAUTICAL DEPARTMENT

OPEN ELECTIVES-1	
A6AE19	Fundamentals of Avionics
A6AE20	Introduction to Aerospace Technology
OPEN ELECTIVES-2	
A6AE25	Introduction to jets and rockets
A6AE26	Non-Destructive Testing Methods
OPEN ELECTIVES-3	
A6AE40	Introduction to Aircraft Industry
A6AE41	Unmanned Aerial Vehicles

FUNDAMENTALS OF AVIONICS

VI Semester : OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6AE19	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Overview on Aviation using Electronics
2. Basic understanding about major electronics systems used for communication
3. Basic understanding about major devices, display and flight controls used in aircraft

UNIT-I BASICS & FLIGHT DECK AND DISPLAY SYSTEMS

BASICS: Basic principles of Avionics, Typical avionics sub system in civil/ military aircraft and space vehicles.

FLIGHT DECK AND DISPLAY SYSTEMS: Flight deck display technologies, CRT, LED, LCD, Touch screen, Head up display, electronic instrumentation systems.

UNIT-II COMMUNICATION SYSTEMS

AUDIO AND COMMUNICATION SYSTEMS: Aircraft audio systems, basic audio transmitter and receiver principles, VHF communication system, UHF communication systems.

UNIT-III FREQUENCY RANGING SYSTEM

RANGING AND LANDING SYSTEMS: VHF Omnirange, VOR receiver principles, distance maturity equipment, principles of operation, Instrument landing system, and localizer and glide slope.

POSITIONING SYSTEM: Global positioning system principles, triangulation, position accuracy, applications in aviation

UNIT-IV NAVIGATION SYSTEM

INERTIAL NAVIGATION SYSTEM: Principle of Operation of INS, navigation over earth, components of inertial Navigation systems, accelerometers, gyros and stabilized platform.

SURVEILLANCE SYSTEM: ATC surveillance systems principles and operation interrogation and replay standards, Collision avoidance system, ground proximity warning system

UNIT-V AUTO FLIGHT SYSTEM

AUTO FLIGHT SYSTEM: Automatic flight control systems fly by wire and fly by light technologies, flight director systems, flight management systems.

Text Books:

1. N. S. Nagaraja(1996),Elements of electronic navigation, 2nd edition, Tata McGraw Hill, New Delhi.
2. Janes W. Wasson, Jeppesen Sandersen(1994), Avionic systems Operation and maintenance,

Reference Books:

1. Albert Hel Frick (2010), Principle of Avionics, 6th edition, Avionics Communications Inc, India.
2. H. J. Pallet (2010), Aircraft Instrumentation and Integrated systems, Pearson Education, New Delhi.
3. J. Powell (1998), Aircraft Radio Systems, Pitman publishers, London

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 To explain the instrumentation used in avionics.
- 2 To classify various ranges of the communication techniques used in aircraft.
- 3 To distinguish between network systems, controlling parts & surfaces
- 4 To compare various principles of navigation systems
- 5 To build phenomena of auto pilot control system

INTRODUCTION TO AEROSPACE TECHNOLOGY

VI Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6AE20	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

1. To introduce the basic concepts of Aerospace technology and the current developments in the field.
2. To provide knowledge on the basic principles on which the development of aerodynamics, Structures, propulsion and satellite systems.

UNIT-I HISTORY OF FLIGHT- THE AEROSPACE ENVIRONMENT

Balloons and dirigibles, heavier than air aircraft, commercial air transport, introduction of jet aircraft, helicopters, missiles, conquest of space, commercial use of space, exploring solar system and beyond. Earth's atmosphere, the temperature extremes of space, laws of gravitation, low earth orbit, microgravity, benefits of microgravity. The near earth radiative environment. The magnetosphere. Environmental impact on spacecraft. Meteoroids and micrometeoroids, space debris. Planetary environments

UNIT-II AERODYNAMICS AND FLIGHT VEHICLE PROPULSION

Anatomy of the airplane, helicopter, launch vehicles and missiles, space vehicles. Static forces and moments on the vehicle. Understanding engineering models. Aerodynamics of wings and bodies. Generation of lift. Sources of drag. Force and moment coefficients, centre of pressure. Thrust for flight, the propeller, the jet engine, rocket engines- description, principles of operation. Governing equations.

UNIT-III FLIGHT VEHICLE PERFORMANCE AND STABILITY

Performance parameters. Performance in steady flight, cruise, climb, range, endurance; accelerated flight- symmetric manoeuvres, turns, sideslips, take off and landing. Flight vehicle stability- longitudinal, lateral and directional- static, dynamic; trim, control. Handling qualities of airplanes

UNIT-IV SATELLITE SYSTEMS ENGINEERING- HUMAN SPACE EXPLORATION

Satellite missions, an operational satellite system, elements of satellite, satellite subsystems. Satellite structures, mechanisms and materials. Power systems. Communication and telemetry. Thermal control. Attitude determination and control. Propulsion and station keeping. Space missions. Mission objectives. Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo (manned flight to the moon), Skylab, Apollo-Soyuz, Space Shuttle. International Space Station, extravehicular activity

UNIT-V INTRODUCTION TO ENGINEERING DESIGN, AIR TRANSPORTATION

Design as a critical component of engineering education- as a skill- the process, design thinking, design drawing. Design for mission, performance and safety requirements. Concurrent engineering. Computer aided engineering, design project. Example: the lighter-than – air vehicle student design project at MIT. Air Transportation Systems- civil, military- objectives- principal constituents- the vehicle, the ground facilities, the organization- role. Regulation- national and international. Indian effort- civil and military- in the field of Aerospace Engineering.

Text Books:

1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD), McGraw-Hill, 2002, ISBN 0-07-112254-0.
2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082

Reference Books:

1. Russell Mikel, Aerospace and Aeronautical Engineering, Willford press, 2017.
2. Ajoykumar Kundu, Mark A Price and David Riordan, Conceptual Design: An Industrial Approach, Wiley-Blackwell, 2019.

COURSE OUTCOMES:

Students should be able to

1. Compare the atmosphere conditions of different altitudes for spacecraft system
2. Analyze how lift, drag and thrust are generated and understand which components constitute them
3. Analyze the flight performance parameters with respective stability condition
4. Distinguish the different systems used in a satellite mission
5. Design lighter than air vehicle using Catia software

INTRODUCTION TO JETS AND ROCKETS

VII Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6AE25	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The course is intended to serve as an introduction to air breathing propulsion systems and Rocket Propulsion Systems.

1. Illustrate an overview of aerospace propulsion system.
2. Identify the foundation in fundamentals of thermodynamics.
3. Compare the ideal components and characteristics of jet engine
4. Interpret the performance of nozzles
5. Simplify the ideal performance analysis of rocket engines.
6. Select appropriate fuel for aerospace application.

UNIT-I

INTRODUCTION TO AEROSPACE PROPULSION

Propulsion system, Propulsive Systems – Evolution, Development, Growth and Challenges. Fundamentals of Thermodynamics – Variables, Thermodynamic Process, Introduction to IC Engines and Reciprocating Engines, Propellers and Working of Propellers.

UNIT-II

PRINCIPLES OF JET PROPULSION

Fundamentals of jet propulsion, Working Principle, Analysis of Ideal Jet Engine cycle, Engine components- merit-significance- ideal component characteristics, Classification – turbo jet, turbo fan, turbo prop and Ramjet engines. Basic Problems based on Engine Cycle.

UNIT-III

RAMJET, SCRAMJET ENGINES AND NOZZLES

Speed limitations of gas turbines, Basics of Ramjets, Combustors for liquid fuel ramjet engines, Combustion Instability and its Suppression, Solid fuel Ramjet Engines, SCRAM jet engines, Applications of RAM Jet and SCRAM Jet Engines to Missiles with Examples, Nozzles- Types of Nozzles, Converging-Diverging Nozzle, Variable Nozzle and Effects of Pressure Ratios on Engine Performance.

UNIT-IV

ROCKET THEORY

Applications of Rockets, Types of Rockets, Basics of Thermal Rocket Engine-Thermodynamics and Ideal Performance Analysis, Equations of motion-Rocket Motion in free space, Tsiolkovsky's equation, Rocket Parameters, Burnout range, Burnout Velocity. Practical Problems

UNIT-V

PROPELLANT ROCKETS

Solid Propulsion-Solid Propellant Rockets, Basic Configuration and Performance, Propellant Grain and Configuration, Propellant Characteristics Combustion Chamber, Ignition Process Liquid Propulsion - Design consideration of liquid rocket combustion chamber, injector, and propellant feed lines, valves, propellant tank outlet and helium pressurized and turbine feed systems-.BIO Fuels and Impact on the Atmosphere, Aviation turbine fuels - Requirements of aviation fuels of kerosene type.

Text Books:

1. Mechanics and Thermodynamics of Propulsion – Philip G Hill & Carl R Peterson , Pearson Publication – 2ndEdt
2. Rocket Propulsion Elements, Sutton, G.P., John Wiley, 1993.

Reference Books:

1. The Jet Engine – Rolls Royce
2. Gas Turbines and Jet and Rocket Propulsion, M. L. Mathur, R. P. Sharma, Standard Publishers Distributors.

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Explain the complexity in working of various engines
- 2 Interpret the elementary principles of thermodynamic cycles as applied to propulsion analysis
- 3 Analyze the process involved in individual components
- 4 Compare the nozzles with various operating conditions.
- 5 Determine Equations of motion in free space, Tsiolkovsky's equation.
- 6 Classify the types of fuel in aviation and aerospace engineering.

NON-DESTRUCTIVE TESTING METHODS

VII Semester : OPEN ELECTIVE - II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6AE26	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. To impart knowledge about the non-destructive testing methods.
2. To provide knowledge on the selection of NDT methods for application in engineering industries.
3. Classify the various NDT methods for detecting defects in the structural components.
4. Judge defects basing on data representation of testing

UNIT-I SURFACE TECHNIQUES

Introduction to non-destructive testing (NDT) - importance of NDT techniques - types of NDT techniques - visual testing (direct and remote visual inspection) - principle and types of liquid penetrant tests (LPT) - advantages and limitations of LPT - applications of LPT.

UNIT-II MAGNETIC PARTICLE TESTING

Introduction to magnetic particle testing (MPT) – principle, magnetization methods - demagnetization – advantages, limitations and applications of MPT, eddy current testing (ECT) method- principle advantages, limitations and applications

UNIT-III ULTRASONIC TESTING

Introduction to ultrasonic testing (UT) -- principle of UT – UTprobes - UT inspection methods (pulse echo, transmission and phased array techniques) - advantages, limitations and applications

UNIT-IV RADIOGRAPHY TESTING

Introduction to radiography testing (RT) - sources of X-rays and Gamma rays - characteristics of Xrays and Gamma rays (absorption, scattering) - filters and screens - film radiography and digital radiography (shadow formation, exposure factors, film handling and storage) --exposure charts - penetrameters - safety issues.

UNIT-V SPECIAL TECHNIQUES

Acoustic emission testing (AET) principle, advantages, limitations - instrumentation and application of AET - infra red thermography (IRT) - contact and non-contact inspection methods - acoustic holography.

Text Books:

1. Baldev Raj, T. Jayakumar, M. Thavasimuthu, "Practical Non-Destructive Testing", Narosa Publishing, London, 2012.
2. Paul E. Mix, "Introduction to Non Destructive Testing", A Training Guide, Wiley- Interscience, New Jersey, USA, June 2005.

Reference Books:

1. ASM Metals Handbook, V-17, "Non-Destructive Evaluation and Quality Control", American Society of Metals, Metals Park, Ohio, USA, 2001
2. W.T. Mc Gonnagle, "Non-Destructive Testing", McGraw Hill Book Co., USA, 2013.

COURSE OUTCOMES:

1. Recognize various non-destructive techniques for engineering industries.
2. Select appropriate non-destructive technique for defects detection in manufactured/operating parts.
3. Perform inspection using major non-destructive testing methods.
4. Understand the importance and application of NDT in Aerospace structural analysis
5. Determine the defects basing on the principal of radiography

INTRODUCTION TO AIRCRAFT INDUSTRY

VIII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6AE40	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

The aim is to introduce students the overview of the Aircraft Industry. The course covers basic design process, Aircraft Stability & control and different mechanical systems.

- 1 Familiarize students with major types of aircraft industry.
- 2 Illustrate the role of regulatory bodies and its business context
- 3 Exemplify the organisational structure of airlines and airports
- 4 Illustrate the importance of air navigation services and its types.
- 5 Familiarize students with air safety and security as a means for ensuring passenger safety.

UNIT-I

AIRCRAFT INDUSTRY OVERVIEW

Flying machine to airline alliances, Types of Aerospace Industry, Introduction to ages of engineering, Aerospace Manufacturing, Introduction to the space environment & human space exploration.

UNIT-II

REGULATORY AND BUSINESS CONTEXT

Regulatory bodies, Introduction to standardisation, Standards and recommended practices (SARPS), Freedoms of the air, Role of governments, Major agreements and treaties, Business side of the industry – Key performance indicators (revenue passenger kilometre, yield etc.), Industry characteristics of passenger airlines.

UNIT-III

AIRLINES AND AIRPORTS

Introduction to airlines, Organisational structure, Types of airline personnel, Flight crew and cabin crew, Role of training with airlines, Organisational culture of an airline

Introduction to airports – Airport personnel, Processing passengers and freight, Security

UNIT-IV

AIR NAVIGATION SERVICES

Introduction to air navigation services, Types of navigation services, Air traffic control (ATC) services, Structure of controlled airspace, Air traffic and control services, Classifications of airspace, Flight information services and alerting services, Air navigation facilities and air navigation service (ANS) providers

UNIT-V

AIR SAFETY AND SECURITY

Introduction to air safety and security, Manufacturer responsibilities, Maintenance procedures, Airside safety, Safety in the air, Issues in air safety, Accident and incident investigation, Security – Response of International Civil Aviation Organization (ICAO).

Text Books:

1. Peter Belobaba, Amedeo Odoni and Cynthia Barnhart (2015), The Global Airline Industry (Aerospace Series), John Wiley & Sons, Ltd, United Kingdom.
2. Connor R Walsh, Airline Industry – Strategies, Operations and Safety, Nova Science Publishers, New York.

Reference Books:

1. Andrew R Thomas (2011), Soft Landing: Airline Industry Strategy, Service and Safety, Apress, New York
2. Ahmed Abdelghany, Khaled Abdelghany (2009), Modeling Applications in the Airline Industry, Ashgate Publishing Ltd, England.

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Discuss the types of aerospace industry and evolution of flying machine to airline alliances.
- 2 Explain standardisation and the importance of regulatory bodies.
- 3 Outline the organisational structure of airlines and airports.
- 4 Demonstrate major types of air navigation services and air traffic control services.
- 5 Illustrate the major processes involved in air safety and security.

UNMANNED AERIAL VEHICLES

VIII Semester : OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6AE41	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES

1. To understand the evolution and applications of unmanned aerial vehicles.
2. To illustrate the subsystems of UAVs.
3. To explain the process involved in design of UAVs.

UNIT-I INTRODUCTION TO UAV

Historical Development, Overview of UAV Systems and sub-systems, Classification, Informal Categories, The Tier System, Classification Change.

UNIT-II BASIC AERODYNAMICS AND PERFORMANCE of UAV

Basic Aerodynamic Equations, Air foils, lift, drag, moments, Aircraft Polar, The Real Wing and Airplane, Induced Drag, Total Air-Vehicle Drag, Flapping Wings, Rotary wings. Climbing Flight, Range, Endurance, Gliding Flight

UNIT-III PROPULSIVE SYSTEMS, STRUCTURES AND LOADS, PAYLOAD

Thrust Generation and basic thrust equation, Sources of Power, Loads, types of loads, Materials, construction techniques. Payloads-Reconnaissance/Surveillance Payloads, Weapon Payloads, Other payloads

UNIT-IV UAV SUBSYSTEMS

Mission Planning and Control Station- Types, Physical Configuration, Planning and Navigation, MPCS Interfaces, Modes of control, piloting and controlling mission, Autopilot system Launch Systems- Basic Considerations, launch Methods for Fixed, rotary wing UAV Recovery Systems

UNIT-V BASICS DESIGN AND CASE

Introduction to Design and Selection of the System - Conceptual Phase, Preliminary Design, Detail Design, Selection of the System Case study on Indian UAVs(Rustom, Lakshya, AURA) Case study on Israeli-Heron, US- MQ9 Reaper

Text Books:

1. Paul Fahlstrom, Thomas Gleason - Introduction to UAV Systems-Wiley (2012)
2. Reg Austin, "Unmanned Air Systems: UAV Design, Development and Deployment", First Edition, Wiley Publishers, 2015.

Reference Books:

1. Miroslaw Adamski, "Power units and power supply systems in UAV", New Edition, Taylor and Francis Group publishers, 2014.
2. Skafidas, "Microcontroller Systems for a UAV", KTH, TRITA-FYS 2002:51 ISSN 0280-316X. 34, 2002.

COURSE OUTCOMES

1. Classify the Unmanned Aerial Vehicles.
2. Calculate the basic performance parameters for aircraft.
3. Identify and illustrate various payloads and propulsive systems.
4. Explain the functioning of subsystems in UAVs.
5. Illustrate the design process for a UAV.

OPEN ELECTIVES OFFERED BY COMPUTER SCIENCE AND ENGINEERING

OE1		OE2	
A6CS48	Software Project Management Fundamentals	A6CS09	Database Management System
A6CS37	Semantic Web	A6CS43	Data Warehousing and Business Intelligence
OE3			
A6CS51	Artificial Intelligence and Applications		
A6CS36	Cyber Security & Cyber laws		

SOFTWARE PROJECT MANAGEMENT FUNDAMENTALS

(OPEN ELECTIVE-I)

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS48	OEC	3	-	-	3	40	60	100

COURSE OBJECTIVE

1. Understanding the specific roles within a software organization as related to project and process management.
2. Understanding the basic infrastructures competences (e.g., process modeling and measurements.)
3. Describe the principles, techniques, methods & tools for model based management of software projects, assurance of product quality and process adherence (quality assurance), as well as experienced based creation and improvements of models (process management)
4. Understanding the basic steps of project planning, project management, quality assurance, and process management and their relationships.
5. To provide basic project management skills with a strong emphasis on issues and problems associated with delivering successful IT projects.

COURSE OUTCOMES

1. To describe and determine the purpose and importance of project management from the perspectives of planning, tracking and completion of project.
2. To compare and differentiate organization structures and project structures.
3. To implement a project to manage project schedule, expenses and resources with the application of suitable project management tools.
4. Understand and practice the process of project management and its application in delivering successful IT projects.
5. Conduct Case Study on “The Command Center Processing and Display System-Replacement” (CCPDS-R)

UNIT - I CONVENTIONAL SOFTWARE MANAGEMENT

CLASSES: 12

The waterfall model, conventional software Management performance. Evolution of Software Economics: Software Economics, pragmatic software cost estimation.

UNIT - II IMPROVING SOFTWARE ECONOMICS

CLASSES: 12

Reducing Software product size, improving of tware processes, improving team effectiveness, improving automation, Achieving required quality, peer in spections.

The old way and the new: The principles of conventional software engineering, principles of moderns of tware management, transitioning to an iterative process.

UNIT - III LIFE CYCLE PHASES

CLASSES: 12

Engineering and production stages, inception, Elaboration, construction, transition phases. **Artifacts of the process:** The artifactsets, Management artifacts, Engineering artifacts, programmatic artifacts. Model based software architectures: A management perspective and technical perspective.

UNIT - IV WORK FLOWS OF THE PROCESS

CLASSES: 14

Software process workflows, Inter Trans workflows. Checkpoints of the Process: Major Mile Stones, Minor Milestones, Periodic status assessments. Iterative Process **Planning:** Work break down structures, planning guidelines, cost and schedule estimating, Interaction planning process, Pragmatic planning.

Project Organizations and Responsibilities: Line-of-Business Organizations, Project Organizations, evolution of Organizations.

Process Automation: Automation Building Blocks, The Project Environment.

UNIT - V PROJECT CONTROL AND PROCESS INSTRUMENTATION

CLASSES: 15

The server care Metrics, Management indicators, quality indicators, life cycle expectations pragmatic Software Metrics, Metrics automation. Tailoring the Process: Process discriminates, Example. Future Software Project Management: Modern Project Profiles, Next generation Software economics, modern Process transitions.

TEXT BOOKS

1. Software Project Management, Walker Royce, Pearson Education, 2012
2. Software Project Management, Bob Hughes & Mike Cotterell, Fifth edition, TataMc-GrawHill, 2006

REFERENCE BOOKS

1. Applied Software Project Management, Andrew Stellman & Jennifer Greene, O'Reilly, 2006
2. Head First PMP, Jennifer Greene & Andrew Stellman, O'Reilly, 2007
3. Software Engineering Project Management, Richard H. Thayer & Edward Yourdon, second edition, Wiley India, 2004.
4. Agile Project Management, Jim Highsmith, Pearson education, 2004
5. The art of Project management, Scott Berkun, O'Reilly, 2005.
6. Software Project Management in Practice, Pankaj Jalote, Pearson Education, 2002

WEB LINKS

1. https://www.tutorialspoint.com/software_engineering/software_project_management.htm
2. https://ebooks.lpude.in/management/mba/term_4/DCAP304_DCAP515_SOFTWARE_PROJECT_MANAGEMENT.pdf

MOOCS: https://onlinecourses.nptel.ac.in/noc19_cs70/preview

EBOOK: <https://www.edulearn.net.in/wp-content/uploads/2019/09/MG6088-Software-Project-Management.pdf>

SEMANTIC WEB**(OPEN ELECTIVE - I)****III B. TECH- II SEMESTER**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS37	OEC	3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the semantic web and its importance.
2. Understand the attribute languages and how to use in semantic web
3. Understand the Rule based languages
4. Understand the different methods to development in internet
5. Learn different types of semantic web tools.

COURSE OUTCOMES

1. Classify the on tologies towards to the internet.
2. Apply the attribute languages on internet.
3. Apply the rule based languages for improving the usage of semantic web.
4. Analyze the methods to development of the internet.
5. Use semantic web tools for efficient development and usage of web.

UNIT - I THE FUTURE OF THE INTERNET**CLASSES: 12**

Introduction, The Syntactic Web, The Semantic Web, How the Semantic Web Will Work. Ontology in Computer Science: Defining the Term Ontology, Differences Among Taxonomies, Thesauri, and Ontologies, Classifying Ontologies, Web Ontologies, Web Ontology Description Languages, Ontology, Categories, and Intelligence.

UNIT - II KNOWLEDGE REPRESENTATION IN DESCRIPTION LOGIC**CLASSES: 12**

Introduction, An Informal Example, The Family of Attributive Languages, Inference Problems. RDF and RDF Schema: Introduction, XML Essentials, RDF, RDF Schema, A Summary of the RDF/RDF Schema Vocabulary. OWL: Introduction, Requirements for Web Ontology Description Languages, Header Information, Versioning, and Annotation Properties, Properties, Classes, Individuals, Data types, A Summary of the OWL Vocabulary.

UNIT - III RULE LANGUAGES**CLASSES: 14**

Introduction, Usage Scenarios for Rule Languages, Datalog, RuleML, SWRL, TRIPLE. Semantic Web Services: Introduction, Web Service Essentials, OWL-S Service Ontology, An OWL-S Example.

UNIT - IV METHODS FOR ONTOLOGY DEVELOPMENT**CLASSES: 15**

Introduction, Uschold and King Ontology Development Method, Toronto Virtual Enterprise Method, Methontology, KACTUS Project Ontology Development Method, Lexicon-Based Ontology Development Method, Simplified Methods. Ontology Sources: Introduction, Metadata, Upper Ontologies, Other Ontologies of Interest, Ontology Libraries.

UNIT - V SEMANTIC WEB SOFTWARE TOOLS & SOFTWARE AGENTS**CLASSES: 12**

Introduction, Metadata and Ontology Editors, Reasoners, Other tools. Introduction, Agent Forms, Agent Architecture, Agents in the Semantic web Context. Semantic Desktop: Introduction, Semantic Desktop Metadata, Semantic Desktop Ontologies, Semantic Desktop Architecture, Semantic Desktop Related Applications. Ontology Application in Art: Introduction, Ontologies for the Description of Works of Art, Metadata Schemas for The Description of Works of Art, Semantic Annotation of Art Images.

TEXT BOOKS

1. Semantic Web- Concepts, Technologies and applications, 2007th edition (24 January 2007) K.Breitman, Marco Antonio Casanova and Walter Truszkowski, Springer..

REFERENCE BOOKS

1. InformationSharingontheSemantingWeb,HeinerStuckenschmidt, Frankvan Harmelen, Springer.
2. Semantic Web Primer, Grigoris Antoniou, FrankVan
3. Semantic Web Services: Concepts, Technologies and Applications, Rudi Studer, Stephan Grimm, Andrees Abeker, Springer
4. Towards the Semantic Web: Ontology Driven Knowledge Management, John Davis, Dieter Fensal, Frank Van Harmelen, J.Wiley.

WEB LINKS

1. <https://www.techtarget.com/searchcio/definition/Semantic-Web>
2. <https://www.w3.org/RDF/Metalog/docs/sw-easy>
MOOC:1. <https://www.coursera.org/learn/web-data>
EBOOK:1. <https://open.hpi.de/courses/semanticweb>
3. [http://prof.mau.ac.ir/images/Uploaded_files/A%20Semantic%20Web%20Primer-The%20MIT%20Press%20\(2012\)%5B7460174%5D.PDF](http://prof.mau.ac.ir/images/Uploaded_files/A%20Semantic%20Web%20Primer-The%20MIT%20Press%20(2012)%5B7460174%5D.PDF)

DATABASE MANAGEMENT SYSTEMS

(OPEN ELECTIVE-II)

IV B. TECH- I SEMESTER

Course	Code	Category	Hours / Week			Credits	Maximum Marks		
			L	T	P		CIE	SEE	Total
A6CS09		OEC	3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Discuss the basic database concepts, applications, data models, schemas and instances.
2. Design Entity Relationship model for a database.
3. Demonstrate the use of constraints and relational algebra operations.
4. Describe the basics of SQL and construct queries using SQL
5. Understand the importance of normalization in databases.
6. Demonstrate the basic concepts of transaction processing and concurrency control.
7. Understand the concepts of database storage structures and identify the access techniques.

COURSE OUTCOMES

At the end of the course the students are able to:

1. Use the basic concepts of Database Systems in Database design
2. Design a Database using ER Modeling
3. Apply relational algebra and calculus concepts AND SQL for a wide range of database operations
4. Apply SQL queries and PL/SQL queries to interact with Database
5. Grasp transaction management principles including transaction states and concurrency control techniques

UNIT - I INTRODUCTION

CLASSES: 09

Introduction and applications of DBMS, Purpose of data base, Data Independence, Database System architecture- Levels, Database users and DBA.

DATABASE DESIGN: Database Design Process, ER Diagrams - Entities, Attributes, Relationships, Constraints, keys, extended ER features, Generalization, Specialization, Aggregation, Conceptual design with the E-R model.

UNIT - II RELATIONAL MODEL & SCHEMA REFINEMENT

CLASSES: 09

RELATIONAL MODEL: Introduction to the relational model, Integrity constraints over relations, Enforcing integrity Constraints, Querying relational data, Logical database design: E-R to relational, Introduction to views, Destroying/altering tables and views.

SCHEMAREFINEMENT AND NORMAL FORMS: Introduction to schema refinement, functional dependencies, reasoning about FDs. Normal forms: 1NF, 2NF, 3NF, BCNF, properties of decompositions, normalization, schema refinement in database design.

UNIT - III RELATIONAL ALGEBRA CALCULUS & SQL

CLASSES: 11

RELATIONAL ALGEBRA AND CALCULUS: Relational algebra operators, relational calculus – Tuple and domain relational calculus.

SQL: Basics of SQL, DDL,DML,DCL, structure - creation, alteration, defining constraints - Primary key, foreign key, unique, not null, check, IN operator, Functions - aggregate functions, Built-in functions — numeric, date, string functions, set operations, sub-queries, correlated sub-queries

UNIT - IV SQL and PL/SQL

CLASSES: 11

SQL: Use of group by, having, order by clauses, join and its types, Exist, Any, All classes. Transaction control commands - Commit, Rollback, Save point,

PL/SOL: Environment, block structure, variables, operators, data types, control structures; cursors, stored procedures, Triggers.

UNIT - V TRANSACTION & CONCURRENCY CONTROL

CLASSES: 10

TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, concurrent executions, Serializability, recoverability, testing for serializability.

CONCURRENCY CONTROL AND RECOVERY SYSTEM: Concurrency control, lock based protocols, time-stamp based protocols, validation based protocols, multiple granularity and deadlock handling. Recovery system - failure classification, storage structure, recovery and atomicity, log based recovery.

TEXT BOOKS

1. Abraham SilbersGhatz, Henry F. Korth, S. Sudharshan, "Database System Concepts". Sixth Edition, Tata McGraw Hill, 2011
2. Raghurama Krishnan, Johannes Gehrke, "Data base Management Systems", TATA McGraw Hill. 3rd Edition, 2007.
3. R.P. Mahapatra & Govind Verma, Database Management Systems, Khanna Publishing House, 2013.
4. Michael McLaughlin, Oracle Database 11g PL/SOL Programming, Oracle press.

REFERENCE BOOKS

1. Peter Rob, Carlos Coronet, Database Systems Design Implementation and Management, 7th edition, 2009.
2. Scott UnTian. Michael McLaughlin, Ron Hardman. "Oracle database 10g PL/SQL programming ". 6th edition, Tata McGraw Hill, 2010
3. S.K Singh, "Database Systems Concepts. Design and Applications", First edition, Pearson Education, 2006.
4. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems". Fourth Edition. Pearson / Addition wesley, 2007

WEB REFERENCES

1. <https://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database>
2. https://www.w3schools.com/sql/sql_create_table.asp
3. https://www.ibm.com/docs/en/zos-basic-skills?topic=zos-what-is-database-management-system&external_link=true
4. <https://www.cs.ucdavis.edu/~green/courses/ecs165a-w11/>

E-TEXT BOOKS

1. <https://www.freebookcentre.net/Database/Free-Database-Systems-Books-Download.html>
2. <https://www.ddegjust.ac.in/studymaterial/mca-3/ms-11.pdf>

MOOC COURSE

1. <https://www.mooc-list.com/tags/dbms-extensions>

DATA WAREHOUSING AND BUSINESS INTELLIGENCE

(OPEN ELECTIVE-II)

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS43	OEC	3	-	-	3	40	60	100

COURSE OBJECTIVES:

1. This course is concerned with extracting data from the information systems that deal with the day-to-day operations and transforming it into data that can be used by businesses to drive high-level decision making
2. Students will learn how to design and create a data warehouse, and how to utilize the process of extracting, transforming, and loading (ETL) data into data warehouses.

COURSE OUTCOMES:

1. Understand architecture of data warehouse and OLAP operations.
2. Understand Fundamental concepts of BI and Analytics
3. Application of BI Key Performance indicators and Design of Dashboards, Implementation of Web Analytics
4. Understand Utilization of Advanced BI Tools and their Implementation.
5. Implementation of BI Techniques and BI Ethics.

UNIT - I DATA WAREHOUSE

CLASSES:12

DATA WAREHOUSE: Data Warehouse-Data Warehouse Architecture- Multidimensional Data Model-Data cube and OLAP Technology-Data Warehouse Implementation -DBMS schemas for Decision support - Efficient methods for Data cube computation.

UNIT - II BUSINESS INTELLIGENCE

CLASSES: 10

Business Intelligence: Introduction – Definition, Leveraging Data and Knowledge for BI, BI Components, BI Dimensions, Information Hierarchy, Business Intelligence and Business Analytics. BI Life Cycle. Data for BI - Data Issues and Data Quality for BI.

UNIT - III BI IMPLEMENTATION

CLASSES: 10

BI Implementation - Key Drivers, Key Performance Indicators and Performance Metrics, BI Architecture/Framework, Best Practices, Business Decision Making, Styles of BI-vent-Driven alerts-A cyclic process of Intelligence Creation. The value of Business intelligence -Value driven and Information use.

UNIT - IV ADVANCED BI

CLASSES:10

Advanced BI – Big Data and BI, Social Networks, Mobile BI, emerging trends, Description of different BI-Tools (Pentaho, KNIME)

UNIT - V BI IMPLEMENTATION

CLASSES: 12

Business intelligence implementation-Business Intelligence and integration implementation-connecting in BI systems- Issues of legality- Privacy and ethics- Social networking and BI.

TEXT BOOKS

1. Data Mining – Concepts and Techniques - JIAWEI HAN & MICHELINE KAMBER, Elsevier.
2. Rajiv Sabherwal "Business Intelligence" Wiley Publications, 2012.

REFERENCE BOOKS

1. Efraim Turban, Ramesh Sharda, Jay Aronson, David King, Decision Support and Business Intelligence Systems, 9th Edition, Pearson Education, 2009.
2. David Loshin, Business Intelligence - The Savy Manager's Guide Getting Onboard with Emerging IT, Morgan Kaufmann Publishers, 2009
3. Philo Janus, Stacia Misner, Building Integrated Business Intelligence Solutions with SQL Server, 2008 R2 & Office 2010, TMH, 2011.
4. Business Intelligence Data Mining and Optimization for decision making [Author: Carlo-Verellis] [Publication: (Wiley)]
5. Data Warehousing, Data Mining & OLAP- Alex Berson and Stephen J. Smith- Tata McGrawHill Edition, Tenth reprint 2007
6. Building the Data Warehouse- W. H. Inmon, Wiley Dreamtech India Pvt. Ltd.
7. Data Mining Introductory and Advanced topics –MARGARET H DUNHAM, PEA.

WEB LINKS

1. <https://www.coursera.org/learn/data-warehousing-business-intelligence>
2. https://www.vssut.ac.in/lecture_notes/lecture1428550844.pdf

ARTIFICIAL INTELLIGENCE AND APPLICATIONS

(OPEN ELECTIVE-III)

IV B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS51	OEC	3	-	-	3	40	60	100

COURSE OBJECTIVES

To learn

1. The difference between optimal reasoning vs human like reasoning
2. Understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. Apply the Expert and rule based systems in knowledge representation techniques
4. Understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning.
2. Formulate an efficient problem space for problem solving and Represent knowledge using the appropriate technique
3. Design and develop expert systems using Prolog and solve uncertainty problems.
4. Analyze real world problems and implement using ANN, CNN and AI techniques.
5. Implement the concepts of AI in Healthcare, Robotics and Agriculture.

UNIT - I INTRODUCTION AND PROBLEM SOLVING

CLASSES:
14

Introduction: History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A*, Constraint Satisfaction, Game Playing, Alpha-Beta Pruning.

UNIT - II LOGIC AND KNOWLEDGE REPRESENTATION

CLASSES:
14

Logic Concepts and Logic Programming: Introduction, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.

UNIT - III EXPERT SYSTEM AND APPLICATIONS

CLASSES:
14

Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs Traditional Systems, rule based expert systems black board systems truth maintenance systems, Application of Expert Systems.

UNIT - IV ANN, CNN

CLASSES:
2

Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks. Convolutional Neural Networks: Pooling, Padding, Layers used to build ConvNets, Working of CNN, Python Libraries used for AI: Tensor flow, Keras, Theano, Pytorch, Scikit Learn, Pandas, Image Classification using CNN with Python.

UNIT - V APPLICATIONS & CASE STUDIES OF AI

CLASSES:
11

Applications of AI in real world, Future of AI, Benefits and Risks of AI.

Case Studies: AI in Healthcare: Medical Imaging and Diagnostic, AI in Agriculture: Autonomous Tractors, Controlling Pest Infestations, Soil and Crops Health Monitoring, AI in Robotics: Use of Artificial Intelligence in Robotics

TEXT BOOKS

1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011
2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004

REFERENCE BOOKS

1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009.
2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson.
3. Introduction to Artificial Intelligence and expert systems Dan W.Patterson. PHI.
4. Artificial Intelligence by George Flugerrearson fifth edition.

WEB LINKS

1. [https://www.ibm.com/topics/artificial-intelligence#:~:text=At%20its%20simplest%20form%2C%20artificial,in%20conjunction%20with%20artificial%20intelligence2. https://www.techtarget.com/searchenterpriseai/definition/AI-Artificial-Intelligence](https://www.ibm.com/topics/artificial-intelligence#:~:text=At%20its%20simplest%20form%2C%20artificial,in%20conjunction%20with%20artificial%20intelligence2.https://www.techtarget.com/searchenterpriseai/definition/AI-Artificial-Intelligence)
2. MOOCS:<https://www.mygreatlearning.com/great-lakes-artificial-intelligence-and-machine-learning?>
3. E BOOK: <https://www.freebookcentre.net/CompuScience/Free-Artificial-Intelligence-Books-Download.html>

CYBER SECURITY AND CYBER LAWS

(OPEN ELECTIVE-III)

IV B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6CS36	OEC	3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To learn cybercrime cyber security and cyber law.
2. To understand the cyber attacks.
3. To understand information gathering.
4. To learn how to detect and prevent a cyber attack and tools for mitigating them.
5. Develop Competencies For Dealing With Frauds And Deceptions (Confidence Tricks, Scams) And Other Cyber Crimes

COURSE OUTCOMES

1. Explain the basics of cyber security, cyber crime and cyber law.
2. Classify various types of attacks and learn the tools to launch the attacks.
3. Apply various tools to perform information gathering.
4. Understand Cyber crimes and cyber laws.
5. Explore The Legal And Policy Developments In Various Countries To Regulate Cyberspace

UNIT - I INTRODUCTION

CLASSES: 16

Cyber Security-definition, Need for Cyber Security, Protection, Security, risk, threat, vulnerability, exploit, attack, confidentiality, integrity, availability, non-repudiation, authentication, authorization, codes, plain text, encryption, decryption, cipher text, key, ciphers, Symmetric and asymmetric cryptography, Public key, private key, Crypt analysis, Cyber forensics.

UNIT - II CYBER ATTACKS

CLASSES:14

CYBER ATTACKS (definitions and examples): Denial-of-service attacks, Man-in-the middle attack, Phishing, spoofing and spam attacks, Drive by attack, Password attack, SQL injection attack, Cross-site scripting attack, Eavesdropping attack, Birthday attack, Malware attacks, Social Engineering attacks

UNIT - III HANDLING THE ATTACKS

CLASSES: 11

Firewalls, logging and intrusion detection systems, e-mail security, security issues in operating systems, ethics of hacking and cracking.

UNIT - IV CYBER CRIMES

CLASSES: 12

Cyber Crime and Information Security – History of Cyber Crime, classifications of Cyber Crimes – Tools and Methods –Password Cracking, Key loggers, Spywares, SQL Injection – Network Access Control – Cloud Security – Web Security – Wireless Security

UNIT - V CYBER SPACE AND CYBER LAWS

CLASSES: 16

Cyber Space- Fundamental definitions -Interface of Technology and Law – Jurisprudence and-Jurisdiction in Cyber Space - Indian Context of Jurisdiction - Enforcement agencies – Need for IT act - UNCITRAL – E-Commerce basics .Information Technology Act, 2000 - Aims and Objects — Overview of the Act – Jurisdiction Cyber Law, A Global Perspective on Cyber Crimes; Cyber Laws – The Indian IT Act – Cybercrime and Punishment.

TEXT BOOKS

1. Anand Shinde, "Introduction to Cyber Security Guide to the World of Cyber Security", Notion Press, 2021.
2. Nina Godbole, Sunit Belapure, "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Wiley Publishers, 2011.
3. Cyber Law- The Indian Perspective, Pavan Duggal
4. Guide to Cyber Law, Rodney D. Ryder

REFERENCE BOOKS

1. David Kim, Michael G. Solomon, "Fundamentals of Information Systems Security", Jones & Bartlett Learning Publishers, 2013
2. Patrick Engebretson, "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made easy", Elsevier, 2011
3. Justice Yatindra Singh: Cyber Laws, Universal Law Publishing Co., New Delhi
4. Farouq Ahmed, Cyber Law in India, New Era publications, New Delhi

OPEN ELECTIVES OFFERED BY DEPARTMENT OF ECE

OPEN ELECTIVE COURSE-I			
S. No.	Course Code	Course Name	Offering Department
1.	A6EC55	Microprocessors and Interfacing	Electronics & Communication Engineering
2.	A6EC56	Principles of Communications	
OPEN ELECTIVE COURSE-II			
3.	A6EC57	Microcontrollers and Applications	Electronics & Communication Engineering
4.	A6EC58	Fundamentals of Image processing	
OPEN ELECTIVE COURSE-III			
5.	A6EC59	Introduction to Sensors and Actuators	Electronics & Communication Engineering
6.	A6EC60	Introduction to Computer Vision	

MICROPROCESSORS AND INTERFACING

OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits		Maximum Marks	
		L	T	P	C	CIE	SEE	Total
A6EC55	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the basic of 8, 16 bit microprocessor architectures and its functionalities.
2. Develop an assembly language programming skills of various processors.
3. Interface different peripheral devices with microprocessors and microcontrollers.
4. Interface memory devices to 8086 processor.
5. Analyze Serial communication schemes of microprocessor based systems.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the architecture of micro processor.
2. Understand the programming model of micro processors.
3. Interface different external peripherals with microprocessors.
4. Analyze a problem and formulate appropriate computing solution for processor based application.
5. Develop an assembly language program for specified application in communication.

UNIT-I MICROPROCESSORS ARCHITECTURE

Classes: 09

Overview of 8085, 8086 architecture- functional diagram, Register organization, memory segmentation, Memory addresses, physical memory organization.

UNIT-II SIGNAL DESCRIPTION OF 8086

Classes: 09

Signal description of 8086, timing diagrams, Interrupt structure of 8086, Vector interrupt table, Interrupts of 8086 Processor, Interrupt service routine.

UNIT-III INSTRUCTION SET AND ASSEMBLY LANGUAGE PROGRAMMING OF 8086

Classes: 09

Programming model, Addressing modes, Instruction set, Assembler directives, Programs involving logical, branch and call instructions, Sorting, evaluating arithmetic expressions, and string manipulations.

UNIT-IV I/O INTERFACE

Classes: 09

Introduction to 8255 PPI, various modes of operation of 8255 PPI, interfacing 8255 to 8086, Stepper motor interfacing, D/A & A/D converter, Memory interfacing to 8086.

UNIT-V SERIAL COMMUNICATION INTERFACE

Classes: 09

Serial communication standards, serial data transfer schemes, 8251 USART architecture, Interfacing of 8251/ USART to 8086 Processor.

Text Books:

Hall, Microprocessors and Interfacing. TMGH, 2nd edition 2006.

1. Advanced microprocessors and peripherals-A.K ray and K.M.Bhurchandani, TMH, 2nd edition 2006.
3. R.S.Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996.

Reference Books:

1. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.
2. Micro computer system 8086/8088 family architecture, programming and design- By Liu and GA Gibson, PHI, 2nd Ed.,

Web References:

[/www.freebookcentre.net/electronics-ebooks-download/Microprocessor-and-Microcontroller.html](http://www.freebookcentre.net/electronics-ebooks-download/Microprocessor-and-Microcontroller.html)
[/coen.boisestate.edu/smluo/smluo-courses/ece-332-microprocessors-fall07/lecture-notes/](http://coen.boisestate.edu/smluo/smluo-courses/ece-332-microprocessors-fall07/lecture-notes/)
[/www.freebookcentre.net/electronics-ebooks-download/Introduction-to-Microcontrollers-Lecture-Notes.html](http://www.freebookcentre.net/electronics-ebooks-download/Introduction-to-Microcontrollers-Lecture-Notes.html)

E-Text Books:

[/gen.lib.rus.ec/book/index.php?md5=67C5AC79DC8180A7F0641609D0C7800C](http://gen.lib.rus.ec/book/index.php?md5=67C5AC79DC8180A7F0641609D0C7800C)
[//www.faadooengineers.com/threads/9039-8085-microprocessor-by-RAMESH-GANOKAR-ebook-pdf-download](http://www.faadooengineers.com/threads/9039-8085-microprocessor-by-RAMESH-GANOKAR-ebook-pdf-download)
[://e.edim.co/123389964/The_8051_Microcontroller_Architecture_Programming_And_Applications.pdf](http://e.edim.co/123389964/The_8051_Microcontroller_Architecture_Programming_And_Applications.pdf)
https://e.edim.co/123389964/A.K._Ray_and_K.M._BhurchandiAdvanced_Microprocessors_and_Peripherals_3e-Tata_Mcgraw_Hill.pdf

MOOC Course:

[://www.mooc-list.com/tags/microprocessors](http://www.mooc-list.com/tags/microprocessors)
2. <https://www.coursera.org/courses?query=microprocessor>

PRINCIPLES OF COMMUNICATIONS

OPEN ELECTIVE – I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EC56	OEC	3	0	0	3	40	60	100

PREREQUISITE: Knowledge on Electron devices and Electronic circuits

COURSE OBJECTIVES:

The students are able to:

1. Describe the basic concepts of analog communication
2. Significance of digital & analog communication
3. Compare the digital modulation techniques
4. Learn the various multiple access techniques
5. Understanding the Satellite and Optical fiber concepts.

COURSE OUTCOMES:

Upon completion of the course, students will be able to:

1. Understand the concepts of analog modulations
2. Illustrate the basic concepts of digital modulations.
3. Analyse the various digital modulation methodologies.
4. Distinguish the concepts of multiple access techniques.
5. Differentiate the various orbits and associated satellite launches.

UNIT I FUNDAMENTALS OF ANALOG COMMUNICATION

Classes: 09

Modulation, Principles of Amplitude Modulation, AM modulator and Demodulator, Types of Amplitude modulation (DSBSC, SSBSC, VSB). Angle modulation FM and PM waveforms, phase deviation and modulation index, frequency deviation and percent modulation.

UNIT II DIGITAL TRANSMISSION

Classes: 09

Introduction, Sampling theorem, Types of pulse modulations, PCM, companding, differential pulse code modulation, delta modulation, adaptive delta modulation, Inter symbol interference.

UNIT III DIGITAL MODULATION TECHNIQUES

Classes: 09

Introductions, Shannon limit for information capacity, ASK FSK, BPSK QPSK and DPSK modulators and demodulators, comparison of digital modulation techniques.

UNIT IV SPREAD SPECTRUM AND MULTIPLE ACCESS TECHNIQUES

Classes: 09

Introduction, Pseudo noise sequence, DS spread spectrum with coherent binary PSK, FH spread spectrum, multiple access techniques-FDMA, TDMA and CDMA.

UNIT V SATELLITE AND OPTICAL COMMUNICATION

Classes: 09

Satellite Communication Systems, Keplers Law, LEO, MEO and GEO Orbits, Link model Optical Communication Systems-Fiber losses-scattering, absorption attenuation and bending losses, types of optical Sources and Detectors.

TEXTBOOKS:

1. Wayne Tomasi, —Advanced Electronic Communication Systems, 6/e, Pearson Education, 2007.
2. Simon Haykin, —Communication Systems, 4th Edition, John Wiley & Sons., 2001.

REFERENCES:

1. H.Taub, D L Schilling, G Saha, |Principles of Communication|3/e, 2007.
2. B.P.Lathi, |Modern Analog And Digital Communication systems|, 3/e, Oxford University Press, 2007
3. Blake, —Electronic Communication Systems, Thomson Delmar Publications, 2002.
4. Martin S.Roden, —Analog and Digital Communication System, 3rd Edition, PHI, 2002.
5. B.Sklar, |Digital Communication Fundamentals and Applications|2/e Pearson Education 2007.

Web References:

1. <https://personal.utdallas.edu/~torlak/courses/ee4367/lectures/FIBEROPTICS.pdf>
2. Bricker G (2012) 2-D bar codes, Journal of Computing Sciences in Colleges, **28**:1, (25-32), Online publication date: 1-Oct-2012.

E-Text Books:

1. https://books.google.co.in/books/about/Principles_Of_Communication.html?id=6Zunu4Acf8C

MOOC Course:

1. https://onlinecourses.nptel.ac.in/noc18_ee26/preview

MICROCONTROLLER & APPLICATIONS

OPEN ELECTIVE - II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EC57	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. To introduce the students about architectural features of microcontrollers and its registers
2. To introduce about the instruction set of 8051
3. To know about the I/O Ports and Interrupts of 8051
4. To know about Timers/Counters of 8051
5. To introduce Arduino programming and interfacing of sensors

COURSE OUTCOMES:

Upon completion of the course, students will be able to:

1. Understand the architectural features of MCS-51 and select a suitable microcontroller to suit the application.
2. Develop programs for control applications using assembly language and embedded C
3. Use timers and counters for delay generation and event counting and illustrate the use of interrupts and service routines
4. Write algorithms and develop programs for serial data communication applications.
5. Design microcontroller based-applications for simple real-world applications

UNIT I**8051 ARCHITECTURE****Classes: 9**

Introduction to microcontrollers, RISC, CISC, Harvard and Von Neumann architectures. Architecture of 8051, Signal descriptions of 8051, General purpose registers of 8051, register banks, Memory organization.

UNIT II**INSTRUCTION SET OF 8051****Classes: 9**

Assembly language Instruction format, 8051 Addressing modes, Instruction set of 8051: Classification, syntax and function of instructions, Simple programs.

UNIT III**I/O PORT AND INTERRUPT PROGRAMMING****Classes: 9**

I/O Ports of 8051, Features of I/O ports, configuring I/O ports, interface I/O devices such as LED, buzzer, push-button switch, example programs with assembly & C. Interrupts of 8051 and its priorities, IE and IP registers, Interrupt enabling/disabling and priority setting, example programs in assembly and C.

UNIT IV**TIMERS /COUNTERS AND SERIAL I/O****Classes: 9**

Bit structure and function of TMOD and TCON registers, Timer/Counter modes of operations, Timer/Counter programs in assembly and C. Bit structure and function of SCON, PCON registers, SBUF register, Serial Communication modes in 8051, programs on serial communication.

UNIT V**INTRODUCTION TO ARDUINO****Classes: 9**

Introduction to Arduino-uno board, Analog and Digital pins, programming structure of Arduino, introduction to sensors and actuators, Sensor interfacing, programming to sensors, Motor interfacing, LCD interfacing.

TEXTBOOKS:

1. The 8051 Microcontroller(3rd edition) - Kenneth J Ayala
2. The 8051 Microcontroller & Embedded systems using assembly and C (2ndEdition) –M.A.Mazidi , J.C. Mazidi & R.D.McKinlay ISBN: 81-317-1026-2

REFERENCES:

1. The 8051 Microcontroller(4th Edition)- MacKenzie , ISBN:81-317-2018-7
2. The 8051 Microcontroller(1st Edition) – Dr.Uma Rao & Andhe Paallavi, ISBN: 81-317-3252-5
3. Microcontrollers & applications, Ramani Kalpathi, & Ganesh Raja , ISBN: 81-888-4918-9
4. Programming Arduino: Getting Started with Sketches, Second Edition (Tab) 2nd Edition – Simon Monk

Web References:<https://www.the8051microcontroller.com/web-references>

E-Text Books:

1. <https://www.freebookcentre.net/Electronics/Microcontroller-Books.html>
2. <https://www.freebookcentre.net/Electronics/Microcontroller-Application-Books.html>

MOOC Course 1.<https://nptel.ac.in/courses/117/104/117104072/>

FUNDAMENTALS OF IMAGE PROCESSING OPEN ELECTIVE – II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EC58	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The students are able to:

1. To become familiar with digital image fundamentals
2. To get exposed to image enhancement techniques in Spatial and Frequency domain.
3. To learn concepts of degradation function and restoration techniques.
4. To study the image segmentation and representation techniques.
5. To become familiar with image compression and recognition methods

COURSE OUTCOMES:

1. Know and understand the basics and fundamentals of digital image processing.
2. Operate on images using the techniques of smoothing, sharpening and enhancement.
3. Apply the restoration concepts and filtering techniques on digital images.
4. Learn the basics of segmentation and features extraction.
5. Analyze the compression and recognition methods for color images.

UNIT I INTRODUCTION OF IMAGE PROCESSING

Classes: 09

Digital Image Fundamentals-Elements of visual perception, image sensing and acquisition, image sampling and quantization, basic relationships between pixels – neighbourhood, adjacency, connectivity, distance measures.

UNIT II IMAGE ENHANCEMENT

Classes: 09

Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering–Smoothing and Sharpening Spatial Filtering, Image Enhancement in Frequency Domain: Low Pass and High Pass Filters in Frequency Domain, Homomorphic filtering

UNIT III IMAGE TRANSFORMS

Classes: 09

Image Transforms: 2 D- Discrete Fourier Transform and its properties, Discrete Cosine Transform (DCT), Haar Transform, Hadamard Transform, Hotelling Transform and Slant transform.

UNIT IV IMAGE SEGMENTATION

Classes: 09

Introduction-Edge detection, Edge linking and boundary detection – Thresholding – Region based segmentation – Region growing – Region splitting and merging- Watershed segmentation algorithm.

UNIT V IMAGE COMPRESSION

Classes: 09

Image Compression Models, Huffman and Arithmetic Coding, Error Free Compression, Lossy Compression, Lossy and Lossless Predictive Coding, Transform Based Compression, JPEG 2000 Standards.

TEXTBOOKS:

1. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Second Edition, Pearson Education 3rd edition 2008
2. Digital Image Processing- S Jayaraman, S Esakkirajan, T Veerakumar- Mc Graw Hill Edn., 2010.

REFERENCES:

1. Anil Kumar Jain, Fundamentals of Digital Image Processing, Prentice Hall of India.2nd edition 2004
- 2.Murat Tekalp , Digital Video Processing" Prentice Hall, 2nd edition 2015.
3. S. Sridhar , Digital Image Processing, Oxford University Press, 2nd Ed, 2016.

Web References:

1. <http://homepages.inf.ed.ac.uk/rbf/BOOKS/VERNON/Chap004.pdf>

E-Text Books:

1. https://books.google.co.in/books/about/Digital_Image_Processing.html?id=a62xQ2r_f8wC

MOOC Course:

1. <https://nptel.ac.in/courses/117105079/>

INTRODUCTION TO SENSORS AND ACTUATORS

OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EC59	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understanding basic laws and phenomena on which operation of sensors and Actuators - transformation of energy.
2. Create analytical design and development solutions for sensors and actuators.
3. To know the basic laws of behaviour of sensors and actuators.
4. To able to know about the Standards for Smart Sensor Interface
5. Analyse the development and application of sensors and actuators.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply the fundamental physical and technical base of sensors and actuators,
2. Analyse various premises, approaches, procedures and results related to sensors and actuators
3. Analyse basic laws and phenomena that define behaviour of sensors and actuators.
4. Apply the Smart Sensor Interface in various applications
5. Develop the application of sensors and actuators

UNIT-I SENSORS & TRANSDUCERS

Classes: 9

Sensors / Transducers: Principles, Classification, Parameters, Characteristics, Environmental Parameters (EP), Characterization. Mechanical and Electromechanical Sensors: Introduction, Resistive Potentiometer, Resistance Strain Gauge, Semiconductor Strain Gauges, Capacitive Sensors, Electrostatic Transducer, Ultrasonic Sensors.

UNIT-II THERMAL SENSORS

Classes: 9

Thermal Sensors: Introduction, Gas thermometric Sensors, Thermal Expansion Type Thermometric Sensors, Acoustic Temperature Sensor, Dielectric Constant and Refractive Index Thermo-sensors, Nuclear Thermometer, Thermo-EMF Sensors, Thermal Radiation Sensors, Quartz Crystal Thermo-electric Sensors, Spectroscopic Thermometry, Noise Thermometry..

UNIT-III RADIATION SENSORS

Classes: 9

Introduction – Basic Characteristics – Types of Photosensistors– X-ray and Nuclear Radiation Sensors– Fiber Optic Sensors. Electro Analytical Sensors: Introduction – The Electrochemical Cell – The Cell Potential – Standard Hydrogen Electrode (SHE) – Liquid Junction and Other Potentials – Sensor Electrodes.

UNIT-IV SMART SENSORS

Classes: 9

Introduction, Primary Sensors, Excitation, Converters, Compensation, Information Coding/Processing, Data Communication, Standards for Smart Sensor Interface, the Automation. Sensors Applications: Introduction, On-board Automobile Sensors (Automotive Sensors), Home Appliance Sensors.

UNIT-V ACTUATORS

Classes: 9

Actuators: Pneumatic and Hydraulic Actuation Systems- Actuation systems, Pneumatic and hydraulic systems, Directional Control valves, Pressure control valves, Cylinders, Servo and proportional control valves.

Text Books:

1. D. Patranabis, "Sensors and Transducers", PHI Learning Private Limited.
- 2.W. Bolton, "Mechatronics", Pearson Education Limited.

Reference Books:

1. Patranabis, "Sensors and Actuators", 2nd Edition, PHI, 2013.

Web Link:

1. <https://www.journals.elsevier.com/sensors-and-actuators>

E Text books:

1. <https://www.sciencedirect.com/handbook/handbook-of-sensors-and-actuators>

Moocs:

1. <https://www.classcentral.com/course/swayam-sensors-and-actuators-14285>

INTRODUCTION TO COMPUTER VISION**OPEN ELECTIVE - III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EC60	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. To review image processing techniques for computer vision.
2. To understand the concepts of Image Enhancement.
3. To understand the basics of segmentation and its applications.
4. To know about the feature extraction and Hough Transform.
5. To understand three-dimensional image analysis techniques.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Implement fundamental image processing techniques required for computer vision.
2. Analyze the various spatial and frequency domain filtering.
3. Understand the idea about segmentation
4. Apply Hough Transform for line, circle, and ellipse detections.
5. Apply 3D vision techniques and motion related techniques.

UNIT-I IMAGE PROCESSING FOUNDATIONS**Classes: 09**

Fundamental steps in Digital Image Processing, Components of Digital Image Processing Image sensing and acquisition, Image formation model, Pixels, Basic relationship between pixels, Sampling and Quantization in Digital Image Processing

UNIT-II IMAGE ENHANCEMENT**Classes: 09**

Intensity transformations, contrast stretching, histogram equalization, Spatial filtering: Smoothing filters, sharpening filters, Frequency domain filtering, Homomorphic filtering,

UNIT-III BAICS OF SEGMENTATION**Classes: 09**

Edge detection, Thresholding, Region growing, Fuzzy clustering, Watershed algorithm, Active contour models, Texture feature based segmentation, Graph based segmentation, Wavelet based Segmentation - Applications of image segmentation.

UNIT-IV FEATURE EXTRACTION**Classes: 09**

First and second order edge detection operators, Phase congruency, Localized feature extraction -detecting image curvature, shape features, Introduction to Hough transform, shape skeletonization, Boundary descriptors, Moments, Texture descriptors.

UNIT-V 3D IMAGE VISUALIZATION**Classes: 09**

Sources of 3D Data sets, Slicing the Data set, Arbitrary section planes, Volumetric display, Stereo Viewing, Ray tracing, Reflection, Surfaces, Multiple connected surfaces, Image processing in 3D, Measurements on 3D images.

Text Books:

1. Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing', Pearson, Education, Inc., Second Edition, 2004.
2. Mark Nixon, Alberto Aguado, "Feature Extraction and Image Processing", Academic Press, 2008.
3. D. L. Baggio et al., "Mastering OpenCV with Practical Computer Vision Projects", Packt Publishing, 2012.

Reference Books:

1. E. R. Davies, "Computer & Machine Vision", Fourth Edition, Academic Press, 2012.
2. Jan Erik Solem, "Programming Computer Vision with Python: Tools and algorithms for analyzing images", O'Reilly Media, 2012.
3. R. Szeliski, "Computer Vision: Algorithms and Applications", Springer 2011.
4. Simon J. D. Prince, "Computer Vision: Models, Learning, and Inference", Cambridge University Press, 2012.

Web References:

1. <https://machinelearningmastery.com/what-is-computer-vision/>

E-Text Books:

1. <https://machinelearningmastery.com/computer-vision-books/>

MOOC Course

1. https://onlinecourses.nptel.ac.in/noc21_ee23/preview

OPEN ELECTIVES OFFERED BY DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

OPEN ELECTIVE COURSE-I			
S. No.	Course Code	Course Name	Offering Department
1	A6EE55	Electrical Wiring and Safety Measures	Electrical &Electronics Engineering
2	A6EE56	Electrical Materials	
OPEN ELECTIVE COURSE-II			
1	A6EE57	Neural Networks and Fuzzy Logic	Electrical &Electronics Engineering
2	A6EE58	Energy Storage Systems	
OPEN ELECTIVE COURSE-III			
1	A6EE59	Solar Energy and Applications	Electrical &Electronics Engineering
2	A6EE65	Instrumentation and Control	

ELECTRICAL WIRING AND SAFETY MEASURES (OPEN ELECTIVE – I)

III Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE55	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

1. To Study the wiring diagram of residential.
2. To understand the Safety measures of Electrical wiring
3. Apply the Different types of safety rules in Electrical Systems.
4. Design the Different types of Residential Electrification Methods.
5. Identifying Different types of Substations.

Course Outcomes:

The students should be able to

1. To Analyze the safety measures & state safety precautions.
2. To Understand the Different Methods of Earthlings.
3. Apply the Different types of safety rules in Electrical Systems.
4. Design the Different types of Residential Electrification Methods.
5. Identifying Different types of Substations.

SYLLABUS

UNIT-I

BASICS OF ELECTRICAL INSTALLATIONS

Classes: 12

Electric Supply System, Three phase four wire distribution system, Protection of Electric Installation against over load, short circuit and Earth fault, General requirements of electrical installations, testing of installations, Types of loads, Systems of wiring, Service connections, Service Mains, Sub-Circuits, Location of Outlets, Location of Control Switches, Location of Main Board and Distribution board, Guide lines for Installation of Fittings, Load Assessment, Permissible voltage drops and sizes of wires, estimating and costing of Electric installations.

UNIT-II

EARTHING

Classes: 08

Introduction & importance, Factors affecting Earth Resistance, Methods of earthing Substation and Transmission tower earthing, Neutral and Earth wire, Transformer Neutral Earthing.

UNIT-III

SAFETY & PREVENTION OF ACCIDENTS

Classes: 08

Definition of terminology used in safety, I.E. Act & statutory regulations for safety of persons & equipments working with electrical installation. Dos & don'ts for substation operators as listed in IS. Meaning & causes of electrical accidents factors on which severity of shock depends.

UNIT-IV

RESIDENTIAL BUILDING ELECTRIFICATION

Classes: 10

General rules guidelines for wiring of Residential Installation and positioning of equipments. Principles of circuit design in lighting and power circuits. Procedures for designing the circuits and deciding the number of sub-circuits. Method of drawing single line diagram & wiring diagram.

UNIT-V

RULES : SUBSTATION AND METERS

Classes: 12

Rule 28: Voltage level definitions. Rule 30: Service lines & apparatus on consumer premises.
Rule 31: Cut-out on consumer's premises.
Rule 46: Periodical inspection & testing of consumer's installation.
Rule 47: Testing of consumer's installation.
Rule 54: Declared voltage of supply to consumer.
Rule 55: Declared frequency of supply to consumer.

Rule 56: Sealing of meters & cut-outs.
Rule 77: Clearances above ground of the lowest conductor.
Rule 79: Clearances between conductors & trolley wires.
Rule 87: Lines crossing or approaching each other.
Rule 88: Guarding.

Text Books:

1. K.B. Raina, S.K. Bhattacharya Electrical Design; Estimating and costing New Age International (p) Limited, New Delhi Surjit Singh.
2. Electrical Estimating and costing Dhanpat Rai and company, New Delhi .J.B. Gupta
3. A course in Electrical Installation, Estimating & costing S.K. Kataria & sons, S.L. Uppal .
4. Electrical wiring Estimating and costing Khanna Publication. , A.K. Sawhney

Reference Books:

1. Electrical Machine Design Danpat Rai & co.
2. The Electricity Rule 2005 Universal Law Publishing Co. Pvt. Ltd. N. Alagapan S. Ekambaram
3. Electrical Estimating and costing Tata Mc Graw Hill Publication, New Delhi , Surjit Singh
4. Tarlok Singh Installation, Commissioning & Maintenance of Electrical Equipment S.K. Kataria & Sons
5. B.V.S. Rao Operation & Maintenance of Electrical Machines Vol I & II Media Promoters & Publisher Ltd. Mumbai

Web References:

1. <https://electrical-engineering-portal.com> › Technical Articles
2. <https://www.st-andrews.ac.uk/staff/policy/healthandsafety/publications/electricalsafety/>
3. <https://www.cpwd.gov.in/Publication/Internal2013.pdf>

E-Text Books:

1. <https://books.google.co.in/books?isbn=0323170064>
2. <https://www.jove.com/science.../electrical-safety-precautions-and-basic-equipment>

MOOC Course

1. <https://nptel.ac.in/courses/103106071/5>
2. <https://nptel.ac.in/courses/108108099/28>
3. <https://nptel.ac.in/courses/124107001/>

ELECTRICAL MATERIALS (OPEN ELECTIVE – I)

III Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIA	SEE	Total
A6EE56	OEC	3	0	0	3	40	60	100

Course Objectives:

1. To Analyze the various Conducting materials.
2. To Explain the various Semi conducting materials.
3. To Explain The Dielectrics and Insulators.
4. To Explain the Different Magnetic Materials.
5. To Describe the Optical properties of Fibers

Course Outcomes:

The students should be able to

1. Analyze the various Conducting materials.
2. Explain the various Semi conducting materials.
3. Explain The Dielectrics and Insulators.
4. Explain the Different Magnetic Materials.
5. Describe the Optical properties of Fibers.

SYLLABUS

UNIT-I

CONDUCTORS

Classes: 10

Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.

UNIT-II

SEMICONDUCTORS

Classes: 08

General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semi-conductors, extrinsic Semi-conductors, hall effect, drift, mobility, diffusion in Semiconductors, and their applications, superconductors.

UNIT-III

DIELECTRICS AND INSULATORS

Classes: 12

Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials, mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bounding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.

UNIT-IV

MAGNETIC MATERIALS

Classes: 10

Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials,

electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magneto-striction.

UNIT-V**OPTICAL PROPERTIES OF SOLIDS****Classes: 10**

Photo emission, photo emission materials, electro luminous junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).

Text Books:

1. "Electrical Engineering Materials", Dekker, PHI Pbs.
2. "Electrical Engineering Materials", Indulkar, S.Chand

Reference Books:

1. "Electrical Engineering Materials", Tareev
2. "Electrical Engineering Materials", Yu. Koritsky.
3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs

Web References:

1. <https://physics.info/dielectrics/>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265>
3. web.mit.edu/course/6/6.732/www/6.732-pt2.pdf

E-Text Books:

1. <https://easyengineering.net/electrical-engineering-materials-by-dekker/>
2. <https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/>

MOOC Course

1. <https://nptel.ac.in/courses/108108076/>
2. <https://nptel.ac.in/courses/112104203/3>
3. https://onlinecourses.nptel.ac.in/noc18_ee14/

NEURAL NETWORKS AND FUZZY LOGIC

(OPEN ELECTIVE – II)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EE57	OEC	3	0	0	3	40	60	100

Course Objectives:

1. To understand the fundamental theory and concepts of neural networks, neuro-modeling, several neural network paradigms and its applications.
2. To understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic control and other machine intelligence applications of fuzzy logic.
3. To understand the basics of an evolutionary computing paradigm known as genetic algorithms and its application to engineering optimization problems.

Course Outcomes:

The students should be able to

1. Comprehend the concepts of feed forward neural networks
2. Analyze the various feedback networks.
3. Understand the concept of fuzziness involved in various systems and fuzzy set theory.
4. Comprehend the fuzzy logic control and adaptive fuzzy logic and to design the fuzzy control using genetic algorithm.
5. Analyze the application of fuzzy logic control to real time systems.

SYLLABUS

UNIT-I INTRODUCTION & ESSENTIALS TO NEURAL NETWORKS Classes: 12

Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCulloch-PiUs Model, Historical Developments, Potential Applications of ANN, Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN — Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application

UNIT-II SINGLE & MULTI LAYER FEED FORWARD NEURAL NETWORKS Classes: 10

Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training.

Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications. Credit Assignment Problem, Generalized Delta Rule, and Derivation of Back-propagation (BP) Training, Summary of Back-propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.

UNIT-III ASSOCIATIVE MEMORIES-I Classes: 08

Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).

UNIT-IV**ASSOCIATIVE MEMORIES-II****Classes: 10**

Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hopfield

Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications

UNIT-V**FUZZY LOGIC****Classes: 10**

Classical & Fuzzy Sets: Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions.

Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, Defuzzification to crisp sets, De-fuzzification methods.

Text Books:

1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications, Rajasekharan and Pal, PHI.
2. Neural Networks and Fuzzy Logic, C. Naga Bhaskar, G. Vijay Kumar, BS Publication-is.

Reference Books:

1. Artificial Neural Networks, B. Yegnanarayana, PHI.
2. Artificial Neural Networks, Zaruda, PHI.
3. Neural Networks and Fuzzy Logic System, Bail Kosko, PHI.
4. Fuzzy Logic and Neural Networks, M. Amirthavalli, Scitech Publications India Pvt. Ltd.
5. Neural Networks, James A Freeman and Davis Skapura, Pearson Education.
6. Neural networks by satish Kumar, TIVIH, 2004
7. Neural Networks, Simon Hakens, Pearson Education.
8. Neural Engineering, C. Elasmith and CH. Anderson, PHI.

Web References:

1. users.monash.edu/~app/CSE5301/Lnts/LaD.pdf
2. <https://engineering.purdue.edu/~tsoukala/rational.html>
3. <https://pdfs.semanticscholar.org/5e31/c55a00eb3945e3e483caa2e146a95c12f5aa.pdf>

E-Text Books:

1. <https://www.mheducation.co.in/computer.../neural-networks-fuzzy-systems/text-book>
2. www.crectirupati.com/sites/default/files/lecture_notes/NNFL.pdf
3. www.vssut.ac.in/lecture_notes/lecture1423723637.pdf

Course:

1. https://nptel.ac.in/noc/individual_course.php?id=noc19-ge07
2. <https://nptel.ac.in/courses/108104049/16>
3. <https://nptel.ac.in/courses/117105084/>
4. <https://nptel.ac.in/syllabus/127105006/>

ENERGY STORAGE SYSTEMS

(OPEN ELECTIVE – II)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6EE58	OEC	3	0	0	3	40	60	100

Course Objectives:

1. To enable the student to understand the need for energy storage, devices and technologies available and their applications.
2. To Analyze the characteristics of energy from various sources and need for storage
3. To Classify various types of energy storage and various devices used for the purpose
4. To Explain the Pumped hydro Storage System.

Course Outcomes:

The students should be able to

1. Analyze the characteristics of energy from various sources and need for storage
2. Classify various types of energy storage and various devices used for the purpose
3. Explain the Pumped hydro Storage System.
4. Identify various real time applications.

SYLLABUS

UNIT-I ELECTRICAL ENERGY STORAGE TECHNOLOGIES Classes: 12

Characteristics of electricity, Electricity and the roles of EES, High generation cost during peak-demand periods, Need for continuous and flexible supply, Long distance between generation and consumption, Congestion in power grids, Transmission by cable.

UNIT-II NEEDS FOR ELECTRICAL ENERGY STORAGE Classes: 08

Emerging needs for EES, More renewable energy, less fossil fuel, Smart Grid uses, The roles of electrical energy storage technologies, The roles from the viewpoint of a utility, The roles from the viewpoint of consumers, The roles from the viewpoint of generators of renewable energy.

UNIT-III FEATURES OF ENERGY STORAGE SYSTEMS Classes: 08

Classification of EES systems , Mechanical storage systems, Pumped hydro storage (PHS), Compressed air energy storage (CAES), Flywheel energy storage (FES), Electrochemical storage systems, Secondary batteries, Flow batteries, Chemical energy storage, Hydrogen (H₂), Synthetic natural gas (SNG).

UNIT-IV TYPES OF ELECTRICAL ENERGY STORAGE SYSTEMS Classes: 10

Electrical storage systems, Double-layer capacitors (DLC) , Superconducting magnetic energy storage (SMES), Thermal storage systems, Standards for EES, Technical comparison of EES technologies.

UNIT-V APPLICATIONS Classes: 12

Present status of applications, Utility use (conventional power generation, grid operation & service) , Consumer use (uninterruptable power supply for large consumers), New trends in applications ,Renewable energy generation, Smart Grid, Smart Micro grid, Smart House, Electric vehicles, Management and control hierarchy of storage systems, Internal configuration of battery storage systems, External connection of EES systems , Aggregating EES systems and distributed generation (Virtual Power Plant), Battery SCADA– aggregation of many dispersed batteries.

Text Books:

1. “James M. Eyer, Joseph J. Iannucci and Garth P. Corey “ , “Energy Storage Benefits and Market Analysis”, Sandia National Laboratories, 2004.
2. The Electrical Energy Storage by IEC Market Strategy Board.

Reference Books:

1. “Jim Eyer, Garth Corey”, Energy Storage for the Electricity Grid: Benefits and Market Potential Assessment Guide, Report, Sandia National Laboratories, Feb 2010.

Web References:

1. <https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118991978.hces212>

E-Text Books:

1. https://www.pewtrusts.org/~media/.../energy_storage-backs_up_power_supply.pdf
2. <https://energy.mit.edu/wp-content/uploads/2018/04/Energy-Storage-for-the-Grid.pdf>
3. <https://www.adb.org/sites/default/files/.../handbook-battery-energy-storage-system.pdf>

MOOC Course:

1. nptel.ac.in/courses/112105221/56
2. nptel.ac.in/courses/108108036/9
3. <https://nptel.ac.in/courses/108102047/7>

SOLAR ENERGY AND APPLICATIONS

(OPEN ELECTIVE – III)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6EE59	OEC	3	0	0	3	40	60	100

Course Objectives:

1. To Study about solar modules and PV system design and their applications
2. To Deal with grid connected PV systems
3. To Discuss about different energy storage systems

Course Outcomes:

The students should be able to

1. understanding of principles and technologies for solar thermal energy collection, conversion and utilization
2. Understanding of solar heating systems, liquid based solar heating systems for buildings.
3. Identify, formulate and solve simple to complex problems of solar thermal energy conversion and storage.
4. Identify and understand solar thermal systems components and their function.
5. Analyze hot water load and solar resource data and use this information to properly size a solar thermal system

SYLLABUS

UNIT-I

INTRODUCTION

Classes: 10

Characteristics of sunlight – semiconductors and P-N junctions –behavior of solar cells – cell properties – PV cell interconnection

UNIT-II

STAND ALONE PV SYSTEM

Classes: 10

Solar modules – storage systems – power conditioning and regulation - MPPT- protection – stand alone PV systems design – sizing

UNIT-III

GRID CONNECTED PV SYSTEMS

Classes: 10

PV systems in buildings – design issues for central power stations – safety – Economic aspect – Efficiency and performance - International PV programs

UNIT-IV

ENERGY STORAGE SYSTEMS

Classes: 10

Impact of intermittent generation – Battery energy storage – solar thermal energy storage – pumped hydroelectric energy storage

UNIT-V

APPLICATIONS

Classes: 10

Water pumping – battery chargers – solar car – direct-drive applications –Space – Telecommunications.

Text Books:

1. Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd.,2015.
2. Stuart R.Wenham, Martin A.Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007,Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progensa,1994

Reference Books:

1. Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011.
2. McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 5 S.P. Sukhatme, "Solar Energy", Tata McGraw Hill,1987.

Web References:

1. <https://www.loc.gov/rr/scitech/tracer-bullets/solar-updatetb.html>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100516798>
3. <https://link.springer.com/journal/11949>

E-Text Books:

1. https://courses.edx.org/c4x/DelftX/ET.3034TU/asset/solar_energy_v1.1.pdf
2. bookstore.teri.res.in/books/9788179935736

MOOC Course:

1. <https://nptel.ac.in/courses/112105051/>
2. <https://nptel.ac.in/courses/121106014/18>
3. <https://nptel.ac.in/courses/112105050/>
4. <https://nptel.ac.in/syllabus/112105051/>

INSTRUMENTATION AND CONTROL

(OPEN ELECTIVE – III)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6EE65	OEC	3	0	0	3	40	60	100

Course Objectives:

1. To introduce the basic principles of all measuring instruments
2. To deal with the measurement of voltage, current measurements
3. To understand students how different types of meters work and their construction
4. To understand the different ways of system representations such as Transfer function representation and state space representations and to assess the system dynamic response
5. To assess the system performance using time domain analysis and methods for improving it

Course Outcomes:

The students should be able to

1. Identify the instruments suitable for typical measurements.
2. Apply the knowledge about transducers and instrument transformers to use them effectively.
3. Improve the system performance by selecting a suitable controller for a specific application
4. Apply various time domain techniques to assess the system performance

SYLLABUS

UNIT-I INTRODUCTION TO MEASURING INSTRUMENTS Classes: 10

Classification —deflecting, control and damping torques — Ammeters and Voltmeters — PMMC, moving iron type instruments — expression for the deflecting torque and control torque — Errors and compensations, extension of range using shunts and series resistance. Electrostatic Voltmeters-electrometer type and attracted disc type — Extension of range of E.S. Voltmeters.

UNIT-II POTENTIOMETERS & INSTRUMENT TRANSFORMERS Classes: 08

Principle and operation of D.C. Crompton's potentiometer—standardisation — Measurement of unknown resistance, current, voltage. A.C. Potentiometers: polar and coordinate types standardisation — applications. CT and PT — Ratio and phase angle errors.

UNIT-III TRANSDUCERS & OSCILLOSCOPES Classes: 12

Definition of transducers, Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers; Principle operation of LVDT and capacitor transducers; LVDT Applications, Strain gauge and its principle of operation, gauge factor, Thermistors, Thermo couples, Piezo electric transducers, photovoltaic, photo conductive cells, photo diodes.

CRO: Cathode ray oscilloscope-Cathode ray tube-time base generator- horizontal and vertical amplifiers-CRO probes-applications of CRO- Measurement of phase and frequency-lissajous patterns.

UNIT-IV CLASSIFICATION OF CONTROL SYSTEMS Classes: 10

Concepts of Control Systems- Open Loop and closed loop control systems and their differences- Different examples of control systems- Classification of control systems

Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations – Impulse Response and transfer functions .Block diagram algebra – Representation by Signal flow graph – Reduction using mason's gain formula.

UNIT-V TIME RESPONSE ANALYSIS Classes: 10

Standard test signals – Time response of first order systems –Characteristic Equation of Feedback control systems, Transient response of second order systems – Time domain specifications .Effects of proportional derivative, proportional integral systems.

Text Books:

1. Electrical and Electronic Measurements and Instrumentation, R. K. Rajput, S. Chand & Company Ltd.
2. Electrical Measuring Instruments and Measurements, S. C. Bhargava, BS Publications.
3. I. J. Nagrath and M. Gopal", "Control Systems Engineering", New Age International (P) Limited, Publishers, 5th edition, 2009
4. "B. C. Kuo", "Automatic Control Systems", John wiley and sons, 8th edition, 2003.

Reference Books:

1. Electrical & Electronic Measurement & Instruments, A.K.Sawhney Dhanpat Rai & Co. Publications.
2. Electrical and Electronic Measurements, G. K. Banerjee, PHI Learning Pvt. Ltd.
3. Electrical Measurements and Measuring Instruments, Golding and Widdis, Reem Publications.
4. N. K. Sinha", "Control Systems", New Age International (P) Limited Publishers, 3rdEdition, 1998.
5. "NISE", "Control Systems Engineering", John wiley, 6th Edition, 2011.
6. "Katsuhiko Ogata", "Modern Control Engineering", Prentice Hall of India Pvt. Ltd., 3rd edition, 1998.

Web References:

1. home.mit.bme.hu/~virosztek/docs/mt.../Principles_of_electrical_measurement.pdf
2. https://www.mccdaq.com/handbook/chapt_4.aspx

E-Text Books:

1. www.vssut.ac.in/lecture_notes/lecture1423813026.pdf
2. www.vssut.ac.in/lecture_notes/lecture1423904331.pdf
3. www.ent.mrt.ac.lk/~rohan/teaching/EN5001/Reading/DORFCH1.pdf

MOOC Course:

1. <https://nptel.ac.in/syllabus/108106070/>
2. <https://nptel.ac.in/courses/108105064/>
3. <https://nptel.ac.in/courses/108101037/>

**OPEN ELECTIVES
OFFERED BY
DEPARTMENT OF INFORMATION TECHNOLOGY**

OPEN ELECTIVE – I		
S. No.	Course Code	Course Name
1	A6IT28	Adhoc & Sensor Networks
2	A6IT35	Object Oriented Programming
OPEN ELECTIVE – II		
1	A6IT30	Security Testing
2	A6IT36	Human Computer Interaction
OPEN ELECTIVE – III		
1	A6IT37	Introduction to Computer Networks
2	A6IT31	High Performance Computing

AD-HOC & SENSOR NETWORKS

III B. TECH- II SEMESTER – OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT28	OEC	3	0	0	3	40	60	100

Course Objectives:

1. To understand the concepts of sensor networks
2. To understand the MAC and transport protocols for ad hoc networks
3. To understand the security of sensor networks
4. To understand the applications of adhoc and sensor networks

Course Outcomes:

At the end of the course, student will be able to:

1. Know the principles and characteristics of wireless sensor networks.
2. Use proactive routing protocols function and their implications on data transmission delay and bandwidth consumption.
3. Find issues in MAC Layer Protocols, transport layer protocols in MANETS and WSNs.
4. Find security issues and familiar with security key management mechanisms MANETs and WSNs.
5. Practice network simulations on various simulators.

UNIT – I

Classes: 10

Introduction to Ad Hoc Networks - Characteristics of MANETs, Applications of MANETs and Challenges of MANETs.

Routing in MANETs - Criteria for classification, Taxonomy of MANET routing algorithms, Topology- based routing algorithms-**Proactive**: DSDV; **Reactive**: DSR, AODV; Hybrid: ZRP; Position-based routing algorithms-**Location Services**-DREAM, Quorum-based; **Forwarding Strategies**: Greedy Packet, Restricted Directional Flooding-DREAM, LAR.

UNIT – II

Classes: 10

Data Transmission - Broadcast Storm Problem, **Rebroadcasting Schemes**-Simple-flooding, Probability-based Methods, Area-based Methods, Neighbor Knowledge-based: SBA, Multipoint Relaying, AHBP. **Multicasting**: **Tree-based**: AMRIS, MAODV; **Mesh-based**: ODMRP, CAMP; **Hybrid**: AMRoute, MCEDAR.

UNIT – III

Classes: 10

Geocasting: Data-transmission Oriented-LBM; Route Creation Oriented-GeoTORA, MGR. TCP over Ad Hoc TCP protocol overview, TCP and MANETs, Solutions for TCP over Ad hoc

UNIT – IV

Classes: 10

Basics of Wireless, Sensors and Lower Layer Issues: Applications, Classification of sensornetworks, Architecture of sensor network, Physical layer, MAC layer, Link layer, Routing Layer.

UNIT – V

Classes: 10

Upper Layer Issues of WSN: Transport layer, High-level application layer support, Adapting to the inherent dynamic nature of WSNs, Sensor Networks and mobile robots.

TEXT BOOKS:

1. Ad Hoc and Sensor Networks – Theory and Applications, Carlos Corderio Dharma P. Aggarwal, World Scientific Publications, March 2006, ISBN – 981–256–681–3.

2. Wireless Sensor Networks: An Information Processing Approach, Feng Zhao, Leonidas Guibas, Elsevier Science, ISBN – 978-1-55860-914-3 (Morgan Kaufman).

REFERENCE BOOKS:

1. Advanced Selenium Web Accessibility Testing, Software Automation Testing Secrets Revealed, Narayanan Palani(You?)|2019
2. Automated Software Testing A Complete Guide - 2020 Edition-by Gerardus Blokdyk (Author).
3. Software Automation Testing Secrets Revealed Part 2 Selenium Webdriver
Mastering Web Automated Testing -Use of Selenium with Practical Answers to Testing Interview Questions
Narayanan Palani(You?)|2016.

WEB LINKS

1. <https://www.guru99.com/selenium-tutorial.html>.
2. <https://www.browserstack.com/guide/automation-testing-tutorial>

OBJECT ORIENTED PROGRAMMING

III B.Tech II Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6IT35	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the basic object oriented programming concepts and apply them in problem solving.
2. Illustrate inheritance and polymorphism concepts for reusing the program.
3. Demonstrate on the exception handling mechanism
4. Use multithreading concepts to develop inter process communication
5. Understand the basics of java collection framework

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Understand the basic programming concepts and apply them in problem solving
2. Implement object-oriented principles, such as encapsulation, inheritance, polymorphism, and abstraction.
3. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally).
4. Develop multithreaded applications with synchronization.
5. Build real world applications using Collection framework

UNIT-I JAVA BASICS

Classes: 12

JAVA BASICS: Review of Object oriented concepts, History of Java, Java buzzwords, JVM architecture, Data types, Variables, arrays, operators, control statements, type conversion and casting, constructors, methods, Static block, Static Method, String Handling, Scanner class

UNIT-II INHERITANCE, POLYMORPHISM, PACKAGES AND INTERFACES

Classes: 11

INHERITANCE AND POLYMORPHISM: Basic concepts, Types of inheritance, Member access rules, Usage of this and Super key word, Polymorphism, Abstract classes, Encapsulation, getter and setter methods, Dynamic method dispatch, Usage of final keyword.

PACKAGES AND INTERFACES: Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces

UNIT-III EXCEPTION HANDLING AND FILES

Classes: 10

EXCEPTION HANDLING: Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes.

I / O STREAMS AND FILES: Concepts of streams, Stream classes- Byte and Character stream, Reading console Input and Writing Console output, IO/Serialization, File Handling.

UNIT-IV MULTITHREADING

Classes: 10

MULTI THREADING: Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication, Concurrency.

UNIT-V COLLECTION FRAMEWORK

Classes: 10

COLLECTION FRAMEWORK: Introduction to Java Collections, Overview of Java Collection frame work, Generics, Commonly used Collection classes- Array List, Vector, Hash table, Stack, Enumeration, Iterator.

Text Books:

1. Herbert Schildt and Dale Skrien, ||Java Fundamentals – A comprehensive Introduction||, McGraw Hill, 1st Edition, 2013.
2. Herbert Schildt, —Java the complete referencell, McGraw Hill, Osborne, 7th Edition, 2011.
3. T.Budd, —Understanding Object- Oriented Programming with Javall, Pearson Education, Updated Edition (New Java 2 Coverage), 1999.

Reference Books:

1. P.J.Dietel and H.M.Dietel , —Java How to program||, Prentice Hall, 6th Edition, 2005.
2. P.Radha Krishna , —Object Oriented programming through Javall, CRC Press, 1st Edition, 2007.
3. S.Malhotra and S. Choudhary, —Programming in Javall, Oxford University Press, 2nd Edition, 2014.

SECURITY TESTING

IV Year I Semester: OPEN ELECTIVE – II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT30	OEC	3	0	0	3	40	60	100

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Explain the basic principles and techniques of how attackers can enter computer systems.
2. Perform analyses of data breaches and audits of information technology security.
3. Evaluate the strengths and weaknesses of various information technology solutions regarding data security.
4. Present and perform demonstrations of pen tests for educational purposes.
5. evaluate the societal role of hacking from a social, ethical and economic standpoint

UNIT – I Introduction to OWASP

Classes: 12

OWASP: Introduction to web applications security, threats and OWASP principles, introduction to secure design, web server: introduction a secure setup of apache, firewalling a server Browser: general concepts, functionalities, browsers war, configuration (HTTP-cookies, contents, scripting etc. attack to browsers, and users tracking/profiling (third party cookies, super cookies, XSS, CSFR, Command Injection), browser security (add-ons, plugins, same-origin policy etc.) & secure browsing.

UNIT – II OWASP Privacy preserving

Classes: 12

OWASP Privacy preserving: attacks to privacy, (spyware & backdoors, browser, email etc.) Tracking techniques: (HTTP cookies, third party cookies, browser fingerprinting, CSP) Advanced browser configuration, anonymity and onion routing (Tor). Internet E-mail: Architecture and infrastructure, functions, agents and standards, MIME & PGP, phishing, spamming & spoofing, DKIM, SPF, introduction to email forensics.

UNIT – III Introduction to Ethical Hacking Terminology

Classes: 10

Introduction to Ethical Hacking Terminology-Five stages of hacking-Vulnerability Research-Legal implication of hacking Impact of hacking. Foot printing & Social engineering.

UNIT – IV Information gathering methodologies

Classes: 10

Information gathering methodologies- Competitive Intelligence- DNS Enumerations- Social Engineering attacks. Scanning & Enumeration Port Scanning-Network Scanning- Vulnerability Scanning- NMAP scanning tool- OS Finger printing Enumeration. System Hacking Password.

UNIT – V Sniffers & SQL Injection

Classes: 10

Sniffers & SQL Injection Active and passive sniffing- ARP Poisoning- Session Hijacking- DNS Spoofing- Conduct SQL Injection attack – Counter measures. Cracking techniques- Key loggers- Escalating privileges- Hiding Files-Steganography technologies- Countermeasures.

Text Books:

1. Kimberly Graves, “CEH: Official Certified Ethical Hacker Review Guide”, Wiley Publishing Inc., 2007.ISBN: 978-0-7821-4437-6.

Reference Books:

1. Shakeel Ali & Tedi Heriyanto, “Backtrack -4: Assuring security by penetration testing”, PACKT Publishing., 2011. ISBN: 978-1-849513-94-4.

HUMAN COMPUTER INTERACTION

IV B.TECH - I Semester: (OPEN ELECTIVE – II)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT36	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

To learn

1. To learn the foundations of Human Computer Interaction
2. To become familiar with the design technologies for individuals and persons with disabilities.
3. To be aware of mobile HCI.
4. To learn the guidelines for user interface
5. To get an understanding of web interface design

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to learn

1. Design effective dialog for HCI
2. Design effective HCI for individuals and persons with disabilities.
3. Assess the importance of user feedback.
4. Explain the HCI implications for designing multimedia/ ecommerce/ e-learning Web sites.
5. Develop meaningful user interface

UNIT – I Foundation of HCI

Classes: 12

The Human: I/O channels, Memory, Reasoning and problem solving;

The computer: Devices, Memory, processing and networks;

Interaction: Models, frameworks, Ergonomics, styles, elements, interactivity- Paradigms.

UNIT – II Design & Software process

Classes: 12

Interactive Design: Basics, process, scenarios, navigation, screen design, Iteration and prototyping.

HCI in software process: Software life cycle, usability engineering, Prototyping in practice, design rationale. Design rules: principles, standards, guidelines, rules. Evaluation Techniques ,Universal Design.

UNIT – III Models and Theories

Classes: 10

HCI Models: Cognitive models: Socio-Organizational issues and stakeholder requirements – Communication and collaboration models-Hypertext, Multimedia and WWW.

UNIT – IV Mobile HCI

Classes: 12

Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools. - Case Studies.

UNIT – V Web Interface Design

Classes: 09

Designing Web Interfaces: Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow - Case Studies

Text Books:

1. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, —Human Computer InteractionII, 3rd Edition, Pearson Education, 2004 (UNIT I, II & III)
2. Brian Fling, —Mobile Design and DevelopmentII, First Edition, O'Reilly Media Inc., 2009 (UNIT – IV)

Reference Books:

1. Bill Scott and Theresa Neil, —Designing Web InterfacesII, First Edition, O'Reilly, 2009. (UNIT-V)
2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech,
3. User Interface Design, Soren Lauesen , Pearson Education.

INTRODUCTION TO COMPUTER NETWORKS

IV B. TECH- II SEMESTER: OPEN ELECTIVE – III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT37	OEC	3	0	0	3	40	60	100

Course Objectives:

1. Understand and apply different network commands.
2. Analyse different networking functions and features for implementing optimal solutions Apply different networking concepts for implementing network solution
3. Implement different network protocols.

Course Outcomes:

1. Apply the basics of Physical layer in real time applications
2. Apply data link layer concepts, design issues, and protocols
3. Apply Network layer routing protocols and IP addressing
4. Implement the functions of Application layer and Presentation layer paradigms and Protocols

UNIT I:

Classes: 10

Introduction:

Network Types, LAN, MAN, WAN, Network Topologies Reference models- The OSI Reference Model- the TCP/IP Reference Model - A Comparison of the OSI and TCP/IP Reference Models, OSI Vs TCP/IP, Lack of OSI models success

Physical Layer –Introduction to Guided Media- Twisted-pair cable, Coaxial cable and Fiber optic cable and unguided media: Wireless-Radio waves, microwaves, infrared.

Modern Networks – History of Internet, Wi-Fi Networks, Blue-Tooth and Edge Computing

UNIT II:

Classes: 10

Data link layer: Design issues, Framing: fixed size framing, variable size framing, flow control, error control, error detection and correction codes, CRC, Checksum: idea, one's complement internet checksum, services provided to Network Layer

Elementary Data Link Layer protocols: simplex protocol, Simplex stop and wait, Simplex protocol for Noisy Channel.

Sliding window protocol: One bit, Go back N, Selective repeat-Stop and wait protocol, Data link layer in HDLC: configuration and transfer modes, frames, control field, point to point protocol (PPP): framing transition phase, multiplexing, multi link PPP.

UNIT – III:

Classes: 10

Media Access Control: Random Access: ALOHA, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance, Controlled Access: Reservation, Polling, Token Passing, **Channelization**: frequency division multiple Access(FDMA), time division multiple access(TDMA), code division multiple access(CDMA).

Wired LANs: Ethernet, Ethernet Protocol, Standard Ethernet, Fast Ethernet (100 Mbps), Gigabit Ethernet, 10 Gigabit Ethernet.

UNIT – IV:

Classes: 10

The Network Layer Design Issues – Store and Forward Packet Switching-Services Provided to the Transport layer- Implementation of Connectionless Service-Implementation of Connection Oriented Service- Comparison of Virtual Circuit and Datagram Networks, Routing Algorithms-The Optimality principle-Shortest path, Flooding, Distance vector, Link state, Hierarchical, Congestion Control algorithms-General principles of congestion

control, Congestion prevention policies, Approaches to Congestion Control-Traffic Aware Routing- Admission Control-Traffic Throttling-Load Shedding. Traffic Control Algorithm: Leaky bucket & Token bucket. Internet Working: How networks differ- How networks can be connected- Tunnelling, internetwork routing-, Fragmentation, network layer in the internet – IP protocols-IP Version 4 protocol-IPV4 Header Format, IP addresses, Class full Addressing, CIDR, NAT-, Subnets-IP Version 6-The main IPV6 header, Transition from IPV4 to IPV6, Comparison of IPV4 & IPV6- Internet control protocols- ICMP-ARPDHCP

UNIT –V:

Classes: 10

The Transport Layer: Transport layer protocols: Introduction-services- port number-User data gram protocol- User datagram-UDP services-UDP applications-Transmission control protocol: TCP services- TCP features- Segment- A TCP connection- windows in TCP- flow control-Error control, Congestion control in TCP. Application Layer: World Wide Web: HTTP, Electronic mail-Architecture- web based mail- email security-TELENET-local versus remote Logging-Domain Name System: Name Space, DNS in Internet, Resolution- Caching- Resource Records- DNS messages- Registrars-security of DNS Name Servers, SNMP.

Text Books:

1. Computer Networks — Andrew S Tanenbaum, Fifth Edition. Pearson Education/PHI
2. Data Communications and Networks – Behrouz A. Forouzan, Fifth Edition TMH.

References Books:

1. Data Communications and Networks- Achut S Godbole, Atul Kahate
2. Computer Networks, Mayank Dave, CENGAGE

HIGH PERFORMANCE COMPUTING

IV B. TECH- II SEMESTER – OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6IT31	OEC	3	0	0	3	40	60	100

Course Objectives:

1. To learn about Modern Processors and concepts
2. To understand the concepts of Optimizations
3. To learn about Parallel Computers and Programming
4. To study about Memory Parallel Programming using Open MP and MPI

Course Outcomes:

At the end of the course, student will be able to:

1. Transform algorithms in the computational area to efficient programming code for modern computer architectures
2. Write, organise and handle programs for scientific computations.
3. Use tools for performance optimisation and debugging and Analyse code with respect to performance.
4. Suggest and implement performance improvements
5. Write, organise and handle programs for MIP

UNIT – I MODERN PROCESSORS

Classes: 09

Stored-program computer architecture, General-purpose cache-based microprocessor architecture, Memory hierarchies, Multicore processors, Multi-threaded processors, Vector processors.

UNIT – II BASIC OPTIMIZATION TECHNIQUES FOR SERIAL CODE

Classes: 09

Scalar profiling, Common sense optimizations, Simple measures, large impact, The role of compilers, C++ optimizations

UNIT – III PARALLEL COMPUTERS

Classes: 09

Taxonomy of parallel computing paradigms, Shared-memory computers, Distributed-memory computers, Hierarchical (hybrid) systems, Networks, Need for parallelization, Parallelism, Parallel scalability

UNIT- IV SHARED MEMORY PARALLEL PROGRAMMING WITH OPENMP

Classes: 08

introduction to Open MP, Parallel execution, Data scoping, Open MP work sharing for loops, Synchronization, Reductions, Loop scheduling, Tasking, Miscellaneous, Open MP-parallel Jacobi algorithm, Open MP: Wave front parallelization.

UNIT – V DISTRIBUTED-MEMORY PARALLEL PROGRAMMING WITH MPI

Classes: 09

Message passing, introduction to MPI, A simple example, Messages and point-to-point communication, Collective communication, Nonblocking point-to-point communication, Virtual topologies, MPI parallelization of a Jacobi solver, MPI implementation, Performance properties

TEXT BOOK:

1. Georg Hager, Gerhard Wellein, "Introduction to High Performance Computing for Scientists and Engineers", Chapman & Hall / CRC Computational Science series, 2011.

REFERENCE BOOKS:

1. Charles Severance, Kevin Dowd, "High Performance Computing", O'Reilly Media, 2nd Edition, 1998.
2. Kai Hwang, Faye Alaye Briggs, "Computer Architecture and Parallel Processing", McGraw Hill, 1984

OPEN ELECTIVES
OFFERED BY
DEPARTMENT OF MECHANICAL ENGINEERING

III B. Tech. II Semester: OPEN ELECTIVE - I		
Sl. No.	Course Code	Subject Name
1	A6ME55	Basics of Mechanical Engineering
2	A6ME60	Fundamentals of Engineering Materials
IV B. Tech. I Semester: OPEN ELECTIVE - II		
Sl. No.	Course Code	Subject Name
1	A6ME65	Fundamentals of Operations Research
2	A6ME70	Introduction to Industrial Engineering
IV B. Tech. II Semester: OPEN ELECTIVE - III		
Sl. No.	Course Code	Subject Name
1	A6ME75	Renewable Energy Sources
2	A6ME80	Elements of Electric and Hybrid Vehicles

BASICS OF MECHANICAL ENGINEERING

III B. Tech. II Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-I)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME55	OEC	3	0	0	3	40	60	100

COURSE OVERVIEW:

This course deals with fundamental concepts of design, manufacturing and thermal engineering.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Demonstrate basic concepts of mechanics of rigid and deformable bodies.
2. Explain fundamentals of metallurgy and material science.
3. Discuss basic concepts and principles of thermodynamics.
4. Demonstrate the elements of fabrication processes and equipment.
5. Explain basic concepts of CAD/CAM.

UNIT-I MECHANICS

MECHANICS OF RIGID BODIES: Force, Force system, Classification of force systems, triangle law, parallelogram law – derivation and problems, polygon law, resolution of forces in coplanar concurrent force system – problems; Definition of friction, laws of friction, coefficient of friction, types of friction, angle of friction, angle of repose, Definition of centroid and centre of gravity, problems on centroid of plane areas; Definition of area moment of inertia, parallel axis theorem, perpendicular axis theorem, simple problems on area moment of inertia.

MECHANICS OF DEFORMABLE BODIES: Stress, Strain, Types of stresses and strains, Poisson's ratio, stress-strain curve of mild steel, Hooke's Law, factor of safety, mechanical properties of materials, bulk modulus, shear modulus, bars of uniform and varying sections-simple problems.

UNIT-II METALLURGY & MATERIAL SCIENCE

Bonds in Solids, crystallization of metals, Classification of steels, Necessity of alloying, types of solid solutions, lever rule, phase rule, cooling curve of pure iron, Iron-Iron carbide equilibrium diagram.

Heat treatment – Annealing, quenching, normalizing, Hardening, tempering; Ceramic materials-classification, properties and applications.

UNIT-III THERMO DYNAMICS

System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic view points, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry.

First law of Thermodynamics – Corollaries, First law applied to a Closed System, applied to a flow system, Steady Flow Energy Equation and Limitations of the First Law, Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements, Carnot cycle and its specialties.

UNIT-IV MANUFACTURING

Casting-Types of casting, advantage of casting and its applications, defects in castings; Patterns - Pattern making, Types, Materials used for patterns, pattern allowances; Welding-classification, Heat affected zone in welding, welding defects-causes and remedies.

Basic extrusion process and its characteristics, Forging operations and principles; Basic machining operations-drilling, milling, turning, step turning, threading, grinding and shaping; Limits, fits and

tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system.

UNIT-V CAD/CAM

Fundamentals of CAD/CAM – Benefits of CAD, Computer configuration for CAD applications, Computer peripherals for CAD; Wire frame modeling: Definition, advantages, disadvantages, wire frame entities – analytic entities and synthetic entities.

Surface modeling: Definition, advantages, disadvantages, surface entities – analytic entities and synthetic entities; Solid Modeling: Definition, constructive solid geometry, advantages, modeling entities.

Text Books:

1. K. Vijay Kumar Reddy and J. Suresh Kumar, Singer's Engineering Mechanics, B.S Publications
2. Avner, Introduction to physical metallurgy, Tata McGraw Hill Edition.
3. P.K Nag, Thermodynamics, McGraw Hill Education.
4. R.K Jain, Production Technology, Khanna Publishers.
5. Ibrahim Zeid & R. Sivasubramanian, CAD/CAM Theory & Practice, McGraw Hill Education.

Reference Books:

1. R.K. Bansal, Engineering Mechanics, Laxmi Publications
2. O.P. Khanna, Material science and metallurgy, Dhanpat Ray Publications
3. Pakirappa, V.Naveen Kumar, V.Naresh, Engineering Thermodynamics, Durga Publishing House
4. P.C Sharma, Production Technology, S. CHAND.
5. P. Radha Krishnan, S. Subramanyan and V. Raju, CAD/CAM, New Age International Publishers.
6. Sadhu Singh, Strength of Materials, Khanna Publishers.

FUNDAMENTALS OF ENGINEERING MATERIALS

III B. Tech. II Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-I)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME60	OEC	3	0	0	3	40	60	100

COURSE OVERVIEW:

This course will enable students to understand basic structure and crystal arrangement of materials, the phase diagrams, advantages of heat treatment, various heat treatment processes, the need and application of composite materials.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Explain basic concepts of crystal structure such as unit cells, crystal systems of metals etc.
2. Demonstrate the concept of alloying and formation of different types of phases in alloys.
3. Differentiate ferrous and non-ferrous alloys.
4. Explain various heat treatment processes.
5. Classify and explain polymers, ceramics and composites.

UNIT-I CRYSTAL STRUCTURE

Unit cells, crystal systems of metals, Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, determination of grain size, effect of grain size on the properties of alloys.

UNIT-II ALLOYS & PHASE DIAGRAMS

Alloys- substitutional and interstitial solid solutions.

Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructure of ledeburite, austenite, ferrite and cementite.

UNIT-III FERROUS AND NON FERROUS ALLOYS

Alloying of steel, properties of stainless steel and tool steels, maraging steels; cast irons-grey, white, malleable and spheroidal cast irons.

Copper and copper alloys- brass, bronze and cupro-nickel; Aluminium and Aluminium alloys.

UNIT-IV HEAT TREATMENT OF STEEL

Annealing, tempering, normalizing and spheroidising, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.

UNIT-V POLYMERS, CERAMICS AND COMPOSITES

Classification, properties and applications of polymers, ceramics, composites and nano materials.

Text Books:

1. V. Raghavan, "Material Science and Engineering", Prentice Hall of India Private Limited, 1999.
2. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011.
3. Sidney H. Avener, Introduction to Physical Metallurgy, 2nd edition, Tata McGraw hill education

(P) Ltd, New Delhi, India, 2007.

Reference Books:

1. W. D. Callister, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley India, 2006.
2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.
3. V. D. Kodgire, Material Science and Metallurgy for engineers, 1st Edition, Everest, Pune, India, 2006.

FUNDAMENTALS OF OPERATIONS RESEARCH

IV B. Tech. I Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-II)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME65	OEC	3	0	0	3	40	60	100

COURSE OVERVIEW:

This course aims to introduce the basic concepts of Operational Research such as linear programming, transportation and assignment problem, sequencing, game theory and inventory to the students.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Describe types of models and solve linear programming problem.
2. Solve transportation and assignment problems.
3. Analyze sequencing and replacement models and apply them for optimization.
4. Apply gaming theory for optimal decision making.
5. Analyze inventory models to optimize the cost

UNIT-I INTRODUCTION & ALLOCATION

Development – Definition – Characteristics and Phases – Types of models – Operations Research models – applications.

ALLOCATION: Linear Programming Problem – Formulation – Graphical solution – Simplex method – Artificial variables techniques: Two-phase method, Big-M method; Duality Principle.

UNIT-II TRANSPORTATION & ASSIGNMENT PROBLEMS

TRANSPORTATION PROBLEM: Formulation – Optimal solution, unbalanced transportation problem – Degeneracy.

ASSIGNMENT PROBLEM: Formulation – Optimal solution – Variants of Assignment Problem; Traveling Salesman problem.

UNIT-III SEQUENCING & REPLACEMENT

SEQUENCING: Introduction – Flow –Shop sequencing – n jobs through two machines – n jobs through three machines.

REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely.

UNIT-IV THEORY OF GAMES

THEORY OF GAMES: Introduction –Terminology – Solution of games with saddle points and without saddle points – 2×2 games – $m \times 2$ & $2 \times n$ games – graphical method – $m \times n$ games - dominance principle.

UNIT-V INVENTORY

INVENTORY: Introduction – Single item, Deterministic models – Types – Purchase inventory models with one price break and multiple price breaks – Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.

Text Books:

1. J.K. Sharma, Operation Research: Theory and Applications, 5th Edition, MacMillan India Ltd., 2012.
2. A.C.S. Kumar, Operations Research: Quantitative Analysis for Business Decisions, 1st Edition,

Yes Dee Publications, 2015.

Reference Books:

1. A.M. Natarajan, P. Balasubramaniam, A. Tamilarasi, Operations Research, 2nd Edition, Pearson Education India, 2014.
2. Harvey M. Wagner, Principles of Operations Research, 2nd Edition, Prentice Hall India Learning Pvt. Ltd., 1980.
3. Frederick S. Hillier, Gerald J. Lieberman, Bodhibrata Nag, Preetam Basu, Introduction to Operations Research, 11th Edition, Tata McGraw Hill India, 2021.

INTRODUCTION TO INDUSTRIAL ENGINEERING

IV B. Tech. I Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-II)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME70	OEC	3	0	0	3	40	60	100

COURSE OVERVIEW:

This course focuses on production systems that produce goods or provide services for customers. Industrial engineers define, design, build, operate and improve production processes that convert resources to high quality products or services effectively, efficiently and safely.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Describe the tools and concepts of industrial engineering principles.
2. Differentiate various types of plant layouts and maintenance.
3. Apply the principles of method study and time study to analyze and optimize work processes for increased effectiveness.
4. Apply Statistical Methods for Quality Control and explain Total Quality Management principles.
5. Use the techniques like human resource management, industrial relations, wage incentive plans, Value engineering for the effective enterprise resource planning.

UNIT-I INTRODUCTION

Definition of industrial engineering (I.E), development, applications, role of an industrial engineer, differences between production management and industrial engineering, quantitative tools of IE and productivity measurement. concepts of management, importance, functions of management, scientific management, Taylor's principles, theory X and theory Y, Fayol's principles of management.

UNIT-II PLANT LAYOUT

Factors governing plant location, types of production layouts, advantages and disadvantages of process layout and product layout, applications, quantitative techniques for optimal design of layouts, plant maintenance, preventive and breakdown maintenance.

UNIT-III WORK STUDY

Importance, types of production, applications, workstudy, method study and time study, work sampling, PMTS.

Micro-motion study, rating techniques, MTM, work factor system, principles of Ergonomics, flow process charts, string diagrams and Therbligs.

UNIT-IV STATISTICAL QUALITY CONTROL

Quality control, Queuing assurance and its importance, SQC, attribute sampling inspection with single and double sampling, Control charts – X and R – charts X and S charts and their applications, numerical examples.

TOTAL QUALITY MANAGEMENT: zero defect concept, quality circles, implementation, applications, ISO quality systems. six sigma – definition, basic concepts.

UNIT-V RESOURCE MANAGEMENT

Concept of human resource management, personnel management and industrial relations, functions of personnel management, Job-evaluation, its importance and types, merit rating, quantitative methods, wage incentive plans, types.

VALUE ANALYSIS: Value engineering, implementation procedure, enterprise resource planning and supply chain management.

Text Books:

1. O.P Khanna, Industrial Engineering and management, 2nd Edition, Dhanpat Rai publications, 2018.
2. Martand Telsang, Industrial Engineering and Production Management, 3rd Edition, S.Chand & Company Ltd. New Delhi, 2013.

Reference Books:

1. Bhattacharya D. K., Industrial Management, 2nd Edition, Vikas publishers, 2011.
2. J.G Monks, Operations Management, 3rd Edition, McGraw Hill Publishers, 2007.
3. T.R. Banga, S.C. Sharma, N. K. Agarwal, Industrial Engineering and Management Science, 12th Edition, Khanna Publishers, 2008.

RENEWABLE ENERGY SOURCES

IV B. Tech. II Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-III)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME75	OEC	3	0	0	3	40	60	100

COURSE OVERVIEW:

The energy has become an important and one of the basic infrastructures for the economic development of the country. It is imperative for the sustained growth of the economy. This course envisages the new and renewable source of energy, available in nature and to expose the students on sources of energy crisis and the alternates available, also stress up on the application of non-conventional energy technologies.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Demonstrate various energy sources and their availability.
2. Demonstrate the knowledge of solar radiation and its applications.
3. Analyse the performance of wind energy conversion systems.
4. Explain Biomass conversion techniques and Geothermal sources and resources
5. Explain various methods for harvesting the ocean energy sources.

UNIT-I ENERGY SOURCES & THEIR AVAILABILITY

Energy Sources and their Availability - Importance of Renewable Energy Resources – Present energy scenario – World energy status – Energy scenario in India – Classification of RES. Introduction to Solar, Wind, Geothermal, Bio-mass, Ocean Energy Sources, Comparison of these Energy Sources, Advantages and Limitations of RES.

UNIT-II INTRODUCTION TO SOLAR RADIATION & ITS APPLICATIONS

Introduction to Solar radiation - beam and diffuse radiation, solar constant, measurement of solar radiation. Solar Energy Storage – Collectors: Flat plate and Concentrating collectors, Focusing and non-focusing type.

Applications of Solar Energy: Solar water heater – Solar dryer – Solar distillation – Solar Photo-Voltaic Cell.

UNIT-III PRINCIPLE OF WIND ENERGY CONVERSION

Principle of Wind Energy Conversion – Basic components of Wind Energy Conversion Systems, Wind mill Components, Various Types and their Constructional Features.

Effect of Density, Frequency Variances, Angle of attack, and Wind Speed – Design Considerations of Horizontal and Vertical Axis Wind Machines.

UNIT-IV BIOMASS CONVERSION TECHNIQUES, GEOTHERMAL SOURCES AND RESOURCES

Biomass conversion techniques – Biogas Generation – Types of biogas plants – Advantages and disadvantages of biogas plants – Types of Bio-digester – Advantages and disadvantages.

Geothermal Sources like hydrothermal, geo-pressured hot dry rock, magma – Simple geothermal Power plant.

UNIT-V TIDAL ENERGY & OCEAN THERMAL ENERGY CONVERSION

Tidal Energy – Principle of working, performance and limitations. Wave Energy – Principle of working. Ocean Thermal Energy Conversion – OTEC power plants – working principle and Limitations.

Text Books:

1. G. D. Rai, Non-Conventional Energy Sources, 4th Edition, Khanna Publishers, 2000.
2. B H Khan, Non-Conventional Energy Resources, 2nd Edition, Tata McGraw Hill Education Pvt. Ltd, 2011

Reference Books:

1. S. Hasan Saeed and D.K. Sharma, Non-Conventional Energy Resources, 3rd Edition, S.K. Kataria & Sons, 2012.
2. N.K. Bansal, M. Kleemann, M. Heliss, Renewable energy sources and conversion technology, Tata McGraw Hill, 1990.
3. S. P. Sukhatme, Solar Energy Principles and Application, Tata McGraw Hill Education Pvt. Ltd, 2009.

ELEMENTS OF ELECTRIC AND HYBRID VEHICLES

IV B. Tech. II Semester: MECHANICAL ENGINEERING (OPEN ELECTIVE-III)

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME80	OEC	3	0	0	3	40	60	100

COURSE OVERVIEW:

This course introduces the fundamental concepts, principles, analysis, design, and control systems of electric and hybrid vehicles.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Analyze the impact of conventional vehicles on the society and different types of drive train topologies
2. Evaluate the performance and energy consumption in electric vehicles
3. Assess the configuration and architecture in hybrid vehicles
4. Choose the type of motor used in electric and hybrid electric vehicles
5. Design the drive train suitable for electric and hybrid vehicles

UNIT-I INTRODUCTION TO HYBRID ELECTRIC VEHICLES

History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies

UNIT-II ELECTRIC VEHICLES

Configurations of EVs, Performance of EVs, Tractive Effort in Normal Driving, Energy Consumption.

UNIT-III HYBRID ELECTRIC VEHICLES

Concept of Hybrid Electric Drive Trains.
Architectures of Hybrid Electric Drive Trains.

UNIT-IV ELECTRIC PROPULSION SYSTEMS

Introduction, DC Motor Drives, Induction Motor Drives, Permanent Magnetic BLDC Motor Drives, SRM Drives

UNIT-V DESIGN PRINCIPLE OF SERIES AND PARALLEL HYBRID ELECTRIC DRIVE TRAIN

Design Principles of a Series (Electrical Coupling) Hybrid Drive Train: Operation Patterns, Control Strategies, Design Example.

Design Principles of Parallel (Mechanically Coupled) Hybrid Electric Drive Train: Drive Train Configuration and Design Objectives, Control Strategies, Parametric Design of a Drive Train.

Text Books:

1. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 2004.
2. Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.

Reference Books:

1. James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003.
2. Allen Fuhs, Hybrid Vehicles and the future of personal transportation, CRC Press, 2011.

**OPEN ELECTIVES
OFFERED BY
HSM**

OPEN ELECTIVE COURSE- I			
S. No.	Course Code	Course Name	Offering Department
1	A6HS08	Business Economics and Financial Analysis	HSM
2	A6HS09	Basics of Entrepreneurship	HSM
OPEN ELECTIVE COURSE- II			
3	A6HS11	Advanced Entrepreneurship	HSM
OPEN ELECTIVE COURSE- III			
4	A6HS12	Management Science	HSM

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

VI Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6HS08	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.

COURSE OUTCOMES:

At the end of the course, the student will

1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures.
2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis.
3. Develop an understanding of Analyse how capital budgeting decisions are carried out.
4. Understanding the framework for both manual and computerised accounting process
5. Know how to analyse and interpret the financial statements through ratio analysis.

UNIT-I

INTRODUCTION & DEMAND ANALYSIS

Classes: 12

Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.

UNIT-II

PRODUCTION & COST ANALYSIS

Classes: 12

Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance.

UNIT-III

MARKETS & NEW ECONOMIC ENVIRONMENT

Classes: 12

Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.

UNIT-IV

CAPITAL BUDGETING

Classes: 12

Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).

UNIT-V INTRODUCTION TO FINANCIAL ACCOUNTING & FINANCIAL ANALYSIS Classes: 12

Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, Trial Balance - Final Accounts (Tracing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.

Text Books:

1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009.
2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013.
3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.

Reference Books:

1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012.
2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012.
3. Lipsey & Chrystel, Economics, Oxford University Press, 2012.
4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012.
5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012.
6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012.
7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012.
8. Dwivedi: Managerial Economics, Vikas, 2012.
9. Shailaja & Usha: MEFA, University Press, 2012.
10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012.
11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011.
12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.

BASICS OF ENTREPRENEURSHIP

VI Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6HS09	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

The curriculum helps students

1. Understand and **discover** entrepreneurship
2. Build a strong foundation for students to **start, build, and grow** a viable and sustainable venture
3. Develop an entrepreneurial outlook and **mindset**, critical **skills** and **knowledge**
4. Mitigate three types of **risks**: Customer, Business Model, and Product/Technical

COURSE OUTCOMES:

1. Entrepreneurship and Innovation minors will be able to sell themselves and their ideas. Students master oral and visual presentation skills and establish a foundation of confidence in the skills necessary to cause others to act.
2. The ability to imagine, recognize, and seize opportunities for innovation and new venture creation in the arts and culture sector
3. The know-how to bring capital to ideas in both commercial and nonprofit domains.

UNIT-I DISCOVER YOURSELF AND IDENTIFY PROBLEMS WORTH SOLVING

Classes: 10

Discover Yourself: Find your flow, Effectuation, Case Study: Tristan Walker: The extroverted introvert, Identify your entrepreneurial style,
Identify Problems Worth Solving: What is a business opportunity and how to identify it, Find problems around you that are worth solving, Methods for finding and understanding problems - (Observation, Questioning, DT, Jobs to be done (JTBD), How to run problem interviews to understand the customer's worldview, Introduction to Design Thinking – Process and Examples,

Generate ideas that are potential solutions to the problem identified – DISRUPT, Class Presentation: Present the problem you "love"

UNIT-II CUSTOMER, BUSINESS MODEL, VALIDATION

Classes: 10

CUSTOMER :Identify Your Customer Segments and Early Adopters - The difference between a consumer and a customer (decision maker), Market Types, Segmentation and Targeting, Defining the personas; Understanding Early Adopters and Customer Adoption Patterns, Identify the innovators and early adopters for your startup; Craft Your Value Proposition - Come up with creative solutions for the identified problems, Deep dive into Gains, Pains and "Jobs-To- Be-Done" (using Value Proposition Canvas, or VPC), Identify the UVP of your solution using the Value Proposition section of the VPC, Outcome-Driven Innovation, Class Presentation: Communicating the Value Proposition- 1 min Customer Pitch .

BUSINESS MODEL: Get Started with Lean Canvas.

VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.

UNIT-III**MONEY AND TEAM****Classes: 12**

MONEY: Revenue Streams - Basics of how companies make money, Understand income, costs, gross and net margins, Identify primary and secondary revenue, streams ; Pricing and Costs - Value, price, and costs; Different pricing, Understand product costs and operations costs; Basics of unit costing strategies; Financing Your New Venture - How to finance business ideas, Various sources of funds available to an entrepreneur and pros and cons of each, What investors expect from you, Practice Pitching to Investors and Corporates.

TEAM: Team Building - Shared Leadership, Role of a good team in a venture's success; What to look for in a team; How do you ensure there is a good fit? Defining clear roles and responsibilities, How to pitch to candidates to join your startup, Explore collaboration tools and techniques - Brainstorming, Mind mapping, Kanban Board, Slack.

UNIT-IV**MARKETING & SALES****Classes: 10**

MARKETING & SALES: Positioning - Understand the difference between product and brand and the link between them, Define the positioning statement for your product/service and how it should translate into what your customers should see about that brand in the market place.

Channels & Strategy : • Building Digital Presence and leveraging Social media, Creating your company profile page, Measuring the effectiveness of selected channels, Budgeting and planning.

UNIT-V**PLANNING & TRACKING****Classes: 08**

Sales Planning : Understanding why customers buy and how buying decisions are made; Listening, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up and closing a sale; Asking for the sale

Planning & Tracking: Importance of project management to launch and track progress, Understanding time management, workflow, and delegation of tasks.

Business Regulation : Basics of business regulations of starting and operating a business; Importance of being compliant and keeping proper documentation, How to find help to get started.

Text Books:

1. S. R. Bhowmik & M. Bhowmik, Entrepreneurship, New Age International, 2007.
2. Steven Fisher, Ja-nae' Duane, The Startup Equation -A Visual Guidebook for Building Your Startup, Indian Edition, Mc Graw Hill Education India Pvt. Ltd, 2016.
3. D F Kuratko and T V Rao —Entrepreneurship-A South-Asian Perspective —Cengage Learning, 2012. (For PPT, Case Solutions Faculty may visit : login.cengage.com)

Reference Books:

1. Vasant Desai —Small Scale industries and entrepreneurship|| Himalaya publishing 2012.
2. Rajeev Roy —Entrepreneurship|| 2e, Oxford, 2012.
3. B.Janakiramand M.Rizwan|| Entrepreneurship Development: Text & Cases, Excel Books, 2011.
4. Stuart Read, Effectual Entrepreneurship, Routledge, 2013.
5. Nandan H, Fundamentals of Entrepreneurship, PHI, 2013

Web References:

1. <http://freevideolectures.com/Course/3641/Entrepreneurship-Through-the-Lens-of-Venture-Capital>
2. <http://www.onlinevideolecture.com/?course=mba-programs&subject=entrepreneurship>
3. http://nptel.ac.in/courses/122106032/Pdf/7_4.pdf
4. <https://www.scribd.com/doc/21516826/Entrepreneurship-Notes>
5. <http://freevideolectures.com/Course/3514/Economics-/Management-/Entrepreneurship/50>
6. Journal of Entrepreneurship & Organization Management, Vikalpa, IIMA, IIMB Review, Decision, IIMC, Vision, HBR.

ADVANCED ENTREPRENEURSHIP

VII Semester: OPEN ELECTIVE-II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6HS11	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

The curriculum helps students

1. Understand and **discover** entrepreneurship
2. Build a strong foundation for students to **start, build, and grow** a viable and sustainable venture
3. Develop an entrepreneurial outlook and **mindset**, critical **skills** and **knowledge**
4. Mitigate three types of **risks**: Customer, Business Model, and Product/Technical

COURSE OUTCOMES:

1. The knowledge and mastery of organizational and management techniques in the arts, including leadership, organizational design, human resources, legal issues, and finance
2. The knowledge of the legal and ethical environment impacting business organizations and exhibit an understanding and appreciation of the ethical implications of decisions.
3. An ability to engage in critical thinking by analyzing situations and constructing and selecting viable solutions to solve problems.

UNIT-I

ORIENTATION TO GROWTH

Classes: 10

ORIENTATION TO GROWTH: Getting Ready for Growth-Why growth stage is different compared to startup phase, Why Product-Market fit is not enough, Case study, To assess readiness for growth, To chart a growth path.

UNIT-II

CUSTOMER, BUSINESS MODEL, VALIDATION

Classes: 10

CUSTOMERS :Expanding Customer Base-Revisit your business model and develop few variants (more business model types),Identify additional customer segments that your solution can address, Evaluate business models for the new customer segments, Relook at the Problem Statement (can you expand the scope and scalability of your business by repositioning your problem statement?), Explore additional ways to monetize.

BUSINESS MODEL: Get Started with Lean Canvas.

VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.

UNIT-III

TRACTION AND TEAM

Classes: 12

TRACTION : Scaling - How to gain traction beyond early customers, Defining traction (in quantifiable terms) and identifying the most important metrics to measure traction, Calculate cost of new customer acquisition, Estimate your customer lifetime value(LTV),Identifying waste in your operations and focusing your team on what is important for traction.

Channels and Strategy - The Bullseye framework, Identify Channels using Bulls Eye Frame work, Measuring the effectiveness of selected channels, Budgeting and planning.

UNIT-IV

MONEY & SALES

Classes: 10

MONEY : Growing Revenues - Stabilizing key revenue streams, Developing additional revenue streams(licensing, franchising),Exploring new channels and partnerships, Sales Planning - Understanding why customers buy and how buying decisions are made; Listening skills, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up and closing a sale; Asking for the sale.

Strengthening Sales - Follow-up and closing a sale; Asking for the sale, Building a professional sales team, Sales compensation and incentives, Sales planning, setting targets.

Improving Margins - Testing price elasticity, Optimizing costs and operational expenses,

Advanced concepts of unit costing. Financial Modeling - Financial modeling of your venture's growth, Analyzing competitor and peer's financial models.

UNIT-V

SUPPORT

Classes: 08

SUPPORT: Legal - Overview of legal issues and their impact on entrepreneurs, Importance of getting professional help (legal and accounting), Importance of being compliant and keeping proper documentation, Patents and Intellectual property, Trademarks.

Mentors, Advisors, and Experts - The importance of a Mentor and how to find one, Role of business advisors and experts for specific targets in your growth plan

Text Books:

1. Bygrave, W., & Zacharakis, A. (2017) Entrepreneurship, 4th Edition (3rd Edition is ok too) Wiley.
2. [Carree, M. A., Thurik, A. R. "The impact of entrepreneurship on economic growth" In: Audretsch, D. B., Acs, Z. J. \(eds\). Handbook of Entrepreneurship Research. Berlin: Springer Verlag, 2010.](#)

Reference Books:

1. Eric, Reis (2017) The Startup Way: How Entrepreneurial Management Transforms Culture and Drives Growth
2. Thurik, A. R., Audretsch, D. B., Stam, E. "The rise of the entrepreneurial economy and the future of dynamic capitalism" Technovation
3. [Audretsch, D. B., Grilo, I., Thurik, A. R. \(eds\). The Handbook of Research on Entrepreneurship Policy. Cheltenham, UK: Edward Elgar, 2007.](#)

Web References:

1. <https://www.ediindia.org/EDIILibrary.aspx>
2. <https://www.entrepreneurshipsecret.com/sources-of-information-for-entrepreneurship-development-contd>

MANAGEMENT SCIENCE

VIII Semester: OPEN ELECTIVE-III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6HS12	OEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. Familiarize & obtain Knowledge with the process of management and to provide basic insights into management practices.
2. Understand the structure & Designing of an Organization.
3. Knowledge on the aspects of Production.
4. Analyse the market and the strategies involved in Marketing.
5. Knowledge on concepts related to Human Resources.
6. Understand the techniques used in Project Management.
7. Familiarize with strategies used for analysis of an Organization.
8. Understand the Contemporary Management Issues.
9. Familiarize with the management skills which can be applied in the Organizational context to achieve Organizational goals.

COURSE OUTCOMES:

1. Knowledge on management theories and practices.
2. Understanding designing organizational structure.
3. Understanding on the methods & charts used in operations management.
4. Ability to understand the market and its environment.
5. Understand the processes, functions etc in Human Resources Management.
6. Ability to solve problems in managing the Project.
7. Knowledge on Strategic alternatives.
8. Familiar with the practices implemented in management.
9. Understand the social responsibilities of Management.
10. Understand the basic concepts of Management.

UNIT-I INTRODUCTION TO MANAGEMENT AND ORGANIZATION

Classes: 12

Concepts of Management and organization- nature, importance and Functions of Management, Systems Approach to Management-Taylor's Scientific Management Theory- Fayal's Principles of Management-Maslow's theory of Hierarchy of Human Needs- Douglas McGregor's Theory X and Theory Y - Hertzberg Two Factor Theory of Motivation - Leadership Styles, Social responsibilities of Management, Designing Organizational Structures: Basic concepts related to Organization - Departmentation and Decentralization, Types and Evaluation of mechanistic and organic structures of organization and suitability.

UNIT-II OPERATIONS AND MARKETING MANAGEMENT

Classes: 12

Principles and Types of Plant Layout-Methods of Production(Job, batch and Mass Production), Work Study - Basic procedure involved in Method Study and Work Measurement - Business Process Reengineering(BPR) - Statistical Quality Control: control charts for Variables and Attributes (simple Problems) and Acceptance Sampling, TQM, Six Sigma, Deming's contribution to quality, Objectives of Inventory control, EOQ, ABC Analysis, Purchase Procedure, Stores Management and Store Records - JIT System, Supply Chain Management, Functions of Marketing, Marketing Mix, and Marketing Strategies based on Product Life Cycle, Channels of distribution.

UNIT-III HUMAN RESOURCES MANAGEMENT(HRM)

Classes: 12

Concepts of HRM, HRD and Personnel Management and Industrial Relations (PMIR), HRM vs PMIR, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating - Capability Maturity Model (CMM) Levels - Performance Management System.

UNIT-IV

PROJECT MANAGEMENT (PERT/ CPM)

Classes: 12

Network Analysis, Programme Evaluation and Review Technique (PERT), Critical Path Method (CPM), Identifying critical path, Probability of Completing the project within given time, Project Cost Analysis, Project Crashing (simple problems)

UNIT-V

STRATEGIC MANAGEMENT AND CONTEMPORARY STRATEGIC ISSUES

Classes: 12

Mission, Goals, Objectives, Policy, Strategy, Programmes, Elements of Corporate Planning Process, Environmental Scanning, Value Chain Analysis, SWOT Analysis, Steps in Strategy Formulation and Implementation, Generic Strategy alternatives. Bench Marking and Balanced Score Card as Contemporary Business Strategies.

TEXT BOOKS:

- 1.A.R.Aryasri : Management Science, TMH, (Latest edition)
- 2.Stoner, Freeman, Gilbert, Management, 6th Ed, Pearson Education, New Delhi, 2004.

REFERENCE BOOKS:

1. Kotler Philip and Keller Kevin Lane: Marketing Management, Pearson, 2012.
2. Koontz and Weihrich: Essentials of Management, McGraw Hill, 2012.
3. Thomas N. Duening and John M. Ivancevich Management - Principles and Guidelines, Biztantra, 2012.
4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2012.
5. Samuel C. Certo: Modern Management, 2012.
6. Schermerhorn, Capling, Poole and Wiesner: Management, Wiley, 2012.
7. Parnell: Strategic Management, Cengage, 2012.
8. Lawrence R Jauch, R. Gupta and William F. Glueck: Business Policy and Strategic Management Science, McGraw Hill, 2012.

Web References:

- 1.<https://www.smartworld.com/notes/management-science-pdf-notes-ms-notes-pdf>
2. <https://www.allintuworld.in/download/management-science-notes/>